



BSI Technical Guideline 03125 Preservation of Evidence of Cryptographically Signed Documents

Annex TR-ESOR-C.2:

Conformity Test Specification (Level 2 Technical Conformity)

Designation Technical Conformity Test Specification (Level 2)

Abbreviation BSI TR-ESOR-C.2

Version 1.3 update 2

(on base of the eIDAS-Regulation and ETSI Preservation Standards with a new certification scheme including a new version of C.2-Testbed with support for ETSI TS 119 512))

Date 06.05.2025



Document history

<i>Version</i>	<i>Date</i>	<i>Editor</i>	<i>Description</i>
1.3	31.03.2022	BSI	TR-ESOR-C.2
1.3 (update 1)	22.03.2023	BSI	First update of TR-ESOR-C.2 v. 1.3
1.3 (update 2)	06.05.2025	BSI	Third update of TR-ESOR-C.2 v.1.3 (TR-S.512)

Table 1: Document history

Federal Office for Information Security
P.O. Box 20 03 63
53133 Bonn
Tel.: +49 22899 9582- 0
E-Mail: tresor@bsi.bund.de
Internet: <https://www.bsi.bund.de>
© Federal Office for Information Security 2025

Table of Contents

1	Introduction.....	5
2	Overview	8
3	Web Service Interfaces	11
3.1	Test suite for TR-S.4-interface and additional TR-S.512-interface	11
3.1.1	Test environment	11
3.2	Standard Test Configurations	12
3.2.1	CONFIG_Common.....	12
3.2.2	CONFIG_ArchiSafe_C.2.....	13
3.2.3	CONFIG_LXAIP (conditional).....	13
3.2.4	CONFIG_S.4 (conditional).....	13
3.2.5	CONFIG_S.512 (conditional).....	14
3.3	Standard Test Objects	14
3.4	Mandatory Tests.....	26
3.4.1	Tests of the TR-S.4-interface	26
3.4.1.1	Function RetrieveInfo of S.4-interface	26
3.4.1.2	Function ArchiveSubmission of S.4-interface.....	30
3.4.1.3	Function ArchiveUpdate of S.4-interface.....	49
3.4.1.4	Function ArchiveRetrieval of S.4-interface	60
3.4.1.5	Function ArchiveEvidence of S.4-interface	66
3.4.1.6	Function ArchiveDeletion of S.4-interface	75
3.4.1.7	Function Verify of S.4-interface.....	81
3.4.1.8	Additional tests of S.4-interface	108
3.4.2	Tests of the TR-S.512-interface	119
3.4.2.1	Function RetrieveInfo of S.512-interface	119
3.4.2.2	Function PreservePO of S.512-interface	122
3.4.2.3	Function UpdatePOC of S.512-interface.....	142
3.4.2.4	Function RetrievePO of S.512-interface.....	155
3.4.2.5	Function DeletePO of S.512-interface.....	172
3.4.2.6	Function ValidateEvidence of S.512-interface	178
3.4.2.7	Additional tests of S.512-interface	203
4	TR-ESOR-C.2-Testbed how-to	215
4.1	TR-ESOR-C.2-Testbed Prerequisites.....	215
4.2	TR-ESOR-C.2-Testbed introduction.....	215
4.2.1	Subdirectory DOCKERS.....	216
4.2.2	Subdirectory ERVT-UPLOADS.....	216
4.2.3	Subdirectory LXAIP-UPLOADS.....	216

4.2.4	Subdirectory PROPERTIES	216
4.2.5	Subdirectory RUN-DATA	216
4.2.6	Subdirectory TESTDATA	216
4.2.7	Subdirectory XSVT-UPLOADS	217
4.2.8	File TR-ESOR-1.3.0-C2-Testbed-soapui-project.xml	217
4.2.9	File BSI TR-03125_C.2_V1.3.pdf	217
4.3	TR-ESOR-C.2-Testbed configuration.....	217
4.4	TR-ESOR-C.2-Testbed: configuration and start of XSVT and ERVT	219
4.5	TR-ESOR-C.2-Testbed usage	221
4.6	TR-ESOR-C.2-Testbed – final documentation	222
4.7	TR-ESOR-C.2-Testbed results	224

Table of Figures

Figure 1: Schematic Depiction of the IT Reference Architecture with TR-S.4	6
Figure 2: Schematic Depiction of the IT Reference Architecture with TR-S.512	6
Figure 3: Overview of the BSI TR-ESOR-C.2-Testbed environment (version 1.3 update 3)	12
Figure 4: TR-ESOR-C.2-Testbed – project directory structure.....	215
Figure 5: Example output of dbuild.cmd/dbuild.sh.	219
Figure 6: Final statement of initialization of XSVT.....	220
Figure 7: Final statement of initialization of ERVT.	220
Figure 8: Sample output of dstatus.cmd/dstatus.sh	221
Figure 9: Sample output of dstart.cmd/dstart.sh.	221

1 Introduction

The goal of the Technical Guideline “Preservation of Evidence of Cryptographically Signed Documents” is to specify technical security requirements for the long-term preservation of evidence of cryptographically signed electronic documents and data along with associated electronic administrative data (meta data).

A Middleware defined for this purpose (TR-ESOR-Middleware) in the sense of this Guideline includes all of the modules (M) and interfaces (S) [for the German “Schnittstellen”] used for securing and preserving the authenticity and proving the integrity of the stored documents and data.

The Reference Architecture introduced in the Main Document of this Technical Guideline consists of the interfaces, functions and logical units described in the following:

- the TR-S.4 and the TS119512 input interface TR-S.512 according to profile **[TR-ESOR-TRANS]** of the TR-ESOR-Middleware, which serves to embed the TR-ESOR-Middleware in the existing IT and infrastructure landscape;
- the “ArchiSafe-Module” (**[TR-ESOR-M.1]**), which regulates the flow of information in the Middleware, implements the security requirements for the interfaces with the IT applications and ensures that the application systems are decoupled from the ECM-/Long-term storage;
- the “Cryptographic”-Module (**[TR-ESOR-M.2]**) and the associated interfaces TR-S.1 and TR-S.3 that provide the required functions for creating hash values and verifying electronic signatures or seals or time-stamps, verifying electronic certificates, and for obtaining qualified electronic time stamps or (optional) electronic signatures or seals for the Middleware. Furthermore, it can provide the functions for the encryption and decryption of data and documents;
- the “ArchiSig”-Module (**[TR-ESOR-M.3]**) with the TR-S.6 interface that provides the required functions for the preservation of evidence of the digitally signed documents;
- an ECM-/Long-Term Storage with the TR-S.2 and TR-S.5 interfaces that assumes the physical archiving/storage and also the storage of the meta data that preserve evidence.

This ECM-/Long-Term Storage is no longer directly a part of the Technical Guideline, but requirements will be set for it through the two interfaces that are still part of the TR-ESOR-Middleware.

The application layer that can include an XML-Adapter is not a direct part of this Technical Guideline, either, even though this XML-Adapter can be implemented as part of a Middleware.

The IT Reference Architecture depicted in Figure 1 and Figure 2 is based on the ArchiSafe Reference Architecture and is supposed to enable and support the logical (functional) interoperability of future products with the objectives and requirements of the Technical Guideline. In principle, the upper interface of the TR-ESOR-Middleware is either the S.4-Interface (TR-S.4) pursuant to **[TR-ESOR-E]**, according to Figure 1, or the S.512-Interface (TR-S.512) pursuant to **[ETSI TS 119 512]** according to the profile **[TR-ESOR-TRANS]**, according to Figure 2.

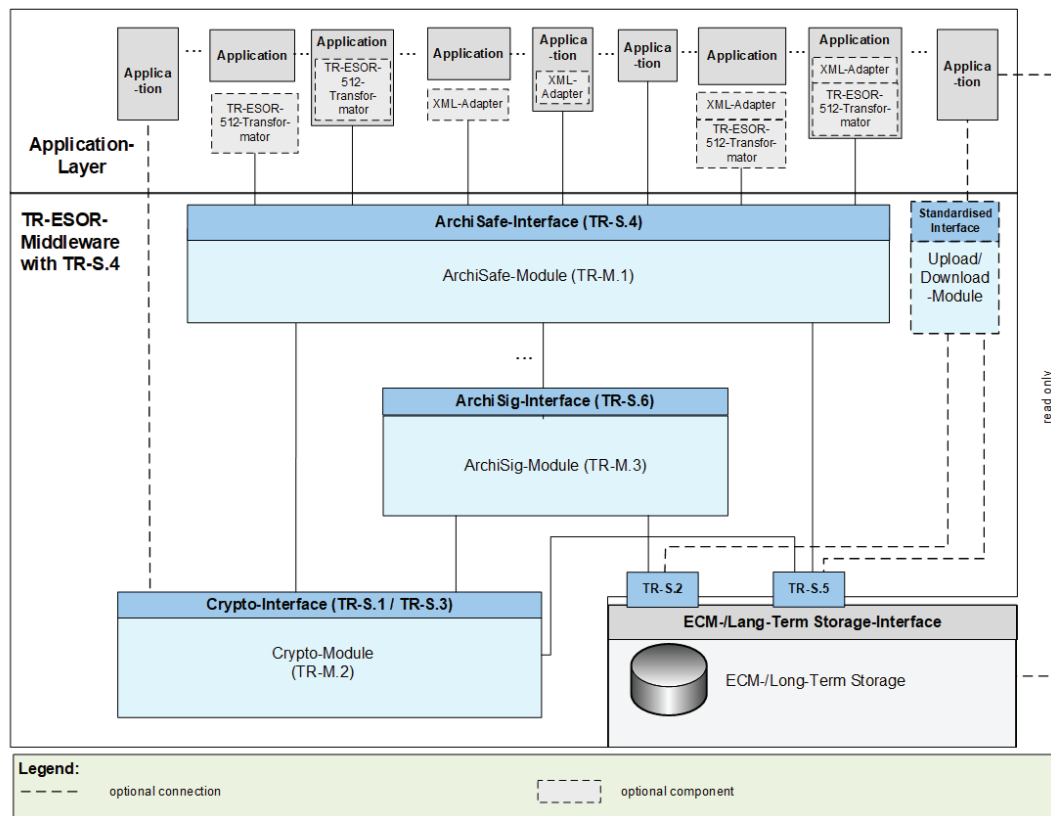


Figure 1: Schematic Depiction of the IT Reference Architecture with **TR-S.4**

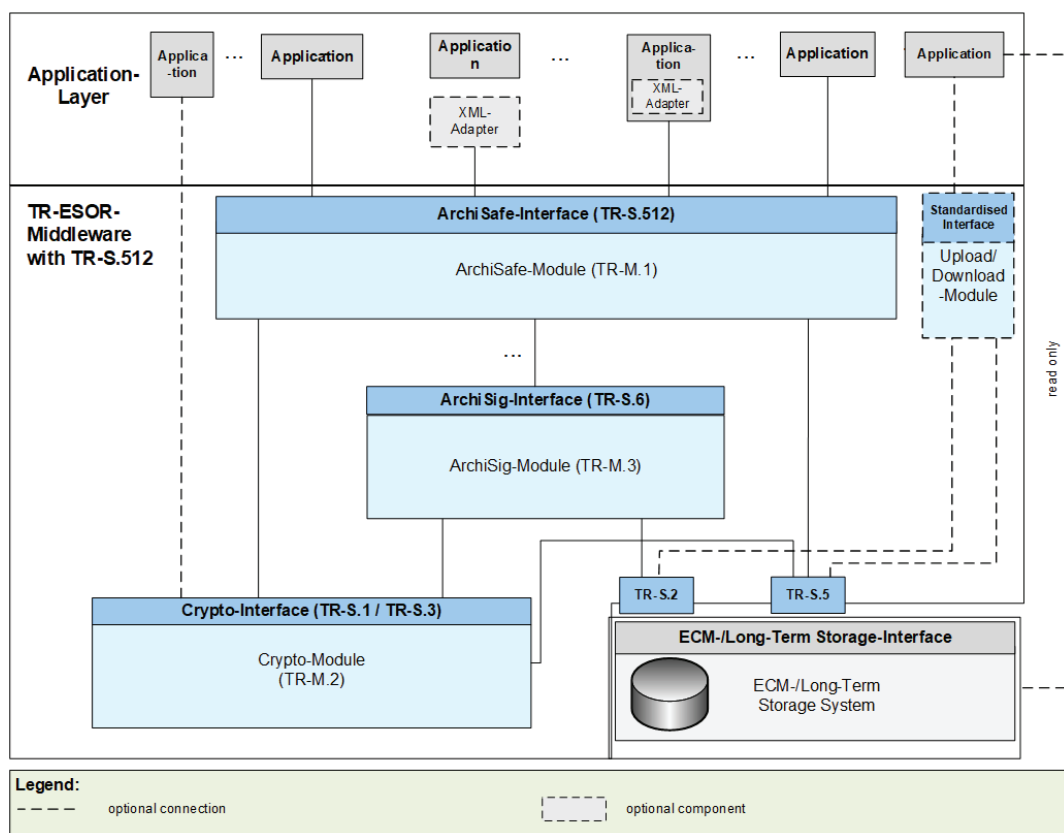


Figure 2: Schematic Depiction of the IT Reference Architecture with **TR-S.512**

The optional XML-Adapter and/or the optional TR-ESOR-512-Transformer¹ may be integrated under the following circumstances:

- Both as separate components with two interfaces to the IT-application and the ArchiSafe-Module,
- Both as separate components, but as parts of the IT-application with interfaces to the ArchiSafe-Module
- One common component, separate from the IT-application and including the XML-Adapter and TR-ESOR-512-Transformer as parts, with two interfaces to the IT-application and the ArchiSafe-Module,
- One common component including the XML-Adapter and TR-ESOR-512-Transformer as parts, which is part of the IT-application, with an interface to the ArchiSafe-Module.

The “ETSI TS119512 TR-ESOR Transformer” enables preservation services pursuant to [eIDAS-VO] to transform incoming ETSI TS119512 (V1.1.2) messages² into TR-S4 messages. These messages can be sent to an attached <http://www.bsi.bund.de/EN/tr-esor> system without any need for changing the system.

The usage of the “ETSI TS119512 TR-ESOR Transformer” is recommended in case that a TR-ESOR-Product with a TR-S.4-Interface should be used in Europe supporting interoperability with other European (qualified) Preservation Services or Preservation Products.

This Technical Guideline is modularly structured, and the individual annexes to the Main Document specify the functional and technological security requirements for the needed IT components and interfaces of the TR-ESOR-Middleware. The specifications are strictly platform, product and manufacturer independent.

The document at hand bears the designation “Annex TR-ESOR-C.2” and describes and specifies the conformity tests for the Conformity Level 2 (Technical Conformity) for TR-ESOR V1.3 supporting also the archive information package LXAIP, ASiC-AIP (not yet ready for certification), the Upload- and Download-interface and the upper ETSI TS 119 512-interface TR-S.512.

Products that want to be certified pursuant to the Technical Guideline 03125 TR-ESOR shall prove their conformity pursuant to this document [TR-ESOR-C.2] and [TR-ESOR-C.1].

¹ See [ETSI TS 119512 TR-ESOR Transformer under an Open Source Licence](#)

² In the profiling of [TR-ESOR-TRANS]

2 Overview

Products or systems, which want to become certified according to this Technical Guideline, have to demonstrate their conformance to the specifications. There are two defined conformance levels, which mainly differ in the technical detail specifications of interfaces and data formats used:

- Conformity Level 1 – Functional Conformity,
- Conformity Level 2 – Technical Conformity.

The functional test criteria (tests) for reaching “Conformity Level 1 – Functional Conformity” are described in [TR-ESOR-C.1].

This document specifies the technical test criteria (tests) for reaching Conformity Level 1 (Technical Conformity), which are derived from the requirements and web service interfaces specified in Annex [TR-ESOR-E], [TR-ESOR-F], [TR-ESOR-ERS] and [TR-ESOR-VR] and complement the mandatory (red) tests for interface functions specified in section 5.5 of Annex [TR-ESOR-C.1] of the present Technical Guideline.

The tests for Conformity Level 1 (Technical Conformity) aim at verifying technical interoperability between different components within the reference architecture as depicted in Figure 1 and Figure 2. For this purpose the present document specifies test cases for the different web service based interfaces (see section 3) and verification report structures for central data structures (see section [TR-ESOR-VR]).

Each test case is identified by a unique ID and specified in a semi-formal pseudo-code, which provides the basis for the technical implementation of the test cases in an appropriate test environment.

In general, the optional elements of the request – and response structures of the functional calls according Annex E are omitted. The usage or omission of these optional elements has no impact on the fulfilling on the conformance requirements of this specification.

The tests of the web service based interfaces can be performed using the freely available SoapUI³ test tool.

All tests, which are required to be passed are marked with the color **red**. Tests, which are only applicable in certain situations are marked **grey**. All tests use the same basic configuration, which is denoted by CONFIG_ArchiSafe_C.2 in the following. In case, the TOT does support some additional features (e.g. LXAIP pursuant to [TR-ESOR-F], chapter 3.2), corresponding additional configuration has to be taken into consideration (e.g. CONFIG_LXAIP).

In order to become certified according to Conformity Level 1 – Functional Conformity and Conformity Level 2 – Technical Conformity, a product or system has to pass

- all mandatory (red) conformity criteria (tests) for this Conformity Level 1 – Functional Conformity pursuant to [TR-ESOR-C.1] and
- all mandatory (red) conformity criteria (tests) for Conformity Level 2 – technical Conformity pursuant to this document, [TR-ESOR-C.2].

If one or more tests, marked with the color red, are not successful, the conformity cannot be certified.

Because [TR-ESOR-C.2] extends [TR-ESOR-C.1], it is necessary for the testing body to assess the assessment criteria of [TR-ESOR-C.1] beforehand.

NOTICE 1:

In TR-ESOR-Version1.3 the three terms „Archive Information Package (AIP)“, “Archive Information Container” and „Archive Data Object“ are used synonymously.

³ See <http://www.soapui.org> .

NOTICE 2:

The TR-ESOR input interface TR-S.4 or the TS119512 input interface TR-S.512 pursuant to the preservation-API of [ETSI TS 119 512] in the profiling of [TR-ESOR-TRANS] shall be used which logically corresponds to the input-interface TR-S.4 of the TR-ESOR-Middleware [TR-ESOR-E], as shown in the table 1 of [TR-ESOR-E], **clause 4.1**. Another input interface instead of TR-S.4 or TR-S.512 is not allowed. To improve readability, the insertion of the TS119512-function calls, equivalent to TR-S.4, is renounced, at several places in this document and the footnote "Either TR-S.4 or TR-S.512 shall be supported." is added to each test case concerning "TR-S.4" or "TR-S.512".

NOTICE 3:

In this TR-ESOR- Version, the word "AIP" means in all TR-ESOR-Annexes :

- a) the XML-based archival information package "XAIP" pursuant to [TR-ESOR-F], **clause 3.1** as well as
- b) the logical XAIP "LXAIP" pursuant to [TR-ESOR-F], **clause 3.2** as well as
- c) the "ASiC-AIP" pursuant to [TR-ESOR-F], **clause 3.3** on base of [ETSI EN 319162-1].

In general, this TR-ESOR Version differentiates in detail between "XAIP", "LXAIP" and "ASiC-AIP".

To improve readability, at several places in this document, the insertion of "LXAIP" and "ASiC-AIP" is renounced, and the footnote "'XAIP" shall be supported, "LXAIP" and "ASiC-AIP, if implemented" may be supported" is added to each "XAIP".

NOTICE 4:

In this TR-ESOR- Version, **BIN** is restricted to the following preservation object formats:

- ASiC-ERS (in TR-ESOR v1.3 called ASiC-AIP) pursuant to [ETSI TS 119 512], Annex A.3.1 and A.3.1.3 (<http://uri.etsi.org/ades/ASiC/type/ASiC-ERS>) and pursuant to [TR-ESOR-F], Clause 3.3);
- CAdES pursuant to [ETSI TS 119 512] Annex A.1.1 (<http://uri.etsi.org/ades/CAdES>). If there is no MIME type filled, then the default application/cms is used;
- XAdES pursuant to [ETSI TS 119 512] Annex A.1.2 (<http://uri.etsi.org/ades/XAdES>). If there is no MIME type filled, then the default application/xml is used;
- PAdES pursuant to [ETSI TS 119 512] Annex A.1.3 (<http://uri.etsi.org/ades/PAdES>). If there is no MIME Type filled, then the default application/pdf is used;
- ASiC-E pursuant to [ETSI TS 119 512] Annex A.1.4 (<http://uri.etsi.org/ades/ASiC/type/ASiC-E>). If there is no MIME type filled, then the default [application/vnd.etsi.asic-e+zip](http://uri.etsi.org/ades/ASiC/type/ASiC-E) is used;
- ASiC-S pursuant to [ETSI EN 319 162] (<http://uri.etsi.org/ades/ASiC/type/ASiC-S>). If there is no MIME type filled, then the default [application/vnd.etsi.asic-s+zip](http://uri.etsi.org/ades/ASiC/type/ASiC-S) is used;
- DigestList pursuant to [ETSI TS 119 512] Annex A.1.6 (<http://uri.etsi.org/19512/format/DigestList>). If there is no MIME Type filled, then the default application/xml is used.

In case of Upload- and Download-function the following format is allowed additionally:

- Binary Data (BIN) as Octet Stream, which is stored in the ECM-/Long-Term Storage by an "Upload-Request", – but only,
- if connected with a corresponding LXAIP and referenced there acc. to [TR-ESOR-F], Sec. 3.2,
- if appropriate, retrieved by a "Download-Request" – linked to a corresponding LXAIP or embedded in a XAIP and retrieved by an ArchiveRetrieval-Request – and
- no Upload of XAIP, nor LXAIP, nor ASiC-AIP is possible.

ATTENTION 1:

If implemented, an ASiC-AIP-Container pursuant to clause 3.3 in [TR-ESOR-F] shall be inserted in the TS119512-intefrace as a base64Binary-coded ASiC-AIP-Container with the Type=<http://uri.etsi.org/ades/ASiC/type/ASiC-ERS> Attribute (see <https://www.w3.org/TR/xmlschema-2/#base64Binary>).

In TR-ESOR V1.3, actually, ASiC-AIP is announced but it is still not released and does not lead to a certification.

NOTICE 5: In the following text the term “digital signature” covers “advanced electronic signatures” pursuant to **[eIDAS-VO], Article 3(11)**, “qualified electronic signatures” pursuant to **[eIDAS-VO], Article 3(12)**, “advanced electronic seals” pursuant to **[eIDAS-VO], Article 3(26)** and “qualified electronic seals” pursuant to **[eIDAS-VO], Article 3(27)**. Insofar, the term “digital signed document” covers as well documents signed by advanced electronic signatures or seals as documents signed by qualified electronic signatures or seals.

In this TR the term “**cryptographic signed documents**” covers not only qualified signed documents pursuant to **[eIDAS-VO], Article 3(12)** or qualified sealed documents pursuant to **[eIDAS-VO], Article 3(27)** or qualified time-stamped documents pursuant to **[eIDAS-VO], Article 3(34)** (within the meaning of the **eIDAS** regulation)) but also documents with advanced electronic signatures pursuant to **[eIDAS-VO], Article 3(11)** or with advanced electronic seals pursuant to **[eIDAS-VO], Article 3(26)** or with electronic time-stamps pursuant to **[eIDAS-VO], Article 3(33)**, as they are often used in the internal communication of public authorities. The documents with simple signatures or seals based on other (e.g. non-cryptographic) technologies are not meant here.

3 Web Service Interfaces

As shown in Figure 1 and Figure 2 the reference architecture comprises different interfaces, which can be implemented by a TR-ESOR-compliant middleware product. The present specification defines test suites for the interface TR-S.4 (see section 3.4.1) and the interface TR-S.512 (see section 3.4.2), which comprise test cases for different functions (see section 3.1) using the different standard test objects as defined in section 3.3.

3.1 Test suite for TR-S.4-interface and additional TR-S.512-interface

The test suite for TR-S.4 contains the test cases for the following functions pursued to [TR-ESOR-E], chapter 3:

- RetrieveInfo (see section 3.4.1.1)
- ArchiveSubmission (see section 3.4.1.2)
- ArchiveUpdate (see section 3.4.1.3)
- ArchiveRetrieval (see section 3.4.1.4)
- ArchiveEvidence (see section 3.4.1.5)
- ArchiveDeletion (see section 3.4.1.6)
- Verify (see section 3.4.1.7)
- Additional tests of S.4-interface (see section 3.4.1.8).

The test suite for additional TR-S.512-interface contains the test cases for the following functions pursued to [TR-ESOR-E], chapter 4:

- RetrieveInfo (see section 3.4.2.1)
- PreservePO (see section 3.4.2.2)
- Update POC (see section 3.4.2.3)
- RetrievePO (see section 3.4.2.4)
- DeletePO (see section 3.4.2.5)
- ValidateEvidence (see section 3.4.2.6)
- Additional tests of S.512-interface (see section 3.4.2.7)

3.1.1 Test environment

Requirement:

(A3.1.1-1) Following figure describes the structure of the BSI-TR-ESOR-C.2-Testbed Environment (also called TR-C.2-Testbed Environment) in the version 1.3 (update 3), which **shall** be used in order to perform the test cases defined in this document, incl. TR-ESOR-Testclient (incl. implementing test cases and provided test data), tr-esor-AIP-eIDAS-SigValidator (XSV) and BSI-ErVerifyTool (ERVTool).

The results of the execution of all test cases **shall** be documented and provided to certification body (BSI).

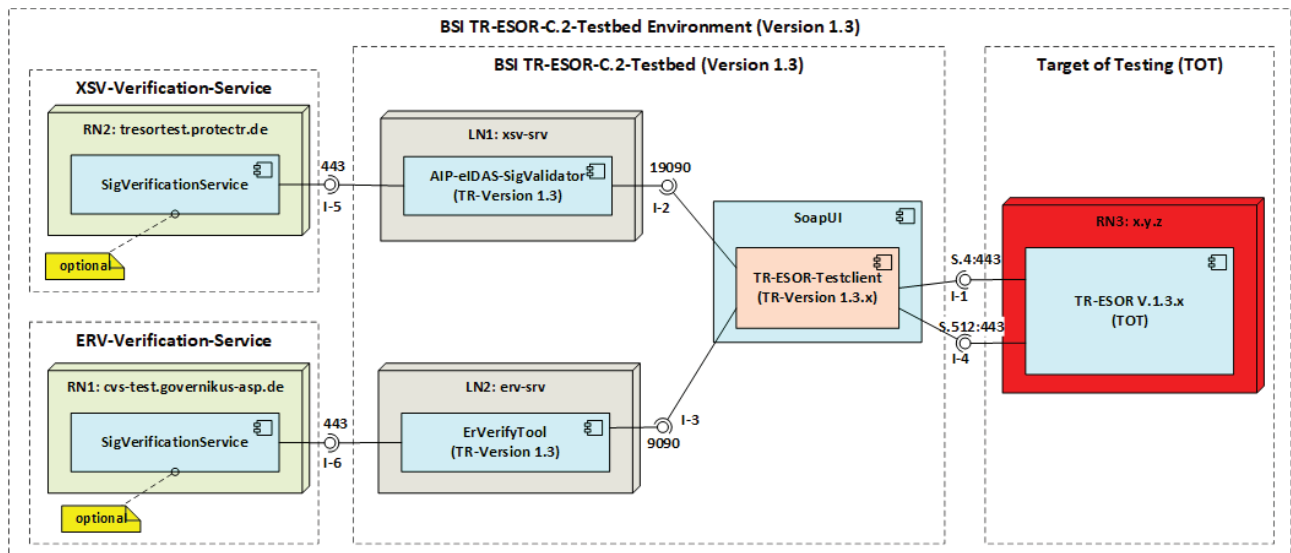


Figure 3: Overview of the BSI TR-ESOR-C.2-Testbed environment (version 1.3 update 3)

The TR-ESOR-C.2-Testbed environment defines four separated domains:

1. **Target of Testing (ToT)** – it is the TR-ESOR version 1.3 (update 3) complied product to be tested against this test specification,
2. **BSI TR-ESOR-C.2-Testbed** – covers the implementation of the test cases as well as the integration of the BSI-Testtools,
3. **XSV-Verification-Service** – a verification service used by the **tr-esor-AIP-eIDAS-SigValidator** (AIP-eIDAS-SigValidaor or simply XSV) in order to verify the supported signatures, seals and time-stamps (cf. <https://github.com/de-bund-bsi-tr-esor/tr-esor-AIP-eIDAS-SigValidator>),
4. **ERV-Verification-Service** – a verification service used by the BSI-ErVerifyTool (ERVerifyTool or simply ERV) in order to verify the archive time stamps obtained from provided evidence records (cf. <https://github.com/de-bund-bsi-tr-esor/ERVerifyTool>).

The optional parts have been marked by yellow comments boxes containing the text “optional”. The both verification tools XSVT and ERVT are able to work online, with activated optional verification service corresponding for the digital signatures/seals or timestamps, or offline. The verification services have to be integrated by the user itself, please refer to the corresponding documentation for more information, how to do that.

3.2 Standard Test Configurations

Here, a set of standard configurations of the test setup will be described. These setups are referenced in the test cases and should be used to actually perform the tests.

3.2.1 CONFIG_Common

This standard configuration shall apply for all tests.

- The test setup shall contain the product to be tested (Target of Testing, TOT).
- The user manual in the test setup shall refer at least to one preservation profile pursuant to [ETSI TS 119 512], clause 5.4.7, to one actual preservation evidence policy pursuant to [ETSI TS 119 511], clause 6.5 and to one actual signature validation policy pursuant to [ETSI TS 119 511], clause 6.6.
- The test setup shall contain all other modules of the reference architecture (including the storage) functionally not covered by the TOT.

- The TOT and all other modules required shall be installed and configured according to the respective guidance including all security recommendations.
- The TOT and all other modules shall be physically and logically interconnected. The connections shall be secured as described in the respective guidance documents (e.g. enabling encryption, explicit physical connection).
- The BSI-ERVerifyTool shall be installed and running in a proper working mode (see chapter 3.1.1).
- The BSI-AIP-eIDAS-SigValidator testing tool shall be installed and running in a proper working mode (see chapter 3.1.1).
- The TOT shall provide at least one active Protection Profile and at least one inactive Protection Profile.
- The TOT shall support Evidence Record pursuant to [RFC4998].

The complete test setup shall be up and running and in an operational and working mode.

3.2.2 CONFIG_ArchiSafe_C.2

This configuration is based on CONFIG_Common and introduces following additional points:

Additionally, the ArchiSafe-Module shall be configured as follows:

- XAIP shall be mandatory.
- The default value in case of “no-input of the PO-Format” shall be “XAIP”.
- A XSD defining the XAIPs or LXAIps pursuant to [TR-ESOR-F] shall be configured.
- TOT shall support either TR-S.4-interface or TR-S.512-interface.
- The validation of digital signatures/seals and electronic time stamps⁴ during ArchiveSubmission/PreservePO and ArchiveUpdate/UpdatePOC shall be enabled.

3.2.3 CONFIG_LXAIP (conditional)

This configuration is based on CONFIG_ArchiSafe_C.2.

If LXAIP is implemented and CONFIG_LXAIP is configurable, then TOT does fullfill following requirements in addition to level CONFIG_ArchiSafe_C.2:

- TOT shall support LXAIP-format pursuant to [TR-ESOR-F], chapter 3.2.
- TOT shall support the Upload/Download-module.
- TOT shall support the retrieval of submitted LXAIP as transformed equivalent XAIP.

3.2.4 CONFIG_S.4 (conditional)

This configuration is based on CONFIG_ArchiSafe_C.2.

If TR-S.512 is implemented and CONFIG_S.4 is configurable, then TOT shall fullfill following requirements in addition to level CONFIG_ArchiSafe_C.2:

- TOT shall support the TR-S.4-interface pursuant to [TR-ESOR-E], chapter 3.
- The TOT shall support the optional input parameter `pres:POFormat` while executing the function ArchiveRetrieval

⁴ The validation of digital signatures/seals or electronic time stamps of documents included in the XAIP/LXAIP, DXAIP/DLXAIP or passed over as binary.

- The XSD verification of XAIP/DXAIP containers during `ArchiveSubmission` and `ArchiveUpdate` shall be enabled.
- The XSD verification of LXAIP/DLXAIP containers during `ArchiveSubmission` and `ArchiveUpdate` shall be enabled. (conditional, if `CONFIG_LXAIP` applies)
- TOT shall support the following URI <http://www.bsi.bund.de/tr-esor/lxaip/1.3> as value of `pres:POFormat` as optional input of `ArchiveRetrievalRequest` input parameter of `ArchiveRetrieval` function (c.f. [TR-ESOR-E], chapter 3.3.2). (conditional, if `CONFIG_LXAIP` applies)

3.2.5 CONFIG_S.512 (conditional)

This configuration is based on `CONFIG_ArchiSafe_C.2`.

If TR-S.512 is implemented and `CONFIG_S.512` is configurable, then TOT shall fullfill following requirenments in addition to level `CONFIG_ArchiSafe_C.2`:

- TOT shall support the TR-S.512-interface pursued to [TR-ESOR-E], chapter 4.
- The XSD verification of XAIP/DXAIP containers during `PreservePO` and `UpdatePOC` shall be enabled.
- The XSD verification of LXAIP/DLXAIP containers during `PreservePO` and `UpdatePOC` shall be enabled. (conditional, if `CONFIG_LXAIP` applies)
- TOT shall support the following URI <http://www.bsi.bund.de/tr-esor/lxaip/1.3> as value of `pres:POFormat` as optional input of `RetrievePO` input parameter of `RetrievePO` function (c.f. [TR-ESOR-E], chapter 3.3.2 and [ETSI TS 119 512], chapter 5.3.4.1.1). (conditional, if `CONFIG_LXAIP` applies)

3.3 Standard Test Objects

For most of the tests test data is required. In order to make the tests repeatable, this section defines some standard test objects.

Notice

The used test data will be published by BSI as part of the open source application TR-ESOR-C.2-Testbed. It is strongly recommended to use the testbed and the corresponding test data while TR-ESOR product development for test purposes.

<i>Container or Object Name</i>	<i>Description</i>
ACTIVE_PROFILE	An active Preservation Profile supported by TOT.
ACTIVE_PROFILE_URI	The Profile Identifier as URI of ACTIVE_PROFILE.
INACTIVE_PROFILE	An inactive Preservation Profile supported by TOT.
INACTIVE_PROFILE_URI	The Profile Identifier as URI of INACTIVE_PROFILE.
UNKNOWN_PROFILE_URI	A Profile Identifier of an unknown profile.
UNKNOWN_OPT_INPUT	An unknown control as optional input.
XAIP_OK	The XAIP is syntactically correct and passes the defined consistency checks. The XAIP contains following objects: • TXT_DATA.txt and • PDF_DATA.pdf as well as • one metadata object XML_MDO.xml embedded as binary metadata.
TXT_DATA.txt	A reference text-based test data (CRLF and CP1252).
PDF_DATA.pdf	A reference PDF test data.
XML_MDO.xml	A reference XML-based test data (LF and UTF-8).
XAIP_OK_V1_ER_OK	An evidence record provided by TOT for XAIP_OK.

<i>Container or Object Name</i>	<i>Description</i>
XAIP_OK_SIG_OK	The XAIP is syntactically correct and passes the defined consistency checks. The XAIP contains following objects: • data object TXT_DATA.txt and • corresponding detached CADES TXT_DATA.txt.p7s, • data object PDF_DATA.pdf and • corresponding detached CADES PDF_DATA.pdf.p7s and • a metadata object XML_MDO.xml.
TXT_DATA.txt.p7s	A detached CADES of TXT_DATA.txt.
PDF_DATA.pdf.p7s	A detached CADES of PDF_DATA.pdf.
XAIP_OK_SIG_OK_V1_ER_OK	An evidence record provided by TOT for XAIP_OK_SIG_OK.
LXAIP_OK	The LXAIP is syntactically correct and passes the defined consistency checks. The LXAIP references following data: • TXT_DATA.txt⁵ as data object, • PDF_DATA.pdf as data object, • XML_MDO.xml as metadata object.
ASiC_AIP_OK	The ASiC-AIP is syntactically and passes the defined consistency checks. The ASiC-AIP contains an LXAIP with two data objects and two corresponding digital signatures and one metadata object.
REF_TXT_DATA_50	A reference to TXT_DATA.txt, which has been uploaded in advance as TXT_DATA_50.txt.
REF_PDF_DATA_50	A reference to PDF_DATA.pdf, which has been uploaded in advance as PDF_DATA_50.pdf.
REF_XML_MDO_50	A reference to XML_MDO.xml, which has been uploaded in advance as MDO_XML_50.xml.
LXAIP_OK_V1_ER_OK	An evidence record provided by TOT for LXAIP_OK.
LXAIP_OK_SIG_OK	The LXAIP is syntactically correct and passes the defined consistency checks. The LXAIP references following data: • TXT_DATA.txt as data object, • TXT_DATA.txt.p7s as credential, • PDF_DATA.pdf as data object, • PDF_DATA.pdf.p7s as credential and • XML_MDO.xml as metadata object.
REF_TXT_DATA_51	A reference to TXT_DATA.txt, which has been uploaded in advance as TXT_DATA_51.txt.
REF_CADES_DET_TXT_DATA_51	A reference to TXT_DATA.txt.p7s, which has been uploaded in advance as TXT_DATA_51.txt.p7s.
REF_PDF_DATA_51	A reference to PDF_DATA.pdf, which has been uploaded in advance as PDF_DATA_51.pdf.
REF_CADES_DET_PDF_DATA_51	A reference to PDF_DATA.pdf.p7s, which has been uploaded in advance as PDF_DATA_51.pdf.p7s.
REF_XML_MDO_51	A reference to XML_MDO.xml, which has been uploaded in advance as XML_MDO_51.xml.
LXAIP_OK_SIG_OK_V1_ER_OK	An evidence record provided by TOT for LXAIP_OK_SIG_OK.
XAIP_NOK	The schema validation of the XAIP fails. (unexpected element xaip:dataObject in element xaip:packageInfoUnit)
XAIP_NOK_VERSION	The fix attribute XAIPVersion in the element xaip:XAIP has wrong value.
LXAIP_NOK	The schema validation of the LXAIP fails. (missing element ds:DigestValue in element asic:DataObjectReference)
XAIP_NOK_EXPIRED	The schema validation for the XAIP succeeds, but the xaip:preservationInfo-element indicates a preservation date, which is already exceeded.

⁵ Every instance of LXAIP or DLXAIP has to contain its own references. There is no possibility to share the references between two or more LXAIP or DLXAIP instances (c.f. M-ADD-03a in Chapter 3.4.1.8.5.).

<i>Container or Object Name</i>	<i>Description</i>
XAIP_NOK_SUBMTIME	The schema validation for the XAIP succeeds, but the <code>xaip:submissionTime</code> -element deviates from the current time beyond a reasonable tolerance range. The documentation of the middleware or the module, which shall be tested, shall contain some assertions and related conditions or constraints indicating when the <code>xaip:submissionTime</code> contained in the provided XAIP deviates too much from the current time.
XAIP_NOK_SIG	The XAIP is syntactically correct and passes the defined consistency checks, but the XAIP contains an invalid digital signature. Invalid digital signature means that the digital signature is syntactically not correct or at least one of the evidence relevant data, for example a signature, time-stamp, certificate, certificate revocation list or OCSP-response, etc., is wrong.
XAIP_NOK_ER	The XAIP is syntactically correct and passes the defined consistency checks, but the XAIP contains an invalid Evidence Record: ER_NOK_XAIP_OK.
ER_NOK_XAIP_OK	An invalid Evidence Record for XAIP_OK means, that the Evidence Record is syntactically not correct or does not fulfil the requirements according to Annex ERS.
XAIP_NOK_ER_VR	A verification report produced by the TOT for XAIP_NOK_ER.
LXAIP_OK_ER_NOK	The LXAIP is based on LXAIP_OK, syntactically correct and passes the defined consistency checks, but the LXAIP contains an invalid Evidence Record: ER_NOK_LXAIP_OK.
LXAIP_OK_ER_NOK_VR	A verification report produced by the TOT for LXAIP_OK_ER_NOK.
REF_TXT_DATA_52	A reference to TXT_DATA.txt, which has been uploaded in advance as TXT_DATA_52.txt.
REF_PDF_DATA_52	A reference to PDF_DATA.pdf, which has been uploaded in advance as PDF_DATA_52.pdf.
REF_XML_MDO_52	A reference to XML_MDO.xml, which has been uploaded in advance as XML_MDO_52.xml.
ER_NOK_LXAIP_OK	An invalid Evidence Record for LXAIP_OK means, that the Evidence Record is syntactically not correct or does not fulfil the requirements according to Annex ERS.
CADES_ATT_OK	Alias for TXT_DATA.txt.p7m.
TXT_DATA.txt.p7m	A valid detached CADES for TXT_DATA.txt.
XADES_ENVELOPED_OK	A valid attached enveloped XAdES.
XADES_ENVELOPING_OK	A valid attached enveloping XAdES.
PADES_OK	A valid visible PAdES.
ASiC_E_CADES_OK	A valid ASiC-E with a valid CADES.
ASiC_E_XADES_OK	A valid ASiC-E with a valid XAdES.
ASiC_E_TST_OK	A valid ASiC-E with a valid timestamp.
ASiC_E_ER_OK	A valid ASiC-E with a valid evidence record.
DIGEST_LIST	A valid digest list.
ASiC_S_CADES_OK	A valid ASiC-S with a valid CADES.
ASiC_S_XADES_OK	A valid ASiC-S with a valid XAdES.
ASiC_S_TST_OK	A valid ASiC-S with a valid timestamp.
ASiC_S_ER_OK	A valid ASiC-S with a valid evidence record.
CADES_ATT_NOK	An invalid attached CADES.
CADES_ATT_NOK_VR	A verification report produced by TOT of CADES_ATT_NOK.
XADES_ENVELOPED_NOK	An invalid attached enveloped XAdES.
XADES_ENVELOPING_NOK	An invalid attached enveloping XAdES.
PADES_NOK	An invalid visible PAdES.ASiC_E_CADES_NOK
ASiC_E_CADES_NOK	A valid ASiC-E with invalid CADES.
ASiC_E_XADES_NOK	A valid ASiC-E with invalid XAdES.
ASiC_E_TST_NOK	A valid ASiC-E with invalid timestamp.
ASiC_E_ER_NOK	A valid ASiC-E with invalid evidence record.
ASiC_S_CADES_NOK	A valid ASiC-S with invalid CADES.
ASiC_S_XADES_NOK	A valid ASiC-S with invalid XAdES.

<i>Container or Object Name</i>	<i>Description</i>
ASIC_S_TST_NOK	A valid ASiC-S with invalid timestamp.
ASIC_S_ER_NOK	A valid ASiC-S with invalid evidence record.
DXAIP_OK	The DXAIP is syntactically correct and represents a valid update container ("Delta XAIP" – with version 2) for XAIP_OK (version 1), which contains the corresponding AOID. Additional data: TXT_DATA.txt.p7m.
DXAIP_OK_2	The DXAIP is syntactically correct and represents a valid update container ("Delta XAIP" – with version 3) for XAIP_OK (in version 2), which contains the corresponding AOID. Additional data: - XML_DATA.xml and - XML_DATA.xml.p7s.
XML_DATA.xml	A reference xml-based test data (CRLF and UTF-8).
XML_DATA.xml.p7s	A valid attached CADES of XML_DATA.xml.
002-test-document.xml	Some other xml reference data for XAdES.
XADES_DET_OK	A valid detached XAdES of 002-test-document.xml.
XADES_DET_NOK_VR	A verification report of XADES_DET_OK, created by the TOT.
XADES_DET_NOK	An invalid detached XAdES of 002-test-document.xml.
XADES_DET_NOK_VR	A verification report of XADES_DET_NOK, created by TOT.
DXAIP_OK_SIG_ONLY	The DXAIP is syntactically correct and represents a valid update container ("Delta XAIP" – with version 2) for XAIP_OK (version 1), which contains the corresponding AOID. Additional data: TXT_DATA_QESI.txt.p7s.
TXT_DATA_QESI.txt.p7s	A valid detached CADES of TXT_DATA.txt made with a test digital seal certificate.
DLXAIP_OK	The DXAIP is syntactically correct and represents a valid update container ("Delta LXAIP") for LXAIP_OK (with version 1), which contains the corresponding AOID. DLXAIP references following data additionally: TXT_DATA.txt.p7m.
REF_CADES_ATT_TXT_DATA_53	A reference to TXT_DATA.txt.p7m, which has been uploaded in advance as TXT_DATA_53.txt.p7m.
DLXAIP_OK_2	The DXAIP is syntactically correct and represents a valid update container ("Delta LXAIP") for LXAIP_OK (with version 2), which contains the corresponding AOID. DLXAIP references following data additionally: - XML_DATA.xml and - XML_DATA.xml.p7s .
REF_XML_DATA_54	A reference to XML_DATA.xml, which has been uploaded in advance as XML_DATA_54.xml.
REF_CADES_DET_XML_DATA_54	A reference to XML_DATA.xml.p7s, which has been uploaded in advance as XML_DATA_54.xml.p7s.
DXAIP_NOK_AOID	The schema validation for the Delta XAIP succeeds, but the update container ("Delta XAIP") contains a not yet assigned AOID.
DXAIP_NOK	The Delta XAIP is syntactically not correct, the schema validation fails.
DLXAIP_NOK	The Delta LXAIP is syntactically not correct, the schema validation fails.
REF_TXT_DATA_55	A reference to TXT_DATA.txt, which has been uploaded in advance as TXT_DATA_55.txt.
REF_PDF_DATA_55	A reference to PDF_DATA.pdf, which has been uploaded in advance as PDF_DATA_55.pdf.
REF_XML_MDO_55	A reference to XML_MDO.xml, which has been uploaded in advance as XML_MDO_55.xml.
REF_XML_DATA_56	A reference to XML_DATA.xml, which has been uploaded in advance as XML_DATA_56.xml.
REF_CADES_DET_XML_DATA_56	A reference to XML_DATA.xml.p7s, which has been uploaded in advance as XML_DATA_56.xml.p7s.
DXAIP_NOK_EXPIRED	The schema validation for the Delta XAIP succeeds, but the xaip:preservationInfo-element indicates a point in time in the past.

<i>Container or Object Name</i>	<i>Description</i>
DXAIP_NOK_SUBMTIME	The schema validation for the Delta XAIP succeeds, but the <code>xaip:submissionTime</code> -element deviates from the current time beyond a reasonable tolerance range. The documentation of the middleware or the module, which shall be tested, shall contain some assertions and related conditions or constraints indicating when the submission time contained in the provided XAIP deviates too much from the current time
DXAIP_NOK_SIG	The schema validation for the Delta XAIP succeeds, but the XAIP contains an invalid digital signature.
DXAIP_NOK_SIG_VR	A verification report produced by TOT for DXAIP_NOK_SIG.
DXAIP_NOK_ER	The schema validation for the Delta LXAIP succeeds, but the Delta LXAIP contains an invalid Evidence Record (invalid format).
DXAIP_NOK_ER_VR	A verification report produced by TOT for DXAIP_NOK_ER.
DLXAIP_NOK_ER	The schema validation for the Delta LXAIP succeeds, but the Delta LXAIP contains an invalid Evidence Record (invalid format).
DLXAIP_NOK_ER_VR	A verification report produced by TOT of DLXAIP_NOK_ER.
ER_NOK_LXAIP_OK_V2	Incorrect evidence record (invalid format) of DLXAIP_NOK_ER.
REF_CADES_ATT_TXT_DATA_57	A reference to TXT_DATA.txt.p7m, which has been uploaded in advance as TXT_DATA_57.txt.p7m.
DXAIP_NOK_VERSION	The schema validation for the XAIP succeeds, but there is a syntactical collision with the original XAIP such that the schema validation for the compound XAIP fails, for example, the element <code>xaip:prevVersion</code> in the <code>xaip:updateSection</code> of the DXAIP is not the latest version of this XAIP.
REF_TXT_DATA_58	A reference to TXT_DATA.txt, which has been uploaded in advance as TXT_DATA_58.txt.
REF_PDF_DATA_58	A reference to PDF_DATA.pdf, which has been uploaded in advance as PDF_DATA_58.pdf.
REF_XML_MDO_58	A reference to XML_MDO.xml, which has been uploaded in advance as XML_MDO_58.xml.
REF_CADES_ATT_TXT_DATA_59	A reference to TXT_DATA.txt.p7m, which has been uploaded in advance as TXT_DATA_59.txt.p7m.
REF_CADES_ATT_TXT_DATA_60	A reference to TXT_DATA.txt.p7m, which has been uploaded in advance as TXT_DATA_60.txt.p7m.
REF_XML_DATA_60-1	A reference to XML_DATA.xml, which has been uploaded in advance as XML_DATA_60-1.xml.
REF_CADES_DET_XML_DATA_60-1	A reference to XML_DATA.xml.p7s, which has been uploaded in advance as XML_DATA_60-1.xml.p7s.
XAIP_OK_V3_ER_OK	An evidence record for DXAIP_OK_2, created by TOT.
XAIP_OK_V2_ER_OK	An evidence record for DXAIP_OK, created by TOT.
LXAIP_OK_V3_ER_OK	An evidence record for DLXAIP_OK_2, created by TOT.
LXAIP_OK_V2_ER_OK	An evidence record for DLXAIP_OK, created by TOT.
XAIP_OK_SIG_OK_VR	A verification report for XAIP_OK_SIG_OK, created by TOT.
XAIP_OK_SIG_OK_ER	An evidence record for XAIP_OK_SIG_OK, created by TOT.
REF_TXT_DATA_61	A reference to TXT_DATA.txt, which has been uploaded in advance as TXT_DATA_61.txt.
REF_CADES_DET_TXT_DATA_61	A reference to TXT_DATA.txt.p7s, which has been uploaded in advance as TXT_DATA_61.txt.p7s.
REF_PDF_DATA_61	A reference to PDF_DATA.pdf, which has been uploaded in advance as PDF_DATA_61.pdf.
REF_CADES_DET_PDF_DATA_61	A reference to PDF_DATA.pdf.p7s, which has been uploaded in advance as PDF_DATA_61.pdf.p7s.
REF_XML_MDO_61	A reference to XML_MDO.xml, which has been uploaded in advance as XML_MDO_61.xml.
LXAIP_OK_SIG_OK_ER	An evidence record of LXAIP_OK_SIG_OK, created by TOT.
DXAIP_OK_SIG_OK	A syntactically correct Delta-XAIP consisting of second version of XAIP_OK_SIG_OK. DXAIP contains following additional data: - XML_DATA.xml and - XML_DATA.xml.p7s.

<i>Container or Object Name</i>	<i>Description</i>
DLXAIP_OK_SIG_OK	A syntactically correct Delta-XAIP consisting of second version of LXAIP_OK_SIG_OK. DXAIP references following additional data: - XML_DATA.xml and - XML_DATA.xml.p7s.
REF_XML_DATA_62	A reference to XML_DATA.xml, which has been uploaded in advance as XML_DATA_62.xml.
REF_CADES_DET_XML_DATA_62	A reference to XML_DATA.xml.p7s, which has been uploaded in advance as XML_DATA_62.xml.p7s.
REF_XML_DATA_63	A reference to XML_DATA.xml, which has been uploaded in advance as XML_DATA_63.xml.
REF_CADES_DET_XML_DATA_63	A reference to XML_DATA.xml.p7s, which has been uploaded in advance as XML_DATA_63.xml.p7s.
REF_XML_DATA_64	A reference to XML_DATA.xml, which has been uploaded in advance as XML_DATA_64.xml.
REF_CADES_DET_XML_DATA_64	A reference to XML_DATA.xml.p7s, which has been uploaded in advance as XML_DATA_64.xml.p7s.
XAIP_OK_SIG_NOK	An XAIP_OK_SIG_OK containing an invalid digital signature (TXT_DATA_NOK.txt.p7s).
TXT_DATA_NOK.txt.p7s	An invalid digital signature of TXT_DATA.txt.
XAIP_OK_SIG_NOK_VR	A verification report of XAIP_OK_SIG_NOK created by the TOT.
XAIP_OK_SIG_OK_ER_NOK	An invalid evidence record of XAIP_OK_SIG_OK.
XAIP_OK_SIG_OK_ER_NOK_VR	A verification report of XAIP_OK_SIG_OK_ER_NOK created by the TOT.
LXAIP_OK_SIG_NOK	An LXAIP_OK_SIG_OK referencing a single invalid digital signature (TXT_DATA_NOK.txt.p7s).
REF_TXT_DATA_65	A reference to TXT_DATA.txt, which has been uploaded in advance as TXT_DATA_65.txt.
REF_CADES_DET_NOK_TXT_DATA_65	A reference to TXT_DATA_NOK.txt.p7s, which has been uploaded in advance as TXT_DATA_65_NOK.p7s.
REF_PDF_DATA_65	A reference to PDF_DATA.pdf, which has been uploaded in advance as PDF_DATA_65.pdf.
REF_CADES_DET_PDF_DATA_65	A reference to PDF_DATA.pdf.p7s, which has been uploaded in advance as PDF_DATA_65.pdf.p7s.
REF_XML_MDO_65	A reference to XML_MDO.xml, which has been uploaded in advance as XML_MDO_65.xml.
LXAIP_OK_SIG_OK_VR	A verification report of LXAIP_OK_SIG_OK produced by TOT.
LXAIP_OK_SIG_NOK_VR	A verification report of LXAIP_OK_SIG_NOK created by TOT.
LXAIP_OK_SIG_OK_ER_NOK	A LXAIP_OK_SIG_OK containing additionally an invalid evidence record (ER_NOK_XAIP_OK_SIG_OK_00).
LXAIP_OK_SIG_OK_ER_NOK_VR	A verification report of LXAIP_OK_SIG_OK_ER_NOK, created by the TOT.
REF_TXT_DATA_66	A reference to TXT_DATA.txt, which has been uploaded in advance as TXT_DATA_66.txt.
REF_CADES_DET_TXT_DATA_66	A reference to TXT_DATA_NOK.txt.p7s, which has been uploaded in advance as TXT_DATA_66.txt.p7s.
REF_PDF_DATA_66	A reference to PDF_DATA.pdf, which has been uploaded in advance as PDF_DATA_66.pdf.
REF_CADES_DET_PDF_DATA_66	A reference to PDF_DATA.pdf.p7s, which has been uploaded in advance as PDF_DATA_66.pdf.p7s.
REF_XML_MDO_66	A reference to XML_MDO.xml, which has been uploaded in advance as XML_MDO_66.xml.
ER_NOK_XAIP_OK_SIG_OK_00	An invalid evidence record of LXAIP_OK_SIG_OK.
REF_TXT_DATA_67	A reference to TXT_DATA.txt, which has been uploaded in advance as TXT_DATA_67.txt.
REF_CADES_DET_NOK_TXT_DATA_67	A reference to TXT_DATA_NOK.txt.p7s, which has been uploaded in advance as TXT_DATA_NOK_67.txt.p7s.
REF_PDF_DATA_67	A reference to PDF_DATA.pdf, which has been uploaded in advance as PDF_DATA_67.pdf.
REF_CADES_DET_PDF_DATA_67	A reference to PDF_DATA.pdf.p7s, which has been uploaded in advance as PDF_DATA_67.pdf.p7s.

<i>Container or Object Name</i>	<i>Description</i>
REF_XML_MDO_67	A reference to XML_MDO.xml, which has been uploaded in advance as XML_MDO_67.xml.
REF_TXT_DATA_68	A reference to TXT_DATA.txt, which has been uploaded in advance as TXT_DATA_68.txt.
REF_CADES_DET_TXT_DATA_68	A reference to TXT_DATA_NOK.txt.p7s, which has been uploaded in advance as TXT_DATA_68.txt.p7s.
REF_PDF_DATA_68	A reference to PDF_DATA.pdf, which has been uploaded in advance as PDF_DATA_68.pdf.
REF_CADES_DET_PDF_DATA_68	A reference to PDF_DATA.pdf.p7s, which has been uploaded in advance as PDF_DATA_68.pdf.p7s.
REF_XML_MDO_68	A reference to XML_MDO.xml, which has been uploaded in advance as XML_MDO_68.xml.
DXAIP_OK_SIG_NOK	A DXAIP_OK_SIG_OK containing an invalid digital signature (XML_DATA_NOK.xml.p7s).
XML_DATA_NOK.xml.p7s	An invalid detached electronic signature of XML_DATA.xml.
DXAIP_OK_SIG_NOK_VR	A verification report of DXAIP_OK_SIG_NOK created by the TOT.
DLXAIP_OK_SIG_NOK	A DLXAIP_OK_SIG_OK referencing an invalid digital signature (XML_DATA_NOK.xml.p7s).
DXAIP_OK_SIG_NOK_VR	A verification report of DLXAIP_OK_SIG_NOK, created by the TOT.
REF_XML_DATA_69	A reference to XML_DATA.xml, which has been uploaded in advance as XML_DATA_69.xml.
REF_CADES_DET_NOK_XML_DATA_69	A reference to XML_DATA.xml.p7s, which has been uploaded in advance as XML_DATA_NOK_69.xml.p7s.
REF_XML_DATA_70	A reference to XML_DATA.xml, which has been uploaded in advance as XML_DATA_70.xml.
REF_CADES_DET_NOK_XML_DATA_70	A reference to XML_DATA.xml.p7s, which has been uploaded in advance as XML_DATA_NOK_70.xml.p7s.
CADES_DET_OK	A valid digital signature of TXT_DATA.txt.
CADES_DET_NOK	An invalid digital signature of TXT_DATA.txt.
CADES_DET_NOK_VR	A verification report of CADES_DET_NOK created by the TOT.
XADES_ENVELOPED_NOK_VR	A verification report of XADES_ENVELOPED_NOK created by the TOT.
XADES_ENVELOPING_NOK_VR	A verification report of XADES_ENVELOPING_NOK created by the TOT.
PADES_NOK_VR	A verification report of PADES_NOK created by the TOT.
ASIC_E_CADES_NOK_VR	A verification report of ASIC_E_CADES_NOK created by the TOT.
ASIC_E_XADES_NOK_VR	A verification report of ASIC_E_XADES_NOK created by the TOT.
ASIC_E_TST_NOK_VR	A verification report of ASIC_E_TST_NOK created by the TOT.
ASIC_E_ER_NOK_VR	A verification report of ASIC_E_ER_NOK created by the TOT.
ASIC_S_CADES_NOK_VR	A verification report of ASIC_S_CADES_NOK created by the TOT.
ASIC_S_XADES_NOK_VR	A verification report of ASIC_S_XADES_NOK created by the TOT.
ASIC_S_TST_NOK_VR	A verification report of ASIC_S_TST_NOK created by the TOT.
ASIC_S_ER_NOK_VR	A verification report of ASIC_S_ER_NOK created by the TOT.
ER_OK_SHA-1_XAIP_OK_SIG_OK	A valid evidence record of XAIP_OK_SIG_OK using a SHA-1 for building the hash tree.
ER_OK_SHA-1_XAIP_OK_SIG_OK_VR	A verification report of ER_OK_SHA-1_XAIP_OK_SIG_OK created by the TOT.
ER_NOK_SHA-1_XAIP_OK_SIG_OK	A invalid evidence record of XAIP_OK_SIG_OK using a SHA-1 for building the hash tree.
ER_NOK_SHA-1_XAIP_OK_SIG_OK_VR	A verification report of ER_NOK_SHA-1_XAIP_OK_SIG_OK created by the TOT.
ER_OK_SHA-224_XAIP_OK_SIG_OK	A valid evidence record of XAIP_OK_SIG_OK using a SHA-224 for building the hash tree.
ER_OK_SHA-224_XAIP_OK_SIG_OK_VR	A verification report of ER_OK_SHA-224_XAIP_OK_SIG_OK created by the TOT.
ER_NOK_SHA-224_XAIP_OK_SIG_OK	A invalid evidence record of XAIP_OK_SIG_OK using a SHA-224 for building the hash tree.
ER_NOK_SHA-224_XAIP_OK_SIG_OK_VR	A verification report of ER_NOK_SHA-224_XAIP_OK_SIG_OK created by the TOT.

<i>Container or Object Name</i>	<i>Description</i>
ER_OK_SHA-256_XAIP_OK_SIG_OK	A valid evidence record of XAIP_OK_SIG_OK using a SHA-256 for building the hash tree.
ER_OK_SHA-256_XAIP_OK_SIG_OK_VR	A verification report of ER_OK_SHA-256_XAIP_OK_SIG_OK created by the TOT.
ER_NOK_SHA-256_XAIP_OK_SIG_OK	A invalid evidence record of XAIP_OK_SIG_OK using a SHA-256 for building the hash tree.
ER_NOK_SHA-256_XAIP_OK_SIG_OK_VR	A verification report of ER_NOK_SHA-256_XAIP_OK_SIG_OK created by the TOT.
ER_OK_SHA-384_XAIP_OK_SIG_OK	A valid evidence record of XAIP_OK_SIG_OK using a SHA-384 for building the hash tree.
ER_OK_SHA-384_XAIP_OK_SIG_OK_VR	A verification report of ER_OK_SHA-384_XAIP_OK_SIG_OK created by the TOT.
ER_NOK_SHA-384_XAIP_OK_SIG_OK	A invalid evidence record of XAIP_OK_SIG_OK using a SHA-384 for building the hash tree.
ER_NOK_SHA-384_XAIP_OK_SIG_OK_VR	A verification report of ER_NOK_SHA-384_XAIP_OK_SIG_OK created by the TOT.
ER_OK_SHA-512_XAIP_OK_SIG_OK	A valid evidence record of XAIP_OK_SIG_OK using a SHA-512 for building the hash tree.
ER_OK_SHA-512_XAIP_OK_SIG_OK_VR	A verification report of ER_OK_SHA-512_XAIP_OK_SIG_OK created by the TOT.
ER_NOK_SHA-512_XAIP_OK_SIG_OK	A invalid evidence record of XAIP_OK_SIG_OK using a SHA-512 for building the hash tree.
ER_NOK_SHA-512_XAIP_OK_SIG_OK_VR	A verification report of ER_NOK_SHA-512_XAIP_OK_SIG_OK created by the TOT.
ER_OK_SHA3-224_XAIP_OK_SIG_OK	A valid evidence record of XAIP_OK_SIG_OK using a SHA3-224 for building the hash tree.
ER_OK_SHA3-224_XAIP_OK_SIG_OK_VR	A verification report of ER_OK_SHA3-224_XAIP_OK_SIG_OK created by the TOT.
ER_NOK_SHA3-224_XAIP_OK_SIG_OK	A invalid evidence record of XAIP_OK_SIG_OK using a SHA3-224 for building the hash tree.
ER_NOK_SHA3-224_XAIP_OK_SIG_OK_VR	A verification report of ER_NOK_SHA3-224_XAIP_OK_SIG_OK created by the TOT.
ER_OK_SHA3-256_XAIP_OK_SIG_OK	A valid evidence record of XAIP_OK_SIG_OK using a SHA3-256 for building the hash tree.
ER_OK_SHA3-256_XAIP_OK_SIG_OK_VR	A verification report of ER_OK_SHA3-256_XAIP_OK_SIG_OK created by the TOT.
ER_NOK_SHA3-256_XAIP_OK_SIG_OK	A invalid evidence record of XAIP_OK_SIG_OK using a SHA3-256 for building the hash tree.
ER_NOK_SHA3-256_XAIP_OK_SIG_OK_VR	A verification report of ER_NOK_SHA3-256_XAIP_OK_SIG_OK created by the TOT.
ER_OK_SHA3-384_XAIP_OK_SIG_OK	A valid evidence record of XAIP_OK_SIG_OK using a SHA3-384 for building the hash tree.
ER_OK_SHA3-384_XAIP_OK_SIG_OK_VR	A verification report of ER_OK_SHA3-384_XAIP_OK_SIG_OK created by the TOT.
ER_NOK_SHA3-384_XAIP_OK_SIG_OK	A invalid evidence record of XAIP_OK_SIG_OK using a SHA3-384 for building the hash tree.
ER_NOK_SHA3-384_XAIP_OK_SIG_OK_VR	A verification report of ER_NOK_SHA3-384_XAIP_OK_SIG_OK created by the TOT.
ER_OK_SHA3-512_XAIP_OK_SIG_OK	A valid evidence record of XAIP_OK_SIG_OK using a SHA3-512 for building the hash tree.
ER_OK_SHA3-512_XAIP_OK_SIG_OK_VR	A verification report of ER_OK_SHA3-512_XAIP_OK_SIG_OK created by the TOT.
ER_NOK_SHA3-512_XAIP_OK_SIG_OK	A invalid evidence record of XAIP_OK_SIG_OK using a SHA3-512 for building the hash tree.
ER_NOK_SHA3-512_XAIP_OK_SIG_OK_VR	A verification report of ER_NOK_SHA3-512_XAIP_OK_SIG_OK created by the TOT.
XAIP_OK_V1_ER_OK_RESG	A valid evidence record of the first version of XAIP_OK, after successful resigning operation on the hash tree has been finished.
XAIP_OK_V2_ER_OK_RESG	A valid evidence record of the second version of XAIP_OK and DXAIP_OK, after successful resigning operation on the hash tree has been finished.

<i>Container or Object Name</i>	<i>Description</i>
XAIP_OK_V3_ER_OK_RESG	A valid evidence record of the third version of XAIP_OK and DXAIP_OK and DXAIP_OK_2, after successful resigning operation on the hash tree has been finished.
LXAIP_OK_V1_ER_OK_RESG	A valid evidence record of the first version of LXAIP_OK, after successful resigning operation on the hash tree has been finished.
LXAIP_OK_V2_ER_OK_RESG	A valid evidence record of the second version of LXAIP_OK and DLXAIP_OK, after successful resigning operation on the hash tree has been finished.
LXAIP_OK_V3_ER_OK_RESG	A valid evidence record of the third version of LXAIP_OK and DLXAIP_OK and DLXAIP_OK_2, after successful resigning operation on the hash tree has been finished.
XAIP_OK_V1_ER_OK_RESG_REH	The XAIP_OK_V1_ER_OK_RESG, after successful rehashing operation on the hash tree has been finished.
XAIP_OK_V2_ER_OK_RESG_REH	The XAIP_OK_V2_ER_OK_RESG, after successful rehashing operation on the hash tree has been finished.
XAIP_OK_V3_ER_OK_RESG_REH	The XAIP_OK_V3_ER_OK_RESG, after successful rehashing operation on the hash tree has been finished.
LXAIP_OK_V1_ER_OK_RESG_REH	The LXAIP_OK_V1_ER_OK_RESG, after successful rehashing operation on the hash tree has been finished.
LXAIP_OK_V2_ER_OK_RESG_REH	The LXAIP_OK_V2_ER_OK_RESG, after successful rehashing operation on the hash tree has been finished.
LXAIP_OK_V3_ER_OK_RESG_REH	The LXAIP_OK_V3_ER_OK_RESG, after successful rehashing operation on the hash tree has been finished.
REF_TXT_DATA_71	A reference to TXT_DATA.txt, which has been uploaded in advance as TXT_DATA_71.txt.
REF_PDF_DATA_71	A reference to PDF_DATA.pdf, which has been uploaded in advance as PDF_DATA_71.pdf.
REF_XML_MDO_71	A reference to XML_MDO.xml, which has been uploaded in advance as XML_MDO_71.xml.
REF_TXT_DATA_72	A reference to TXT_DATA.txt, which has been uploaded in advance as TXT_DATA_72.txt.
REF_PDF_DATA_72	A reference to PDF_DATA.pdf, which has been uploaded in advance as PDF_DATA_72.pdf.
REF_XML_MDO_72	A reference to XML_MDO.xml, which has been uploaded in advance as XML_MDO_72.xml.
ER_NOK_LXAIP_OK_72	An invalid evidence record of LXAIP_OK.
REF_CADES_ATT_TXT_DATA_73	A reference to valid attached digital signature of TXT_DATA.txt, which has been uploaded in advance as TXT_DATA_73.txt.p7m.
REF_TXT_DATA_74	A reference to TXT_DATA.txt, which has been uploaded in advance as TXT_DATA_74.txt.
REF_PDF_DATA_74	A reference to PDF_DATA.pdf, which has been uploaded in advance as PDF_DATA_74.pdf.
REF_XML_MDO_74	A reference to XML_MDO.xml, which has been uploaded in advance as XML_MDO_74.xml.
XAIP_OK_SO	A valid XAIP containing only single data object: • TXT_DATA.txt.
ER_SO-1_OK_XAIP_OK_SO	A valid evidence record of XAIP_OK_SO, the hash value of the data object is directly included in the message imprint of the timestamp: • ER(TSP(H(DO-01))).
ER_SO-2_OK_XAIP_OK_SO	A valid evidence record of XAIP_OK_SO, the hash value of the data object is placed in the single reduced hash tree and the same hash value is included in the message imprint of the timestamp: • ER(H(DO-01),TSP(H(DO-01))).
ER_SO-3_OK_XAIP_OK_SO	A valid evidence record of XAIP_OK_SO, the hash value of the data object is placed in the single reduced hash tree and the hash value of those hash value (computet hash tree root) is included in the message imprint of the timestamp: • ER(H(DO-01),TSP(H(H(DO-01))).
REF_TXT_DATA_50b	A reference to TXT_DATA.txt, which has been uploaded in advance as TXT_DATA_50.txt.

<i>Container or Object Name</i>	<i>Description</i>
REF_PDF_DATA_50b	A reference to PDF_DATA.pdf, which has been uploaded in advance as PDF_DATA_50.pdf.
REF_XML_MDO_50b	A reference to XML_MDO.xml, which has been uploaded in advance as MDO_XML_50.xml.
REF_TXT_DATA_51b	A reference to TXT_DATA.txt, which has been uploaded in advance as TXT_DATA_51.txt.
REF_CADES_DET_TXT_DATA_51b	A reference to TXT_DATA.txt.p7s, which has been uploaded in advance as TXT_DATA_51.txt.p7s.
REF_PDF_DATA_51b	A reference to PDF_DATA.pdf, which has been uploaded in advance as PDF_DATA_51.pdf.
REF_CADES_DET_PDF_DATA_51b	A reference to PDF_DATA.pdf.p7s, which has been uploaded in advance as PDF_DATA_51.pdf.p7s.
REF_XML_MDO_51b	A reference to XML_MDO.xml, which has been uploaded in advance as XML_MDO_51.xml.
REF_TXT_DATA_52b	A reference to TXT_DATA.txt, which has been uploaded in advance as TXT_DATA_52.txt.
REF_PDF_DATA_52b	A reference to PDF_DATA.pdf, which has been uploaded in advance as PDF_DATA_52.pdf.
REF_XML_MDO_52b	A reference to XML_MDO.xml, which has been uploaded in advance as XML_MDO_52.xml.
REF_CADES_ATT_TXT_DATA_53b	A reference to TXT_DATA.txt.p7m, which has been uploaded in advance as TXT_DATA_53.txt.p7m.
REF_XML_DATA_54b	A reference to XML_DATA.xml, which has been uploaded in advance as XML_DATA_54.xml.
REF_CADES_DET_XML_DATA_54b	A reference to XML_DATA.xml.p7s, which has been uploaded in advance as XML_DATA_54.xml.p7s.
REF_TXT_DATA_55b	A reference to TXT_DATA.txt, which has been uploaded in advance as TXT_DATA_55.txt.
REF_PDF_DATA_55b	A reference to PDF_DATA.pdf, which has been uploaded in advance as PDF_DATA_55.pdf.
REF_XML_MDO_55b	A reference to XML_MDO.xml, which has been uploaded in advance as XML_MDO_55.xml.
REF_XML_DATA_56b	A reference to XML_DATA.xml, which has been uploaded in advance as XML_DATA_56.xml.
REF_CADES_DET_XML_DATA_56b	A reference to XML_DATA.xml.p7s, which has been uploaded in advance as XML_DATA_56.xml.p7s.
REF_CADES_ATT_TXT_DATA_57b	A reference to TXT_DATA.txt.p7m, which has been uploaded in advance as TXT_DATA_57.txt.p7m.
REF_TXT_DATA_58b	A reference to TXT_DATA.txt, which has been uploaded in advance as TXT_DATA_58.txt.
REF_PDF_DATA_58b	A reference to PDF_DATA.pdf, which has been uploaded in advance as PDF_DATA_58.pdf.
REF_XML_MDO_58b	A reference to XML_MDO.xml, which has been uploaded in advance as XML_MDO_58.xml.
REF_CADES_ATT_TXT_DATA_59b	A reference to TXT_DATA.txt.p7m, which has been uploaded in advance as TXT_DATA_59.txt.p7m.
REF_CADES_ATT_TXT_DATA_60b	A reference to TXT_DATA.txt.p7m, which has been uploaded in advance as TXT_DATA_60.txt.p7m.
REF_XML_DATA_60-1b	A reference to XML_DATA.xml, which has been uploaded in advance as XML_DATA_60-1.xml.
REF_CADES_DET_XML_DATA_60-1b	A reference to XML_DATA.xml.p7s, which has been uploaded in advance as XML_DATA_60-1.xml.p7s.
REF_TXT_DATA_61b	A reference to TXT_DATA.txt, which has been uploaded in advance as TXT_DATA_61.txt.
REF_CADES_DET_TXT_DATA_61b	A reference to TXT_DATA.txt.p7s, which has been uploaded in advance as TXT_DATA_61.txt.p7s.
REF_PDF_DATA_61b	A reference to PDF_DATA.pdf, which has been uploaded in advance as PDF_DATA_61.pdf.
REF_CADES_DET_PDF_DATA_61b	A reference to PDF_DATA.pdf.p7s, which has been uploaded in advance as PDF_DATA_61.pdf.p7s.

<i>Container or Object Name</i>	<i>Description</i>
REF_XML_MDO_61b	A reference to XML_MDO.xml, which has been uploaded in advance as XML_MDO_61.xml.
REF_XML_DATA_62b	A reference to XML_DATA.xml, which has been uploaded in advance as XML_DATA_62.xml.
REF_CADES_DET_XML_DATA_62b	A reference to XML_DATA.xml.p7s, which has been uploaded in advance as XML_DATA_62.xml.p7s.
REF_XML_DATA_63b	A reference to XML_DATA.xml, which has been uploaded in advance as XML_DATA_63.xml.
REF_CADES_DET_XML_DATA_63b	A reference to XML_DATA.xml.p7s, which has been uploaded in advance as XML_DATA_63.xml.p7s.
REF_TXT_DATA_65b	A reference to TXT_DATA.txt, which has been uploaded in advance as TXT_DATA_65.txt.
REF_CADES_DET_NOK_TXT_DATA_65b	A reference to TXT_DATA_NOK.txt.p7s, which has been uploaded in advance as TXT_DATA_65_NOK.p7s.
REF_PDF_DATA_65b	A reference to PDF_DATA.pdf, which has been uploaded in advance as PDF_DATA_65.pdf.
REF_CADES_DET_PDF_DATA_65b	A reference to PDF_DATA.pdf.p7s, which has been uploaded in advance as PDF_DATA_65.pdf.p7s.
REF_XML_MDO_65b	A reference to XML_MDO.xml, which has been uploaded in advance as XML_MDO_65.xml.
REF_TXT_DATA_66b	A reference to TXT_DATA.txt, which has been uploaded in advance as TXT_DATA_66.txt.
REF_CADES_DET_TXT_DATA_66b	A reference to TXT_DATA_NOK.txt.p7s, which has been uploaded in advance as TXT_DATA_66.txt.p7s.
REF_PDF_DATA_66b	A reference to PDF_DATA.pdf, which has been uploaded in advance as PDF_DATA_66.pdf.
REF_CADES_DET_PDF_DATA_66b	A reference to PDF_DATA.pdf.p7s, which has been uploaded in advance as PDF_DATA_66.pdf.p7s.
REF_XML_MDO_66b	A reference to XML_MDO.xml, which has been uploaded in advance as XML_MDO_66.xml.
REF_TXT_DATA_67b	A reference to TXT_DATA.txt, which has been uploaded in advance as TXT_DATA_67.txt.
REF_CADES_DET_NOK_TXT_DATA_67b	A reference to TXT_DATA_NOK.txt.p7s, which has been uploaded in advance as TXT_DATA_NOK_67.txt.p7s.
REF_PDF_DATA_67b	A reference to PDF_DATA.pdf, which has been uploaded in advance as PDF_DATA_67.pdf.
REF_CADES_DET_PDF_DATA_67b	A reference to PDF_DATA.pdf.p7s, which has been uploaded in advance as PDF_DATA_67.pdf.p7s.
REF_XML_MDO_67b	A reference to XML_MDO.xml, which has been uploaded in advance as XML_MDO_67.xml.
REF_TXT_DATA_68b	A reference to TXT_DATA.txt, which has been uploaded in advance as TXT_DATA_68.txt.
REF_CADES_DET_TXT_DATA_68b	A reference to TXT_DATA_NOK.txt.p7s, which has been uploaded in advance as TXT_DATA_68.txt.p7s.
REF_PDF_DATA_68b	A reference to PDF_DATA.pdf, which has been uploaded in advance as PDF_DATA_68.pdf.
REF_CADES_DET_PDF_DATA_68b	A reference to PDF_DATA.pdf.p7s, which has been uploaded in advance as PDF_DATA_68.pdf.p7s.
REF_XML_MDO_68b	A reference to XML_MDO.xml, which has been uploaded in advance as XML_MDO_68.xml.
REF_XML_DATA_69b	A reference to XML_DATA.xml, which has been uploaded in advance as XML_DATA_69.xml.
REF_CADES_DET_NOK_XML_DATA_69b	A reference to XML_DATA.xml.p7s, which has been uploaded in advance as XML_DATA_NOK_69.xml.p7s.
REF_XML_DATA_70b	A reference to XML_DATA.xml, which has been uploaded in advance as XML_DATA_70.xml.
REF_CADES_DET_NOK_XML_DATA_70b	A reference to XML_DATA.xml.p7s, which has been uploaded in advance as XML_DATA_NOK_70.xml.p7s.
REF_TXT_DATA_71b	A reference to TXT_DATA.txt, which has been uploaded in advance as TXT_DATA_71.txt.

<i>Container or Object Name</i>	<i>Description</i>
REF_PDF_DATA_71b	A reference to PDF_DATA.pdf, which has been uploaded in advance as PDF_DATA_71.pdf
REF_XML_MDO_71b	A reference to XML_MDO.xml, which has been uploaded in advance as XML_MDO_71.xml.
REF_TXT_DATA_72b	A reference to TXT_DATA.txt, which has been uploaded in advance as TXT_DATA_72.txt.
REF_PDF_DATA_72b	A reference to PDF_DATA.pdf, which has been uploaded in advance as PDF_DATA_72.pdf.
REF_XML_MDO_72b	A reference to XML_MDO.xml, which has been uploaded in advance as XML_MDO_72.xml.
REF_CADES_ATT_TXT_DATA_73b	A reference to valid attached digital signature of TXT_DATA.txt, which has been uploaded in advance as TXT_DATA_73.txt.p7m.
REF_TXT_DATA_74b	A reference to TXT_DATA.txt, which has been uploaded in advance as TXT_DATA_74.txt.
REF_PDF_DATA_74b	A reference to PDF_DATA.pdf, which has been uploaded in advance as PDF_DATA_74.pdf.
REF_XML_MDO_74b	A reference to XML_MDO.xml, which has been uploaded in advance as XML_MDO_74.xml.

Table 2: *Definition of test objects*⁶

Furthermore, data elements of the type “anyType” are defined in the namespace “urn:oasis:names:tc:dss:1.0:core:schema” which can be found at <http://ws.openecard.org/schema/oasis-dss-core-schema-v1.0-os.xsd>.

⁶ The test objects are listed in order of appearance in the particular test cases.

3.4 Mandatory Tests

3.4.1 Tests of the TR-S.4-interface

3.4.1.1 Function RetrieveInfo of S.4-interface

All test cases defined in this section are derived from the general requirement A2.0-2 together with the interface specification of the RetrieveInfo function in section 3.8 of [TR-ESOR-E].

At least one active Preservation Profile is available:

<http://www.bsi.bund.de/tr-esor/V1.3.0/profile/S.4/v1.0>.

with URI= "<http://www.bsi.bund.de/tr-esor/V1.3.0/profile/S.4/V1.0>

At least one inactive Preservation Profile is available:

<http://www.bsi.bund.de/tr-esor/V1.3.0/profile/S.4/v0.05>

with URI = <http://www.bsi.bund.de/tr-esor/V1.3.0/profile/S.4/v0.5>

3.4.1.1.1 M-RI-01 – Retrieve Preservation Profile

Identifier	M-RI-01
Test Purpose	The test shall retrieve the actual Preservation Profile with URI= " http://www.bsi.bund.de/tr-esor/V1.3.0/profile/S.4/V1.0 "
Configuration	• CONFIG_S4
Pre-test conditions	• Authenticated connection to middleware exists. • At least one active Preservation Profile is available (ACTIVE_PROFILE) and one not-active (INACTIVE_PROFILE).

Step	Test sequence of M-RI-01	Expected Results
1	RetrieveInfoRequest (ProfileIdentifier(ACTIVE_PROFILE_URI))	RetrieveInfoResponse (dss:Result(resultmajor#ok), ACTIVE_PROFILE)
2	RetrieveInfoRequest (Status(inactive))	RetrieveInfoResponse (dss:Result(resultmajor#ok), INACTIVE_PROFILE)
3	RetrieveInfoRequest (Status(all))	RetrieveInfoResponse (dss:Result(resultmajor#ok), ACTIVE_PROFILE, INACTIVE_PROFILE)
4	RetrieveInfoRequest (Status(active))	RetrieveInfoResponse (dss:Result(resultmajor#ok), ACTIVE_PROFILE)
5	RetrieveInfoRequest (ProfileIdentifier(ACTIVE_PROFILE_URI), Status(active))	RetrieveInfoResponse (dss:Result(resultmajor#ok), ACTIVE_PROFILE)
6	RetrieveInfoRequest (ProfileIdentifier(ACTIVE_PROFILE_URI), Status(inactive))	RetrieveInfoResponse (dss:Result(resultmajor#ok))
7	RetrieveInfoRequest (ProfileIdentifier(ACTIVE_PROFILE_URI), Status(all))	RetrieveInfoResponse (dss:Result(resultmajor#ok), ACTIVE_PROFILE)
8	RetrieveInfoRequest ()	RetrieveInfoResponse (dss:Result(resultmajor#ok), ACTIVE_PROFILE)

Verdict of M-RI-01
Observations:
Verdict:

3.4.1.1.2 M-RI-02 – RetrieveInfo with unknown URI

Identifier	M-RI-02
Test Purpose	The test shall verify that it will yield an error, if <code>RetrieveInfo</code> is called without providing a known URI. As the present call also contains the <code>UNKNOWN_PROFILE_URI</code> , the <code>dss:Result</code> needs to indicate “/resultminor/ar/unknownURI”.

Identifier	M-RI-02
Configuration	CONFIG_S.4
Pre-test conditions	Authenticated connection to middleware exists.

Step	Test sequence of M-RI-02	Expected Results
1	RetrieveInfoRequest (UNKNOWN_PROFILE_URI)	RetrieveInfoResponse (dss:Result(resultmajor#error, resultminor/ar1/unknownURI))
2	RetrieveInfoRequest (ACTIVE_PROFILE_URI)	RetrieveInfoResponse (dss:Result(resultmajor#ok), ACTIVE_PROFILE)

Verdict of M-RI-02
Observations:
Verdict:

3.4.1.1.3 M-RI-03 – RetrieveInfo with unknown control in OptionalInputs

Identifier	M-RI-03
Test Purpose	The test shall verify that there will be an error, if the request contains unknown controls in the <code>OptionalInputs</code> -element and the <code>RetrieveInfoRequest</code> will not be successful.
Configuration	CONFIG_S.4
Pre-test conditions	- Authenticated connection to middleware exists. - At least one active Preservation Profile is available with URI= "http://www.bsi.bund.de/tr-esor/V1.3.0/profile/S.4/V1.0" - The <code>OptionalInputs</code>-element and the <code>RetrieveInfoRequest</code> contains an unknown item.

Step	Test sequence of M-RI-03	Expected Results
1	RetrieveInfoRequest (dss:OptionalInputs(SomethingUnknown), ACTIVE_PROFILE_URI)	RetrieveInfoResponse (dss:Result(resultmajor#error, resultminor/ar1/notSupported))
2	RetrieveInfoRequest (ACTIVE_PROFILE_URI)	RetrieveInfoResponse (dss:Result(resultmajor#ok), ACTIVE_PROFILE)

Verdict of M-RI-03
Observations:
Verdict:

3.4.1.2 Function ArchiveSubmission of S.4-interface

All test cases defined in this section are derived from the requirement [TR ESOR M.1], (A4.1 1) and the general requirement [TR-ESOR-E], (A2.0-2) together with the interface specification of the ArchiveSubmission function in section 3.1 of [TR-ESOR-E].

3.4.1.2.1 M-SU-01 – XAIP_OK without AOID

Identifier	M-SU-01
Test Purpose	The test shall verify that it is possible to submit XAIP_OK without AOID.
Configuration	CONFIG_S.4
Pre-test conditions	Authenticated connection to middleware exists.

Step	Test sequence of M-SU-01	Expected Results
1	ArchiveSubmissionRequest (XAIP_OK)	ArchiveSubmissionResponse (dss:Result(resultmajor#ok), AOID-01)
2	ArchiveRetrievalRequest (AOID-01)	ArchiveRetrievalResponse (dss:Result(resultmajor#ok), XAIP_OK(AOID-01, VID-01))
3	ArchiveEvidenceRequest ⁷ (AOID-01)	ArchiveEvidenceResponse (dss:Result(resultmajor#ok), XAIP_ERS(AOID-01,VID-01, XAIP_OK_V1_ER_OK))
4	Test of correctness of the obtained XAIP_OK(AOID-01, VID-01) by using AIP-eIDAS-SigValidator VerifyRequest (dss:InputDocuments(dss:Document(dss:InlineXML(XAIP_OK(AOID-01, VID-01)))))	VerifyResponse(dss:Result(resultmajor#ok), dss:OptionalOutputs(dssvr:VerificationReport(dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:XAIPReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid) [...]))))

⁷ Make sure the tested TR-ESOR-system has already obtained the archive time stamp.

Step	Test sequence of M-SU-01	Expected Results
5	Test of correctness of the obtained XAIP_OK_V1_ER_OK by using ERVerifyTool. <pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:InlineXML(XAIP_OK(AOID-01, VID-01)))) dss:SignatureObject(dss:Other(XAIP_ERS(AOID-01,VID-01, XAIP_OK_V1_ER_OK))))) </pre>	<pre> VerifyResponse(dss:Result(resultmajor#ok), dss:OptionalOutputs(dssvr:VerificationReport(dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:EvidenceRecordReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid)) [...]))))) </pre>
6	<pre> ArchiveSubmissionRequest (XAIP_OK_SIG_OK) </pre>	<pre> ArchiveSubmissionResponse (dss:Result(resultmajor#ok), AOID-02) </pre>
7	<pre> ArchiveRetrievalRequest (AOID-02) </pre>	<pre> ArchiveRetrievalResponse (dss:Result(resultmajor#ok), XAIP_OK_SIG_OK(AOID-02, VID-01)) </pre>
8	<pre> ArchiveEvidenceRequest⁷ (AOID-02) </pre>	<pre> ArchiveEvidenceResponse (dss:Result(resultmajor#ok), XAIP_ERS(AOID-02,VID-01, XAIP_OK_SIG_OK_V1_ER_OK)) </pre>

Verdict of M-SU-01

Observations:

Verdict:

3.4.1.2.2 M-SU-01a – LXAIP_OK without AOID

Identifier	M-SU-01a
Test Purpose	The test shall verify that it is possible to submit XAIP_OK without AOID.
Configuration	- CONFIG_S4 - CONFIG_LXAIP
Pre-test conditions	Authenticated connection to middleware exists.

Step	Test sequence of M-SU-01a	Expected Results
1	To be done manually! Following files have to be uploaded into middleware via upload/download-module (cf. chapters 4.2 and 4.3): 1) TXT_DATA_50.txt 2) XML_MDO_50.xml 3) PDF_DATA_50.pdf 4) TXT_DATA_51.txt 5) TXT_DATA_51.txt.p7s 6) PDF_DATA_51.pdf 7) PDF_DATA_51.pdf.p7s 8) XML_MDO_51.xml 9) TXT_DATA_52.txt 10) PDF_DATA_52.pdf 11) XML_MDO_52.xml 12) TXT_DATA_53.txt.p7m 13) XML_DATA_54.xml 14) XML_DATA_54.xml.p7s 15) TXT_DATA_55.txt 16) PDF_DATA_55.pdf 17) XML_MDO_55.xml 18) XML_DATA_56.xml 19) XML_DATA_56.xml.p7s 20) TXT_DATA_57.txt.p7m 21) TXT_DATA_58.txt 22) PDF_DATA_58.pdf 23) XML_MDO_58.xml 24) TXT_DATA_59.txt.p7m	As a result following data structures (proper instances of <code>basic:DataObjectReferences</code>) will be delivered: 1) REF_TXT_DATA_50 2) REF_XML_MDO_50 3) REF_PDF_DATA_50 4) REF_TXT_DATA_51 5) REF_CADES_DET_TXT_DATA_51 6) REF_PDF_DATA_51 7) REF_CADES_DET_PDF_DATA_51 8) REF_XML_MDO_51 9) REF_TXT_DATA_52 10) REF_PDF_DATA_52 11) REF_XML_MDO_52 12) REF_CADES_ATT_TXT_DATA_53 13) REF_XML_DATA_54 14) REF_CADES_DET_XML_DATA_54 15) REF_TXT_DATA_55 16) REF_PDF_DATA_55 17) REF_XML_MDO_55 18) REF_XML_DATA_56 19) REF_CADES_DET_XML_DATA_56 20) REF_CADES_ATT_TXT_DATA_57 21) REF_TXT_DATA_58 22) REF_PDF_DATA_58 23) REF_XML_MDO_58 24) REF_CADES_ATT_TXT_DATA_59

Step	Test sequence of M-SU-01a	Expected Results
1	To be done manually! Following files have to be uploaded into middleware via upload/download-module (cf. chapters 4.2 and 4.3): 25) TXT_DATA_60.txt.p7m 26) XML_DATA_60-1.xml 27) XML_DATA_60-1.xml.p7s 28) TXT_DATA_61.txt 29) TXT_DATA_61.txt.p7s 30) PDF_DATA_61.pdf 31) PDF_DATA_61.pdf.p7s 32) XML_MDO_61.xml 33) XML_DATA_62.xml 34) XML_DATA_62.xml.p7s 35) XML_DATA_63.xml 36) XML_DATA_63.xml.p7s 37) XML_DATA_64.xml 38) XML_DATA_64.xml.p7s 39) TXT_DATA_65.txt 40) TXT_DATA_65_NOK.p7s 41) PDF_DATA_65.pdf 42) PDF_DATA_65.pdf.p7s 43) XML_MDO_65.xml 44) TXT_DATA_66.txt 45) TXT_DATA_66.txt.p7s 46) PDF_DATA_66.pdf 47) PDF_DATA_66.pdf.p7s 48) XML_MDO_66.xml 49) TXT_DATA_67.txt 50) TXT_DATA_NOK_67.txt.p7s 51) PDF_DATA_67.pdf 52) PDF_DATA_67.pdf.p7s 53) XML_MDO_67.xml 54) TXT_DATA_68.txt 55) TXT_DATA_68.txt.p7s 56) PDF_DATA_68.pdf 57) PDF_DATA_68.pdf.p7s 58) XML_MDO_68.xml 59) XML_DATA_69.xml 60) XML_DATA_NOK_69.xml.p7s	As a result following data structures (proper instances of <code>basic:DataObjectReferences</code>) will be delivered: 25) REF_CADES_ATT_TXT_DATA_60 26) REF_XML_DATA_60-1 27) REF_CADES_DET_XML_DATA_60-1 28) REF_TXT_DATA_61 29) REF_CADES_DET_TXT_DATA_61 30) REF_PDF_DATA_61 31) REF_CADES_DET_PDF_DATA_61 32) REF_XML_MDO_61 33) REF_XML_DATA_62 34) REF_CADES_DET_XML_DATA_62 35) REF_XML_DATA_63 36) REF_CADES_DET_XML_DATA_63 37) REF_XML_DATA_64 38) REF_CADES_DET_XML_DATA_64 39) REF_TXT_DATA_65 40) REF_CADES_DET_NOK_TXT_DATA_65 41) REF_PDF_DATA_65 42) REF_CADES_DET_PDF_DATA_65 43) REF_XML_MDO_65 44) REF_TXT_DATA_66 45) REF_CADES_DET_TXT_DATA_66 46) REF_PDF_DATA_66 47) REF_CADES_DET_PDF_DATA_66 48) REF_XML_MDO_66 49) REF_TXT_DATA_67 50) REF_CADES_DET_NOK_TXT_DATA_67 51) REF_PDF_DATA_67 52) REF_CADES_DET_PDF_DATA_67 53) REF_XML_MDO_67 54) REF_TXT_DATA_68 55) REF_CADES_DET_TXT_DATA_68 56) REF_PDF_DATA_68 57) REF_CADES_DET_PDF_DATA_68 58) REF_XML_MDO_68 59) REF_XML_DATA_69 60) REF_CADES_DET_NOK_XML_DATA_69
1	To be done manually! Following files have to be uploaded into middleware via upload/download-module (cf. chapters 4.2 and 4.3): 61) XML_DATA_70.xml 62) XML_DATA_NOK_70.xml.p7s 63) TXT_DATA_71.txt 64) PDF_DATA_71.pdf 65) XML_MDO_71.xml 66) TXT_DATA_72.txt 67) PDF_DATA_72.pdf 68) XML_MDO_72.xml 69) TXT_DATA_73.txt.p7m 70) TXT_DATA_74.txt 71) PDF_DATA_74.pdf 72) XML_MDO_74.xml The references in the particular LXAIP* and DLXAIP* test objects will be adjusted according to the results in step a) automatically, the references have to be inserted into the configuration file of the testbed.	As a result following data structures (proper instances of <code>basic:DataObjectReferences</code>) will be delivered: 61) REF_XML_DATA_70 62) REF_CADES_DET_NOK_XML_DATA_70 63) REF_TXT_DATA_71 64) REF_PDF_DATA_71 65) REF_XML_MDO_71 66) REF_TXT_DATA_72 67) REF_PDF_DATA_72 68) REF_XML_MDO_72 69) REF_CADES_ATT_TXT_DATA_73 70) REF_TXT_DATA_74 71) REF_PDF_DATA_74 72) REF_XML_MDO_74

Step	Test sequence of M-SU-01a	Expected Results
2	To be done manually! Try to upload the XAIP_OK by using the upload/download interface of the middleware.	The upload-operation failed. Some kind of the machine-readable error code is delivered back. No data has been stored in the Storage.
3	To be done manually! Try to upload the LXAIP_OK by using the upload/download interface of the middleware.	The upload-operation failed. Some kind of the machine-readable error code is delivered back. No data has been stored in the Storage.
4	To be done manually! Try to upload the ASiC_AIP_OK by using the upload/download interface of the middleware.	The upload-operation failed. Some kind of the machine-readable error code is delivered back. No data has been stored in the Storage.
5	ArchiveSubmissionRequest (LXAIP_OK(REF_TXT_DATA_50, REF_PDF_DATA_50, REF_XML_MDO_50)))	ArchiveSubmissionResponse (dss:Result(resultmajor#ok), AOID-01a)
6	ArchiveRetrievalRequest (dss:OptionalInputs(pres:POFormat(http://www.bsi.bund.de/tr-esor/lxaip/1.3)), AOID-01a)	ArchiveRetrievalResponse (dss:Result(resultmajor#ok), LXAIP_OK(AOID-01a, VID-01))
7	ArchiveEvidenceRequest ⁸ (AOID-01a)	ArchiveEvidenceResponse (dss:Result(resultmajor#ok), XAIP_ERS(AOID-01a, VID-01, LXAIP_OK_V1_ER_OK))
8	Make sure all in LXAIP_OK(AOID-01a, VID-01) referenced files have been copied into corresponding LXAIP-directory of the AIP-eIDAS-SigValidator adjust the given references in in LXAIP_OK(AOID-01a, VID-01) in order to get them resolveable by the AIP-eIDAS-SigValidator.	LXAIP_OK_XSV(AOID-01a, VID-01)

⁸ Make sure the tested TR-ESOR-system has already obtained the archive time stamp.

Step	Test sequence of M-SU-01a	Expected Results
9	Test of correctness of the obtained LXAIP_OK(AOID-01, VID-01) by using AIP-eIDAS-SigValidator <pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:InlineXML(LXAIP_OK_XSV(AOID-01a, VID-01))))) </pre>	<pre> VerifyResponse(dss:Result(resultmajor#ok), dss:OptionalOutputs(dssvr:VerificationReport(dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid </pre>
10	Test of correctness of the obtained LXAIP_OK_V1_ER_OK by using ERVerifyTool. <pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:InlineXML(LXAIP_OK_XSV(AOID-01a, VID-01)))) dss:SignatureObject(dss:Other(XAIP_ERS(AOID-01a, VID-01, LXAIP_OK_V1_ER_OK))))) </pre>	<pre> VerifyResponse(dss:Result(resultmajor#ok), dss:OptionalOutputs(dssvr:VerificationReport(dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid </pre>
11	<pre> ArchiveSubmissionRequest (LXAIP_OK_SIG_OK(REF_TXT_DATA_51, REF_CADES_DET_TXT_DATA_51, REF_PDF_DATA_51, REF_CADES_DET_PDF_DATA_51, REF_XML_MDO_51)) </pre>	<pre> ArchiveSubmissionResponse (dss:Result(resultmajor#ok), AOID-02a) </pre>
12	<pre> ArchiveRetrievalRequest (dss:OptionalInputs(pres:POFormat(http://www.bsi.bund.de/tr-esor/lxaip/1.3)), AOID-02a) </pre>	<pre> ArchiveRetrievalResponse (dss:Result(resultmajor#ok), LXAIP_OK_SIG_OK(AOID-02a, VID-01)) </pre>

Step	Test sequence of M-SU-01a	Expected Results
13	ArchiveEvidenceRequest ⁸ (AOID-02a)	ArchiveEvidenceResponse (dss:Result(resultmajor#ok), XAIP_ERS(AOID-02a, VID-01, LXAIP_OK_SIG_OK_V1_ER_OK))

Verdict of M-SU-01a
Observations:
Verdict:

3.4.1.2.3 M-SU-02 – XAIP_NOK

Identifier	M-SU-02
Test Purpose	The test shall verify that there will be an error, if a (syntactically incorrect) XAIP_NOK is submitted.
Configuration	CONFIG_S.4
Pre-test conditions	Authenticated connection to middleware exists.

Step	Test sequence M-SU-02	Expected Results
1	ArchiveSubmissionRequest (XAIP_NOK)	ArchiveSubmissionResponse (dss:Result(resultmajor#error, resultminor/ar1/XAIP_NOK)) Alternatively response to the one above could be following or similar: <soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/" > <soap:Body> <soap:Fault> <faultcode>soap:Client</faultcode> <faultstring>Problems creating SAAJ object model</faultstring> </soap:Fault> </soap:Body> </soap:Envelope>

Step	Test sequence M-SU-02	Expected Results
2	ArchiveSubmissionRequest (XAIP_NOK_VERSION)	ArchiveSubmissionResponse (dss:Result(resultmajor#error, resultminor/ar1/XAIP_NOK)) Alternatively response to the one above could be following or similar: <soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/" > <soap:Body> <soap:Fault> <faultcode>soap:Client</faultcode> <faultstring>Problems creating SAAJ object model</faultstring> </soap:Fault> </soap:Body> </soap:Envelope>

Verdict of M-SU-02
Observations:
Verdict:

3.4.1.2.4 M-SU-02a – LXAIP_NOK

Identifier	M-SU-02a
Test Purpose	The test shall verify that there will be an error, if a (syntactically incorrect, missing element <code>ds:DigestValue</code> in element <code>asic:DataObjectReference</code>) LXAIP_NOK is submitted.
Configuration	- CONFIG_S.4 - CONFIG_LXAIP
Pre-test conditions	Authenticated connection to middleware exists.

Step	Test sequence M-SU-02a	Expected Results
1	ArchiveSubmissionRequest (LXAIP_NOK)	ArchiveSubmissionResponse (dss:Result(resultmajor#error, resultminor/ar1/XAIP_NOK)) Alternatively response to the one above could be following or similar: <soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/" > <soap:Body> <soap:Fault> <faultcode>soap:Client</faultcode> <faultstring>Problems creating SAAJ object model</faultstring> </soap:Fault> </soap:Body> </soap:Envelope>

Verdict of M-SU-02a
Observations:
Verdict:

3.4.1.2.5 M-SU-03 – XAIP_NOK_EXPIRED

Identifier	M-SU-03
Test Purpose	The test shall verify that there will be an error, if the semantically incorrect XAIP_NOK_EXPIRED is submitted.
Configuration	CONFIG_S.4
Pre-test conditions	Authenticated connection to middleware exists.

Step	Test sequence	Expected Results
1	ArchiveSubmissionRequest (XAIP_NOK_EXPIRED)	ArchiveSubmissionResponse (dss:Result(resultmajor#error, resultminor/ar1/XAIP_NOK_EXPIRED))

Verdict of M-SU-03
Observations:
Verdict:

3.4.1.2.6 M-SU-04 – XAIP_NOK_SUBMTIME

Identifier	M-SU-04
Test Purpose	The test shall verify that there will be an error, if the semantically incorrect XAIP_NOK_SUBMTIME is submitted.
Configuration	CONFIG_S.4
Pre-test conditions	Authenticated connection to middleware exists.

Step	Test sequence	Expected Results
1	Prepare XAIP_NOK_SUBMTIME, set the value of the element <code>xaip:submissionTime</code> in <code>xaip:submissionInfo</code> to the time in the past	XAIP_NOK_SUBMTIME(past)
2	ArchiveSubmissionRequest (XAIP_NOK_SUBMTIME(past))	ArchiveSubmissionResponse (dss:Result(resultmajor#error, resultminor/ar1/XAIP_NOK_SUBMTIME))
3	Prepare XAIP_NOK_SUBMTIME, set the value of the element <code>xaip:submissionTime</code> in <code>xaip:submissionInfo</code> to the time in the future	XAIP_NOK_SUBMTIME(future)
4	ArchiveSubmissionRequest (XAIP_NOK_SUBMTIME(future))	ArchiveSubmissionResponse (dss:Result(resultmajor#error, resultminor/ar1/XAIP_NOK_SUBMTIME))

Verdict of M-SU-04

Observations:

Verdict:

3.4.1.2.7 M-SU-05 – XAIP_NOK_SIG

Identifier	M-SU-05
Test Purpose	The test shall verify that there will be an error, if the submitted XAIP_NOK_SIG contains digital signatures, which do not verify correctly ⁹ .
Configuration	CONFIG_S.4
Pre-test conditions	- Authenticated connection to middleware exists. - The TOT is configured only to accept valid signatures.

Step	Test sequence	Expected Results
1	ArchiveSubmissionRequest (XAIP_NOK_SIG)	ArchiveSubmissionResponse (dss:Result(resultmajor#error, resultminor/ar1/XAIP_NOK_SIG) dss:OptionalOutputs(XAIP_NOK_SIG_VR))

Verdict of M-SU-05

Observations:

Verdict:

⁹ See also Annex E (Version 1.3, Chapter 3.7.1 (VerifyUnderSignaturePolicy)).

3.4.1.2.8 M-SU-06 – XAIP_NOK_ER

Identifier	M-SU-06
Test Purpose	The test shall verify that there will be an error, because the submitted XAIP_NOK_ER contains an incorrect Evidence Record.
Configuration	CONFIG_S4
Pre-test conditions	Authenticated connection to middleware exists.

Step	Test sequence	Expected Results
1	ArchiveSubmissionRequest (XAIP_NOK_ER)	ArchiveSubmissionResponse (dss:Result(resultmajor#error, resultminor/arl/XAIP_NOK_ER) dss:OptionalOutputs(XAIP_NOK_ER_VR))

Verdict of M-SU-06
Observations:
Verdict:

3.4.1.2.9 M-SU-06a – LXAIP_OK_ER_NOK

Identifier	M-SU-06a
Test Purpose	The test shall verify that there will be an error, because the submitted LXAIP_OK_ER_NOK contains an incorrect Evidence Record.
Configuration	- CONFIG_S4 - CONFIG_LXAIP
Pre-test conditions	- Authenticated connection to middleware exists. - Test case M-SU-01a has been performed successfully.

Step	Test sequence	Expected Results
1	ArchiveSubmissionRequest (LXAIP_OK_ER_NOK(REF_TXT_DATA_52, REF_PDF_DATA_52, REF_XML_MDO_52, ER_NOK_LXAIP_OK))	ArchiveSubmissionResponse (dss:Result(resultmajor#error, resultminor/arl/XAIP_NOK_ER) dss:OptionalOutputs(LXAIP_NOK_ER_VR))

Verdict of M-SU-06a
Observations:

Verdict of M-SU-06a

Verdict:

3.4.1.2.10 M-SU-07 – XAIP_OK and unknown control in OptionalInputs

Identifier	M-SU-07
Test Purpose	The test shall verify that there will be an error, if the request contains unknown controls in the <code>OptionalInputs</code> -element and the archive data object will not be imported.
Configuration	CONFIG_S.4
Pre-test conditions	Authenticated connection to middleware exists.

Step	Test sequence	Expected Results
1	ArchiveSubmissionRequest (dss:OptionalInputs(SomethingUnknown), XAIP_OK)	ArchiveSubmissionResponse (dss:Result(resultmajor#error, resultminor/ar1/notSupported),)

Verdict of M-SU-07

Observations:

Verdict:

3.4.1.2.11 M-SU-08 – BIN_OK

Identifier	M-SU-08
Test Purpose	The test shall validate that it is possible to submit as BIN defined data. BIN means here: - CADES_ATT_OK, - XADES_ENVELOPED_OK, XADES_ENVELOPING_OK, - PADES_OK, - ASiC_S_CADES_OK, ASiC_S_XADES_OK, ASiC_S_TST_OK, ASiC_S_ER_OK, - ASiC_E_CADES_OK, ASiC_E_XADES_OK, ASiC_E_TST_OK, ASiC_E_ER_OK, - DIGEST_LIST.
Configuration	CONFIG_S.4
Pre-test conditions	Authenticated connection to middleware exists.

Step	Test sequence	Expected Results
1	ArchiveSubmissionRequest (ArchiveData(http://uri.etsi.org/ades/CADES , AD-01, CADES_ATT_OK))	ArchiveSubmissionResponse (dss:Result(resultmajor#ok), AOID-200)
2	ArchiveRetrievalRequest (AOID-200)	ArchiveRetrievalResponse (dss:Result(resultmajor#ok), XAIP(AOID-200, VID-01, credential(AD-01, CADES_ATT_OK))))
3	ArchiveSubmissionRequest (ArchiveData(http://uri.etsi.org/ades/XAdES , AD-01, XADES_ENVELOPED_OK))	ArchiveSubmissionResponse (dss:Result(resultmajor#ok), AOID-201)
4	ArchiveRetrievalRequest (AOID-201)	ArchiveRetrievalResponse (dss:Result(resultmajor#ok), XAIP(AOID-201, VID-01, dataObject(AD-01, xmlData(XADES_ENVELOPED_OK)))) OR ¹⁰ ArchiveRetrievalResponse (dss:Result(resultmajor#ok), XAIP(AOID-201, VID-01, dataObject(AD-01, binaryData(XADES_ENVELOPED_OK))))
5	ArchiveSubmissionRequest (ArchiveData(http://uri.etsi.org/ades/XAdES , AD-01, XADES_ENVELOPING_OK))	ArchiveSubmissionResponse (dss:Result(resultmajor#ok), AOID-202)

¹⁰ An enveloped XAdES could be placed in XAIP as binary data stream in an element `<xaip:binaryData>` (as recommended in [TR-ESOR-F], section 6.2), or as xml data stream in an element `<xaip:xmlData>` in the `<xaip:dataObjectsSection>`-element.

Step	Test sequence	Expected Results
6	ArchiveRetrievalRequest (AOID-202)	ArchiveRetrievalResponse (dss:Result(resultmajor#ok), XAIP(AOID-202, VID-01, credential(AD-01, SignatureObject(XADES_ENVELOPING_OK))))) OR ¹¹ ArchiveRetrievalResponse (dss:Result(resultmajor#ok), XAIP(AOID-202, VID-01, credential(AD-01, Base64Signature(XADES_ENVELOPING_OK)))))
7	ArchiveSubmissionRequest (ArchiveData(http://uri.etsi.org/ades/PADES , AD-01, PADES_OK))	ArchiveSubmissionResponse (dss:Result(resultmajor#ok), AOID-203)
8	ArchiveRetrievalRequest (AOID-203)	ArchiveRetrievalResponse (dss:Result(resultmajor#ok), XAIP(AOID-203, VID-01, dataObject(AD-01, PADES_OK))))
9	ArchiveSubmissionRequest (ArchiveData(http://uri.etsi.org/ades/ASiC/type/ASiC-E , AD-01, ASiC_E_CADES_OK))	ArchiveSubmissionResponse (dss:Result(resultmajor#ok), AOID-204)
10	ArchiveRetrievalRequest (AOID-204)	ArchiveRetrievalResponse (dss:Result(resultmajor#ok), XAIP(AOID-204, VID-01, credential(AD-01, Base64Signature(ASiC_E_CADES_OK)))))

¹¹ An enveloping XAdES could be placed in XAIP as binary credential data stream in an element `<dss:base64Signature>` (as recommended in [TR-ESOR-F], section 6.2), or as xml credential data stream directly as element `<dss:Signature>` in the `<dss:SignatureObject>`-element of the `<xaip:credential>` parent element.

Step	Test sequence	Expected Results
11	ArchiveSubmissionRequest (ArchiveData(http://uri.etsi.org/ades/ASiC/type/ASiC-E , AD-01, ASIC_E_XADES_OK))	ArchiveSubmissionResponse (dss:Result(resultmajor#ok), AOID-205)
12	ArchiveRetrievalRequest (AOID-205)	ArchiveRetrievalResponse (dss:Result(resultmajor#ok), XAIP(AOID-205, VID-01, credential(AD-01, Base64Signature(ASIC_E_XADES_OK)))))
13	ArchiveSubmissionRequest (ArchiveData(http://uri.etsi.org/ades/ASiC/type/ASiC-E , AD-01, ASIC_E_TST_OK))	ArchiveSubmissionResponse (dss:Result(resultmajor#ok), AOID-206)
14	ArchiveRetrievalRequest (AOID-206)	ArchiveRetrievalResponse (dss:Result(resultmajor#ok), XAIP(AOID-206, VID-01, credential(AD-01, Base64Signature(ASIC_E_TST_OK)))))
15	ArchiveSubmissionRequest (ArchiveData(http://uri.etsi.org/ades/ASiC/type/ASiC-E , AD-01, ASIC_E_ER_OK))	ArchiveSubmissionResponse (dss:Result(resultmajor#ok), AOID-207)
16	ArchiveRetrievalRequest (AOID-207)	ArchiveRetrievalResponse (dss:Result(resultmajor#ok), XAIP(AOID-207, VID-01, credential(AD-01, Base64Signature(ASIC_E_ER_OK)))))
17	ArchiveSubmissionRequest (ArchiveData(http://uri.etsi.org/19512/format/DigestList , AD-01, DIGEST_LIST))	ArchiveSubmissionResponse (dss:Result(resultmajor#ok), AOID-208)

Step	Test sequence	Expected Results
18	ArchiveRetrievalRequest (AOID-208)	ArchiveRetrievalResponse (dss:Result(resultmajor#ok), XAIP(AOID-208, VID-01, dataObject(AD-01, xmlData(DIGEST_LIST)))) OR ¹² ArchiveRetrievalResponse (dss:Result(resultmajor#ok), XAIP(AOID-208, VID-01, dataObject(AD-01, binaryData(DIGEST_LIST))))
19	ArchiveSubmissionRequest (ArchiveData(http://uri.etsi.org/ades/ASiC/type/ASiC-S , AD-01, ASiC_S_CADES_OK))	ArchiveSubmissionResponse (dss:Result(resultmajor#ok), AOID-209)
20	ArchiveRetrievalRequest (AOID-209)	ArchiveRetrievalResponse (dss:Result(resultmajor#ok), XAIP(AOID-209, VID-01, credential(AD-01, Base64Signature(ASiC_S_CADES_OK)))))
21	ArchiveSubmissionRequest (ArchiveData(http://uri.etsi.org/ades/ASiC/type/ASiC-S , AD-01, ASiC_S_XADES_OK))	ArchiveSubmissionResponse (dss:Result(resultmajor#ok), AOID-210)
22	ArchiveRetrievalRequest (AOID-210)	ArchiveRetrievalResponse (dss:Result(resultmajor#ok), XAIP(AOID-210, VID-01, credential(AD-01, Base64Signature(ASiC_S_XADES_OK)))))

¹² A digest list as XML based data object could be placed in XAIP as binary data stream in an element `<xaip:binaryData>` or as xml data stream in an element `<xaip:xmlData>` in the `<xaip:dataObjectsSection>`-element.

Step	Test sequence	Expected Results
23	ArchiveSubmissionRequest (ArchiveData(http://uri.etsi.org/ades/ASiC/type/ASiC-S , AD-01, ASIC_S_TST_OK))	ArchiveSubmissionResponse (dss:Result(resultmajor#ok), AOID-211)
24	ArchiveRetrievalRequest (AOID-211)	ArchiveRetrievalResponse (dss:Result(resultmajor#ok), XAIP(AOID-211, VID-01, credential(AD-01, Base64Signature(ASIC_S_TST_OK)))))
25	ArchiveSubmissionRequest (ArchiveData(http://uri.etsi.org/ades/ASiC/type/ASiC-S , AD-01, ASIC_S_ER_OK))	ArchiveSubmissionResponse (dss:Result(resultmajor#ok), AOID-212)
26	ArchiveRetrievalRequest (AOID-212)	ArchiveRetrievalResponse (dss:Result(resultmajor#ok), XAIP(AOID-212, VID-01, credential(AD-01, Base64Signature(ASIC_S_ER_OK)))))

Verdict of M-SU-08

Observations:

Verdict:

3.4.1.2.12 M-SU-09 – BIN_NOK

Identifier	M-SU-09
Test Purpose	The test shall validate that there will be an error in case an as BIN defined data containing an invalid signature will be submitted. BIN means here: • CADES_ATT_NOK • XADES_ENVELOPED_NOK, XADES_ENVELOPING_NOK, • PADES_NOK, • ASIC_E_CADES_NOK, ASIC_E_XADES_NOK, ASIC_E_TST_NOK, ASIC_E_ER_NOK, • ASIC_S_CADES_NOK, ASIC_S_XADES_NOK, ASIC_S_TST_NOK, ASIC_S_ER_NOK.
Configuration	• CONFIG_S.4
Pre-test conditions	• Authenticated connection to middleware exists. • The TOT is configured only to accept valid signatures.

Step	Test sequence	Expected Results
1	ArchiveSubmissionRequest (ArchiveData(http://uri.etsi.org/ades/CADES , AD-01, CADES_ATT_NOK))	ArchiveSubmissionResponse (dss:Result(resultmajor#error, resultminor/sal#invalidSignature) dss:OptionalOutputs(CADES_ATT_NOK_VR))
2	ArchiveSubmissionRequest (ArchiveData(http://uri.etsi.org/ades/XAdES , AD-01, XADES_ENVELOPED_NOK))	ArchiveSubmissionResponse (dss:Result(resultmajor#error, resultminor/sal#invalidSignature) dss:OptionalOutputs(XADES_ENVELOPED_NOK_VR))
3	ArchiveSubmissionRequest (ArchiveData(http://uri.etsi.org/ades/XAdES , AD-01, XADES_ENVELOPING_NOK))	ArchiveSubmissionResponse (dss:Result(resultmajor#error, resultminor/sal#invalidSignature) dss:OptionalOutputs(XADES_ENVELOPING_NOK_VR))
4	ArchiveSubmissionRequest (ArchiveData(http://uri.etsi.org/ades/PADES , AD-01, PADES_NOK))	ArchiveSubmissionResponse (dss:Result(resultmajor#error, resultminor/sal#invalidSignature) dss:OptionalOutputs(PADES_NOK_VR))
5	ArchiveSubmissionRequest (ArchiveData(http://uri.etsi.org/ades/ASiC/type/ASiC-E , AD-01, ASIC_E_CADES_NOK))	ArchiveSubmissionResponse (dss:Result(resultmajor#error, resultminor/sal#invalidSignature) dss:OptionalOutputs(ASIC_E_CADES_NOK_VR))
6	ArchiveSubmissionRequest (ArchiveData(http://uri.etsi.org/ades/ASiC/type/ASiC-E , AD-01, ASIC_E_XADES_NOK))	ArchiveSubmissionResponse (dss:Result(resultmajor#error, resultminor/sal#invalidSignature) dss:OptionalOutputs(ASIC_E_XADES_NOK_VR))
7	ArchiveSubmissionRequest (ArchiveData(http://uri.etsi.org/ades/ASiC/type/ASiC-E , AD-01, ASIC_E_TST_NOK))	ArchiveSubmissionResponse (dss:Result(resultmajor#error, resultminor/sal#invalidSignature) dss:OptionalOutputs(ASIC_E_TST_NOK_VR))

Step	Test sequence	Expected Results
8	ArchiveSubmissionRequest (ArchiveData(http://uri.etsi.org/ades/ASiC/type/ASiC-E , AD-01, ASIC_E_ER_NOK))	ArchiveSubmissionResponse (dss:Result(resultmajor#error, resultminor/sal#invalidSignature) dss:OptionalOutputs(ASIC_E_ER_NOK_VR))
9	ArchiveSubmissionRequest (ArchiveData(http://uri.etsi.org/ades/ASiC/type/ASiC-S , AD-01, ASIC_S_CADES_NOK))	ArchiveSubmissionResponse (dss:Result(resultmajor#error, resultminor/sal#invalidSignature) dss:OptionalOutputs(ASIC_S_CADES_NOK_VR))
10	ArchiveSubmissionRequest (ArchiveData(http://uri.etsi.org/ades/ASiC/type/ASiC-S , AD-01, ASIC_S_XADES_NOK))	ArchiveSubmissionResponse (dss:Result(resultmajor#error, resultminor/sal#invalidSignature) dss:OptionalOutputs(ASIC_S_XADES_NOK_VR))
11	ArchiveSubmissionRequest (ArchiveData(http://uri.etsi.org/ades/ASiC/type/ASiC-S , AD-01, ASIC_S_TST_NOK))	ArchiveSubmissionResponse (dss:Result(resultmajor#error, resultminor/sal#invalidSignature) dss:OptionalOutputs(ASIC_S_TST_NOK_VR))
12	ArchiveSubmissionRequest (ArchiveData(http://uri.etsi.org/ades/ASiC/type/ASiC-S , AD-01, ASIC_S_ER_NOK))	ArchiveSubmissionResponse (dss:Result(resultmajor#error, resultminor/sal#invalidSignature) dss:OptionalOutputs(ASIC_S_ER_NOK_VR))

Verdict of M-SU-09

Observations:

Verdict:

3.4.1.3 Function ArchiveUpdate of S.4-interface

All test cases defined in this section are derived from the requirement A4.2-3 and the general requirement A2.0-2 together with the interface specification of the `ArchiveUpdate` function in section 3.2 of [TR-ESOR-E]. According to section 2 of [TR-ESOR-E], `ArchiveUpdateRequest` and `ArchiveUpdateResponse` shall be supported.

3.4.1.3.1 M-UP-01 – DXAIP_OK

Identifier	M-UP-01
Test Purpose	The test shall verify that it is possible to submit a correct DXAIP_OK with <code>ArchiveUpdateRequest</code> .
Configuration	CONFIG_S.4
Pre-test conditions	- Authenticated connection to middleware exists. - Make sure that test case M-SU-01 was successfully performed with AOID-01.

Step	Test sequence	Expected Results
1	Insert AOID-01 into the prepared DXAIP_OK-template.	DXAIP_OK(AOID-01, VID-01, VID-02)
2	ArchiveUpdateRequest (DXAIP_OK(AOID-01, VID-01, VID-02))	ArchiveUpdateResponse (dss:Result(resultmajor#OK), VersionID(VID-02))
3	Insert AOID-01 into the prepared DXAIP_OK_2-template.	DXAIP_OK_2(AOID-01, VID-02, VID-03)
4	ArchiveUpdateRequest (DXAIP_OK_2(AOID-01, VID-02, VID-03))	ArchiveUpdateResponse (dss:Result(resultmajor#OK), VersionID(VID-03))
5	ArchiveSubmissionRequest (XAIP_OK)	ArchiveSubmissionResponse (dss:Result(resultmajor#ok), AOID-01z)
6	Insert AOID-01z into the prepared DXAIP_OK_SIG_ONLY-template.	DXAIP_OK_SIG_ONLY(AOID-01z, VID-01, VID-02)
7	ArchiveUpdateRequest (DXAIP_OK_SIG_ONLY(AOID-01z, VID-01, VID-02)))	ArchiveUpdateResponse (dss:Result(resultmajor#OK), VersionID(VID-02))

Verdict of M-UP-01
Observations:
Verdict:

3.4.1.3.2 M-UP-01a – DLXAIP_OK

Identifier	M-UP-01a
Test Purpose	The test shall verify that it is possible to submit a correct DLXAIP_OK with ArchiveUpdateRequest.
Configuration	- CONFIG_S4 - CONFIG_LXAIP
Pre-test conditions	- Authenticated connection to middleware exists. - Make sure that test case M-UP-01a was successfully performed with AOID-01a.

Step	Test sequence	Expected Results
1	Insert AOID-01a into the prepared DLXAIP_OK-template.	DLXAIP_OK(AOID-01a, VID-01, VID-02)
2	ArchiveUpdateRequest (DLXAIP_OK(AOID-01a, VID-01, VID-02, REF_CADES_ATT_TXT_DATA_53))	ArchiveUpdateResponse (dss:Result(resultmajor#OK), VersionID(VID-02))
3	Insert AOID-01a into the prepared DLXAIP_OK_2-template.	DLXAIP_OK_2(AOID-01a, VID-02, VID-03)
4	ArchiveUpdateRequest (DLXAIP_OK_2(AOID-01a, VID-02, VID-03, REF_XML_DATA_54, REF_CADES_DET_XML_DATA_54))	ArchiveUpdateResponse (dss:Result(resultmajor#OK), VersionID(VID-03))

Verdict of M-UP-01a
Observations:
Verdict:

3.4.1.3.3 M-UP-02 – DXAIP_NOK_AOID

Identifier	M-UP-02
Test Purpose	The test shall verify that there will be an error, if a DXAIP_NOK_AOID with a not yet assigned AOID is submitted.
Configuration	CONFIG_S4
Pre-test conditions	Authenticated connection to middleware exists.

Step	Test sequence	Expected Results
1	Choice an AOID-03 within the possible range and make sure, that it has not been assigned yet. This is realised by the following call: ArchiveRetrievalRequest (AOID-03)	ArchiveRetrievalResponse (dss:Result(resultmajor#error, resultminor/ar1/unknownAOID))
2	Insert AOID-03 into DXAIP_NOK_AOID.	DXAIP_NOK_AOID(AOID-03)
3	ArchiveUpdateRequest (DXAIP_NOK_AOID(AOID-03))	ArchiveUpdateResponse (dss:Result(resultmajor#error, resultminor/ar1/DXAIP_NOK_AOID))

Verdict of M-UP-02

Observations:

Verdict:

3.4.1.3.4 M-UP-03 – DXAIP_NOK

Identifier	M-UP-03
Test Purpose	The test shall verify that there will be an error, if a (syntactically incorrect) DXAIP_NOK is submitted.
Configuration	CONFIG_S.4
Pre-test conditions	Authenticated connection to middleware exists.

Step	Test sequence	Expected Results
1	ArchiveSubmissionRequest (XAIP_OK)	ArchiveSubmissionResponse (dss:Result(resultmajor#ok, AOID-04)
2	Insert AOID-04 into the prepared DXAIP_NOK-template.	DXAIP_NOK(AOID-04, VID-01, VID-02)

Step	Test sequence	Expected Results
3	ArchiveUpdateRequest (DXAIP_NOK(AOID-04, VID-01, VID-02))	ArchiveUpdateResponse (dss:Result(resultmajor#error, resultminor/ar1/DXAIP_NOK)) Alternatively response to the one above could be following or similar: <soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"> <soapenv:Body> <soapenv:Fault> <faultcode>Server</faultcode> <faultstring>Failed to dispatch using script; java.lang.Exception: VALIDATION ERRORS: [line 1: Expected element 'protectedObjectPointer@http://www.bsi.bund.de/tr- esor/xaip' instead of 'AOID@http://www.bsi.bund.de/tr- esor/xaip' here in element packageInfoUnit@http://www.bsi.bund.de/tr- esor/xaip]</faultstring> </soapenv:Fault> </soapenv:Body> </soapenv:Envelope>>

Verdict of M-UP-03

Observations:

Verdict:

3.4.1.3.5 M-UP-03a – DLXAIP_NOK

Identifier	M-UP-03a
Test Purpose	The test shall verify that there will be an error, if a (syntactically incorrect) DLXAIP_NOK is submitted.
Configuration	- CONFIG_S.4 - CONFIG_LXAIP
Pre-test conditions	Authenticated connection to middleware exists.

Step	Test sequence	Expected Results
1	ArchiveSubmissionRequest (LXAIP_OK(REF_TXT_DATA_55, REF_PDF_DATA_55, REF_XML_MDO_55))	ArchiveSubmissionResponse (dss:Result(resultmajor#ok), AOID-04a)
2	Insert AOID-04a into the prepared DLXAIP_NOK-template.	DLXAIP_NOK(AOID-04a, VID-01, VID-02)

Step	Test sequence	Expected Results
3	ArchiveUpdateRequest (DLXAIP_NOK(AOID-04a, VID-01, VID-02 REF_XML_DATA_56, REF_CADES_DET_XML_DATA_56))	ArchiveUpdateResponse (dss:Result(resultmajor#error, resultminor/arl/DXAIP_NOK)) Alternatively response to the one above could be following or similar: <soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"> <soapenv:Body> <soapenv:Fault> <faultcode>Server</faultcode> <faultstring>Failed to dispatch using script; java.lang.Exception: VALIDATION ERRORS: [line 1: Expected element 'protectedObjectPointer@http://www.bsi.bund.de/tr- esor/xaip' instead of 'AOID@http://www.bsi.bund.de/tr- esor/xaip' here in element packageInfoUnit@http://www.bsi.bund.de/tr- esor/xaip]</faultstring> </soapenv:Fault> </soapenv:Body> </soapenv:Envelope>

Verdict of M-UP-03a

Observations:

Verdict:

3.4.1.3.6 M-UP-04 – DXAIP_NOK_EXPIRED

Identifier	M-UP-04
Test Purpose	The test shall verify that there will be an error, if the invalid DXAIP_NOK_EXPIRED is submitted.
Configuration	CONFIG_S.4
Pre-test conditions	- Authenticated connection to middleware exists. - Make sure that test case M-UP-03 was successfully performed with AOID-04.

Step	Test sequence	Expected Results
1	Insert AOID-04 into the prepared DXAIP_NOK_EXPIRED-template.	DXAIP_NOK_EXPIRED(AOID-04, VID-01, VID-02)
2	ArchiveUpdateRequest (DXAIP_NOK_EXPIRED(AOID-04, VID-01, VID-02))	ArchiveUpdateResponse (dss:Result(resultmajor#error, resultminor/arl/DXAIP_NOK_EXPIRED))

Verdict of M-UP-04

Observations:

Verdict of M-UP-04

Verdict:

3.4.1.3.7 M-UP-05 – DXAIP_NOK_SUBMTIME

Identifier	M-UP-05
Test Purpose	The test shall verify that there will be an error, if the invalid DXAIP_NOK_SUBMTIME is submitted.
Configuration	CONFIG_S.4
Pre-test conditions	- Authenticated connection to middleware exists. - Make sure that test case M-UP-03 was successfully performed with AOID-04

Step	Test sequence	Expected Results
1	Insert AOID-04 into the prepared DXAIP_NOK_SUBMTIME-template Prepare DXAIP_NOK_SUBMTIME, set the value of the element <code>xaip:submissionTime</code> in <code>xaip:submissionInfo</code> to the time in the past.	DXAIP_NOK_SUBMTIME(AOID-04, VID-01, VID-02, SubmissionTime(in the past))
2	ArchiveUpdateRequest (DXAIP_NOK_SUBMTIME(AOID-04, VID-01, VID-02, SubmissionTime(in the past)))	ArchiveUpdateResponse (dss:Result(resultmajor#error, resultminor/ar1/DXAIP_NOK_SUBMTIME))
3	Insert AOID-04 into the prepared DXAIP_NOK_SUBMTIME-template Prepare DXAIP_NOK_SUBMTIME, set the value of the element <code>xaip:submissionTime</code> in <code>xaip:submissionInfo</code> to the time in the future	DXAIP_NOK_SUBMTIME(AOID-04, VID-01, VID-02, SubmissionTime(in the future))
4	ArchiveUpdateRequest (DXAIP_NOK_SUBMTIME(AOID-04, VID-01, VID-02, SubmissionTime(in the future)))	ArchiveUpdateResponse (dss:Result(resultmajor#error, resultminor/ar1/DXAIP_NOK_SUBMTIME))

Verdict of M-UP-05

Observations:

Verdict:

3.4.1.3.8 M-UP-06 – DXAIP_NOK_SIG

Identifier	M-UP-06
Test Purpose	The test shall verify that there will be an error, if the submitted DXAIP_NOK_SIG contains digital signatures, which do not verify correctly ¹³ .
Configuration	CONFIG_S.4
Pre-test conditions	- Authenticated connection to middleware exists. - Make sure that test case M-UP-03 was successfully performed with AOID-04 - The TOT is configured only to accept valid signatures.

Step	Test sequence	Expected Results
1	Insert AOID-04 into the prepared DXAIP_NOK_SIG-template.	DXAIP_NOK_SIG(AOID-04, VID-01, VID-02)
2	ArchiveUpdateRequest (DXAIP_NOK_SIG(AOID-04, VID-01, VID-02)))	ArchiveUpdateResponse (dss:Result(resultmajor#error, resultminor/arl/DXAIP_NOK_SIG), dss:OptionalOutputs(DXAIP_NOK_SIG_VR(AOID-04, VID-01, VID-02))))

Verdict of M-UP-06
Observations:
Verdict:

3.4.1.3.9 M-UP-07 – DXAIP_NOK_ER

Identifier	M-UP-07
Test Purpose	The test shall verify that there will be an error, because the submitted DXAIP_NOK_ER contains an invalid Evidence Record.
Configuration	CONFIG_S.4
Pre-test conditions	- Authenticated connection to middleware exists. - Make sure that test case M-UP-03 was successfully performed with AOID-04.

Step	Test sequence	Expected Results
1	Insert AOID-04 into the prepared DXAIP_NOK_ER-template.	DXAIP_NOK_ER(AOID-04, VID-01, VID-02)
2	ArchiveUpdateRequest (DXAIP_NOK_ER(AOID-04, VID-01, VID-02)))	ArchiveUpdateResponse (dss:Result(resultmajor#error, resultminor/arl/DXAIP_NOK_ER), dss:OptionalOutputs(DXAIP_NOK_ER_VR(AOID-04, VID-01, VID-02))))

¹³ See Annex E (Version 1.2, Chapter 3.7.1, (VerifyUnderSignaturePolicy))

Verdict of M-UP-07
Observations:
Verdict:

3.4.1.3.10 M-UP-07a – DLXAIP_NOK_ER

Identifier	M-UP-07a
Test Purpose	The test shall verify that there will be an error, because the submitted DLXAIP_NOK_ER contains an invalid Evidence Record.
Configuration	- CONFIG_S4 - CONFIG_LXAIP
Pre-test conditions	- Authenticated connection to middleware exists. - Make sure that test case M-UP-03a was successfully performed with AOID-04a.

Step	Test sequence	Expected Results
1	Insert AOID-04a into the prepared DLXAIP_NOK_ER-template.	DLXAIP_NOK_ER(AOID-04a, VID-01, VID-02)
2	ArchiveUpdateRequest (DLXAIP_NOK_ER(AOID-04a, VID-01, VID-02,, REF_CADES_ATT_TXT_DATA_57, ER_NOK_LXAIP_OK_V2))	ArchiveUpdateResponse (dss:Result(resultmajor#error, resultminor/ar1/DXAIP_NOK_ER), dss:OptionalOutputs(DLXAIP_NOK_ER_VR(AOID-04a, VID-01, VID-02)))

Verdict of M-UP-07a
Observations:
Verdict:

3.4.1.3.11 M-UP-08 – DXAIP_NOK_VERSION

Identifier	M-UP-08
Test Purpose	The test shall verify that there will be an error, if a DXAIP_NOK_VERSION is submitted with ArchiveUpdateRequest, which produces a collision ¹⁴ with the already existing XAIP.
Configuration	CONFIG_S4

¹⁴ A wrong „version“ (DXAIP_NOK_Version) may be caused by inserting a DXAIP where the value in the element `prevVersion` in the `updateSection` of the DXAIP is not the latest version of this XAIP.

Identifier	M-UP-08
Pre-test conditions	- Authenticated connection to middleware exists. - Make sure that test case M-UP-03 was successfully performed with AOID-04.

Step	Test sequence	Expected Results
1	Insert AOID-04 into the prepared DXAIP_NOK_VERSION-template. Set the previous version to VID-10.	DXAIP_NOK_VERSION(AOID-04, VID-10, VID-02)
2	ArchiveUpdateRequest (DXAIP_NOK_VERSION(AOID-04, VID-10, VID-02)))	ArchiveUpdateResponse (dss:Result(resultmajor#error, resultminor/arl/DXAIP_NOK_Version)))
3	Insert AOID-04 into the prepared DXAIP_OK-template.	DXAIP_OK(AOID-04, VID-01, VID-02)
4	ArchiveUpdateRequest (DXAIP_OK(AOID-04, VID-01, VID-02)))	ArchiveUpdateResponse (dss:Result(resultmajor#OK), VersionID(VID-02)))
5	Insert AOID-04 into the prepared DXAIP_NOK_VERSION-template. Set the previous version to VID-01.	DXAIP_NOK_VERSION(AOID-04, VID-01, VID-03)
6	ArchiveUpdateRequest (DXAIP_NOK_VERSION(AOID-04, VID-01, VID-03)))	ArchiveUpdateResponse (dss:Result(resultmajor#error, resultminor/arl/DXAIP_NOK_Version)))

Verdict of M-UP-08
Observations:
Verdict:

3.4.1.3.12 M-UP-09 – DXAIP_OK with unknown control in OptionalInputs

Identifier	M-UP-09
Test Purpose	The test shall verify that there will be an error, if the request contains unknown controls in the OptionalInputs-element.
Configuration	CONFIG_S4
Pre-test conditions	Authenticated connection to middleware exists.

Step	Test sequence	Expected Results
1	ArchiveSubmissionRequest (XAIP_OK))	ArchiveSubmissionResponse (dss:Result(resultmajor#ok), AOID-05))

Step	Test sequence	Expected Results
2	Insert AOID-05 into the prepared DXAIP_OK-template.	DXAIP_OK(AOID-05, VID-01, VID-02)
3	ArchiveUpdateRequest (dss:OptionalInputs(SomethingUnknown), DXAIP_OK(AOID-05, VID-01, VID-02))	ArchiveUpdateResponse (dss:Result(resultmajor#error, resultminor/ar1/notSupported))
4	ArchiveUpdateRequest (DXAIP_OK(AOID-05, VID-01, VID-02))	ArchiveUpdateResponse (dss:Result(resultmajor#ok), VersionID(VID-02))
5	Insert AOID-05 and VID-02 as previous and VID-03 as new versions into the prepared DXAIP_OK_2-template.	DXAIP_OK_2(AOID-05, VID-02, VID-03)
6	ArchiveUpdateRequest (DXAIP_OK_2(AOID-05, VID-02, VID-03))	ArchiveUpdateResponse (dss:Result(resultmajor#ok), VersionID(VID-03))

Verdict of M-UP-09

Observations:

Verdict:

3.4.1.3.13 M-UP-09a – DLXAIP_OK with unknown control in OptionalInputs

Identifier	M-UP-09a
Test Purpose	The test shall verify that there will be an error, if the request contains unknown controls in the OptionalInputs-element.
Configuration	- CONFIG_S4 - CONFIG_LXAIP
Pre-test conditions	Authenticated connection to middleware exists.

Step	Test sequence	Expected Results
1	ArchiveSubmissionRequest (LXAIP_OK(REF_TXT_DATA_58, REF_PDF_DATA_58, REF_XML_MDO_58))	ArchiveSubmissionResponse (dss:Result(resultmajor#ok), AOID-05a)
2	Insert AOID-05a into the prepared DLXAIP_OK-template.	DLXAIP_OK(AOID-05a, VID-01, VID-02)

Step	Test sequence	Expected Results
3	ArchiveUpdateRequest (dss:OptionalInputs(SomethingUnknown), DLXAIP_OK(AOID-05a, VID-01, VID-02, REF_CADES_ATT_TXT_DATA_59))	ArchiveUpdateResponse (dss:Result(resultmajor#error, resultminor/ar1/notSupported))
4	ArchiveUpdateRequest (DLXAIP_OK(AOID-05a, VID-01, VID-02, REF_CADES_ATT_TXT_DATA_60))	ArchiveUpdateResponse (dss:Result(resultmajor#ok), VersionID(VID-02))
5	Insert AOID-05a into the prepared DLXAIP_OK_2-template.	DLXAIP_OK_2(AOID-05a, VID-02, VID-03)
6	ArchiveUpdateRequest (DLXAIP_OK_2(AOID-05a, VID-02, VID-03, REF_XML_DATA_60-1, REF_CADES_DET_XML_DATA_60-1))	ArchiveUpdateResponse (dss:Result(resultmajor#OK), VersionID(VID-03))

Verdict of M-UP-09a

Observations:

Verdict:

3.4.1.4 Function ArchiveRetrieval of S.4-interface

All test cases defined in this section are derived from the general requirement A2.0-2 together with the interface specification of the ArchiveRetrieval function in section 3.3 of [TR-ESOR-E].

3.4.1.4.1 M-RE-01 – Retrieval of previously archived XAIPs

Identifier	M-RE-01
Test Purpose	The test shall verify that it is possible to retrieve previously archived XAIPs.
Configuration	CONFIG_S4
Pre-test conditions	- Authenticated connection to middleware exists. - Make sure that the test cases M-SU-01, M-UP-01 and M-UP-09 were successfully performed. - Requirements (A3.3.1-1) and (A3.3.1-2) of [TR-ESOR-E];

Step	Test sequence	Expected Results
1	ArchiveRetrievalRequest (AOID-01)	ArchiveRetrievalResponse (dss:Result(resultmajor#ok), XAIP_OK(AOID-01, VID-03))
2	ArchiveRetrievalRequest (AOID-02)	ArchiveRetrievalResponse (dss:Result(resultmajor#ok), XAIP_OK_SIG_OK(AOID-02, VID-01))
3	The fulfilment of the requirement A3.3.1-2 ¹⁵ is checked.	
4	ArchiveRetrievalRequest (AOID-05, VID-02)	ArchiveRetrievalResponse (dss:Result(resultmajor#ok), XAIP_OK(AOID-05, VID-02))
5	ArchiveRetrievalRequest (AOID-05, VID-01)	ArchiveRetrievalResponse (dss:Result(resultmajor#ok), XAIP_OK(AOID-05, VID-01))
6	ArchiveRetrievalRequest (AOID-05, VID-01, VID-02)	ArchiveRetrievalResponse (dss:Result(resultmajor#ok), XAIP_OK(AOID-05, VID-01, VID-02))
7	ArchiveRetrievalRequest (AOID-05, ALL)	ArchiveRetrievalResponse (dss:Result(resultmajor#ok), XAIP_OK(AOID-05, VID-01, VID-02, VID-03))

¹⁵ See Annex E (Section 3.3.1).

Step	Test sequence	Expected Results
8	Test of correctness of the obtained XAIP_OK(AOID-05, VID-01, VID-02, VID-03) by using AIP-eIDAS-SigValidator <pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:InlineXML(XAIP_OK(AOID-05, VID-01, VID-02, VID-03)))))) </pre>	<pre> VerifyResponse(dss:Result(resultmajor#ok), dss:OptionalOutputs(dssvr:VerificationReport(dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:XAIPReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid)) [...]))))) </pre>

Verdict of M-RE-01

Observations:

Verdict:

3.4.1.4.2 M-RE-01a – Retrieval of previously archived LXAIPs

Identifier	M-RE-01a
Test Purpose	The test shall verify that it is possible to retrieve previously archived LXAIPs.
Configuration	- CONFIG_S4 - CONFIG_LXAIP
Pre-test conditions	- Authenticated connection to middleware exists. - Make sure that the test cases M-SU-01a, M-UP-01a and M-UP-09a were successfully performed. - Requirements (A3.3.1-1) and (A3.3.1-2) of [TR-ESOR-E];;

Step	Test sequence	Expected Results
1	<pre> ArchiveRetrievalRequest (dss:OptionalInputs(pres:POFormat(http://www.bsi.bund.de/tr-esor/lxaip/1.3)), AOID-01a) </pre>	<pre> ArchiveRetrievalResponse (dss:Result(resultmajor#ok), LXAIP_OK(AOID-01a, VID-03)) </pre>

Step	Test sequence	Expected Results
2	ArchiveRetrievalRequest (dss:OptionalInputs(pres:POFormat(http://www.bsi.bund.de/tr-esor/lxaip/1.3)), AOID-02a)	ArchiveRetrievalResponse (dss:Result(resultmajor#ok), LXAIP_OK_SIG_OK(AOID-02a, VID-01))
3	The fulfilment of the requirement A3.3.1-2 ¹⁶ is checked.	
4	ArchiveRetrievalRequest (dss:OptionalInputs(pres:POFormat(http://www.bsi.bund.de/tr-esor/lxaip/1.3)), AOID-05a, VID-02)	ArchiveRetrievalResponse (dss:Result(resultmajor#ok), LXAIP_OK(AOID-05a, VID-02))
5	ArchiveRetrievalRequest (dss:OptionalInputs(pres:POFormat(http://www.bsi.bund.de/tr-esor/lxaip/1.3)), AOID-05a, VID-01)	ArchiveRetrievalResponse (dss:Result(resultmajor#ok), LXAIP_OK(AOID-05a, VID-01))
6	ArchiveRetrievalRequest (dss:OptionalInputs(pres:POFormat(http://www.bsi.bund.de/tr-esor/lxaip/1.3)), AOID-05a, VID-01, VID-02)	ArchiveRetrievalResponse (dss:Result(resultmajor#ok), LXAIP_OK(AOID-05a, VID-01, VID-02))
7	ArchiveRetrievalRequest (dss:OptionalInputs(pres:POFormat(http://www.bsi.bund.de/tr-esor/lxaip/1.3)), AOID-05a, ALL)	ArchiveRetrievalResponse (dss:Result(resultmajor#ok), LXAIP_OK(AOID-05a, VID-01, VID-02, VID-03))
8	The all in LXAIP_OK(AOID-05a, VID-01, VID-02, VID-03) referenced files has been copied into corresponding LXAIP-directory of the AIP-eIDAS-SigValidator during test preparation (cf. chapter 4.3) and the given references in LXAIP_OK(AOID-05a, VID-01, VID-02, VID-03) will be adjusted automatically in order to get them resolveable by the AIP-eIDAS-SigValidator.	LXAIP_OK_XSV(AOID-05a, VID-01, VID-02, VID-03)

¹⁶ See Annex E (Section 3.3.1).

Step	Test sequence	Expected Results
9	Test of correctness of the obtained and prepared LXAIP_OK_XSV(AOID-05a, VID-01, VID-02, VID-03) by using AIP-eIDAS-SigValidator <pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:InlineXML(LXAIP_OK_XSV(AOID-05a, VID-01, VID-02, VID-03)))))) </pre>	<pre> VerifyResponse(dss:Result(resultmajor#ok), dss:OptionalOutputs(dssvr:VerificationReport(dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:XaipReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid)) [...]))))) </pre>
10	<pre> ArchiveRetrievalRequest (dss:OptionalInputs(pres:POFormat(http://www.bsi.bund.de/tr-esor/xaip/1.3)), AOID-01a) </pre>	<pre> ArchiveRetrievalResponse (dss:Result(resultmajor#ok), XAIP_OK(AOID-01a, VID-03)) </pre>
11	<pre> ArchiveRetrievalRequest (AOID-01a) </pre>	<pre> ArchiveRetrievalResponse (dss:Result(resultmajor#ok), XAIP_OK(AOID-01a, VID-03)) </pre>

Verdict of M-RE-01a

Observations:

Verdict:

3.4.1.4.3 M-RE-02 – Retrieval of XAIPs for known and unknown AOIDS

Identifier	M-RE-02
Test Purpose	The test shall verify that it is possible to retrieve XAIPs for known AOIDS and there will be an error in case of unknown AOIDS.
Configuration	CONFIG_S4
Pre-test conditions	- Authenticated connection to middleware exists. - Make sure that the test cases M-SU-01 was successfully performed.

Step	Test sequence	Expected Results
1	Choice an AOID-06 within the possible range and make sure, that it has not been assigned yet. This is realised by the following call: ArchiveRetrievalRequest (AOID-06)	ArchiveRetrievalResponse (dss:Result(resultmajor#error, resultminor/ar1/unknownAOID))
2	ArchiveRetrievalRequest (AOID-01)	ArchiveRetrievalResponse (dss:Result(resultmajor#ok), XAIP_OK(AOID-01, VID-03))
3	ArchiveRetrievalRequest (AOID-02)	ArchiveRetrievalResponse (dss:Result(resultmajor#ok), XAIP_OK_SIG_OK(AOID-02, VID-01))
4	ArchiveRetrievalRequest (AOID-06)	ArchiveRetrievalResponse (dss:Result(resultmajor#error, resultminor/ar1/unknownAOID))

Verdict M-RE-02**Observations:****Verdict:**

3.4.1.4.4 M-RE-03 – Retrieval of XAIPs for known and unknown VersionIDs

Identifier	M-RE-03
Test Purpose	The test shall verify that there will be a requestOnlyPartlySuccessfulWarning, if it is not possible to retrieve XAIPs for all indicated VersionID-attributes.
Configuration	CONFIG_S4
Pre-test conditions	- Authenticated connection to middleware exists. - Make sure that the test case M-SU-01 was successful performed.

Step	Test sequence	Expected Results
1	Choice a VID-10 within the possible range and make sure, that it has not been assigned yet. This is realised by the following call: ArchiveRetrievalRequest (AOID-01, VID-10)	ArchiveRetrievalResponse (dss:Result(resultmajor#error, resultminor/ar1/unknownVersionID))

Step	Test sequence	Expected Results
2	ArchiveRetrievalRequest (AOID-01, VID-02)	ArchiveRetrievalResponse (dss:Result(resultmajor#ok), XAIP_OK(AOID-01, VID-02))
3	ArchiveRetrievalRequest (AOID-01, VID-10)	ArchiveRetrievalResponse (dss:Result(resultmajor#error, resultminor/ar1/unknownVersionID))
4	ArchiveRetrievalRequest (AOID-01, VID-02, VID-10)	ArchiveRetrievalResponse (dss:Result(resultmajor#warning resultminor/ar1/requestOnlyPartlySuccessfulWarning), XAIP_OK(AOID-01, VID-02))

Verdict of M-RE-03

Observations:

Verdict:

3.4.1.4.5 M-RE-04 – Unknown control in OptionalInputs

Identifier	M-RE-04
Test Purpose	The test shall verify that there will be an error, if the request contains unknown controls in the <code>OptionalInputs</code> -element.
Configuration	CONFIG_S.4
Pre-test conditions	- Authenticated connection to middleware exists. - Make sure that test case M-SU-01 was successfully performed.

Step	Test sequence	Expected Results
1	ArchiveRetrievalRequest (dss:OptionalInputs(SomethingUnknown), AOID-01)	ArchiveRetrievalResponse (dss:Result(resultmajor#error, resultminor/ar1/notSupported))

Verdict of M-RE-04

Observations:

Verdict:

3.4.1.5 Function ArchiveEvidence of S.4-interface

All test cases defined in this section are derived from the general requirement A2.0-2 together with the interface specification of the ArchiveEvidence function in section 3.4 of [TR ESOR E].

3.4.1.5.1 M-EV-01 – Retrieval of Evidence Records of previously archived XAIPs without specifying the desired ERS Format

Identifier	M-EV-01
Test Purpose	The test shall verify that it is possible to retrieve Evidence Records for previously archived XAIPs.
Configuration	CONFIG_S4
Pre-test conditions	- Authenticated connection to middleware exists. - Make sure that test cases M-SU-01 M-UP-01, M-UP-09 and M-RE-01 were successfully performed.

Step	Test sequence	Expected Results
1	ArchiveEvidenceRequest (AOID-01)	ArchiveEvidenceResponse (dss:Result(resultmajor#ok), XAIP_ERS ¹⁷ (AOID-01, VID-03, XAIP_OK_V3_ER_OK))
2	ArchiveEvidenceRequest (AOID-02)	ArchiveEvidenceResponse (dss:Result(resultmajor#ok), XAIP_ERS(AOID-02, VID-01, XAIP_OK_SIG_OK_V1_ER_OK))
3	ArchiveEvidenceRequest (AOID-01, VID-02)	ArchiveEvidenceResponse (dss:Result(resultmajor#ok), XAIP_ERS(AOID-01, VID-02, XAIP_OK_V2_ER_OK))
4	ArchiveEvidenceRequest (AOID-01, VID-01)	ArchiveEvidenceResponse (dss:Result(resultmajor#ok), XAIP_ERS(AOID-01, VID-01, XAIP_OK_V1_ER_OK))

¹⁷ XAIP-ERS does mean a RFC4998-ER embedded in the element `xaip:asn1EvidenceRecord`, which in turn is embedded in the element `xaip:evidenceRecord` and can contain the both attributes `AOID` and `VersionID`.

Step	Test sequence	Expected Results
5	ArchiveEvidenceRequest (AOID-01, VID-01, VID-03)	ArchiveEvidenceResponse (dss:Result(resultmajor#ok), XAIP_ERS(AOID-01, VID-01, XAIP_OK_V1_ER_OK), XAIP_ERS(AOID-01, VID-03, XAIP_OK_V3_ER_OK))
6	ArchiveEvidenceRequest (AOID-01, ALL)	ArchiveEvidenceResponse (dss:Result(resultmajor#ok), XAIP_ERS(AOID-01, VID-01, XAIP_OK_V1_ER_OK), XAIP_ERS(AOID-01, VID-02, XAIP_OK_V2_ER_OK), XAIP_ERS(AOID-01, VID-03, XAIP_OK_V3_ER_OK))
7	ArchiveRetrievalRequest (AOID-01, ALL)	ArchiveRetrievalResponse (dss:Result(resultmajor#ok), XAIP_OK(AOID-01, VID-01, VID-02, VID-03))

Step	Test sequence	Expected Results
8	Test of correctness of the obtained XAIP_OK_V1_ER_OK, XAIP_OK_V2_ER_OK and XAIP_OK_V3_ER_OK by using ERVerifyTool. <pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:InlineXML(XAIP_OK(AOID-01, VID-01, VID-02, VID-03)))) dss:SignatureObject(dss:Other(XAIP_ERS(AOID-01,VID-01, XAIP_OK_V1_ER_OK), XAIP_ERS(AOID-01,VID-02, XAIP_OK_V2_ER_OK), XAIP_ERS(AOID-01,VID-03, XAIP_OK_V3_ER_OK)))) </pre>	<pre> VerifyResponse(dss:Result(resultmajor#ok), dss:OptionalOutputs(dssvr:VerificationReport(dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:EvidenceRecordReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid) [...])), dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:EvidenceRecordReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid) [...]))), dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:EvidenceRecordReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid) [...]))))))) </pre>

Verdict of M-EV-01

Observations:

Verdict:

3.4.1.5.2 M-EV-01a – Retrieval of Evidence Records of previously archived LXAIPIs without specifying the desired ERS Format

Identifier	M-EV-01a
Test Purpose	The test shall verify that it is possible to retrieve Evidence Records for previously archived LXAIPIs.

Identifier	M-EV-01a
Configuration	- CONFIG_S4 - CONFIG_LXAIP
Pre-test conditions	- Authenticated connection to middleware exists. - Make sure that test cases M-SU-01a, M-UP-01a and M-UP-09a were successfully performed.

Step	Test sequence	Expected Results
1	ArchiveEvidenceRequest (AOID-01a)	ArchiveEvidenceResponse (dss:Result(resultmajor#ok), XAIP_ERS ¹⁸ (AOID-01a, VID-03, LXAIP_OK_V3_ER_OK))
2	ArchiveEvidenceRequest (AOID-02a)	ArchiveEvidenceResponse (dss:Result(resultmajor#ok), XAIP_ERS(AOID-02a, VID-01, LXAIP_OK_SIG_OK_V1_ER_OK))
3	ArchiveEvidenceRequest (AOID-01a, VID-02)	ArchiveEvidenceResponse (dss:Result(resultmajor#ok), XAIP_ERS(AOID-01a, VID-02, LXAIP_OK_V2_ER_OK))
4	ArchiveEvidenceRequest (AOID-01a, VID-01)	ArchiveEvidenceResponse (dss:Result(resultmajor#ok), XAIP_ERS(AOID-01a, VID-01, LXAIP_OK_V1_ER_OK))
5	ArchiveEvidenceRequest (AOID-01a, VID-01, VID-03)	ArchiveEvidenceResponse (dss:Result(resultmajor#ok), XAIP_ERS(AOID-01a, VID-01, LXAIP_OK_V1_ER_OK), XAIP_ERS(AOID-01a, VID-03, LXAIP_OK_V3_ER_OK))

¹⁸ XAIP-ERS does mean a RFC4998-ER embedded in the element `xaip:asn1EvidenceRecord`, which in turn is embedded in the element `xaip:evidenceRecord` and can contain the both attributes `AOID` and `VersionID`.

Step	Test sequence	Expected Results
6	ArchiveEvidenceRequest (AOID-01a, ALL)	ArchiveEvidenceResponse (dss:Result(resultmajor#ok), XAIP_ERS(AOID-01a, VID-01, LXAIP_OK_V1_ER_OK), XAIP_ERS(AOID-01a, VID-02, LXAIP_OK_V2_ER_OK), XAIP_ERS(AOID-01a, VID-03, LXAIP_OK_V3_ER_OK))
7	ArchiveRetrievalRequest (dss:OptionalInputs(pres:POFormat(http://www.bsi.bund.de/tr-esor/lxaip/1.3)), AOID-01a, ALL)	ArchiveRetrievalResponse (dss:Result(resultmajor#ok), LXAIP_OK(AOID-01a, VID-01, VID-02, VID-03))

Step	Test sequence	Expected Results
8	Test of correctness of the obtained LXAIP_OK_V1_ER_OK, LXAIP_OK_V2_ER_OK and LXAIP_OK_V3_ER_OK by using ERVerifyTool. The particular object references in the obtained LXAIP_OK will be replaced by the references compatible with the ERVerifyTool in advance. <pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:InlineXML(LXAIP_OK(AOID-01a, VID-01, VID-02, VID-03)))) dss:SignatureObject(dss:Other(XAIP_ERS(AOID-01a, VID-01, LXAIP_OK_V1_ER_OK), XAIP_ERS(AOID-01a, VID-02, LXAIP_OK_V2_ER_OK), XAIP_ERS(AOID-01a, VID-03, LXAIP_OK_V3_ER_OK)))) </pre>	<pre> VerifyResponse(dss:Result(resultmajor#ok), dss:OptionalOutputs(dssvr:VerificationReport(dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:EvidenceRecordReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid) [...]))), dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:EvidenceRecordReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid) [...]))), dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:EvidenceRecordReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid) [...])))))) </pre>

Verdict of M-EV-01a

Observations:

Verdict:

3.4.1.5.3 M-EV-02 – Retrieval of Evidence Records for known and unknown AOIDS

Identifier	M-EV-02
Test Purpose	The test case verifies that that it is possible to retrieve an EvidenceRecord for known AOIDS and there will be an error in case of unknown AOIDS.
Configuration	CONFIG_S4
Pre-test conditions	- Authenticated connection to middleware exists. - Make sure that the test cases M-SU-01 and M-RE-02 were successfully performed.

Step	Test sequence	Expected Results
1	Make sure, that the previous chosen AOID-06 is within the possible range and it is still not assigned yet. This is realised by the following call: <pre>ArchiveRetrievalRequest (AOID-06)</pre>	<pre>ArchiveRetrievalResponse (dss:Result(resultmajor#error, resultminor/ar1/unknownAOID))</pre>
2	<pre>ArchiveEvidenceRequest (AOID-01)</pre>	<pre>ArchiveEvidenceResponse (dss:Result(resultmajor#ok), XAIP_ERS(AOID-01, VID-03, XAIP_OK_V3_ER_OK))</pre>
3	<pre>ArchiveEvidenceRequest (AOID-02)</pre>	<pre>ArchiveEvidenceResponse (dss:Result(resultmajor#ok), XAIP_ERS(AOID-02, VID-01, XAIP_OK_SIG_OK_V1_ER_OK))</pre>
4	<pre>ArchiveEvidenceRequest (AOID-06)</pre>	<pre>ArchiveEvidenceResponse (dss:Result(resultmajor#error, resultminor/ar1/unknownAOID))</pre>

Verdict of M-EV-02
Observations:
Verdict:

3.4.1.5.4 M-EV-03 – Retrieval of Evidence Records for partly known VersionIDs

Identifier	M-EV-03
Test Purpose	The test case verifies that there is a requestOnlyPartlySuccessfulWarning, if an Evidence Record for an unknown VersionID is requested.

Identifier	M-EV-03
Configuration	CONFIG_S.4
Pre-test conditions	- Authenticated connection to middleware exists. - Make sure that the test cases M-SU-01 and M-RE-03 were successfully performed.

Step	Test sequence	Expected Results
1	Make sure, that concerning AOID-01, the chosen version VID-10 is within the possible range but is an unknown VersionID. This is realised by the following call: <pre>ArchiveRetrievalRequest (AOID-01, VID-10)</pre>	<pre>ArchiveRetrievalResponse (dss:Result(resultmajor#error, resultminor/ar1/unknownVersionID))</pre>
2	<pre>ArchiveEvidenceRequest (AOID-01)</pre>	<pre>ArchiveEvidenceResponse (dss:Result(resultmajor#ok), XAIP_ERS(AOID-01, VID-03, XAIP_OK_V3_ER_OK))</pre>
3	<pre>ArchiveEvidenceRequest (AOID-01, VID-10)</pre>	<pre>ArchiveEvidenceResponse (dss:Result(resultmajor#error, resultminor/ar1/unknownVersionID))</pre>
4	<pre>ArchiveEvidenceRequest (AOID-01, VID-02, VID-10)</pre>	<pre>ArchiveEvidenceResponse (dss:Result(resultmajor#warning, resultminor/ar1/requestOnlyPartlySuccessfulWarning), XAIP_ERS(AOID-01, VID-02, XAIP_OK_V2_ER_OK))</pre>

Verdict of M-EV-03
Observations:
Verdict:

3.4.1.5.5 M-EV-04 – Unknown control in OptionalInputs

Identifier	M-EV-04
Test Purpose	The test shall verify that there will be an error, if the request contains unknown controls in the <code>OptionalInputs</code> -element.
Configuration	CONFIG_S.4
Pre-test conditions	- Authenticated connection to middleware exists. - Make sure that the test case M-SU-01 was successfully performed.

Step	Test sequence	Expected Results
1	ArchiveEvidenceRequest (dss:OptionalInputs(SomethingUnknown), AOID-01)	ArchiveEvidenceResponse (dss:Result(resultmajor#error, resultminor/ar1/notSupported))

Verdict of M-EV-04
Observations:
Verdict:

3.4.1.6 Function ArchiveDeletion of S.4-interface

All test cases defined in this section are derived from the general requirement A2.0-2 together with the interface specification of the ArchiveDeletion function in section 3.5 of [TR ESOR E].

3.4.1.6.1 M-DE-01 – Deletion of XAIP without ReasonOfDeletion

Identifier	M-DE-01
Test Purpose	The test shall verify that it will yield an error, if <code>ArchiveDeletionRequest</code> is called without providing a <code>ReasonOfDeletion</code> , if the element <code>retentionPeriod</code> in the XAIP contains a predetermined future date.
Configuration	CONFIG_S4
Pre-test conditions	- In one test case the element <code>retentionPeriod</code> in the XAIP to be deleted contains a predetermined future date. In the other test case the element <code>retentionPeriod</code> contains a date in the past. - Authenticated connection to middleware exists. - Requirement (A3.5.1-1) of [TR-ESOR-E]: - The <code>ArchiveDeletionRequest</code> of XAIP with AOID-02 with the end of the <code>retentionPeriod</code> in the future must not be successful without a reason for the deletion. - Make sure that the test case M-SU-01 was successfully performed.

Step	Test sequence	Expected Results
1	ArchiveDeletionRequest (AOID-02)	ArchiveDeletionResponse (dss:Result(resultmajor#error, resultminor/ar1/missingReasonOfDeletion))
2	ArchiveRetrievalRequest (AOID-02)	ArchiveRetrievalResponse (dss:Result(resultmajor#ok), XAIP_OK_SIG_OK(AOID-02, VID-01))
3	ArchiveRetrievalRequest (AOID-00 ¹⁹)	ArchiveRetrievalResponse (dss:Result(resultmajor#ok), XAIP_OK(AOID-00, VID-01))
4	ArchiveDeletionRequest (AOID-00)	ArchiveDeletionResponse (dss:Result(resultmajor#ok))
5	ArchiveRetrievalRequest (AOID-00)	ArchiveRetrievalResponse (dss:Result(resultmajor#error, resultminor/ar1/unknownAOID))

¹⁹ The XAIP with AOID-00 and LXAIP with AOID-00a were imported some days ago. The retention period is expired.

Verdict of M-DE-01
Observations:
Verdict:

3.4.1.6.2 M-DE-01a – Deletion of LXAIP without ReasonOfDeletion

Identifier	M-DE-01a
Test Purpose	The test shall verify that it will yield an error, if <code>ArchiveDeletionRequest</code> is called without providing a <code>ReasonOfDeletion</code> if the element <code>retentionPeriod</code> in the LXAIP contains a predetermined future date.
Configuration	- CONFIG_S4 - CONFIG_LXAIP
Pre-test conditions	- In one test case the element <code>retentionPeriod</code> in the LXAIP to be deleted contains a predetermined future date. In the other test case the element <code>retentionPeriod</code> contains a date in the past. - Authenticated connection to middleware exists. - Requirement (A3.5.1-1) of [TR-ESOR-E];: - The <code>ArchiveDeletionRequest</code> of LXAIP with AOID-02a with the end of the <code>retentionPeriod</code> in the future must not be successful without a reason for the deletion. - Make sure that the test case M-SU-01a was successfully performed.

Step	Test sequence	Expected Results
1	ArchiveDeletionRequest (AOID-02a)	ArchiveDeletionResponse (dss:Result(resultmajor#error, resultminor/ar1/missingReasonOfDeletion))
2	ArchiveRetrievalRequest (dss:OptionalInputs(pres:POFormat(http://www.bsi.bund.de/tr-esor/lxaip/1.3)), AOID-02a)	ArchiveRetrievalResponse (dss:Result(resultmajor#ok), LXAIP_OK_SIG_OK(AOID-02a, VID-01))
3	ArchiveRetrievalRequest (dss:OptionalInputs(pres:POFormat(http://www.bsi.bund.de/tr-esor/lxaip/1.3)), AOID-00a ¹⁹)	ArchiveRetrievalResponse (dss:Result(resultmajor#ok), LXAIP_OK(AOID-00a, VID-01))
4	ArchiveDeletionRequest (AOID-00a)	ArchiveDeletionResponse (dss:Result(resultmajor#ok))

Step	Test sequence	Expected Results
5	ArchiveRetrievalRequest (dss:OptionalInputs(pres:POFormat(http://www.bsi.bund.de/tr-esor/lxaip/1.3)), AOID-00a)	ArchiveRetrievalResponse (dss:Result(resultmajor#error, resultminor/ar1/unknownAOID))

Verdict of M-DE-01a
Observations:
Verdict:

3.4.1.6.3 M-DE-02 – Deletion of XAIP with ReasonOfDeletion

Identifier	M-DE-02
Test Purpose	The test shall verify that it is possible to delete an XAIP by calling <code>ArchiveDeletionRequest</code> with <code>ReasonOfDeletion</code> among the <code>dss:OptionalInputs</code> .
Configuration	CONFIG_S4
Pre-test conditions	- The element <code>retentionPeriod</code> in the XAIP to be deleted contains a predetermined future date. - Make sure that the test case M-SU-01 was successfully performed. - Authenticated connection to middleware exists.

Step	Test sequence	Expected Results
1	ArchiveDeletionRequest (dss:OptionalInputs(ReasonOfDeletion(RequestorName(SomeName), RequestInfo(SomeInfo)), AOID-02)	ArchiveDeletionResponse (dss:Result(resultmajor#ok))
2	ArchiveRetrievalRequest (AOID-02)	ArchiveRetrievalResponse (dss:Result(resultmajor#error, resultminor/ar1/unknownAOID))
3	Check requirement A3.5.1-2 ²⁰ .	

²⁰ See Annex E (Section 3.5.1).

Verdict of M-DE-02
Observations:
Verdict:

3.4.1.6.4 M-DE-02a – Deletion of LXAIP with ReasonOfDeletion

Identifier	M-DE-02a
Test Purpose	The test shall verify that it is possible to delete an LXAIP by calling <code>ArchiveDeletionRequest</code> with <code>ReasonOfDeletion</code> among the <code>dss:OptionalInputs</code> .
Configuration	- CONFIG_S4 - CONFIG_LXAIP
Pre-test conditions	- The element <code>retentionPeriod</code> in the LXAIP to be deleted contains a predetermined future date. - Make sure that the test case M-SU-01a was successfully performed. - Authenticated connection to middleware exists.

Step	Test sequence	Expected Results
1	<pre>ArchiveDeletionRequest (dss:OptionalInputs(ReasonOfDeletion(RequestorName(SomeName), RequestInfo(SomeInfo)), AOID-02a)</pre>	<pre>ArchiveDeletionResponse (dss:Result(resultmajor#ok))</pre>
2	<pre>ArchiveRetrievalRequest (dss:OptionalInputs(pres:POFormat(http://www.bsi.bund.de/tr-esor/lxaip/1.3)), AOID-02a)</pre>	<pre>ArchiveRetrievalResponse (dss:Result(resultmajor#error, resultminor/arl/unknownAOID))</pre>
3	Check requirement A3.5.1-2 ²¹ .	

Verdict of M-DE-02a
Observations:
Verdict:

²¹ See Annex E (Section 3.5.1).

3.4.1.6.5 M-DE-03 – Deletion of unknown XAIPs without ReasonOfDeletion

Identifier	M-DE-03
Test Purpose	The test shall verify that it will yield an error, if <code>ArchiveDeletionRequest</code> is called without providing a <code>ReasonOfDeletion</code> . As the present call also contains the unknown AOID-06, the AOID-specific <code>dss:Result</code> needs to indicate that the AOID is unknown.
Configuration	CONFIG_S4
Pre-test conditions	- The element <code>retentionPeriod</code> in the XAIP to be deleted contains a predetermined future date. - Make sure that the test cases M-UP-09 and M-RE-02 were successfully performed. - Make sure, that the randomly generated AOID-06 is within the possible range but has not been assigned yet. - Authenticated connection to middleware exists.

Step	Test sequence	Expected Results
1	ArchiveDeletionRequest (AOID-05)	ArchiveDeletionResponse (dss:Result(resultmajor#error, resultminor/ar1/missingReasonOfDeletion))
2	ArchiveDeletionRequest (AOID-06)	ArchiveDeletionResponse (dss:Result(resultmajor#error, resultminor/ar1/unknownAOID))
3	ArchiveRetrievalRequest (AOID-05)	ArchiveRetrievalResponse (dss:Result(resultmajor#ok), XAIP_OK(AOID-05, VID-03))

Verdict of M-DE-03
Observations:
Verdict:

3.4.1.6.6 M-DE-04 – Deletion of unknown XAIPs with ReasonOfDeletion

Identifier	M-DE-04
Test Purpose	The test shall verify that it will yield an error, if <code>ArchiveDeletionRequest</code> is called with unknown AOIDs with providing a <code>ReasonOfDeletion</code> .
Configuration	CONFIG_S4
Pre-test conditions	- The element <code>retentionPeriod</code> in the XAIP to be deleted contains a predetermined future date. - Make sure that the test case M-RE-02 was successfully performed - Make sure, that the randomly generated AOID-06 is within the possible range but has not been assigned yet. - Authenticated connection to middleware exists.

Step	Test sequence	Expected Results
1	ArchiveDeletionRequest (dss:OptionalInputs(ReasonOfDeletion(RequestorName(SomeName), RequestInfo(SomeInfo))), AOID-06)	ArchiveDeletionResponse (dss:Result(resultmajor#error, resultminor/ar1/unknownAOID))

Verdict of M-DE-04
Observations:
Verdict:

3.4.1.6.7 M-DE-05 – Unknown control in OptionalInputs

Identifier	M-DE-05
Test Purpose	The test shall verify that there will be an error, if the request contains unknown controls in the OptionalInputs-element and the ArchiveDeletionRequest will not be successful.
Configuration	CONFIG_S4
Pre-test conditions	- Authenticated connection to middleware exists. - The element retentionPeriod in the XAIP to be deleted contains a predetermined future date. - Make sure that the test case M-UP-09 was successfully performed.

Step	Test sequence	Expected Results
1	ArchiveDeletionRequest (dss:OptionalInputs(SomethingUnknown), AOID-05)	ArchiveDeletionResponse (dss:Result(resultmajor#error, resultminor/ar1/notSupported))
2	ArchiveRetrievalRequest (AOID-05)	ArchiveRetrievalResponse (dss:Result(resultmajor#ok, XAIP_OK(AOID-05, VID-03))

Verdict of M-DE-05
Observations:
Verdict:

3.4.1.7 Function Verify of S.4-interface

All test cases defined in this section are derived from the general requirement A2.0-2 together with the interface specification of the Verify function in section 3.7 of [TR ESOR E].

3.4.1.7.1 M-VE-01 –XAIP_OK_SIG_OK and XAIP_OK_SIG_OK_ER

Identifier	M-VE-01
Test Purpose	The test shall verify that it is possible to verify XAIP_OK_SIG_OK and XAIP_OK_SIG_OK_V1_ER_OK with the TR-ESOR-specific signature policy http://www.bsi.bund.de/tr-esor/sigpolicy/verify-xaip .
Configuration	CONFIG_S.4
Pre-test conditions	Authenticated connection to middleware exists.

Step	Test sequence	Expected Results
1	ArchiveSubmissionRequest (XAIP_OK_SIG_OK)	ArchiveSubmissionResponse (dss:Result(resultmajor#ok), AOID-100)
2	ArchiveRetrievalRequest (AOID-100)	ArchiveRetrievalResponse (dss:Result(resultmajor#ok), XAIP_OK_SIG_OK(AOID-100, VID-01))
3	ArchiveEvidenceRequest ⁷ (AOID-100)	ArchiveEvidenceResponse (dss:Result(resultmajor#ok), XAIP_ERS(AOID-100,VID-01, XAIP_OK_SIG_OK_V1_ER_OK)))
4	VerifyRequest (dss:InputDocuments(dss:Document(dss:InlineXML(XAIP_OK_SIG_OK(AOID-100, VID-01)))))	VerifyResponse (dss:Result(resultmajor#ok))
5	VerifyRequest (dss:OptionalInputs(vr:ReturnVerificationReport), dss:InputDocuments(dss:Document(dss:InlineXML(XAIP_OK_SIG_OK(AOID-100, VID-01)))))	VerifyResponse (dss:Result(resultmajor#ok), dss:OptionalOutputs(XAIP_OK_SIG_OK_VR))

Step	Test sequence	Expected Results
6	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64XML(XAIP_OK_SIG_OK(AOID-100, VID-01))))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#ok)) </pre>
7	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64Data(XAIP_OK_SIG_OK(AOID-100, VID-01))))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#ok)) </pre>
8	Prepare XAIP_OK_SIG_OK_ER(AOID-100, VID-01) by inserting the XAIP_ERS(AOID-100,VID-01, XAIP_OK_SIG_OK_V1_ER_OK) obtained in step 3 into <code>credentialsSection</code> element of XAIP_OK_SIG_OK(AOID-100, VID-01) from step 2.	XAIP_OK_SIG_OK_ER(AOID-100, VID-01)
9	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:InlineXML(XAIP_OK_SIG_OK_ER(AOID-100,VID-01))))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#ok)) </pre>
10	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64XML(XAIP_OK_SIG_OK_ER(AOID-100,VID-01))))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#ok)) </pre>
11	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64Data(XAIP_OK_SIG_OK_ER(AOID-100,VID-01))))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#ok)) </pre>

Step	Test sequence	Expected Results
12	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:InlineXML(XAIP_OK_SIG_OK(AOID-100, VID-01)))) dss:SignatureObject(dss:Other(XAIP_ERS(AOID-100,VID-01, XAIP_OK_SIG_OK_V1_ER_OK)))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#ok)) </pre>
13	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64XML(XAIP_OK_SIG_OK(AOID-100, VID-01)))) dss:SignatureObject(dss:Other(XAIP_ERS(AOID-100,VID-01, XAIP_OK_SIG_OK_V1_ER_OK)))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#ok)) </pre>
14	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64Data(XAIP_OK_SIG_OK(AOID-100, VID-01)))) dss:SignatureObject(dss:Other(XAIP_ERS(AOID-100,VID-01, XAIP_OK_SIG_OK_V1_ER_OK)))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#ok)) </pre>

Verdict of M-VE-01

Observations:

Verdict:

3.4.1.7.2 M-VE-01a - LXAIP_OK_SIG_OK and LXAIP_OK_SIG_OK_ER

Identifier	M-VE-01a
Test Purpose	The test shall verify that it is possible to verify LXAIP_OK_SIG_OK and LXAIP_OK_SIG_OK_ER with the TR-ESOR-specific signature policy http://www.bsi.bund.de/tr-esor/sigpolicy/verify-xaip .
Configuration	- CONFIG_S.4 - CONFIG_LXAIP
Pre-test conditions	- Authenticated connection to middleware exists. - Test M-SU-01a has been performed successfully and the references has been set properly in LXAIP_OK_SIG_OK

Step	Test sequence	Expected Results
1	ArchiveSubmissionRequest (LXAIP_OK_SIG_OK(REF_TXT_DATA_61, REF_CADES_DET_TXT_DATA_61, REF_PDF_DATA_61, REF_CADES_DET_PDF_DATA_61, REF_XML_MDO_61))	ArchiveSubmissionResponse (dss:Result(resultmajor#ok), AOID-101)
2	ArchiveRetrievalRequest (AOID-101)	ArchiveRetrievalResponse (dss:Result(resultmajor#ok), LXAIP_OK_SIG_OK(AOID-101, VID-01))
3	ArchiveEvidenceRequest ⁷ (AOID-101)	ArchiveEvidenceResponse (dss:Result(resultmajor#ok), XAIP_ERS(AOID-101,VID-01, LXAIP_OK_SIG_OK_V1_ER_OK)))
4	VerifyRequest (dss:InputDocuments(dss:Document(dss:InlineXML(LXAIP_OK_SIG_OK(AOID-101, VID-01)))))	VerifyResponse (dss:Result(resultmajor#ok))
5	VerifyRequest (dss:InputDocuments(dss:OptionalInputs(vr:ReturnVerificationReport), dss:Document(dss:InlineXML(LXAIP_OK_SIG_OK(AOID-101, VID-01)))))	VerifyResponse (dss:Result(resultmajor#ok), dss:OptionalOutputs(LXAIP_OK_SIG_OK_VR))

Step	Test sequence	Expected Results
6	VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64XML(LXAIP_OK_SIG_OK(AOID-101, VID-01)))))	VerifyResponse (dss:Result(resultmajor#ok))
7	VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64Data(LXAIP_OK_SIG_OK(AOID-101, VID-01)))))	VerifyResponse (dss:Result(resultmajor#ok))
8	Prepare LXAIP_OK_SIG_OK_ER(AOID-101, VID-01) by inserting the XAIP_ERS(AOID-101,VID-01, LXAIP_OK_SIG_OK_V1_ER_OK) obtained in step 3 into credentialsSection element of LXAIP_OK_SIG_OK(AOID-101, VID-01) from step 2.	LXAIP_OK_SIG_OK_ER(AOID-101, VID-01)
9	VerifyRequest (dss:InputDocuments(dss:Document(dss:InlineXML(LXAIP_OK_SIG_OK_ER(AOID-101,VID-01)))))	VerifyResponse (dss:Result(resultmajor#ok))
10	VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64XML(LXAIP_OK_SIG_OK_ER(AOID-101,VID-01)))))	VerifyResponse (dss:Result(resultmajor#ok))
11	VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64Data(LXAIP_OK_SIG_OK_ER(AOID-101,VID-01)))))	VerifyResponse (dss:Result(resultmajor#ok))

Step	Test sequence	Expected Results
12	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:InlineXML(LXAIP_OK_SIG_OK(AOID-101, VID-01)))) dss:SignatureObject(dss:Other(XAIP_ERS(AOID-101,VID-01, LXAIP_OK_SIG_OK_V1_ER_OK)))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#ok)) </pre>
13	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64XML(LXAIP_OK_SIG_OK(AOID-101, VID-01)))) dss:SignatureObject(dss:Other(XAIP_ERS(AOID-101,VID-01, LXAIP_OK_SIG_OK_V1_ER_OK)))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#ok)) </pre>
14	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64Data(LXAIP_OK_SIG_OK(AOID-101, VID-01)))) dss:SignatureObject(dss:Other(XAIP_ERS(AOID-101,VID-01, LXAIP_OK_SIG_OK_V1_ER_OK)))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#ok)) </pre>

Verdict of M-VE-01a

Observations:

Verdict:

3.4.1.7.3 M-VE-02 - DXAIP_OK_SIG_OK

Identifier	M-VE-02
Test Purpose	The test shall verify that it is possible to verify DXAIP_OK_SIG_OK with the TR-ESOR-specific signature policy http://www.bsi.bund.de/tr-esor/sigpolicy/verify-xaip .
Configuration	CONFIG_S.4
Pre-test conditions	- Authenticated connection to middleware exists. - The test M-VE-01 has been performed successfully

Step	Test sequence	Expected Results
1	Prepare DXAIP_OK_SIG_OK by inserting the AOID-100	DXAIP_OK_SIG_OK(AOID-100, VID-01, VID-02)
2	<pre>VerifyRequest (dss:InputDocuments(dss:Document(dss:InlineXML(DXAIP_OK_SIG_OK(AOID-100, VID-01, VID-02))))))</pre>	<pre>VerifyResponse (dss:Result(resultmajor#ok))</pre>
3	<pre>VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64XML(DXAIP_OK_SIG_OK(AOID-100, VID-01, VID-02))))))</pre>	<pre>VerifyResponse (dss:Result(resultmajor#ok))</pre>
4	<pre>VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64Data(DXAIP_OK_SIG_OK(AOID-100, VID-01, VID-02))))))</pre>	<pre>VerifyResponse (dss:Result(resultmajor#ok))</pre>

Verdict of M-VE-02
Observations:
Verdict:

3.4.1.7.4 M-VE-02a - DLXAIP_OK_SIG_OK

Identifier	M-VE-02a
Test Purpose	The test shall verify that it is possible to verify DLXAIP_OK_SIG_OK with the TR-ESOR-specific signature policy http://www.bsi.bund.de/tr-esor/sigpolicy/verify-xaip .
Configuration	• CONFIG_S4 • CONFIG_LXAIP
Pre-test conditions	• Authenticated connection to middleware exists. • The test M-VE-01a has been performed successfully.

Step	Test sequence	Expected Results
1	Prepare DLXAIP_OK_SIG_OK by inserting the AOID-101.	DLXAIP_OK_SIG_OK(AOID-101, VID-01, VID-02)
2	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:InlineXML(DLXAIP_OK_SIG_OK(AOID-101, VID-01, VID-02, REF_XML_DATA_62, REF_CADES_DET_XML_DATA_62)))))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#ok)) </pre>
3	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64XML(DLXAIP_OK_SIG_OK(AOID-101, VID-01, VID-02, REF_XML_DATA_63, REF_CADES_DET_XML_DATA_63)))))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#ok)) </pre>
4	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64Data(DLXAIP_OK_SIG_OK(AOID-101, VID-01, VID-02, REF_XML_DATA_64, REF_CADES_DET_XML_DATA_64)))))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#ok)) </pre>

Verdict of M-VE-02a
Observations:
Verdict:

3.4.1.7.5 M-VE-03 – BIN_OK

Identifier	M-VE-03
Test Purpose	The test shall validate that it is possible to verify as BIN defined data. BIN means here: • CADES_DET_OK, CADES_ATT_OK, • XADES_DET_OK, XADES_ENVELOPED_OK, XADES_ENVELOPING_OK, • PADES_OK, • ASiC_S_CADES_OK, ASiC_S_XADES_OK, ASiC_S_TST_OK, ASiC_S_ER_OK, • ASiC_E_CADES_OK, ASiC_E_XADES_OK, ASiC_E_TST_OK, ASiC_E_ER_OK.
Configuration	• CONFIG_S.4
Pre-test conditions	• Authenticated connection to middleware exists. • The as CONDITIONAL marked test steps shall be executed, in case the TOT does support passing by the XML data as native XML, e.g. <code>dss:InlineXML</code>-element, or <code>ds:Signature</code>-element etc.

Step	Test sequence	Expected Results
1	<pre>VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64Data(TXT_DATA.txt))) dss:SignatureObject(dss:Base64Signature(CADES_DET_OK)))</pre>	<pre>VerifyResponse (dss:Result(resultmajor#ok))</pre>
2	<pre>VerifyRequest (dss:SignatureObject(dss:Base64Signature(CADES_ATT_OK)))</pre>	<pre>VerifyResponse (dss:Result(resultmajor#ok))</pre>
3	<pre>VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64Data(PADES_OK))))</pre>	<pre>VerifyResponse (dss:Result(resultmajor#ok))</pre>

Step	Test sequence	Expected Results
4	[CONDITIONAL] VerifyRequest (dss:InputDocuments(dss:Document(dss:InlineXML(002-test-document.xml))) dss:SignatureObject(ds:Signature(XADES_DET_OK)))	VerifyResponse (dss:Result(resultmajor#ok))
5	[CONDITIONAL] VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64XML(002-test-document.xml))) dss:SignatureObject(ds:Signature(XADES_DET_OK)))	VerifyResponse (dss:Result(resultmajor#ok))
6	VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64XML(002-test-document.xml))) dss:SignatureObject(dss:Base64Signature(XADES_DET_OK)))	VerifyResponse (dss:Result(resultmajor#ok))
7	[CONDITIONAL] VerifyRequest (dss:InputDocuments(dss:Document(dss:InlineXML(XADES_ENVELOPED_OK))))	VerifyResponse (dss:Result(resultmajor#ok))
8	VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64XML(XADES_ENVELOPED_OK))))	VerifyResponse (dss:Result(resultmajor#ok))
9	[CONDITIONAL] VerifyRequest (dss:SignatureObject(ds:Signature(XADES_ENVELOPING_OK)))	VerifyResponse (dss:Result(resultmajor#ok))

Step	Test sequence	Expected Results
10	VerifyRequest (dss:SignatureObject(dss:Base64Signature(XADES_ENVELOPING_OK)))	VerifyResponse (dss:Result(resultmajor#ok))
11	VerifyRequest (dss:SignatureObject(dss:Base64Signature(ASiC_S_CADES_OK)))	VerifyResponse (dss:Result(resultmajor#ok))
12	VerifyRequest (dss:SignatureObject(dss:Base64Signature(ASiC_S_XADES_OK)))	VerifyResponse (dss:Result(resultmajor#ok))
13	VerifyRequest (dss:SignatureObject(dss:Base64Signature(ASiC_S_TST_OK)))	VerifyResponse (dss:Result(resultmajor#ok))
14	VerifyRequest (dss:SignatureObject(dss:Base64Signature(ASiC_S_ER_OK)))	VerifyResponse (dss:Result(resultmajor#ok))
15	VerifyRequest (dss:SignatureObject(dss:Base64Signature(ASiC_E_CADES_OK)))	VerifyResponse (dss:Result(resultmajor#ok))
16	VerifyRequest (dss:SignatureObject(dss:Base64Signature(ASiC_E_XADES_OK)))	VerifyResponse (dss:Result(resultmajor#ok))

Step	Test sequence	Expected Results
17	VerifyRequest (dss:SignatureObject(dss:Base64Signature(ASIC_E_TST_OK)))	VerifyResponse (dss:Result(resultmajor#ok))
18	VerifyRequest (dss:SignatureObject(dss:Base64Signature(ASIC_E_ER_OK)))	VerifyResponse (dss:Result(resultmajor#ok))

Verdict of M-VE-03

Observations:

Verdict:

3.4.1.7.6 M-VE-04 - XAIP_OK_SIG_NOK & XAIP_OK_SIG_OK_ER_NOK

Identifier	M-VE-04
Test Purpose	The test shall verify that there will be a verification error, in case XAIP_OK_SIG_NOK or XAIP_OK_SIG_OK_ER_NOK is submitted.
Configuration	CONFIG_S4
Pre-test conditions	Authenticated connection to middleware exists.

Step	Test sequence	Expected Results
1	VerifyRequest (dss:InputDocuments(dss:Document(dss:InlineXML(XAIP_OK_SIG_NOK))))	VerifyResponse (dss:Result(resultmajor#error, resultminor/ar1/XAIP_NOK_SIG) dss:OptionalOutputs(XAIP_OK_SIG_NOK_VR)) OR VerifyResponse (dss:Result(resultmajor#error, resultminor/ar1/XAIP_NOK_SIG) dss:OptionalOutputs(XAIP_OK_SIG_NOK(XAIP_OK_SIG_NOK_VR)))

Step	Test sequence	Expected Results
2	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:InlineXML(XAIP_OK_SIG_OK_ER_NOK)))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#error, resultminor/arl/XAIP_NOK_ER) dss:OptionalOutputs(XAIP_OK_SIG_OK_ER_NOK_VR)) OR VerifyResponse (dss:Result(resultmajor#error, resultminor/arl/XAIP_NOK_SIG) dss:OptionalOutputs(XAIP_OK_SIG_OK_ER_NOK(XAIP_OK_SIG_OK_ER_NOK_VR)))) </pre>
3	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64XML(XAIP_OK_SIG_NOK)))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#error, resultminor/arl/XAIP_NOK_SIG) dss:OptionalOutputs(XAIP_OK_SIG_NOK_VR)) OR VerifyResponse (dss:Result(resultmajor#error, resultminor/arl/XAIP_NOK_SIG) dss:OptionalOutputs(XAIP_OK_SIG_NOK(XAIP_OK_SIG_NOK_VR)))) </pre>
4	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64XML(XAIP_OK_SIG_OK_ER_NOK)))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#error, resultminor/arl/XAIP_NOK_ER) dss:OptionalOutputs(XAIP_OK_SIG_OK_ER_NOK_VR)) OR VerifyResponse (dss:Result(resultmajor#error, resultminor/arl/XAIP_NOK_ER) dss:OptionalOutputs(XAIP_OK_SIG_OK_ER_NOK(XAIP_OK_SIG_OK_ER_NOK_VR)))) </pre>

Verdict of M-VE-04

Observations:

Verdict:

3.4.1.7.7 M-VE-04a - LXAIP_OK_SIG_NOK & LXAIP_OK_SIG_OK_ER_NOK

Identifier	M-VE-04a
Test Purpose	The test shall verify that there will be a verification error, in case of LXAIP_OK_SIG_NOK or LXAIP_OK_SIG_OK_ER_NOK is submitted. The response will include either the VerificationReport-element only or the LXAIP including the VerificationReport-element.
Configuration	- CONFIG_S4 - CONFIG_LXAIP
Pre-test conditions	- Authenticated connection to middleware exists. - Test M-SU-01a has been performed successfully and the references has been set properly in LXAIP_OK_SIG_OK.

Step	Test sequence	Expected Results
1	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:InlineXML(LXAIP_OK_SIG_NOK(TXT_DATA_65.txt, REF_CADES_DET_NOK_TXT_DATA_65, REF_PDF_DATA_65, REF_CADES_DET_PDF_DATA_65, REF_XML_MDO_65))))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#error, resultminor/arl/XAIP_NOK_SIG) dss:OptionalOutputs(LXAIP_OK_SIG_NOK_VR)) OR VerifyResponse (dss:Result(resultmajor#error, resultminor/arl/XAIP_NOK_SIG) dss:OptionalOutputs(LXAIP_OK_SIG_NOK(LXAIP_OK_SIG_NOK_VR))) </pre>
2	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:InlineXML(LXAIP_OK_SIG_OK_ER_NOK(TXT_DATA_66.txt, REF_CADES_DET_TXT_DATA_66, REF_PDF_DATA_66, REF_CADES_DET_PDF_DATA_66, REF_XML_MDO_66, ER_NOK_XAIP_OK_SIG_OK_00))))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#error, resultminor/arl/XAIP_NOK_ER) dss:OptionalOutputs(LXAIP_OK_SIG_OK_ER_NOK_VR)) OR VerifyResponse (dss:Result(resultmajor#error, resultminor/arl/XAIP_NOK_SIG) dss:OptionalOutputs(LXAIP_OK_SIG_OK_ER_NOK(LXAIP_OK_SIG_OK_ER_NOK_VR))) </pre>

Step	Test sequence	Expected Results
3	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64XML(LXAIP_OK_SIG_NOK(TXT_DATA_67.txt, REF_CADES_DET_NOK_TXT_DATA_67, REF_PDF_DATA_67, REF_CADES_DET_PDF_DATA_67, REF_XML_MDO_67))))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#error, resultminor/arl/XAIP_NOK_SIG) dss:OptionalOutputs(LXAIP_OK_SIG_NOK_VR)) OR VerifyResponse (dss:Result(resultmajor#error, resultminor/arl/XAIP_NOK_SIG) dss:OptionalOutputs(LXAIP_OK_SIG_NOK(LXAIP_OK_SIG_NOK_VR))) </pre>
4	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64XML(LXAIP_OK_SIG_OK_ER_NOK(TXT_DATA_68.txt, REF_CADES_DET_TXT_DATA_68, REF_PDF_DATA_68, REF_CADES_DET_PDF_DATA_68, REF_XML_MDO_68, ER_NOK_XAIP_OK_SIG_OK_00))))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#error, resultminor/arl/XAIP_NOK_ER) dss:OptionalOutputs(LXAIP_OK_SIG_OK_ER_NOK_VR)) OR VerifyResponse (dss:Result(resultmajor#error, resultminor/arl/XAIP_NOK_SIG) dss:OptionalOutputs(LXAIP_OK_SIG_OK_ER_NOK(LXAIP_OK_SIG_OK_ER_NOK_VR))) </pre>

Verdict of M-VE-04a

Observations:

Verdict:

3.4.1.7.8 M-VE-05 - DXAIP_OK_SIG_NOK

Identifier	M-VE-05
Test Purpose	The test shall verify that there will be a verification error, in case DXAIP_OK_SIG_NOK is submitted.
Configuration	CONFIG_S.4
Pre-test conditions	- Authenticated connection to middleware exists. - Test M-VE-01 has been performed successfully.

Step	Test sequence	Expected Results
1	Prepare DXAIP_OK_SIG_NOK by inserting AOID-100.	DXAIP_OK_SIG_NOK (AOID-100, VID-01, VID-02)

Step	Test sequence	Expected Results
2	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:InlineXML(DXAIP_OK_SIG_NOK (AOID-100, VID-01, VID-02)))))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#error, resultminor/arl/DXAIP_NOK_SIG) dss:OptionalOutputs(DXAIP_OK_SIG_NOK_VR)) OR VerifyResponse (dss:Result(resultmajor#error, resultminor/arl/DXAIP_NOK_SIG) dss:OptionalOutputs(DXAIP_OK_SIG_NOK(DXAIP_OK_SIG_NOK_VR))) </pre>
3	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64XML(DXAIP_OK_SIG_NOK(AOID-100, VID-01, VID-02)))))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#error, resultminor/arl/DXAIP_NOK_SIG) dss:OptionalOutputs(DXAIP_OK_SIG_NOK_VR)) OR VerifyResponse (dss:Result(resultmajor#error, resultminor/arl/DXAIP_NOK_SIG) dss:OptionalOutputs(DXAIP_OK_SIG_NOK(DXAIP_OK_SIG_NOK_VR))) </pre>

Verdict of M-VE-05

Observations:

Verdict:

3.4.1.7.9 M-VE-05a - DLXAIP_OK_SIG_NOK

Identifier	M-VE-05a
Test Purpose	The test case shall verify that there will be a verification error, in case DLXAIP_OK_SIG_NOK is submitted.
Configuration	- CONFIG_S.4 - CONFIG_LXAIP
Pre-test conditions	- Authenticated connection to middleware exists. - Test M-VE-01a has been performed successfully.

Step	Test sequence	Expected Results
1	Prepare DLXAIP_OK_SIG_NOK by inserting AOID-101.	DLXAIP_OK_SIG_NOK(AOID-100, VID-01, VID-02)

Step	Test sequence	Expected Results
2	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:InlineXML(DLXAIP_OK_SIG_NOK(AOID-101, VID-01, VID-02, REF_XML_DATA_69, REF_CADES_DET_NOK_XML_DATA_69))))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#error, resultminor/ar1/DXAIP_NOK_SIG) dss:OptionalOutputs(DXAIP_OK_SIG_NOK_VR)) OR VerifyResponse (dss:Result(resultmajor#error, resultminor/ar1/DXAIP_NOK_SIG) dss:OptionalOutputs(DLXAIP_OK_SIG_NOK(AOID-101, VID-01, VID-02, DXAIP_OK_SIG_NOK_VR))) </pre>
3	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64XML(DLXAIP_OK_SIG_NOK(AOID-101, VID-01, VID-02, REF_XML_DATA_70, REF_CADES_DET_NOK_XML_DATA_70))))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#error, resultminor/ar1/DXAIP_NOK_SIG) dss:OptionalOutputs(DXAIP_OK_SIG_NOK_VR)) OR VerifyResponse (dss:Result(resultmajor#error, resultminor/ar1/DXAIP_NOK_SIG) dss:OptionalOutputs(DLXAIP_OK_SIG_NOK(AOID-101, VID-01, VID-02, DXAIP_OK_SIG_NOK_VR))) </pre>

Verdict of M-VE-05a

Observations:

Verdict:

3.4.1.7.10 M-VE-06 – BIN_NOK

Identifier	M-VE-06
Test Purpose	The test case shall validate that there will be an error in case an as BIN defined data object containing an invalid signature will be submitted to verification. BIN means here: • CADES_DET_NOK, CADES_ATT_NOK • XADES_DET_NOK, XADES_ENVELOPED_NOK, XADES_ENVELOPING_NOK, • PADES_NOK, • ASIC_E_CADES_NOK, ASIC_E_XADES_NOK, ASIC_E_TST_NOK, ASIC_E_ER_NOK, • ASIC_S_CADES_NOK, ASIC_S_XADES_NOK, ASIC_S_TST_NOK, ASIC_S_ER_NOK.
Configuration	• CONFIG_S4
Pre-test conditions	• Authenticated connection to middleware exists. • The as CONDITIONAL marked test steps shall be executed, in case the TOT does support passing by the XML data as native XML, e.g. <code>dss:InlineXML</code>-element, or <code>ds:Signature</code>-element etc.

Step	Test sequence	Expected Results
1	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64Data(TXT_DATA.txt))), dss:SignatureObject(dss:Base64Signature(CADES_DET_NOK))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#error, resultminor/sal#invalidSignature), dss:OptionalOutputs(CADES_DET_NOK_VR)) </pre>
2	<pre> VerifyRequest (dss:SignatureObject(dss:Base64Signature(CADES_ATT_NOK))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#error, resultminor/sal#invalidSignature), dss:OptionalOutputs(CADES_ATT_NOK_VR)) </pre>
3	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64Data(PADES_NOK)))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#error, resultminor/sal#invalidSignature), dss:OptionalOutputs(PADES_NOK_VR)) </pre>
4	[CONDITIONAL] <pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:InlineXML(002-test-document.xml))), dss:SignatureObject(ds:Signature(XADES_DET_NOK))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#error, resultminor/sal#invalidSignature), dss:OptionalOutputs(XADES_DET_NOK_VR)) </pre>
5	[CONDITIONAL] <pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64XML(002-test-document.xml))), dss:SignatureObject(ds:Signature(XADES_DET_NOK))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#error, resultminor/sal#invalidSignature), dss:OptionalOutputs(XADES_DET_NOK_VR)) </pre>
6	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64XML(002-test-document.xml))), dss:SignatureObject(dss:Base64Signature(XADES_DET_NOK))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#error, resultminor/sal#invalidSignature), dss:OptionalOutputs(XADES_DET_NOK_VR)) </pre>

Step	Test sequence	Expected Results
7	[CONDITIONAL] VerifyRequest (dss:InputDocuments(dss:Document(dss:InlineXML(XADES_ENVELOPED_NOK))))	VerifyResponse (dss:Result(resultmajor#error, resultminor/sal#invalidSignature), dss:OptionalOutputs(XADES_ENVELOPED_NOK_VR))
8	VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64XML(XADES_ENVELOPED_NOK)))	VerifyResponse (dss:Result(resultmajor#error, resultminor/sal#invalidSignature), dss:OptionalOutputs(XADES_ENVELOPED_NOK_VR))
9	[CONDITIONAL] VerifyRequest (dss:SignatureObject(ds:Signature(XADES_ENVELOPING_NOK)))	VerifyResponse (dss:Result(resultmajor#error, resultminor/sal#invalidSignature), dss:OptionalOutputs(XADES_ENVELOPING_NOK_VR))
10	VerifyRequest (dss:SignatureObject(dss:Base64Signature(XADES_ENVELOPING_NOK)))	VerifyResponse (dss:Result(resultmajor#error, resultminor/sal#invalidSignature), dss:OptionalOutputs(XADES_ENVELOPING_NOK_VR))
11	VerifyRequest (dss:SignatureObject(dss:Base64Signature(ASIC_S_CADES_NOK)))	VerifyResponse (dss:Result(resultmajor#error, resultminor/sal#invalidSignature), dss:OptionalOutputs(ASIC_S_CADES_NOK_VR))
12	VerifyRequest (dss:SignatureObject(dss:Base64Signature(ASIC_S_XADES_NOK)))	VerifyResponse (dss:Result(resultmajor#error, resultminor/sal#invalidSignature), dss:OptionalOutputs(ASIC_S_XADES_NOK_VR))
13	VerifyRequest (dss:SignatureObject(dss:Base64Signature(ASIC_S_TST_NOK)))	VerifyResponse (dss:Result(resultmajor#error, resultminor/sal#invalidSignature), dss:OptionalOutputs(ASIC_S_TST_NOK_VR))

Step	Test sequence	Expected Results
14	VerifyRequest (dss:SignatureObject(dss:Base64Signature(ASIC_S_ER_NOK)))	VerifyResponse (dss:Result(resultmajor#error, resultminor/sal#invalidSignature), dss:OptionalOutputs(ASIC_S_ER_NOK_VR))
15	VerifyRequest (dss:SignatureObject(dss:Base64Signature(ASIC_E_CADES_NOK)))	VerifyResponse (dss:Result(resultmajor#error, resultminor/sal#invalidSignature), dss:OptionalOutputs(ASIC_E_CADES_NOK_VR))
16	VerifyRequest (dss:SignatureObject(dss:Base64Signature(ASIC_E_XADES_NOK)))	VerifyResponse (dss:Result(resultmajor#error, resultminor/sal#invalidSignature), dss:OptionalOutputs(ASIC_E_XADES_NOK_VR))
17	VerifyRequest (dss:SignatureObject(dss:Base64Signature(ASIC_E_TST_NOK)))	VerifyResponse (dss:Result(resultmajor#error, resultminor/sal#invalidSignature), dss:OptionalOutputs(ASIC_E_TST_NOK_VR))
18	VerifyRequest (dss:SignatureObject(dss:Base64Signature(ASIC_E_ER_NOK)))	VerifyResponse (dss:Result(resultmajor#error, resultminor/sal#invalidSignature), dss:OptionalOutputs(ASIC_E_ER_NOK_VR))

Verdict of M-VE-06

Observations:

Verdict:

3.4.1.7.11 M-VE-07 – Hash algorithms

Identifier	M-VE-07
Test Purpose	The test shall verify that it is possible to verify XAIP_OK and corresponding evidence record with the TR-ESOR-specific signature policy http://www.bsi.bund.de/tr-esor/sigpolicy/verify-xaip by using different hash algorithms for underlying hash tree. The hash algorithms to be tested are following SHA-1, SHA-224, SHA-256, SHA-384, SHA-512, SHA3-224, SHA3-256, SHA3-384, SHA3-512.
Configuration	CONFIG_S.4
Pre-test conditions	Authenticated connection to middleware exists.

Step	Test sequence	Expected Results
1	Prepare XAIP_OK_SIG_OK by inserting of AOID-999	XAIP_OK_SIG_OK(AOID-999, VID-01)
2	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64XML(XAIP_OK_SIG_OK(AOID-999, VID-01)))), dss:SignatureObject(dss:Other(XAIP_ERS(AOID-999,VID-01, ER_OK_SHA-1_XAIP_OK_SIG_OK)))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#ok)) OR VerifyResponse (dss:Result(resultmajor#error, resultminor/il/algorithm#hashAlgorithmNotSuitable), dss:OptionalOutputs(XAIP_OK_SIG_OK_VR(AOID-999, VID-01, ER_OK_SHA-1_XAIP_OK_SIG_OK_VR))) </pre>
3	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64XML(XAIP_OK_SIG_OK(AOID-999, VID-01)))), dss:SignatureObject(dss:Other(XAIP_ERS(AOID-999,VID-01, ER_NOK_SHA-1_XAIP_OK_SIG_OK)))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#error, resultminor/arl/XAIP_NOK_ER), dss:OptionalOutputs(XAIP_OK_SIG_OK_VR(AOID-999, VID-01, ER_NOK_SHA-1_XAIP_OK_SIG_OK_VR))) OR VerifyResponse (dss:Result(resultmajor#error, resultminor/arl/XAIP_NOK_ER), dss:OptionalOutputs(XAIP_OK_SIG_OK(AOID-999, VID-01, XAIP_OK_SIG_OK_VR(AOID-999, VID-01, ER_NOK_SHA-1_XAIP_OK_SIG_OK_VR)))) </pre>

Step	Test sequence	Expected Results
4	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64XML(XAIP_OK_SIG_OK(AOID-999, VID-01)))) dss:SignatureObject(dss:Other(XAIP_ERS(AOID-999,VID-01, ER_OK_SHA-224_XAIP_OK_SIG_OK)))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#ok)) OR VerifyResponse (dss:Result(resultmajor#error, resultminor/il/algorithm#hashAlgorithmNotSuitable), dss:OptionalOutputs(XAIP_OK_SIG_OK_VR(AOID-999, VID-01, ER_OK_SHA-224_XAIP_OK_SIG_OK_VR))) </pre>
5	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64XML(XAIP_OK_SIG_OK(AOID-999, VID-01)))) dss:SignatureObject(dss:Other(XAIP_ERS(AOID-999,VID-01, ER_NOK_SHA-224_XAIP_OK_SIG_OK)))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#error, resultminor/ar1/XAIP_NOK_ER), dss:OptionalOutputs(XAIP_OK_SIG_OK_VR(AOID-999, VID-01, ER_NOK_SHA-224_XAIP_OK_SIG_OK_VR))) OR VerifyResponse (dss:Result(resultmajor#error, resultminor/ar1/XAIP_NOK_ER), dss:OptionalOutputs(XAIP_OK_SIG_OK (AOID-999, VID-01, XAIP_OK_SIG_OK_VR (AOID-999, VID-01, ER_NOK_SHA-224_XAIP_OK_SIG_OK_VR))))) </pre>
6	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64XML(XAIP_OK_SIG_OK(AOID-999, VID-01)))) dss:SignatureObject(dss:Other(XAIP_ERS(AOID-999,VID-01, ER_OK_SHA-256_XAIP_OK_SIG_OK)))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#ok)) </pre>

Step	Test sequence	Expected Results
7	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64XML(XAIP_OK_SIG_OK(AOID-999, VID-01)))) dss:SignatureObject(dss:Other(XAIP_ERS(AOID-999,VID-01, ER_NOK_SHA-256_XAIP_OK_SIG_OK)))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#error, resultminor/arl/XAIP_NOK_ER), dss:OptionalOutputs(XAIP_OK_SIG_OK_VR(AOID-999, VID-01, ER_NOK_SHA-256_XAIP_OK_SIG_OK_VR))) </pre> OR <pre> VerifyResponse (dss:Result(resultmajor#error, resultminor/arl/XAIP_NOK_ER), dss:OptionalOutputs(XAIP_OK_SIG_OK(AOID-999, VID-01, XAIP_OK_SIG_OK_VR(AOID-999, VID-01, ER_NOK_SHA-256_XAIP_OK_SIG_OK_VR)))) </pre>
8	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64XML(XAIP_OK_SIG_OK(AOID-999, VID-01)))) dss:SignatureObject(dss:Other(XAIP_ERS(AOID-999,VID-01, ER_OK_SHA-384_XAIP_OK_SIG_OK)))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#ok)) </pre>
9	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64XML(XAIP_OK_SIG_OK(AOID-999, VID-01)))) dss:SignatureObject(dss:Other(XAIP_ERS(AOID-999,VID-01, ER_NOK_SHA-384_XAIP_OK_SIG_OK)))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#error, resultminor/arl/XAIP_NOK_ER), dss:OptionalOutputs(XAIP_OK_SIG_OK_VR(AOID-999, VID-01, ER_NOK_SHA-384_XAIP_OK_SIG_OK_VR))) </pre> OR <pre> VerifyResponse (dss:Result(resultmajor#error, resultminor/arl/XAIP_NOK_ER), dss:OptionalOutputs(XAIP_OK_SIG_OK(AOID-999, VID-01, XAIP_OK_SIG_OK_VR(AOID-999, VID-01, ER_NOK_SHA-384_XAIP_OK_SIG_OK_VR)))) </pre>

Step	Test sequence	Expected Results
10	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64XML(XAIP_OK_SIG_OK(AOID-999, VID-01)))) dss:SignatureObject(dss:Other(XAIP_ERS(AOID-999,VID-01, ER_OK_SHA-512_XAIP_OK_SIG_OK)))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#ok)) </pre>
11	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64XML(XAIP_OK_SIG_OK(AOID-999, VID-01)))) dss:SignatureObject(dss:Other(XAIP_ERS(AOID-999,VID-01, ER_NOK_SHA-512_XAIP_OK_SIG_OK)))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#error, resultminor/arl/XAIP_NOK_ER), dss:OptionalOutputs(XAIP_OK_SIG_OK_VR(AOID-999, VID-01, ER_NOK_SHA-512_XAIP_OK_SIG_OK_VR))) </pre> OR <pre> VerifyResponse (dss:Result(resultmajor#error, resultminor/arl/XAIP_NOK_ER), dss:OptionalOutputs(XAIP_OK_SIG_OK(AOID-999, VID-01, XAIP_OK_SIG_OK_VR(AOID-999, VID-01, ER_NOK_SHA-512_XAIP_OK_SIG_OK_VR)))) </pre>
12	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64XML(XAIP_OK_SIG_OK(AOID-999, VID-01)))) dss:SignatureObject(dss:Other(XAIP_ERS(AOID-999,VID-01, ER_OK_SHA3-224_XAIP_OK_SIG_OK)))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#ok)) </pre> OR <pre> VerifyResponse (dss:Result(resultmajor#error, resultminor/il/algorithm#hashAlgorithmNotSuitable), dss:OptionalOutputs(XAIP_OK_SIG_OK_VR(AOID-999, VID-01, ER_OK_SHA3-224_XAIP_OK_SIG_OK_VR))) </pre>

Step	Test sequence	Expected Results
13	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64XML(XAIP_OK_SIG_OK(AOID-999, VID-01)))) dss:SignatureObject(dss:Other(XAIP_ERS(AOID-999,VID-01, ER_NOK_SHA3-224_XAIP_OK_SIG_OK)))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#error, resultminor/arl/XAIP_NOK_ER), dss:OptionalOutputs(XAIP_OK_SIG_OK_VR(AOID-999, VID-01, ER_NOK_SHA3-224_XAIP_OK_SIG_OK_VR))) </pre> OR <pre> VerifyResponse (dss:Result(resultmajor#error, resultminor/arl/XAIP_NOK_ER), dss:OptionalOutputs(XAIP_OK_SIG_OK(AOID-999, VID-01, XAIP_OK_SIG_OK_VR(AOID-999, VID-01, ER_NOK_SHA3-224_XAIP_OK_SIG_OK_VR)))) </pre>
14	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64XML(XAIP_OK_SIG_OK(AOID-999, VID-01)))) dss:SignatureObject(dss:Other(XAIP_ERS(AOID-999,VID-01, ER_OK_SHA3-256_XAIP_OK_SIG_OK)))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#ok)) </pre>
15	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64XML(XAIP_OK_SIG_OK(AOID-999, VID-01)))) dss:SignatureObject(dss:Other(XAIP_ERS(AOID-999,VID-01, ER_NOK_SHA3-256_XAIP_OK_SIG_OK)))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#error, resultminor/arl/XAIP_NOK_ER), dss:OptionalOutputs(XAIP_OK_SIG_OK_VR(AOID-999, VID-01, ER_NOK_SHA3-256_XAIP_OK_SIG_OK_VR))) </pre> OR <pre> VerifyResponse (dss:Result(resultmajor#error, resultminor/arl/XAIP_NOK_ER), dss:OptionalOutputs(XAIP_OK_SIG_OK(AOID-999, VID-01, XAIP_OK_SIG_OK_VR(AOID-999, VID-01, ER_NOK_SHA3-256_XAIP_OK_SIG_OK_VR)))) </pre>

Step	Test sequence	Expected Results
16	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64XML(XAIP_OK_SIG_OK(AOID-999, VID-01)))) dss:SignatureObject(dss:Other(XAIP_ERS(AOID-999,VID-01, ER_OK_SHA3-384_XAIP_OK_SIG_OK)))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#ok)) </pre>
17	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64XML(XAIP_OK_SIG_OK(AOID-999, VID-01)))) dss:SignatureObject(dss:Other(XAIP_ERS(AOID-999,VID-01, ER_NOK_SHA3-384_XAIP_OK_SIG_OK)))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#error, resultminor/ar1/XAIP_NOK_ER), dss:OptionalOutputs(XAIP_OK_SIG_OK_VR(AOID-999, VID-01, ER_NOK_SHA3-384_XAIP_OK_SIG_OK_VR))) </pre> OR <pre> VerifyResponse (dss:Result(resultmajor#error, resultminor/ar1/XAIP_NOK_ER), dss:OptionalOutputs(XAIP_OK_SIG_OK(AOID-999, VID-01, XAIP_OK_SIG_OK_VR(AOID-999, VID-01, ER_NOK_SHA3-384_XAIP_OK_SIG_OK_VR)))) </pre>
18	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64XML(XAIP_OK_SIG_OK(AOID-999, VID-01)))) dss:SignatureObject(dss:Other(XAIP_ERS(AOID-999,VID-01, ER_NOK_SHA-512_XAIP_OK_SIG_OK)))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#ok)) </pre>

Step	Test sequence	Expected Results
19	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64XML(XAIP_OK_SIG_OK(AOID-999, VID-01)))) dss:SignatureObject(dss:Other(XAIP_ERS(AOID-999,VID-01, ER_OK_SHA3-512_XAIP_OK_SIG_OK)))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#error, resultminor/arl/XAIP_NOK_ER), dss:OptionalOutputs(XAIP_OK_SIG_OK_VR(AOID-999, VID-01, ER_OK_SHA3-512_XAIP_OK_SIG_OK_VR))) </pre> OR <pre> VerifyResponse (dss:Result(resultmajor#error, resultminor/arl/XAIP_NOK_ER), dss:OptionalOutputs(XAIP_OK_SIG_OK(AOID-999, VID-01, XAIP_OK_SIG_OK_VR(AOID-999, VID-01, ER_OK_SHA3-512_XAIP_OK_SIG_OK_VR)))) </pre>

Verdict of M-VE-07

Observations:

Verdict:

3.4.1.8 Additional tests of S.4-interface

3.4.1.8.1 M-ADD-01 – XAIP_OK and Resigning

Identifier	M-ADD-01
Test Purpose	The test shall verify that it is possible to retrieve previously submitted and updated XAIP_OK after the time-stamp resigning procedure has been successfully applied.
Configuration	CONFIG_S.4
Pre-test conditions	- Authenticated connection to middleware exists. - The time-stamp resigning procedure has been successfully performed - Make sure that the test case M-UP-01 was successfully performed.

Step	Test sequence	Expected Results
1	ArchiveRetrievalRequest (AOID-01, ALL)	ArchiveRetrievalResponse (dss:Result(resultmajor#ok), XAIP_OK(AOID-01, VID-01, VID-02, VID-03))
2	ArchiveEvidenceRequest (AOID-01, ALL)	ArchiveEvidenceResponse (dss:Result(resultmajor#ok), XAIP_ERS ²² (AOID-01, VID-01, XAIP_OK_V1_ER_OK_RES), XAIP_ERS(AOID-01, VID-02, XAIP_OK_V2_ER_OK_RES), XAIP_ERS(AOID-01, VID-03, XAIP_OK_V3_ER_OK_RES))

²² XAIP-ERS does mean a RFC4998-ER embedded in the element `xaip:asn1EvidenceRecord`, which in turn is embedded in the element `xaip:evidenceRecord` and can contain the both attributes `AOID` and `VersionID`.

Step	Test sequence	Expected Results
3	Test of correctness of the obtained: • XAIP_OK_V1_ER_OK_RESG, • XAIP_OK_V2_ER_OK_RESG and • XAIP_OK_V3_ER_OK_RESG by using ERVerifyTool. <pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:InlineXML(XAIP_OK(AOID-01, VID-01, VID-02, VID-03)))) dss:SignatureObject(dss:Other(XAIP_ERS(AOID-01, VID-01, XAIP_OK_V1_ER_OK_RESG), XAIP_ERS(AOID-01, VID-02, XAIP_OK_V2_ER_OK_RESG), XAIP_ERS(AOID-01, VID-03, XAIP_OK_V3_ER_OK_RESG)))) </pre>	<pre> VerifyResponse(dss:Result(resultmajor#ok), dss:OptionalOutputs(dssvr:VerificationReport(dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:EvidenceRecordReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid) [...]))), dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:EvidenceRecordReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid) [...]))))), dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:EvidenceRecordReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid) [...])))) </pre>

Verdict of M-ADD-01

Observations:

Verdict:

3.4.1.8.2 M-ADD-01a – LXAIP_OK and Resigning

Identifier	M-ADD-01a
Test Purpose	The test shall verify that it is possible to retrieve previously submitted and updated LXAIP_OK after the time-stamp resigning procedure has been successfully applied.

Identifier	M-ADD-01a
Configuration	- CONFIG_S.4 - CONFIG_LXAIP
Pre-test conditions	- Authenticated connection to middleware exists. - The time-stamp resigning procedure has been successfully performed - Make sure that the test case M-UP-01a was successfully performed.

Step	Test sequence	Expected Results
1	ArchiveRetrievalRequest (AOID-01a, ALL)	ArchiveRetrievalResponse (dss:Result(resultmajor#ok), LXAIP_OK(AOID-01a, VID-01, VID-02, VID-03))
2	ArchiveEvidenceRequest (AOID-01a , ALL)	ArchiveEvidenceResponse (dss:Result(resultmajor#ok), XAIP_ERS ²³ (AOID-01a, VID-01, LXAIP_OK_V1_ER_OK_RESG), XAIP_ERS(AOID-01a, VID-02, LXAIP_OK_V2_ER_OK_RESG), XAIP_ERS(AOID-01a, VID-03, LXAIP_OK_V3_ER_OK_RESG))

²³ XAIP-ERS does mean a RFC4998-ER embedded in the element `xaip:asn1EvidenceRecord`, which in turn is embedded in the element `xaip:evidenceRecord` and can contain the both attributes `AOID` and `VersionID`.

Step	Test sequence	Expected Results
3	Test of correctness of the obtained: • LXAIP_OK_V1_ER_OK_RESg, • LXAIP_OK_V2_ER_OK_RESg and • LXAIP_OK_V3_ER_OK_RESg by using ERVerifyTool. The particular object references in the obtained LXAIP_OK will be replaced by the references compatible with the ERVerifyTool in advance. <pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:InlineXML(LXAIP_OK(AOID-01a, VID-01, VID-02, VID-03))))) dss:SignatureObject(dss:Other(XAIP_ERS(AOID-01a, VID-01, LXAIP_OK_V1_ER_OK_RESg), XAIP_ERS(AOID-01a, VID-02, LXAIP_OK_V2_ER_OK_RESg), XAIP_ERS(AOID-01a, VID-03, LXAIP_OK_V3_ER_OK_RESg)))) </pre>	<pre> VerifyResponse(dss:Result(resultmajor#ok), dss:OptionalOutputs(dssvr:VerificationReport(dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:EvidenceRecordReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid)) [...])), dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:EvidenceRecordReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid)) [...])), dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:EvidenceRecordReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid)) [...]))))) </pre>

Verdict of M-ADD-01a

Observations:

Verdict:

3.4.1.8.3 M-ADD-02 – XAIP_OK and Rehashing

Identifier	M-ADD-02
Test Purpose	The test shall verify that it is possible to retrieve previously submitted and updated XAIP_OK after the hash tree renewal procedure (rehashing) has been successfully applied.
Configuration	CONFIG_S.4
Pre-test conditions	- Authenticated connection to middleware exists. - The rehashing procedure has been successfully performed - Make sure that the test case M-ADD-01 was successfully performed.

Step	Test sequence	Expected Results
1	ArchiveRetrievalRequest (AOID-01, ALL)	ArchiveRetrievalResponse (dss:Result(resultmajor#ok), XAIP_OK(AOID-01, VID-01, VID-02, VID-03))
2	ArchiveEvidenceRequest (AOID-01, ALL)	ArchiveEvidenceResponse (dss:Result(resultmajor#ok), XAIP_ERS ²⁴ (AOID-01, VID-01, XAIP_OK_V1_ER_OK_RESG_REH), XAIP_ERS(AOID-01, VID-02, XAIP_OK_V2_ER_OK_RESG_REH), XAIP_ERS(AOID-01, VID-03, XAIP_OK_V3_ER_OK_RESG_REH))

²⁴ XAIP-ERS does mean a RFC4998-ER embedded in the element `xaip:asn1EvidenceRecord`, which in turn is embedded in the element `xaip:evidenceRecord` and can contain the both attributes `AOID` and `VersionID`.

Step	Test sequence	Expected Results
3	Test of correctness of the obtained: • XAIP_OK_V1_ER_OK_RESG_REH, • XAIP_OK_V2_ER_OK_RESG_REH and • XAIP_OK_V3_ER_OK_RESG_REH by using ERVerifyTool. <pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:InlineXML(XAIP_OK(AOID-01, VID-01, VID-02, VID-03)))) dss:SignatureObject(dss:Other(XAIP_ERS(AOID-01, VID-01, XAIP_OK_V1_ER_OK_RESG_REH), XAIP_ERS(AOID-01, VID-02, XAIP_OK_V2_ER_OK_RESG_REH), XAIP_ERS(AOID-01, VID-03, XAIP_OK_V3_ER_OK_RESG_REH)))) </pre>	<pre> VerifyResponse(dss:Result(resultmajor#ok), dss:OptionalOutputs(dssvr:VerificationReport(dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:EvidenceRecordReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid) [...]))), dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:EvidenceRecordReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid) [...]))), dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:EvidenceRecordReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid) [...])))))) </pre>

Verdict of M-ADD-02

Observations:

Verdict:

3.4.1.8.4 M-ADD-02a – LXAIP_OK and Resigning

Identifier	M-ADD-02a
Test Purpose	The test shall verify that it is possible to retrieve previously submitted and updated LXAIP_OK after the hash tree renewal procedure (rehashing) has been successfully applied.

Identifier	M-ADD-02a
Configuration	• CONFIG_S.4 • CONFIG_LXAIP
Pre-test conditions	• Authenticated connection to middleware exists. • The rehashing procedure has been successfully performed • Make sure that the test case M-ADD-01a was successfully performed.

Step	Test sequence	Expected Results
1	ArchiveRetrievalRequest (AOID-01a, ALL)	ArchiveRetrievalResponse (dss:Result(resultmajor#ok), LXAIP_OK(AOID-01a, VID-01, VID-02, VID-03))
2	ArchiveEvidenceRequest (AOID-01a, ALL)	ArchiveEvidenceResponse (dss:Result(resultmajor#ok), XAIP_ERS ²⁵ (AOID-01a, VID-01, LXAIP_OK_V1_ER_OK_RESG_REH), XAIP_ERS(AOID-01a, VID-02, LXAIP_OK_V2_ER_OK_RESG_REH), XAIP_ERS(AOID-01a, VID-03, LXAIP_OK_V3_ER_OK_RESG_REH))

²⁵ XAIP-ERS does mean a RFC4998-ER embedded in the element `xaip:asn1EvidenceRecord`, which in turn is embedded in the element `xaip:evidenceRecord` and can contain the both attributes `AOID` and `VersionID`.

Step	Test sequence	Expected Results
3	Test of correctness of the obtained: • LXAIP_OK_V1_ER_OK_RESG_REH, • LXAIP_OK_V2_ER_OK_RESG_REH and • LXAIP_OK_V3_ER_OK_RESG_REH by using ERVerifyTool. The particular object references in the obtained LXAIP_OK will be replaced by the references compatible with the ERVerifyTool in advance. <pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:InlineXML(LXAIP_OK (AOID-01a, VID-01, VID-02, VID-03))))) dss:SignatureObject(dss:Other(XAIP_ERS(AOID-01a, VID-01, LXAIP_OK_V1_ER_OK_RESG_REH), XAIP_ERS(AOID-01a, VID-02, LXAIP_OK_V2_ER_OK_RESG_REH), XAIP_ERS(AOID-01a, VID-03, LXAIP_OK_V3_ER_OK_RESG_REH)))) </pre>	<pre> VerifyResponse(dss:Result(resultmajor#ok), dss:OptionalOutputs(dssvr:VerificationReport(dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:EvidenceRecordReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid)) [...]))), dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:EvidenceRecordReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid)) [...]))), dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:EvidenceRecordReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid)) [...]))) </pre>

Verdict of M-ADD-02a

Observations:

Verdict:

3.4.1.8.5 M-ADD-03a – Relation between upload data and LXAIP/DLXAIP

Identifier	M-ADD-03a
Test Purpose	The test shall verify that 1:1-relation between the uploaded data and corresponding LXAIP/DLXAIP is adhered to. It is not possible to use a single reference in two different LXAIP/DLXAIP instances and in such case, an error will be returned.
Configuration	- CONFIG_S4 - CONFIG_LXAIP
Pre-test conditions	- Authenticated connection to middleware exists. - Test case M-SU-01a has been performed successfully.

Step	Test sequence	Expected Results
1	ArchiveSubmissionRequest (LXAIP_OK(REF_TXT_DATA_71, REF_PDF_DATA_71, REF_XML_MDO_71)))	ArchiveSubmissionResponse (dss:Result(resultmajor#ok), AOID-300a)
2	ArchiveSubmissionRequest (LXAIP_OK(REF_TXT_DATA_71, REF_PDF_DATA_71, REF_XML_MDO_71)))	ArchiveSubmissionResponse (dss:Result(resultmajor#error, resultminor/al/common#parameterError))
3	ArchiveSubmissionRequest (LXAIP_OK_ER_NOK(REF_TXT_DATA_72, REF_PDF_DATA_72, REF_XML_MDO_72, ER_NOK_LXAIP_OK_72)))	ArchiveSubmissionResponse (dss:Result(resultmajor#error, resultminor/ar1/XAIP_NOK_ER) dss:OptionalOutputs(LXAIP_OK_ER_NOK_VR))
4	ArchiveSubmissionRequest (LXAIP_OK(REF_TXT_DATA_72, REF_PDF_DATA_72, REF_XML_MDO_72)))	ArchiveSubmissionResponse (dss:Result(resultmajor#error, resultminor/al/common#parameterError))
5	ArchiveUpdateRequest (DLXAIP_OK(AOID-300a, VID-01, VID-02, REF_CADES_ATT_TXT_DATA_73)))	ArchiveUpdateResponse (dss:Result(resultmajor#OK), VersionID(VID-02))
6	ArchiveSubmissionRequest (LXAIP_OK(REF_TXT_DATA_74, REF_PDF_DATA_74, REF_XML_MDO_74)))	ArchiveSubmissionResponse (dss:Result(resultmajor#ok), AOID-301a)

Step	Test sequence	Expected Results
7	ArchiveUpdateRequest (DLXAIP_OK(AOID-301a, VID-01, VID-02, REF_CADES_ATT_TXT_DATA_73))	ArchiveUpdateResponse (dss:Result(resultmajor#error, resultminor/al/common#parameterError))

Verdict of M-ADD-03a

Observations:

Verdict:

3.4.1.8.6 M-ADD-04 – Evidence Record with single hash value

Identifier	M-ADD-04
Test Purpose	The test shall verify that it is possible to verify every variant of the evidence record computed on a single data object: - ER(TSP(H(DO-01))) – hash value of the data object is directly included in the message imprint of the timestamp (ER_SO-1_OK_XAIP_OK_SO), - ER(H(DO-01),TSP(H(DO-01))) – the hash value of the data object is placed in the single reduced hash tree and the same hash value is included in the message imprint of the timestamp (ER_SO-2_OK_XAIP_OK_SO), - ER(H(DO-01),TSP(H(H(DO-01)))) – the hash value of the data object is placed in the single reduced hash tree and the hash value of those hash value (computed hash tree root) is included in the message imprint of the timestamp (ER_SO-3_OK_XAIP_OK_SO).
Configuration	CONFIG_S4
Pre-test conditions	Authenticated connection to middleware exists.

Step	Test sequence	Expected Results
1	VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64Data(XAIP_OK_SO(AOID-100, VID-01)))) dss:SignatureObject(dss:Other(XAIP_ERS(AOID-100,VID-01, ER_SO-1_OK_XAIP_OK_SO))))	VerifyResponse (dss:Result(resultmajor#ok))

Step	Test sequence	Expected Results
2	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64Data(XAIP_OK_SO(AOID-100, VID-01)))) dss:SignatureObject(dss:Other(XAIP_ERS(AOID-100,VID-01, ER_SO-2_OK_XAIP_OK_SO)))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#ok)) </pre>
3	<pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:Base64Data(XAIP_OK_SO(AOID-100, VID-01)))) dss:SignatureObject(dss:Other(XAIP_ERS(AOID-100,VID-01, ER_SO-3_OK_XAIP_OK_SO)))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#ok)) </pre>

Verdict of M-ADD-04

Observations:

Verdict:

3.4.2 Tests of the TR-S.512-interface

The following section introduces the mandatory tests to be passed through in case the TOT does support additionally the TR-S.512-interface (or simply S.512) as well. The test-IDs of the S.512-tests have the small “b” at the tail, e.g. “M-RI-02**b**”.

3.4.2.1 Function RetrieveInfo of S.512-interface

3.4.2.1.1 M-RI-01b – Retrieve Preservation Profile

Identifier	M-RI-01b
Test Purpose	The test shall retrieve the actual preservation profile with the corresponding ProfileIdentifier: “ http://www.bsi.bund.de/tr-esor/V1.3.0/profile/preservation-api/V1.1.2 ”
Configuration	CONFIG_S.512
Pre-test conditions	- Authenticated connection to middleware exists. - At least one active Preservation Profile is available (ACTIVE_PROFILE) and one not-active (INACTIVE_PROFILE).

Step	Test sequence	Expected Results
1	RetrieveInfo (pres ²⁶ :Profile(ACTIVE_PROFILE_URI))	RetrieveInfoResponse (dsb ²⁷ :Result(resultmajor:Success), pres:Profile(ACTIVE_PROFILE))
2	RetrieveInfo (pres:Status(inactive))	RetrieveInfoResponse (dsb:Result(resultmajor:Success), pres:Profile(INACTIVE_PROFILE))
3	RetrieveInfo (pres:Status(all))	RetrieveInfoResponse (dsb:Result(resultmajor:Success), pres:Profile(ACTIVE_PROFILE), pres:Profile(INACTIVE_PROFILE))
4	RetrieveInfo (pres:Status(active))	RetrieveInfoResponse (dsb:Result(resultmajor:Success), pres:Profile(ACTIVE_PROFILE))
5	RetrieveInfo (pres:Profile(ACTIVE_PROFILE_URI), pres:Status(active))	RetrieveInfoResponse (dsb:Result(resultmajor:Success), pres:Profile(ACTIVE_PROFILE))
6	RetrieveInfo (pres:Profile(ACTIVE_PROFILE_URI), pres:Status(inactive))	RetrieveInfoResponse (dsb:Result(resultmajor:Success))

²⁶ Namespace „pres“ means „<http://uri.etsi.org/19512/v1.1.2#>“.

²⁷ Namespace „dsb“ means „<http://docs.oasis-open.org/dss-x/ns/base>“

Step	Test sequence	Expected Results
7	RetrieveInfo (pres:Profile(ACTIVE_PROFILE_URI), pres:Status(all))	RetrieveInfoResponse (dsb:Result(resultmajor:Success), pres:Profile(ACTIVE_PROFILE))
8	RetrieveInfoRequest ()	RetrieveInfoResponse (dsb:Result(resultmajor:Success), pres:Profile(ACTIVE_PROFILE))

Verdict of M-RE-01b

Observations:

Verdict:

3.4.2.1.2 M-RI-02b – RetrieveInfo with unknown URI

Identifier	M-RI-02b
Test Purpose	The test shall verify that it will yield an error, if <code>RetrieveInfo</code> is called without providing a known URI. As the present call also contains the <code>UNKNOWN_PROFILE_URI</code> , the <code>dsb:Result</code> needs to contain “ <code>http://uri.etsi.org/19512/error/notSupported</code> ” as content of <code>dsb:ResultMinor</code> element.
Configuration	CONFIG_S.512
Pre-test conditions	Authenticated connection to middleware exists.

Step	Test sequence	Expected Results
1	RetrieveInfo (pres:Profile(UNKNOWN_PROFILE_URI))	RetrieveInfoResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/notSupported))))
2	RetrieveInfo (pres:Profile(ACTIVE_PROFILE_URI))	RetrieveInfoResponse (dsb:Result(resultmajor:Success), pres:Profile(ACTIVE_PROFILE))

Verdict of M-RI-02b

Observations:

Verdict:

3.4.2.1.3 M-RI-03b – RetrieveInfo with unknown control in OptionalInputs

Identifier	M-RI-03b
Test Purpose	The test shall verify that there will be an error, if the request contains unknown controls in the <code>OptionalInputs</code> -element and the <code>RetrieveInfo</code> request will not be successful.
Configuration	CONFIG_S.512
Pre-test conditions	- Authenticated connection to middleware exists. - At least one active Preservation Profile is available with URI= "http://www.bsi.bund.de/tresor/V1.3/profile/preservation-api/V1.1.2" - The <code>OptionalInputs</code>-element and the <code>RetrieveInfo</code> request contains an unknown item.

Step	Test sequence	Expected Results
1	<pre>RetrieveInfo (pres:OptionalInputs(dsb:Other(dsb:Value(SomethingUnknown))), pres:Profile(ACTIVE_PROFILE_URI))</pre>	<pre>RetrieveInfoResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/notSupported)))</pre>
2	<pre>RetrieveInfo (Pres:Profile(ACTIVE_PROFILE_URI))</pre>	<pre>RetrieveInfoResponse (dsb:Result(resultmajor#ok), pres:Profile(ACTIVE_PROFILE))</pre>

Verdict of M-RI-03b
Observations:
Verdict:

3.4.2.2 Function PreservePO of S.512-interface

3.4.2.2.1 M-SU-01b – XAIP OK without AOID

Identifier	M-SU-01b
Test Purpose	The test shall verify that it is possible to submit XAIP_OK without AOID.
Configuration	CONFIG_S.512
Pre-test conditions	Authenticated connection to middleware exists.

Step	Test sequence	Expected Results
1	<pre> PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3", pres:xmlData(XAIP_OK))) </pre>	<pre> PreservePOResponse (dsb:Result(resultmajor:Success), pres:POID(AOID-01b)) </pre>
2	<pre> RetrievePO (pres:POID(AOID-01b), pres:SubjectOfRetrieval("PO")) </pre>	<pre> RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3", pres:xmlData(XAIP_OK(AOID-01b, VID-01)))) </pre>
3	<pre> RetrievePO²⁸ (pres:POID(AOID-01b), pres:SubjectOfRetrieval("Evidence")) </pre>	<pre> RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="urn:ietf:rfc:4998:EvidenceRecord", pres:xmlData(XAIP_ERS(AOID-01b,VID-01, XAIP_OK_V1_ER_OK)))) </pre>
4	Test of correctness of the obtained XAIP_OK(AOID-01, VID-01) by using AIP-eIDAS-SigValidator <pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:InlineXML(XAIP_OK(AOID-01b, VID-01))))) </pre>	<pre> VerifyResponse(dss:Result(resultmajor#ok), dss:OptionalOutputs(dssvr:VerificationReport(dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:XAIPReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid)) [...]))))) </pre>

²⁸ Make sure the tested TR-ESOR-system has already obtained the archive time stamp.

Step	Test sequence	Expected Results
5	Test of correctness of the obtained XAIP_OK_V1_ER_OK by using ERVerifyTool. <pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:InlineXML(XAIP_OK(AOID-01b, VID-01)))) dss:SignatureObject(dss:Other(XAIP_ERS(AOID-01b,VID-01, XAIP_OK_V1_ER_OK))))) </pre>	<pre> VerifyResponse(dss:Result(resultmajor#ok), dss:OptionalOutputs(dssvr:VerificationReport(dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:EvidenceRecordReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid) [...])))))) </pre>
6	<pre> PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3", pres:xmlData(XAIP_OK_SIG_OK))) </pre>	<pre> PreservePOResponse (dsb:Result(resultmajor:Success), pres:POID(AOID-02b)) </pre>
7	<pre> RetrievePO (pres:POID(AOID-02b), pres:SubjectOfRetrieval("PO")) </pre>	<pre> RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3", pres:xmlData(XAIP_OK_SIG_OK(AOID-02b, VID-01)))) </pre>
8	<pre> RetrievePO (pres:POID(AOID-02b), pres:SubjectOfRetrieval("Evidence")) </pre>	<pre> RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="urn:ietf:rfc:4998:EvidenceRecord", pres:xmlData(XAIP_ERS(AOID-02b,VID-01, XAIP_OK_SIG_OK_V1_ER_OK)))) </pre>

Verdict of M-SU-01b

Observations:

Verdict:

3.4.2.2.2 M-SU-01ab – LX AIP_OK without AOID

Identifier	M-SU-01ab
Test Purpose	The test shall verify that it is possible to submit XAIP_OK without AOID.

Identifier	M-SU-01ab
Configuration	- CONFIG_S.512 - CONFIG_LXAIP
Pre-test conditions	Authenticated connection to middleware exists.

Step	Test sequence	Expected Results
1	To be done manually! a) Following files have to be uploaded into middleware via upload/download-module (cf. chapters 4.2 and 4.3): 1) TXT_DATA_50.txt 2) XML_MDO_50.xml 3) PDF_DATA_50.pdf 4) TXT_DATA_51.txt 5) TXT_DATA_51.txt.p7s 6) PDF_DATA_51.pdf 7) PDF_DATA_51.pdf.p7s 8) XML_MDO_51.xml 9) TXT_DATA_52.txt 10) PDF_DATA_52.pdf 11) XML_MDO_52.xml 12) TXT_DATA_53.txt.p7m 13) XML_DATA_54.xml 14) XML_DATA_54.xml.p7s 15) TXT_DATA_55.txt 16) PDF_DATA_55.pdf 17) XML_MDO_55.xml 18) XML_DATA_56.xml 19) XML_DATA_56.xml.p7s 20) TXT_DATA_57.txt.p7m 21) TXT_DATA_58.txt 22) PDF_DATA_58.pdf 23) XML_MDO_58.xml 24) TXT_DATA_59.txt.p7m 25) TXT_DATA_60.txt.p7m 26) XML_DATA_60-1.xml 27) XML_DATA_60-1.xml.p7s 28) TXT_DATA_61.txt 29) TXT_DATA_61.txt.p7s 30) PDF_DATA_61.pdf 31) PDF_DATA_61.pdf.p7s 32) XML_MDO_61.xml 33) XML_DATA_62.xml 34) XML_DATA_62.xml.p7s 35) XML_DATA_63.xml 36) XML_DATA_63.xml.p7s 37) TXT_DATA_65.txt 38) TXT_DATA_65_NOK.p7s 39) PDF_DATA_65.pdf 40) PDF_DATA_65.pdf.p7s 41) XML_MDO_65.xml 42) TXT_DATA_66.txt 43) TXT_DATA_66.txt.p7s 44) PDF_DATA_66.pdf 45) PDF_DATA_66.pdf.p7s 46) XML_MDO_66.xml 47) TXT_DATA_67.txt 48) TXT_DATA_NOK_67.txt.p7s 49) PDF_DATA_67.pdf 50) PDF_DATA_67.pdf.p7s 51) XML_MDO_67.xml	a) As a result following data structures (proper instances of <code>asic:DataObjectReferences</code>) will be delivered: 1) REF_TXT_DATA_50b 2) REF_XML_MDO_50b 3) REF_PDF_DATA_50b 4) REF_TXT_DATA_51b 5) REF_CADES_DET_TXT_DATA_51b 6) REF_PDF_DATA_51b 7) REF_CADES_DET_PDF_DATA_51b 8) REF_XML_MDO_51b 9) REF_TXT_DATA_52b 10) REF_PDF_DATA_52b 11) REF_XML_MDO_52b 12) REF_CADES_ATT_TXT_DATA_53b 13) REF_XML_DATA_54b 14) REF_CADES_DET_XML_DATA_54b 15) REF_TXT_DATA_55b 16) REF_PDF_DATA_55b 17) REF_XML_MDO_55b 18) REF_XML_DATA_56b 19) REF_CADES_DET_XML_DATA_56b 20) REF_CADES_ATT_TXT_DATA_57b 21) REF_TXT_DATA_58b 22) REF_PDF_DATA_58b 23) REF_XML_MDO_58b 24) REF_CADES_ATT_TXT_DATA_59b 25) REF_CADES_ATT_TXT_DATA_60b 26) REF_XML_DATA_60-1b 27) REF_CADES_DET_XML_DATA_60-1b 28) REF_TXT_DATA_61b 29) REF_CADES_DET_TXT_DATA_61b 30) REF_PDF_DATA_61b 31) REF_CADES_DET_PDF_DATA_61b 32) REF_XML_MDO_61b 33) REF_XML_DATA_62b 34) REF_CADES_DET_XML_DATA_62b 35) REF_XML_DATA_63b 36) REF_CADES_DET_XML_DATA_63b 37) REF_TXT_DATA_65b 38) REF_CADES_DET_NOK_TXT_DATA_65b 39) REF_PDF_DATA_65b 40) REF_CADES_DET_PDF_DATA_65b 41) REF_XML_MDO_65b 42) REF_TXT_DATA_66b 43) REF_CADES_DET_TXT_DATA_66b 44) REF_PDF_DATA_66b 45) REF_CADES_DET_PDF_DATA_66b 46) REF_XML_MDO_66b 47) REF_TXT_DATA_67b 48) REF_CADES_DET_NOK_TXT_DATA_67b 49) REF_PDF_DATA_67b 50) REF_CADES_DET_PDF_DATA_67b 51) REF_XML_MDO_67b

Step	Test sequence	Expected Results
	52) TXT_DATA_68.txt 53) TXT_DATA_68.txt.p7s 54) PDF_DATA_68.pdf 55) PDF_DATA_68.pdf.p7s 56) XML_MDO_68.xml 57) XML_DATA_69.xml 58) XML_DATA_NOK_69.xml.p7s 59) XML_DATA_70.xml 60) XML_DATA_NOK_70.xml.p7s 61) TXT_DATA_71.txt 62) PDF_DATA_71.pdf 63) XML_MDO_71.xml 64) TXT_DATA_72.txt 65) PDF_DATA_72.pdf 66) XML_MDO_72.xml 67) TXT_DATA_73.txt.p7m 68) TXT_DATA_74.txt 69) PDF_DATA_74.pdf 70) XML_MDO_74.xml b) The references in the particular LXAIP* and DLXAIP* test objects will be adjusted according to the results in step a) automatically, the references have to be inserted into the configuration file of the testbed.	52) REF_TXT_DATA_68b 53) REF_CADES_DET_TXT_DATA_68b 54) REF_PDF_DATA_68b 55) REF_CADES_DET_PDF_DATA_68b 56) REF_XML_MDO_68b 57) REF_XML_DATA_69b 58) REF_CADES_DET_NOK_XML_DATA_69b 59) REF_XML_DATA_70b 60) REF_CADES_DET_NOK_XML_DATA_70b 61) REF_XML_MDO_71b 62) REF_PDF_DATA_71b 63) REF_XML_MDO_71b 64) REF_TXT_DATA_72b 65) REF_PDF_DATA_72b 66) REF_XML_MDO_72b 67) REF_CADES_ATT_TXT_DATA_73b 68) REF_TXT_DATA_74b 69) REF_PDF_DATA_74b 70) REF_XML_MDO_74b
2	To be done manually! Try to upload the XAIP_OK by using the upload/download interface of the middleware.	The upload-operation failed. Some kind of the machine-readable error code is delivered back. No data has been stored in the Storage.
3	To be done manually! Try to upload the LXAIP_OK by using the upload/download interface of the middleware.	The upload-operation failed. Some kind of the machine-readable error code is delivered back. No data has been stored in the Storage.
4	To be done manually! Try to upload the ASiC_AIP_OK by using the upload/download interface of the middleware.	The upload-operation failed. Some kind of the machine-readable error code is delivered back. No data has been stored in the Storage.
5	PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID=" http://www.bsi.bund.de/tr-esor/lxaip/1.3 ", pres:xmlData(LXAIP_OK(REF_TXT_DATA_50b, REF_PDF_DATA_50b, REF_XML_MDO_50b)))	PreservePOResponse (dsb:Result(resultmajor:Success), pres:POID(AOID-01ab))
6	RetrievePO (pres:POID(AOID-01ab), pres:SubjectOfRetrieval("PO"), pres:POFormat(http://www.bsi.bund.de/tr-esor/lxaip/1.3))	RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID=" http://www.bsi.bund.de/tr-esor/lxaip/1.3 ", pres:xmlData(LXAIP_OK(AOID-01ab, VID-01))))
7	RetrievePO (pres:POID(AOID-01ab), pres:SubjectOfRetrieval("Evidence"),)	RetrievePOResponse (dss:Result(resultmajor:Success), pres:PO(@FormatID="urn:ietf:rft:4998:EvidenceRecord", pres:xmlData(XAIP_ERS(AOID-01ab, VID-01, LXAIP_OK_V1_ER_OK))))

Step	Test sequence	Expected Results
		)
8	Make sure all in LXAIP_OK(AOID-01ab, VID-01) referenced files have been copied into corresponding LXAIP-directory of the AIP-eIDAS-SigValidator adjust the given references in in LXAIP_OK(AOID-01ab, VID-01) in order to get them resolveable by the AIP-eIDAS-SigValidator.	LXAIP_OK_XSV(AOID-01ab, VID-01)
9	Test of correctness of the obtained LXAIP_OK(AOID-01ab, VID-01) by using AIP-eIDAS-SigValidator VerifyRequest (dss:InputDocuments(dss:Document(dss:InlineXML(LXAIP_OK_XSV(AOID-01ab, VID-01)))))	VerifyResponse(dss:Result(resultmajor#ok), dss:OptionalOutputs(dssvr:VerificationReport(dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:XAIReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid) [...]))))
10	Test of correctness of the obtained LXAIP_OK_V1_ER_OK by using ERVerifyTool. VerifyRequest (dss:InputDocuments(dss:Document(dss:InlineXML(LXAIP_OK_XSV(AOID-01ab, VID-01)))) dss:SignatureObject(dss:Other(XAIP_ERS(AOID-01ab, VID-01, LXAIP_OK_V1_ER_OK))))	VerifyResponse(dss:Result(resultmajor#ok), dss:OptionalOutputs(dssvr:VerificationReport(dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:EvidenceRecordReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid) [...]))))
11	PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID=" http://www.bsi.bund.de/tr-esor/lxaip/1.3 ", pres:xmlData(LXAIP_OK_SIG_OK(REF_TXT_DATA_51b,	PreservePOResponse (dsb:Result(resultmajor:Success), AOID-02ab)

Step	Test sequence	Expected Results
	REF_CADES_DET_TXT_DATA_51b, REF_PDF_DATA_51b, REF_CADES_DET_PDF_DATA_51b, REF_XML_MDO_51b))))	
12	RetrievePO (pres:POID(AOID-02ab), pres:SubjectOfRetrieval("PO"), pres:POFormat(http://www.bsi.bund.de/tr-esor/lxaip/1.3))	RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID=" http://www.bsi.bund.de/tr-esor/lxaip/1.3 ", pres:xmlData(LXAIP_OK_SIG_OK(AOID-02ab, VID-01)))))
13	RetrievePO (pres:POID(AOID-02ab), pres:SubjectOfRetrieval("Evidence"),)	RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="urn:ietf:rfc:4998:EvidenceRecord", pres:xmlData(XAIP_ERS(AOID-02ab, VID-01, LXAIP_OK_SIG_OK_V1_ER_OK)))))

M-SU-01ab
Observations:
Verdict:

3.4.2.2.3 M-SU-02b – XAIP_NOK

Identifier	M-SU-02b
Test Purpose	The test shall verify that there will be an error, if a (syntactically incorrect) XAIP_NOK is submitted.
Configuration	CONFIG_S.512
Pre-test conditions	Authenticated connection to middleware exists.

Step	Test sequence	Expected Results
1	<pre> PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID="http://www.bsi.bund.de/tr- esor/xaip/1.3", pres:xmlData(XAIP_NOK))) </pre>	<pre> PreservePOResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError))) </pre> Alternatively response to the one above could be following or similar: <pre> <soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/" > <soap:Body> <soap:Fault> <faultcode>soap:Client</faultcode> <faultstring>Problems creating SAAJ object model</faultstring> </soap:Fault> </soap:Body> </soap:Envelope> </pre>
2	<pre> PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(pres:xmlData(XAIP_NOK_VERSION))) </pre>	<pre> PreservePOResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError))) </pre> Alternatively response to the one above could be following or similar: <pre> <soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/" > <soap:Body> <soap:Fault> <faultcode>soap:Client</faultcode> <faultstring>Problems creating SAAJ object model</faultstring> </soap:Fault> </soap:Body> </soap:Envelope> </pre>

Verdict of M-SU-02b

Observations:

Verdict:

3.4.2.2.4 M-SU-02ab – LXAIP_NOK

Identifier	M-SU-02ab
Test Purpose	The test shall verify that there will be an error, if a (syntactically incorrect, missing element ds:DigestValue in element asic:DataObjectReference) LXAIP_NOK is submitted.
Configuration	- CONFIG_S.512 - CONFIG_LXAIP
Pre-test conditions	Authenticated connection to middleware exists.

Step	Test sequence	Expected Results
1	<pre> PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/lxaip/1.3", pres.xmlData(LXAIP_NOK))) </pre>	<pre> PreservePOResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError))) </pre> Alternatively response to the one above could be following or similar: <pre> <soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/" > <soap:Body> <soap:Fault> <faultcode>soap:Client</faultcode> <faultstring>Problems creating SAAJ object model</faultstring> </soap:Fault> </soap:Body> </soap:Envelope> </pre>

Verdict of M-SU-02ab

Observations:

Verdict:

3.4.2.2.5 M-SU-03b – XAIP_NOK_EXPIRED

Identifier	M-SU-03b
Test Purpose	The test shall verify that there will be an error, if the semantically incorrect XAIP_NOK_EXPIRED is submitted.
Configuration	CONFIG_S.512
Pre-test conditions	Authenticated connection to middleware exists.

Step	Test sequence	Expected Results
1	<pre> PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3", pres.xmlData(XAIP_NOK_EXPIRED))) </pre>	<pre> PreservePOResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError))) </pre>

Verdict of M-SU-03b

Observations:

Verdict of M-SU-03b

Verdict:

3.4.2.2.6 M-SU-04b – XAIP_NOK_SUBMTIME

Identifier	M-SU-04b
Test Purpose	The test shall verify that there will be an error, if the semantically incorrect XAIP_NOK_SUBMTIME is submitted.
Configuration	CONFIG_S.512
Pre-test conditions	Authenticated connection to middleware exists.

Step	Test sequence	Expected Results
1	Prepare XAIP_NOK_SUBMTIME, set the value of the element <code>xaip:submissionTime</code> in <code>xaip:submissionInfo</code> to the time in the past	XAIP_NOK_SUBMTIME(past)
2	<pre> PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(pres:xmlData(@FormatID="http://www.bsi.bund.de/tr- esor/xaip/1.3", XAIP_NOK_SUBMTIME(past)))) </pre>	<pre> PreservePOResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError))) </pre>
3	Prepare XAIP_NOK_SUBMTIME, set the value of the element <code>xaip:submissionTime</code> in <code>xaip:submissionInfo</code> to the time in the future	XAIP_NOK_SUBMTIME(future)
4	<pre> PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID="http://www.bsi.bund.de/tr- esor/xaip/1.3", pres:xmlData(XAIP_NOK_SUBMTIME(future)))) </pre>	<pre> PreservePOResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError))) </pre>

Verdict of M-SU-04b

Observations:

Verdict:

3.4.2.2.7 M-SU-05b – XAIP_NOK_SIG

Identifier	M-SU-05b
Test Purpose	The test shall verify that there will be an error, if the submitted XAIP_NOK_SIG contains digital signatures, which do not verify correctly ²⁹ .
Configuration	CONFIG_S.512
Pre-test conditions	- Authenticated connection to middleware exists. - The TOT is configured only to accept valid signatures.

Step	Test sequence	Expected Results
1	<pre> PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3", pres:xmlData(XAIP_NOK_SIG))) </pre>	<pre> PreservePOResponse (dsb:OptionalOutputs(dsb:Other(dsb:Value(XAIP_NOK_SIG_VR))), dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError))) </pre>

Verdict of M-SU-05b
Observations:
Verdict:

3.4.2.2.8 M-SU-06b – XAIP_NOK_ER

Identifier	M-SU-06b
Test Purpose	The test shall verify that there will be an error, because the submitted XAIP_NOK_ER contains an incorrect Evidence Record.
Configuration	CONFIG_S.512
Pre-test conditions	Authenticated connection to middleware exists.

Step	Test sequence	Expected Results
1	<pre> PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3", pres:xmlData(XAIP_NOK_ER))) </pre>	<pre> PreservePOResponse (dsb:OptionalOutputs(dsb:Other(dsb:Value(XAIP_NOK_ER_VR))), dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError))) </pre>

²⁹ See also Annex E (Version 1.3, Chapter 3.7.1 (VerifyUnderSignaturePolicy)).

Verdict of M-SU-06b
Observations:
Verdict:

3.4.2.2.9 M-SU-06ab – LXAIP_NOK_ER

Identifier	M-SU-06ab
Test Purpose	The test shall verify that there will be an error, because the submitted LXAIP_OK_ER_NOK contains an incorrect Evidence Record.
Configuration	- CONFIG_S.512 - CONFIG_LXAIP
Pre-test conditions	- Authenticated connection to middleware exists. - Test case M-SU-01ab has been performed successfully.

Step	Test sequence	Expected Results
1	<pre> PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/lxaip/1.3", pres:xmlData(LXAIP_OK_ER_NOK(REF_TXT_DATA_52b, REF_PDF_DATA_52b, REF_XML_MDO_52b, ER_NOK_LXAIP_OK))) </pre>	<pre> PreservePOResponse (dsb:OptionalOutputs(dsb:Other(dsb:Value(LXAIP_NOK_ER_VR))), dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError))) </pre>

Verdict of M-SU-06ab
Observations:
Verdict:

3.4.2.2.10 M-SU-07b – XAIP_OK and unknown control in OptionalInputs

Identifier	M-SU-07b
Test Purpose	The test shall verify that there will be an error, if the request contains unknown controls in the <code>OptionalInputs</code> -element and the archive data object will not be imported.
Configuration	CONFIG_S.512
Pre-test conditions	Authenticated connection to middleware exists.

Step	Test sequence	Expected Results
1	<pre> PreservePO (pres:OptionalInputs(dsb:Other(dsb:Value(SomethingUnknown))), pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3", pres:xmlData(XAIP_OK))) </pre>	<pre> PreservePOResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/notSupported))) </pre>

Verdict of M-SU-07b
Observations:
Verdict:

3.4.2.2.11 M-SU-08b – BIN_OK

Identifier	M-SU-08b
Test Purpose	The test shall validate that it is possible to submit as BIN defined data. BIN means here: - CADES_ATT_OK, - XADES_ENVELOPED_OK, XADES_ENVELOPING_OK, - PADES_OK, - ASiC_S_CADES_OK, ASiC_S_XADES_OK, ASiC_S_TST_OK, ASiC_S_ER_OK, - ASiC_E_CADES_OK, ASiC_E_XADES_OK, ASiC_E_TST_OK, ASiC_E_ER_OK, - DIGEST_LIST.
Configuration	CONFIG_S.512
Pre-test conditions	Authenticated connection to middleware exists.

Step	Test sequence	Expected Results
1	<pre> PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID="http://uri.etsi.org/ades/CAAdES", pres:binaryData(CADES_ATT_OK))) </pre>	<pre> PreservePOResponse (dsb:Result(resultmajor:Success), pres:POID(AOID-200b)) </pre>

Step	Test sequence	Expected Results
	))	
2	RetrievePO (pres:POID(AOID-200b), pres:SubjectOfRetrieval("PO"))	RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID=" http://www.bsi.bund.de/tr-esor/xaip/1.3 ", pres:xmlData(XAIP(AOID-200b, VID-01, credential(CADES_ATT_OK))))
3	PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID=" http://uri.etsi.org/ades/XAdES ", pres:binaryData(XADES_ENVELOPED_OK)))	PreservePOResponse (dsb:Result(resultmajor:Success), pres:POID(AOID-201b))
4	RetrievePO (pres:POID(AOID-201b), pres:SubjectOfRetrieval("PO"))	RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID=" http://www.bsi.bund.de/tr-esor/xaip/1.3 ", pres:xmlData(XAIP(AOID-201b, VID-01, dataObject(xmlData(XADES_ENVELOPED_OK)))))) OR ³⁰ RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID=" http://www.bsi.bund.de/tr-esor/xaip/1.3 ", pres:xmlData(XAIP(AOID-201b, VID-01, dataObject(binaryData(XADES_ENVELOPED_OK))))))
5	PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID=" http://uri.etsi.org/ades/XAdES ", pres:binaryData(XADES_ENVELOPING_OK)))	PreservePOResponse (dsb:Result(resultmajor:Success), pres:POID(AOID-202b))
6	RetrievePO (pres:POID(AOID-202b), pres:SubjectOfRetrieval("PO"))	RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID=" http://www.bsi.bund.de/tr-esor/xaip/1.3 ", pres:xmlData(XAIP(AOID-202b, VID-01, credential(SignatureObject(XADES_ENVELOPING_OK))))

³⁰ An enveloped XAdES could be placed in XAIP as binary data stream in an element `<xaip:binaryData>` (as recommended in [TR-ESOR-F], section 6.2), or as xml data stream in an element `<xaip:xmlData>` in the `<xaip:dataObjectsSection>`-element.

Step	Test sequence	Expected Results
		<pre>))) OR³¹ RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3", pres:xmlData(XAIP(AOID-202b, VID-01, credential(Base64Signature(XADES_ENVELOPING_OK))))))) </pre>
7	<pre> PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID="http://uri.etsi.org/ades/PAdES", pres:binaryData(PADES_OK))) </pre>	<pre> PreservePOResponse (dsb:Result(resultmajor:Success), pres:POID(AOID-203b)) </pre>
8	<pre> RetrievePO (pres:POID(AOID-203b), pres:SubjectOfRetrieval("PO")) </pre>	<pre> RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3", pres:xmlData(XAIP(AOID-203b, VID-01, dataObject(PADES_OK))))) </pre>
9	<pre> PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID="http://uri.etsi.org/ades/ASiC/type/ASiC-E", pres:binaryData(ASiC_E_CADES_OK))) </pre>	<pre> PreservePOResponse (dsb:Result(resultmajor:Success), pres:POID(AOID-204b)) </pre>
10	<pre> RetrievePO (pres:POID(AOID-204b), pres:SubjectOfRetrieval("PO")) </pre>	<pre> RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3", pres:xmlData(XAIP(AOID-204b, VID-01, credential(ASiC_E_CADES_OK))))) </pre>
11	<pre> PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(</pre>	<pre> PreservePOResponse (dsb:Result(resultmajor:Success), pres:POID(AOID-205b)) </pre>

³¹ An enveloping XAdES could be placed in XAIP as binary credential data stream in an element `<dss:base64Signature>` (as recommended in [TR-ESOR-F], section 6.2), or as xml credential data stream directly as element `<dss:Signature>` in the `<dss:SignatureObject>`-element of the `<xaip:credential>` parent element.

Step	Test sequence	Expected Results
	<pre>@FormatID="http://uri.etsi.org/ades/ASiC/type/ASiC-E", pres:binaryData(ASIC_E_XADES_OK)))</pre>	
12	<pre>RetrievePO (pres:POID(AOID-205b), pres:SubjectOfRetrieval("PO"))</pre>	<pre>RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3", pres:xmlData(XAIP(AOID-205b, VID-01, credential(ASIC_E_XADES_OK)))))</pre>
13	<pre>PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID="http://uri.etsi.org/ades/ASiC/type/ASiC-E", pres:binaryData(ASIC_E_TST_OK)))</pre>	<pre>PreservePOResponse (dsb:Result(resultmajor:Success), pres:POID(AOID-206b))</pre>
14	<pre>RetrievePO (pres:POID(AOID-206b), pres:SubjectOfRetrieval("PO"))</pre>	<pre>RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3", pres:xmlData(XAIP(AOID-206b, VID-01, credential(ASIC_E_TST_OK)))))</pre>
15	<pre>PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID="http://uri.etsi.org/ades/ASiC/type/ASiC-E", pres:binaryData(ASIC_E_ER_OK)))</pre>	<pre>PreservePOResponse (dsb:Result(resultmajor:Success), pres:POID(AOID-207b))</pre>
16	<pre>RetrievePO (pres:POID(AOID-207b), pres:SubjectOfRetrieval("PO"))</pre>	<pre>RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3", pres:xmlData(XAIP(AOID-207b, VID-01, credential(ASIC_E_ER_OK)))))</pre>
17	<pre>PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID="http://uri.etsi.org/19512/format/DigestList", pres:binaryData(DIGEST_LIST)))</pre>	<pre>PreservePOResponse (dsb:Result(resultmajor:Success), pres:POID(AOID-208b))</pre>

Step	Test sequence	Expected Results
18	RetrievePO (pres:POID(AOID-208b), pres:SubjectOfRetrieval("PO"))	RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID=" http://www.bsi.bund.de/tr-esor/xaip/1.3 ", pres:xmlData(XAIP(AOID-208b, VID-01, dataObject(xmlData(DIGEST_LIST))))))) OR ³² RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID=" http://www.bsi.bund.de/tr-esor/xaip/1.3 ", pres:xmlData(XAIP(AOID-208b, VID-01, dataObject(binaryData(DIGEST_LIST)))))))
19	PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID=" http://uri.etsi.org/ades/ASiC/type/ASiC-S ", pres:binaryData(ASiC_S_CADES_OK)))	PreservePOResponse (dsb:Result(resultmajor:Success), pres:POID(AOID-209b)))
20	RetrievePO (pres:POID(AOID-209b), pres:SubjectOfRetrieval("PO"))	RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID=" http://www.bsi.bund.de/tr-esor/xaip/1.3 ", pres:xmlData(XAIP(AOID-209b, VID-01, credential(ASiC_S_CADES_OK))))))
21	PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID=" http://uri.etsi.org/ades/ASiC/type/ASiC-S ", pres:binaryData(ASiC_S_XADES_OK)))	PreservePOResponse (dsb:Result(resultmajor:Success), pres:POID(AOID-210b)))
22	RetrievePO (pres:POID(AOID-210b), pres:SubjectOfRetrieval("PO"))	RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID=" http://www.bsi.bund.de/tr-esor/xaip/1.3 ", pres:xmlData(XAIP(AOID-210b, VID-01, credential(ASiC_S_XADES_OK))))))

³² A digest list as XML based data object could be placed in XAIP as binary data stream in an element `<xaip:binaryData>` or as xml data stream in an element `<xaip:xmlData>` in the `<xaip:dataObjectsSection>`-element.

Step	Test sequence	Expected Results
23	<pre> PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID="http://uri.etsi.org/ades/ASiC/type/ASiC-S", pres:binaryData(ASIC_S_TST_OK))) </pre>	<pre> PreservePOResponse (dsb:Result(resultmajor:Success), pres:POID(AOID-211b)) </pre>
24	<pre> RetrievePO (pres:POID(AOID-211b), pres:SubjectOfRetrieval("PO")) </pre>	<pre> RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3", pres:xmlData(XAIP(AOID-211b, VID-01, credential(ASIC_S_TST_OK))))) </pre>
25	<pre> PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID="http://uri.etsi.org/ades/ASiC/type/ASiC-S", pres:binaryData(ASIC_S_ER_OK))) </pre>	<pre> PreservePOResponse (dsb:Result(resultmajor:Success), pres:POID(AOID-212b)) </pre>
26	<pre> RetrievePO (pres:POID(AOID-212b), pres:SubjectOfRetrieval("PO")) </pre>	<pre> RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3", pres:xmlData(XAIP(AOID-212b, VID-01, credential(ASIC_S_ER_OK))))) </pre>

Verdict of M-SU-08b

Observations:

Verdict:

3.4.2.2.12 M-SU-09b – BIN_NOK

Identifier	M-SU-09b
Test Purpose	The test shall validate that there will be an error in case an as BIN defined data containing an invalid signature will be submitted. BIN means here: • CADES_ATT_NOK • XADES_ENVELOPED_NOK, XADES_ENVELOPING_NOK, • PADES_NOK, • ASIC_E_CADES_NOK, ASIC_E_XADES_NOK, ASIC_E_TST_NOK, ASIC_E_ER_NOK, • ASIC_S_CADES_NOK, ASIC_S_XADES_NOK, ASIC_S_TST_NOK, ASIC_S_ER_NOK.
Configuration	• CONFIG_S.512

Identifier	M-SU-09b
Pre-test conditions	- Authenticated connection to middleware exists. - The TOT is configured only to accept valid signatures.

Step	Test sequence	Expected Results
1	<pre> PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID="http://uri.etsi.org/ades/CAAdES", pres:binaryData(CADES_ATT_NOK))) </pre>	<pre> PreservePOResponse (dsb:OptionalOutputs(dsb:Other(dsb:Value(CADES_ATT_NOK_VR))), dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError))) </pre>
2	<pre> PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID="http://uri.etsi.org/ades/XAdES", pres:binaryData(XADES_ENVELOPED_NOK))) </pre>	<pre> PreservePOResponse (dsb:OptionalOutputs(dsb:Other(dsb:Value(XADES_ENVELOPED_NOK_VR))), dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError))) </pre>
3	<pre> PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID="http://uri.etsi.org/ades/XAdES", pres:binaryData(XADES_ENVELOPING_NOK))) </pre>	<pre> PreservePOResponse (dsb:OptionalOutputs(dsb:Other(dsb:Value(XADES_ENVELOPING_NOK_VR))), dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError))) </pre>
4	<pre> PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID="http://uri.etsi.org/ades/PAAdES", pres:binaryData(PADES_NOK))) </pre>	<pre> PreservePOResponse (dsb:OptionalOutputs(dsb:Other(dsb:Value(PADES_NOK_VR))), dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError))) </pre>
5	<pre> PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID="http://uri.etsi.org/ades/ASiC/typ e/ASiC-E", pres:binaryData(ASIC_E_CADES_NOK))) </pre>	<pre> PreservePOResponse (dsb:OptionalOutputs(dsb:Other(dsb:Value(ASIC_E_CADES_NOK_VR))), dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError))) </pre>
6	<pre> PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID="http://uri.etsi.org/ades/ASiC/typ e/ASiC-E", pres:binaryData(ASIC_E_XADES_NOK))) </pre>	<pre> PreservePOResponse (dsb:OptionalOutputs(dsb:Other(dsb:Value(ASIC_E_XADES_NOK_VR))), dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError))) </pre>

Step	Test sequence	Expected Results
7	<pre> PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID="http://uri.etsi.org/ades/ASiC/type/ASiC-E", pres:binaryData(ASIC_E_TST_NOK))) </pre>	<pre> PreservePOResponse (dsb:OptionalOutputs(dsb:Other(dsb:Value(ASIC_E_TST_NOK_VR)), dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)))) </pre>
8	<pre> PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID="http://uri.etsi.org/ades/ASiC/type/ASiC-E", pres:binaryData(ASIC_E_ER_NOK))) </pre>	<pre> PreservePOResponse (dsb:OptionalOutputs(dsb:Other(dsb:Value(ASIC_E_ER_NOK_VR)), dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)))) </pre>
9	<pre> PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID="http://uri.etsi.org/ades/ASiC/type/ASiC-S", pres:binaryData(ASIC_S_CADES_NOK))) </pre>	<pre> PreservePOResponse (dsb:OptionalOutputs(dsb:Other(dsb:Value(ASIC_S_CADES_NOK_VR)), dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)))) </pre>
10	<pre> PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID="http://uri.etsi.org/ades/ASiC/type/ASiC-S", pres:binaryData(ASIC_S_XADES_NOK))) </pre>	<pre> PreservePOResponse (dsb:OptionalOutputs(dsb:Other(dsb:Value(ASIC_S_XADES_NOK_VR)), dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)))) </pre>
11	<pre> PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID="http://uri.etsi.org/ades/ASiC/type/ASiC-S", pres:binaryData(ASIC_S_TST_NOK))) </pre>	<pre> PreservePOResponse (dsb:OptionalOutputs(dsb:Other(dsb:Value(ASIC_S_TST_NOK_VR)), dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)))) </pre>
12	<pre> PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID="http://uri.etsi.org/ades/ASiC/type/ASiC-S", pres:binaryData(ASIC_S_ER_NOK))) </pre>	<pre> PreservePOResponse (dsb:OptionalOutputs(dsb:Other(dsb:Value(ASIC_S_ER_NOK_VR)), dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)))) </pre>

Verdict of M-SU-09b

Observations:

Verdict of M-SU-09b**Verdict:**

3.4.2.3 Function UpdatePOC of S.512-interface

3.4.2.3.1 M-UP-01b – DXAIP_OK

Identifier	M-UP-01b
Test Purpose	The test shall verify that it is possible to submit a correct DXAIP_OK with UpdatePO request.
Configuration	CONFIG_S.512
Pre-test conditions	- Authenticated connection to middleware exists. - Make sure that test case M-SU-01b was successfully performed with AOID-01b.

Step	Test sequence	Expected Results
1	Insert AOID-01b into the prepared DXAIP_OK-template.	DXAIP_OK(AOID-01b, VID-01, VID-02)
2	UpdatePOC (pres:POID(AOID-01b), pres:DeltaPOC(@FormatID(" http://www.bsi.bund.de/tr-esor/dxaip/1.3 "), pres:xmlData(DXAIP_OK(AOID-01b, VID-01, VID-02)))))	UpdatePOCResponse (dsb:Result(resultmajor:Success), VersionID(VID-02))
3	Insert AOID-01b into the prepared DXAIP_OK_2-template.	DXAIP_OK_2(AOID-01b, VID-02, VID-03)
4	UpdatePOC (pres:POID(AOID-01b), pres:DeltaPOC(@FormatID(" http://www.bsi.bund.de/tr-esor/dxaip/1.3 "), pres:xmlData(DXAIP_OK_2(AOID-01b,VID-02,VID-03)))))	UpdatePOCResponse (dsb:Result(resultmajor:Success), VersionID(VID-03))
5	PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID=" http://www.bsi.bund.de/tr-esor/xaip/1.3 ", pres:xmlData(XAIP_OK))))	PreservePOResponse (dsb:Result(resultmajor:Success), pres:POID(AOID-01zb))
6	Insert AOID-01zb into the prepared DXAIP_OK_SIG_ONLY-template.	DXAIP_OK_SIG_ONLY(AOID-01zb, VID-01, VID-02)

Step	Test sequence	Expected Results
7	<pre>UpdatePOC (pres:POID(AOID-01zb), pres:DeltaPOC(@FormatID("http://www.bsi.bund.de/tr- esor/dxaip/1.3"), pres:xmlData(DXAIP_OK_SIG_ONLY(AOID-01zb, VID-01, VID-02)))))</pre>	<pre>UpdatePOCResponse (dsb:Result(resultmajor:Success), VersionID(VID-02))</pre>

Verdict of M-UP-01b

Observations:

Verdict:

3.4.2.3.2 M-UP-01ab – DLXAIP_OK

Identifier	M-UP-01ab
Test Purpose	The test shall verify that it is possible to submit a correct DLXAIP_OK with <code>UpdatePO</code> request.
Configuration	- CONFIG_S.512 - CONFIG_LXAIP
Pre-test conditions	- Authenticated connection to middleware exists. - Make sure that test case M-SU-01ab and M-UP-01ab was successfully performed with AOID-01ab.

Step	Test sequence	Expected Results
1	Insert AOID-01ab into the prepared DLXAIP_OK-template.	DLXAIP_OK(AOID-01ab, VID-01, VID-02)
2	<pre>UpdatePOC (pres:POID(AOID-01ab), pres:DeltaPOC(@FormatID("http://www.bsi.bund.de/tr- esor/dlxaip/1.3"), pres:xmlData(DLXAIP_OK(AOID-01ab,VID-01,VID-02, REF_CADES_ATT_TXT_DATA_53b)))))</pre>	<pre>UpdatePOCResponse (dsb:Result(resultmajor:Success), VersionID(VID-02))</pre>
3	Insert AOID-01ab into the prepared DLXAIP_OK_2-template.	DLXAIP_OK_2(AOID-01ab, VID-02, VID-03)

Step	Test sequence	Expected Results
4	<pre> UpdatePOC (pres:POID(AOID-01ab), pres:DeltaPOC(@FormatID("http://www.bsi.bund.de/tr- esor/dlxaip/1.3"), pres:xmlData(DLXAIP_OK_2(AOID-01ab, VID-02, VID-03, REF_XML_DATA_54b, REF_CADES_DET_XML_DATA_54b))))) </pre>	<pre> UpdatePOCResponse (dsb:Result(resultmajor:Success), VersionID(VID-03)) </pre>

Verdict of A-UP-01ab

Observations:

Verdict:

3.4.2.3.3 M-UP-02b – DXAIP_NOK_AOID

Identifier	M-UP-02b
Test Purpose	The test shall verify that there will be an error, if a DXAIP_NOK_AOID with a not yet assigned AOID is submitted.
Configuration	CONFIG_S.512
Pre-test conditions	Authenticated connection to middleware exists.

Step	Test sequence	Expected Results
1	Choice an AOID-03b within the possible range and make sure, that it has not been assigned yet. This is realised by the following call: <pre> RetrievePO (pres:POID(AOID-03b), pres:SubjectOfRetrieval("PO")) </pre>	<pre> PreservePOResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/unknownPOID))) </pre>
2	Insert AOID-03b into DXAIP_NOK_AOID.	DXAIP_NOK_AOID(AOID-03b)
3	<pre> UpdatePOC (pres:POID(AOID-01ab), pres:DeltaPOC(@FormatID("http://www.bsi.bund.de/tr- esor/dxaip/1.3"), pres:xmlData(DXAIP_NOK_AOID(AOID-03b))))) </pre>	<pre> UpdatePOCResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/unknownPOID))) </pre>

Verdict of M-UP-02b
Observations:
Verdict:

3.4.2.3.4 M-UP-03b – DXAIP_NOK

Identifier	M-UP-03b
Test Purpose	The test shall verify that there will be an error, if a (syntactically incorrect) DXAIP_NOK is submitted.
Configuration	CONFIG_S.512
Pre-test conditions	Authenticated connection to middleware exists.

Step	Test sequence	Expected Results
1	<pre> PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3", pres:xmlData(XAIP_OK))) </pre>	<pre> PreservePOResponse (dsb:Result(resultmajor:Success), pres:POID(AOID-04b)) </pre>
2	Insert AOID-04b into the prepared DXAIP_NOK-template.	DXAIP_NOK(AOID-04b, VID-01, VID-02)
3	<pre> UpdatePOC (pres:POID(AOID-04b), pres:DeltaPOC(@FormatID("http://www.bsi.bund.de/tr-esor/dxaip/1.3"), pres:xmlData(DXAIP_NOK(AOID-04b, VID-01, VID-02)))) </pre>	<pre> UpdatePOResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/DeltaPOCInternalProblem))) </pre> Alternatively, response to the one above could be following or similar: <pre> <soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"> <soapenv:Body> <soapenv:Fault> <faultcode>Server</faultcode> <faultstring>Failed to dispatch using script; java.lang.Exception: VALIDATION ERRORS: [line 1: Expected element 'protectedObjectPointer@http://www.bsi.bund.de/tr-esor/xaip' instead of 'AOID@http://www.bsi.bund.de/tr-esor/xaip' here in element packageInfoUnit@http://www.bsi.bund.de/tr-esor/xaip]</faultstring> </soapenv:Fault> </soapenv:Body> </soapenv:Envelope>> </pre>

Verdict of M-UP-03b
Observations:

Verdict of M-UP-03b

Verdict:

3.4.2.3.5 M-UP-03ab – DLXAIP_NOK

Identifier	M-UP-03ab
Test Purpose	The test shall verify that there will be an error, if a (syntactically incorrect) DLXAIP_NOK is submitted.
Configuration	- CONFIG_S.512 - CONFIG_LXAIP
Pre-test conditions	Authenticated connection to middleware exists.

Step	Test sequence	Expected Results
1	PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID=" http://www.bsi.bund.de/tr-esor/lxaip/1.3 ", pres:xmlData(LXAIP_OK(REF_TXT_DATA_55b, REF_PDF_DATA_55b, REF_XML_MDO_55b))))	PreservePOResponse (dsb:Result(resultmajor:Success), pres:POID(AOID-04ab))
2	Insert AOID-04ab into the prepared DLXAIP_NOK-template.	DLXAIP_NOK(AOID-04ab, VID-01, VID-02)
3	UpdatePOC (pres:POID(AOID-04ab, pres:DeltaPOC(@FormatID(" http://www.bsi.bund.de/tr-esor/dlxaip/1.3 "), pres:xmlData(DLXAIP_NOK(AOID-04ab, VID-01, VID-02 REF_XML_DATA_56b, REF_CADES_DET_XML_DATA_56b))))	UpdatePOResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/DeltaPOCInternalProblem))) Alternatively, response to the one above could be following or similar: <soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"> <soapenv:Body> <soapenv:Fault> <faultcode>Server</faultcode> <faultstring>Failed to dispatch using script; java.lang.Exception: VALIDATION ERRORS: [line 1: Expected element 'protectedObjectPointer@http://www.bsi.bund.de/tr-esor/xaip' instead of 'AOID@http://www.bsi.bund.de/tr-esor/xaip' here in element packageInfoUnit@http://www.bsi.bund.de/tr-esor/xaip]</faultstring> </soapenv:Fault> </soapenv:Body> </soapenv:Envelope>

Verdict of M-UP-03ab
Observations:
Verdict:

3.4.2.3.6 M-UP-04b – DXAIP_NOK_EXPIRED

Identifier	M-UP-04b
Test Purpose	The test shall verify that there will be an error, if the invalid DXAIP_NOK_EXPIRED is submitted.
Configuration	CONFIG_S.512
Pre-test conditions	- Authenticated connection to middleware exists. - Make sure that test case M-UP-03b was successfully performed with AOID-04b.

Step	Test sequence	Expected Results
1	Insert AOID-04b into the prepared DXAIP_NOK_EXPIRED-template.	DXAIP_NOK_EXPIRED(AOID-04b, VID-01, VID-02)
2	<pre> UpdatePOC (pres:POID(AOID-04b), pres:DeltaPOC(@FormatID("http://www.bsi.bund.de/tr- esor/dxaip/1.3"), pres:xmlData(DXAIP_NOK_EXPIRED(AOID-04b, VID-01, VID-02)))) </pre>	<pre> UpdatePOResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/DeltaPOCInternalProblem))) </pre>

Verdict of M-UP-04b
Observations:
Verdict:

3.4.2.3.7 M-UP-05b – DXAIP_NOK_SUBMETIME

Identifier	M-UP-05b
Test Purpose	The test shall verify that there will be an error, if the invalid DXAIP_NOK_SUBMETIME is submitted.
Configuration	CONFIG_S.512
Pre-test conditions	- Authenticated connection to middleware exists. - Make sure that test case M-UP-03b was successfully performed with AOID-04b.

Step	Test sequence	Expected Results
1	Insert AOID-04b into the prepared DXAIP_NOK_SUBMTIME-template Prepare DXAIP_NOK_SUBMTIME, set the value of the element <code>xaip:submissionTime</code> in <code>xaip:submissionInfo</code> to the time in the past.	DXAIP_NOK_SUBMTIME(AOID-04b, VID-01, VID-02, SubmissionTime(in the past))
2	UpdatePOC (pres:POID(AOID-04b), pres:DeltaPOC(@FormatID(" http://www.bsi.bund.de/tr-esor/dxaip/1.3 "), pres:xmlData(DXAIP_NOK_SUBMTIME(AOID-04b, VID-01, VID-02, SubmissionTime(in the past))))))	UpdatePOResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/DeltaPOCInternalProblem))))
3	Insert AOID-04b into the prepared DXAIP_NOK_SUBMTIME-template Prepare DXAIP_NOK_SUBMTIME, set the value of the element <code>xaip:submissionTime</code> in <code>xaip:submissionInfo</code> to the time in the future	DXAIP_NOK_SUBMTIME(AOID-04b, VID-01, VID-02, SubmissionTime(in the future))
4	UpdatePOC (pres:POID(AOID-04b), pres:DeltaPOC(@FormatID(" http://www.bsi.bund.de/tr-esor/dxaip/1.3 "), pres:xmlData(DXAIP_NOK_SUBMTIME(AOID-04b, VID-01, VID-02, SubmissionTime(in the future))))))	UpdatePOResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/DeltaPOCInternalProblem))))

Verdict of M-UP-05b

Observations:

Verdict:

3.4.2.3.8 M-UP-06b – DXAIP_NOK_SIG

Identifier	M-UP-06b
Test Purpose	The test shall verify that there will be an error, if the submitted DXAIP_NOK_SIG contains digital signatures, which do not verify correctly.
Configuration	CONFIG_S.512
Pre-test conditions	- Authenticated connection to middleware exists. - Make sure that test case M-UP-03b was successfully performed with AOID-04b. - The TOT is configured only to accept valid signatures.

Step	Test sequence	Expected Results
1	Insert AOID-04b into the prepared DXAIP_NOK_SIG-template.	DXAIP_NOK_SIG(AOID-04b, VID-01, VID-02)
2	UpdatePOC (pres:POID(AOID-04b), pres:DeltaPOC(@FormatID(" http://www.bsi.bund.de/tr-esor/dxaip/1.3 "), pres:xmlData(DXAIP_NOK_SIG(AOID-04b, VID-01, VID-02))))))	PreservePOResponse (dsb:OptionalOutputs(dsb:Other(dsb:Value(DXAIP_NOK_SIG_VR(AOID-04b, VID-01, VID-02))))), dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/DeltaPOCInternalProblem))))

Verdict of M-UP-06b

Observations:

Verdict:

3.4.2.3.9 M-UP-07b – DXAIP_NOK_ER

Identifier	M-UP-07b
Test Purpose	The test shall verify that there will be an error, because the submitted DXAIP_NOK_ER contains an invalid Evidence Record.
Configuration	CONFIG_S.512
Pre-test conditions	- Authenticated connection to middleware exists. - Make sure that test case M-UP-03b was successfully performed with AOID-04b.

Step	Test sequence	Expected Results
1	Insert AOID-04b into the prepared DXAIP_NOK_ER-template.	DXAIP_NOK_ER(AOID-04b, VID-01, VID-02)
2	UpdatePOC (pres:POID(AOID-04b), pres:DeltaPOC(@FormatID(" http://www.bsi.bund.de/tr-esor/dxaip/1.3 "), pres:xmlData(DXAIP_NOK_ER(AOID-04b, VID-01, VID-02))))))	PreservePOResponse (dsb:OptionalOutputs(dsb:Other(dsb:Value(DXAIP_NOK_ER_VR(AOID-04b, VID-01, VID-02))))), dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/DeltaPOCInternalProblem))))

Verdict of M-UP-07b

Observations:

Verdict:

3.4.2.3.10 M-UP-07ab – DLXAIP_NOK_ER

Identifier	M-UP-07ab
Test Purpose	The test shall verify that there will be an error, because the submitted DLXAIP_NOK_ER contains an invalid Evidence Record.
Configuration	- CONFIG_S.512 - CONFIG_LXAIP
Pre-test conditions	- Authenticated connection to middleware exists. - Make sure that test case M-UP-03ab was successfully performed with AOID-04ab.

Step	Test sequence	Expected Results
1	Insert AOID-04ab into the prepared DLXAIP_NOK_ER-template.	DLXAIP_NOK_ER(AOID-04ab, VID-01, VID-02)
2	<pre>UpdatePOC (pres:POID(AOID-04b), pres:DeltaPOC(@FormatID("http://www.bsi.bund.de/tr-esor/dlxaip/1.3"), pres:xmlData(DLXAIP_NOK_ER(AOID-04ab, VID-01, VID-02, REF_CADES_ATT_TXT_DATA_57b, ER_NOK_LXAIP_OK_V2)))))</pre>	<pre>PreservePOResponse (dsb:OptionalOutputs(dsb:Other(dsb:Value(DLXAIP_NOK_ER_VR(AOID-04ab, VID-01, VID-02))), dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError))))</pre>

Verdict of M-UP-07ab
Observations:
Verdict:

3.4.2.3.11 M-UP-08b – DXAIP_NOK_VERSION

Identifier	M-UP-08b
Test Purpose	The test shall verify that there will be an error, if a DXAIP_NOK_VERSION is submitted with UpdatePO request, which produces a collision ³³ with the already existing XAIP.
Configuration	CONFIG_S.512
Pre-test conditions	- Authenticated connection to middleware exists. - Make sure that test case M-UP-03b was successfully performed with AOID-04b.

³³ A wrong „version“ (DXAIP_NOK_VERSION) may be caused by inserting a DXAIP where the value in the element `prevVersion` in the `updateSection` of the DXAIP is not the latest version of this XAIP.

Step	Test sequence	Expected Results
1	Insert AOID-04b into the prepared DXAIP_NOK_VERSION-template. Set the previous version to VID-10.	DXAIP_NOK_VERSION(AOID-04b, VID-10, VID-02)
2	UpdatePOC (pres:POID(AOID-04b), pres:DeltaPOC(@FormatID(" http://www.bsi.bund.de/tr-esor/dxaip/1.3 "), pres:xmlData(DXAIP_NOK_VERSION(AOID-04b, VID-10, VID-02)))))	UpdatePOResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/DeltaPOCInternalProblem))))
3	Insert AOID-04b into the prepared DXAIP_OK-template.	DXAIP_OK(AOID-04b, VID-01, VID-02)
4	UpdatePOC (pres:POID(AOID-04b), pres:DeltaPOC(@FormatID(" http://www.bsi.bund.de/tr-esor/dxaip/1.3 "), pres:xmlData(DXAIP_OK(AOID-04b, VID-01, VID-02)))))	UpdatePOCResponse (dsb:Result(resultmajor:Success), VersionID(VID-02)))
5	Insert AOID-04b into the prepared DXAIP_NOK_VERSION-template. Set the previous version to VID-01.	DXAIP_NOK_VERSION(AOID-04b, VID-01, VID-03)
6	UpdatePOC (pres:POID(AOID-04b), pres:DeltaPOC(@FormatID(" http://www.bsi.bund.de/tr-esor/dxaip/1.3 "), pres:xmlData(DXAIP_NOK_VERSION(AOID-04b, VID-01, VID-03)))))	UpdatePOResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/DeltaPOCInternalProblem))))

Verdict of M-UP-08b	
Observations:	
Verdict:	

3.4.2.3.12 M-UP-09b – DXAIP_OK with unknown control in OptionalInputs

Identifier	M-UP-09b
Test Purpose	The test shall verify that there will be an error, if the request contains unknown controls in the OptionalInputs-element.
Configuration	CONFIG_S.512

Identifier	M-UP-09b
Pre-test conditions	Authenticated connection to middleware exists.

Step	Test sequence	Expected Results
1	<pre> PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID="http://www.bsi.bund.de/tr- esor/xaip/1.3", pres:xmlData(XAIP_OK))) </pre>	<pre> PreservePOResponse (dsb:Result(resultmajor:Success), pres:POID(AOID-05b)) </pre>
2	Insert AOID-05b into the prepared DXAIP_OK-template.	DXAIP_OK(AOID-05b, VID-01, VID-02)
3	<pre> UpdatePOC (pres:OptionalInputs(dsb:Other(dsb:Value(SomethingUnknown))), pres:POID(AOID-05b, pres:DeltaPOC(@FormatID("http://www.bsi.bund.de/tr- esor/dxaip/1.3"), pres:xmlData(DXAIP_OK(AOID-05b, VID-01, VID-02)))) </pre>	<pre> UpdatePOCResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/notSupported))) </pre>
4	<pre> UpdatePOC (pres:POID(AOID-05b), pres:DeltaPOC(@FormatID("http://www.bsi.bund.de/tr- esor/dxaip/1.3"), pres:xmlData(DXAIP_OK(AOID-05b, VID-01, VID-02)))) </pre>	<pre> UpdatePOCResponse (dsb:Result(resultmajor:Success), VersionID(VID-02)) </pre>
5	Insert AOID-05b and VID-02 as previous and VID-03 as new versions into the prepared DXAIP_OK_2-template.	DXAIP_OK_2(AOID-05b, VID-02, VID-03)
6	<pre> UpdatePOC (pres:POID(AOID-05b, pres:DeltaPOC(@FormatID("http://www.bsi.bund.de/tr- esor/dxaip/1.3"), pres:xmlData(DXAIP_OK_2(AOID-05b, VID-02, VID-03)))) </pre>	<pre> UpdatePOCResponse (dsb:Result(resultmajor:Success), VersionID(VID-03)) </pre>

Verdict of M-UP-09b
Observations:
Verdict:

3.4.2.3.13 M-UP-09ab – DLXAIP_OK with unknown control in OptionalInputs

Identifier	M-UP-09ab
Test Purpose	The test shall verify that there will be an error, if the request contains unknown controls in the OptionalInputs-element.
Configuration	- CONFIG_S.512 - CONFIG_LXAIP
Pre-test conditions	Authenticated connection to middleware exists.

Step	Test sequence	Expected Results
1	<pre> PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID="http://www.bsi.bund.de/tr- esor/lxaip/1.3", pres:xmlData(LXAIP_OK(REF_TXT_DATA_58b, REF_PDF_DATA_58b, REF_XML_MDO_58b)))) </pre>	<pre> PreservePOResponse (dsb:Result(resultmajor:Success), pres:POID(AOID-05ab)) </pre>
2	Insert AOID-05ab into the prepared DLXAIP_OK-template.	DLXAIP_OK(AOID-05ab, VID-01, VID-02)
3	<pre> UpdatePOC (pres:OptionalInputs(dsb:Other(dsb:Value(SomethingUnknown))), pres:POID(AOID-05ab, pres:DeltaPOC(@FormatID("http://www.bsi.bund.de/tr- esor/dlxaip/1.3"), pres:xmlData(DLXAIP_OK(AOID-05ab, VID-01, VID-02, REF_CADES_ATT_TXT_DATA_60b))))) </pre>	<pre> UpdatePOCResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/notSupported)))) </pre>
4	<pre> UpdatePOC (pres:POID(AOID-05ab, pres:DeltaPOC(@FormatID("http://www.bsi.bund.de/tr- esor/dlxaip/1.3"), pres:xmlData(DLXAIP_OK(AOID-05ab, VID-01, VID-02, REF_CADES_ATT_TXT_DATA_59b))))) </pre>	<pre> UpdatePOCResponse (dsb:Result(resultmajor:Success), VersionID(VID-02)) </pre>
5	Insert AOID-05ab into the prepared DLXAIP_OK_2-template.	DLXAIP_OK_2(AOID-05ab, VID-02, VID-03)

Step	Test sequence	Expected Results
6	<pre>UpdatePOC (pres:POID(AOID-05ab, pres:DeltaPOC(@FormatID("http://www.bsi.bund.de/tr- esor/dlxaip/1.3"), pres:xmlData(DLXAIP_OK_2(AOID-05ab, VID-02, VID-03, REF_XML_DATA_60-1b, REF_CADES_DET_XML_DATA_60-1b)))))</pre>	<pre>UpdatePOCResponse (dsb:Result(resultmajor:Success), VersionID(VID-03))</pre>

Verdict of M-UP-09ab**Observations:****Verdict:**

3.4.2.4 Function RetrievePO of S.512-interface

3.4.2.4.1 M-RE-01b - Retrieval of previously archived XAIPs

Identifier	M-RE-01b
Test Purpose	The test shall verify that it is possible to retrieve previously archived XAIPs.
Configuration	• CONFIG_S.512
Pre-test conditions	• Authenticated connection to middleware exists. • Make sure that the test cases M-SU-01b, M-UP-01b and M-UP-09b were successfully performed. • Requirements (A3.3.1-1) and (A3.3.1-2) of [TR-ESOR-E];

Step	Test sequence	Expected Results
1	RetrievePO (pres:POID(AOID-01b), pres:SubjectOfRetrieval("PO"),)	RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID=" http://www.bsi.bund.de/tr-esor/xaip/1.3 ", pres:xmlData(XAIP_OK(AOID-01b, VID-03)))))
2	RetrievePO (pres:POID(AOID-02b), pres:SubjectOfRetrieval("PO"))	RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID=" http://www.bsi.bund.de/tr-esor/xaip/1.3 ", pres:xmlData(XAIP_OK_SIG_OK(AOID-02b, VID-01)))))
3	The fulfilment of the requirement A3.3.1-2 ³⁴ is checked.	
4	RetrievePO (pres:POID(AOID-05b), pres:VersionID(VID-02), pres:SubjectOfRetrieval("PO"))	RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID=" http://www.bsi.bund.de/tr-esor/xaip/1.3 ", pres:xmlData(XAIP_OK(AOID-05b, VID-02)))))
5	RetrievePO (pres:POID(AOID-05b), pres:VersionID(VID-01), pres:SubjectOfRetrieval("PO"))	RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID=" http://www.bsi.bund.de/tr-esor/xaip/1.3 ", pres:xmlData(XAIP_OK(AOID-05b, VID-01)))))
6	RetrievePO (pres:POID(AOID-05b), pres:VersionID(VID-01), pres:VersionID(VID-02), pres:SubjectOfRetrieval("PO"))	RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID=" http://www.bsi.bund.de/tr-esor/xaip/1.3 ", pres:xmlData(XAIP_OK(AOID-05b, VID-01, VID-02)))))
7	RetrievePO (pres:POID(AOID-05b), pres:VersionID("all"), pres:SubjectOfRetrieval("PO"))	RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID=" http://www.bsi.bund.de/tr-esor/xaip/1.3 ", pres:xmlData(XAIP_OK(AOID-05b, VID-01, VID-02, VID-03)))))

³⁴ See Annex E (Section 3.3.1).

8	Test of correctness of the obtained XAIP_OK(AOID-05b, VID-01, VID-02, VID-03) by using AIP-eIDAS-SigValidator <pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:InlineXML(XAIP_OK(AOID-05b, VID-01, VID-02, VID-03)))))) </pre>	<pre> VerifyResponse(dss:Result(resultmajor#ok), dss:OptionalOutputs(dssvr:VerificationReport(dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:XAIPReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid)) [...]))))) </pre>
---	---	--

Verdict of M-RE-01b

Observations:

Verdict:

3.4.2.4.2 M-RE-01ab – Retrieval of previously archived LXAIps

Identifier	M-RE-01ab
Test Purpose	The test shall verify that it is possible to retrieve previously archived LXAIps.
Configuration	- CONFIG_S.512 - CONFIG_LXAIP
Pre-test conditions	- Authenticated connection to middleware exists. - Make sure that the test cases M-SU-01ab, M-UP-01ab and M-UP-09ab were successfully performed. - Requirements (A3.3.1-1) and (A3.3.1-2) of [TR-ESOR-E];;

Step	Test sequence	Expected Results
1	<pre> RetrievePO (pres:POID(AOID-01ab), pres:SubjectOfRetrieval("PO"), pres:POFormat("http://www.bsi.bund.de/tr-esor/lxaip/1.3")) </pre>	<pre> RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/lxaip/1.3", pres.xmlData(LXAIP_OK(AOID-01ab, VID-03)))) </pre>
2	<pre> RetrievePO (pres:POID(AOID-02ab), pres:SubjectOfRetrieval("PO"), pres:POFormat("http://www.bsi.bund.de/tr-esor/lxaip/1.3")) </pre>	<pre> RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/lxaip/1.3", pres.xmlData(LXAIP_OK_SIG_OK(AOID-02ab, VID-01)))) </pre>

Step	Test sequence	Expected Results
3	The fulfilment of the requirement A3.3.1-2 ³⁵ is checked.	
4	RetrievePO (pres:POID(AOID-05ab), pres:VersionID(VID-02), pres:SubjectOfRetrieval("PO"), pres:POFormat(" http://www.bsi.bund.de/tr-esor/lxaip/1.3 "))	RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID=" http://www.bsi.bund.de/tr-esor/lxaip/1.3 ", pres:xmlData(LXAIP_OK(AOID-05ab, VID-02)))))
5	RetrievePO (pres:POID(AOID-05ab), pres:VersionID(VID-01), pres:SubjectOfRetrieval("PO"), pres:POFormat(" http://www.bsi.bund.de/tr-esor/lxaip/1.3 "))	RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID=" http://www.bsi.bund.de/tr-esor/lxaip/1.3 ", pres:xmlData(LXAIP_OK(AOID-05ab, VID-01)))))
6	RetrievePO (pres:POID(AOID-05ab), pres:VersionID(VID-01), pres:VersionID(VID-02), pres:SubjectOfRetrieval("PO"), pres:POFormat(" http://www.bsi.bund.de/tr-esor/lxaip/1.3 "))	RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID=" http://www.bsi.bund.de/tr-esor/lxaip/1.3 ", pres:xmlData(LXAIP_OK(AOID-05ab, VID-01, VID-02)))))
7	RetrievePO (pres:POID(AOID-05ab), pres:VersionID("all"), pres:SubjectOfRetrieval("PO"), pres:POFormat(" http://www.bsi.bund.de/tr-esor/lxaip/1.3 "))	RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID=" http://www.bsi.bund.de/tr-esor/lxaip/1.3 ", pres:xmlData(LXAIP_OK(AOID-05ab, VID-01, VID-02, VID-03)))))
8	The all in LXAIP_OK(AOID-05ab, VID-01, VID-02, VID-03) referenced files has been copied into corresponding LXAIP-directory of the AIP-eIDAS-SigValidator during test preparation (cf. chapter 4.3) and the given references in LXAIP_OK(AOID-05ab, VID-01, VID-02, VID-03) will be adjusted automatically in order to get them resolveable by the AIP-eIDAS-SigValidator.	LXAIP_OK_XSV(AOID-05ab, VID-01, VID-02, VID-03)

³⁵ See Annex E (Section 3.3.1).

Step	Test sequence	Expected Results
9	Test of correctness of the obtained and prepared LXAIP_OK_XSV(AOID-05a, VID-01, VID-02, VID-03) by using AIP-eIDAS-SigValidator <pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:InlineXML(LXAIP_OK_XSV(AOID-05ab, VID-01, VID-02, VID-03)))))) </pre>	<pre> VerifyResponse(dss:Result(resultmajor#ok), dss:OptionalOutputs(dssvr:VerificationReport(dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:XAIReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid)) [...]))))) </pre>
10	<pre> RetrievePO (pres:POID(AOID-01ab), pres:SubjectOfRetrieval("PO"), pres:POFormat("http://www.bsi.bund.de/tr-esor/xaip/1.3")) </pre>	<pre> RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3", pres.xmlData(XAIP_OK(AOID-05ab, VID-03)))) </pre>
11	<pre> RetrievePO (pres:POID(AOID-01ab), pres:SubjectOfRetrieval("PO"),) </pre>	<pre> RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3", pres.xmlData(XAIP_OK(AOID-05ab, VID-03)))) </pre>

Verdict of M-RE-01ab
Observations:
Verdict:

3.4.2.4.3 M-RE-02b – Retrieval of XAIPs for known and unknown AOIDs

Identifier	M-RE-02b
Test Purpose	The test shall verify that it is possible to retrieve XAIPs for known AOIDs and there will be an error in case of unknown AOIDs.
Configuration	CONFIG_S.512
Pre-test conditions	- Authenticated connection to middleware exists. - Make sure that the test cases M-SU-01b was successfully performed.

Step	Test sequence	Expected Results
1	Choice an AOID-06b within the possible range and make sure, that it has not been assigned yet. This is realised by the following call: <pre>RetrievePO (pres:POID(AOID-06b), pres:SubjectOfRetrieval("PO"),)</pre>	<pre>RetrievePOResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/unknownPOID)))</pre>
2	<pre>RetrievePO (pres:POID(AOID-01b), pres:SubjectOfRetrieval("PO"),)</pre>	<pre>RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3", pres:xmlData(XAIP_OK(AOID-01b, VID-03))))</pre>
3	<pre>RetrievePO (pres:POID(AOID-02b), pres:SubjectOfRetrieval("PO"))</pre>	<pre>RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3", pres:xmlData(XAIP_OK_SIG_OK(AOID-02b, VID-01))))</pre>
4	<pre>RetrievePO (pres:POID(AOID-06b), pres:SubjectOfRetrieval("PO"),)</pre>	<pre>RetrievePOResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/unknownPOID)))</pre>

Verdict of M-RE-02b
Observations:
Verdict:

3.4.2.4.4 M-RE-03b – Retrieval of XAIPs for known and unknown VersionIDs

Identifier	M-RE-03b
Test Purpose	The test shall verify that there will be a <code>requestOnlyPartlySuccessful</code> warning, if it is not possible to retrieve XAIPs for all indicated VersionIDs.
Configuration	CONFIG_ArchiSafe_C.2
Pre-test conditions	- Authenticated connection to middleware exists. - Make sure that the test case M-SU-01 was successful performed.

Step	Test sequence	Expected Results
1	Choice a VID-10 within the possible range and make sure, that it has not been assigned yet. This is realised by the following call: <pre>RetrievePO (pres:POID(AOID-01b), pres:VersionID(VID-10), pres:SubjectOfRetrieval("PO"),)</pre>	<pre>RetrievePOResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/unknownVersionID)))</pre>
2	<pre>RetrievePO (pres:POID(AOID-01b), pres:VersionID(VID-02), pres:SubjectOfRetrieval("PO"),)</pre>	<pre>RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3", pres:xmlData(XAIP_OK(AOID-01b, VID-02))))</pre>
3	<pre>RetrievePO (pres:POID(AOID-01b), pres:VersionID(VID-10), pres:SubjectOfRetrieval("PO"),)</pre>	<pre>RetrievePOResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/unknownVersionID)))</pre>
4	<pre>RetrievePO (pres:POID(AOID-01b), pres:VersionID(VID-02), pres:VersionID(VID-10), pres:SubjectOfRetrieval("PO"),)</pre>	<pre>RetrievePOResponse (dsb:Result(dsb:ResultMajor(resultmajor:Success), dsb:ResultMinor(warning/requestOnlyPartlySuccessful)), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3", pres:xmlData(XAIP_OK(AOID-01b, VID-02))))</pre>

Verdict of M-RE-03b

Observations:

Verdict:

3.4.2.4.5 M-RE-04b – Unknown control in OptionalInputs

Identifier	M-RE-04b
Test Purpose	The test shall verify that there will be an error, if the request contains unknown controls in the <code>OptionalInputs</code> -element.
Configuration	CONFIG_S.512
Pre-test conditions	- Authenticated connection to middleware exists. - Make sure that test case M-SU-01b was successfully performed.

Step	Test sequence	Expected Results
1	RetrievePO (pres:OptionalInputs(dsb:Other(dsb:Value(SomethingUnknown))), pres:POID(AOID-01b), pres:SubjectOfRetrieval("PO"),)	RetrievePOResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/notSupported)))

Verdict of M-RE-04b
Observations:
Verdict:

3.4.2.4.6 M-EV-01b – Retrieval of Evidence Records of previously archived XAIPs without specifying the desired ERS Format

Identifier	M-EV-01b
Test Purpose	The test shall verify that it is possible to retrieve Evidence Records for previously archived XAIPs.
Configuration	CONFIG_S.512
Pre-test conditions	- Authenticated connection to middleware exists. - Make sure that test cases M-SU-01b, M-UP-01b, M-UP-09b and M-RE-01b were successfully performed.

Step	Test sequence	Expected Results
1	RetrievePO (pres:POID(AOID-01b), pres:SubjectOfRetrieval("Evidence"))	RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="urn:ietf:rfc:4998:EvidenceRecord", pres:xmlData(XAIP_ERS(AOID-01b,VID-03, XAIP_OK_V3_ER_OK)))))
2	RetrievePO (pres:POID(AOID-02b), pres:SubjectOfRetrieval("Evidence"))	RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="urn:ietf:rfc:4998:EvidenceRecord", pres:xmlData(XAIP_ERS(AOID-02b,VID-01, XAIP_OK_SIG_OK_V1_ER_OK)))))
3	RetrievePO (pres:POID(AOID-01b), pres:VersionID(VID-02), pres:SubjectOfRetrieval("Evidence"))	RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="urn:ietf:rfc:4998:EvidenceRecord", pres:xmlData(XAIP_ERS(AOID-01b,VID-02, XAIP_OK_V2_ER_OK))))

Step	Test sequence	Expected Results
4	RetrievePO (pres:POID(AOID-01b), pres:VersionID(VID-01), pres:SubjectOfRetrieval("Evidence"))	RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="urn:ietf:rfc:4998:EvidenceRecord", pres:xmlData(XAIP_ERS(AOID-01b, VID-01, XAIP_OK_V1_ER_OK))))
5	RetrievePO (pres:POID(AOID-01b), pres:VersionID(VID-01), pres:VersionID(VID-03), pres:SubjectOfRetrieval("Evidence"))	RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="urn:ietf:rfc:4998:EvidenceRecord", pres:xmlData(XAIP_ERS(AOID-01b, VID-01, XAIP_OK_V1_ER_OK))), pres:PO(@FormatID="urn:ietf:rfc:4998:EvidenceRecord", pres:xmlData(XAIP_ERS(AOID-01b, VID-03, XAIP_OK_V3_ER_OK))))
6	RetrievePO (pres:POID(AOID-01b), pres:VersionID("all"), pres:SubjectOfRetrieval("Evidence"))	RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="urn:ietf:rfc:4998:EvidenceRecord", pres:xmlData(XAIP_ERS(AOID-01b, VID-01, XAIP_OK_V1_ER_OK))), pres:PO(@FormatID="urn:ietf:rfc:4998:EvidenceRecord", pres:xmlData(XAIP_ERS(AOID-01b, VID-02, XAIP_OK_V2_ER_OK))), pres:PO(@FormatID="urn:ietf:rfc:4998:EvidenceRecord", pres:xmlData(XAIP_ERS(AOID-01b, VID-03, XAIP_OK_V3_ER_OK))))
7	RetrievePO (pres:POID(AOID-01b), pres:VersionID("all"), pres:SubjectOfRetrieval("PO"))	RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID=" http://www.bsi.bund.de/tr-esor/xaip/1.3 ", pres:xmlData(XAIP_OK(AOID-01b, VID-01, VID-02, VID-03)))))

Step	Test sequence	Expected Results
8	Test of correctness of the obtained XAIP_OK_V1_ER_OK, XAIP_OK_V2_ER_OK and XAIP_OK_V3_ER_OK by using ERVerifyTool. <pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:InlineXML(XAIP_OK(AOID-01b, VID-01, VID-02, VID-03)))) dss:SignatureObject(dss:Other(XAIP_ERS(AOID-01b,VID-01, XAIP_OK_V1_ER_OK), XAIP_ERS(AOID-01b,VID-02, XAIP_OK_V2_ER_OK), XAIP_ERS(AOID-01b,VID-03, XAIP_OK_V3_ER_OK)))) </pre>	<pre> VerifyResponse(dss:Result(resultmajor#ok), dss:OptionalOutputs(dssvr:VerificationReport(dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:EvidenceRecordReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid) [...]))), dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:EvidenceRecordReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid) [...]))), dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:EvidenceRecordReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid) [...])))))) </pre>
9	<pre> RetrievePO (pres:POID(AOID-01b), pres:VersionID("all"), pres:SubjectOfRetrieval("POwithEmbeddedEvidence")) </pre>	<pre> RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3", pres:xmlData(XAIP_OK(AOID-01b, VID-01, VID-02, VID-03, XAIP_OK_V1_ER_OK, XAIP_OK_V2_ER_OK, XAIP_OK_V3_ER_OK)))) </pre>

Step	Test sequence	Expected Results
10	Test of correctness of the obtained XAIP_OK with embedded evidence records XAIP_OK_V1_ER_OK, XAIP_OK_V2_ER_OK and XAIP_OK_V3_ER_OK by using ERVerifyTool. <pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:InlineXML(XAIP_OK(AOID-01b, VID-01, VID-02, VID-03, XAIP_OK_V1_ER_OK, XAIP_OK_V2_ER_OK, XAIP_OK_V3_ER_OK))))) </pre>	<pre> VerifyResponse(dss:Result(resultmajor#ok), dss:OptionalOutputs(dssvr:VerificationReport(dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:EvidenceRecordReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid)) [...])), dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:EvidenceRecordReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid)) [...])), dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:EvidenceRecordReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid)) [...]))))) </pre>

Verdict of M-EV-01b

Observations:

Verdict:

3.4.2.4.7 M-EV-01ab – Retrieval of Evidence Records of previously archived LXAIps without specifying the desired ERS Format

Identifier	M-EV-01ab
Test Purpose	The test shall verify that it is possible to retrieve Evidence Records for previously archived LXAIps.

Identifier	M-EV-01ab
Configuration	- CONFIG_S.512 - CONFIG_LXAIP
Pre-test conditions	- Authenticated connection to middleware exists. - Make sure that test cases M-SU-01ab, M-UP-01ab and M-UP-09ab were successfully performed.

Step	Test sequence	Expected Results
1	<pre>RetrievePO (pres:POID(AOID-01ab), pres:SubjectOfRetrieval("Evidence"),)</pre>	<pre>RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="urn:ietf:rfc:4998:EvidenceRecord", pres:xmlData(XAIP_ERS³⁶(AOID-01ab, VID-03, LXAIP_OK_V3_ER_OK))))</pre>
2	<pre>RetrievePO (pres:POID(AOID-02ab), pres:SubjectOfRetrieval("Evidence"),)</pre>	<pre>RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="urn:ietf:rfc:4998:EvidenceRecord", pres:xmlData(XAIP_ERS(AOID-02ab, VID-01, LXAIP_OK_SIG_OK_V1_ER_OK))))</pre>
3	<pre>RetrievePO (pres:POID(AOID-01ab), pres:VersionID(VID-02), pres:SubjectOfRetrieval("Evidence"),)</pre>	<pre>RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="urn:ietf:rfc:4998:EvidenceRecord", pres:xmlData(XAIP_ERS(AOID-01ab, VID-02, LXAIP_OK_V2_ER_OK))))</pre>
4	<pre>RetrievePO (pres:POID(AOID-01ab), pres:VersionID(VID-01), pres:SubjectOfRetrieval("Evidence"),)</pre>	<pre>RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="urn:ietf:rfc:4998:EvidenceRecord", pres:xmlData(XAIP_ERS(AOID-01ab, VID-01, LXAIP_OK_V1_ER_OK))))</pre>

³⁶ XAIP-ERS does mean a RFC4998-ER embedded in the element `xaip:asn1EvidenceRecord`, which in turn is embedded in the element `xaip:evidenceRecord` and can contain the both attributes `AOID` and `VersionID`.

Step	Test sequence	Expected Results
5	<pre>RetrievePO (pres:POID(AOID-01ab), pres:VersionID(VID-01), pres:VersionID(VID-03), pres:SubjectOfRetrieval("Evidence"),)</pre>	<pre>RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="urn:ietf:rfc:4998:EvidenceRecord", pres:xmlData(XAIP_ERS(AOID-01ab, VID-01, LXAIP_OK_V1_ER_OK))), pres:PO(@FormatID="urn:ietf:rfc:4998:EvidenceRecord", pres:xmlData(XAIP_ERS(AOID-01ab, VID-03, LXAIP_OK_V3_ER_OK))))</pre>
6	<pre>RetrievePO (pres:POID(AOID-01ab), pres:VersionID("all"), pres:SubjectOfRetrieval("Evidence"),)</pre>	<pre>RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="urn:ietf:rfc:4998:EvidenceRecord", pres:xmlData(XAIP_ERS(AOID-01ab, VID-01, LXAIP_OK_V1_ER_OK))), pres:PO(@FormatID="urn:ietf:rfc:4998:EvidenceRecord", pres:xmlData(XAIP_ERS(AOID-01ab, VID-02, LXAIP_OK_V2_ER_OK))), pres:PO(@FormatID="urn:ietf:rfc:4998:EvidenceRecord", pres:xmlData(XAIP_ERS(AOID-01ab, VID-03, LXAIP_OK_V3_ER_OK))))</pre>
7	<pre>RetrievePO (pres:POID(AOID-01ab), pres:VersionID("all"), pres:SubjectOfRetrieval("PO"), pres:POFormat(http://www.bsi.bund.de/tr-esor/lxaip/1.3))</pre>	<pre>RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/lxaip/1.3", pres:xmlData(LXAIP_OK(AOID-01ab,VID-01,VID-02,VID-03))))</pre>

Step	Test sequence	Expected Results
8	Test of correctness of the obtained LXAIP_OK_V1_ER_OK, LXAIP_OK_V2_ER_OK and LXAIP_OK_V3_ER_OK by using ERVerifyTool. The particular object references in the obtained LXAIP_OK will be replaced by the references compatible with the ERVerifyTool in advance. <pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:InlineXML(LXAIP_OK(AOID-01ab, VID-01, VID-02, VID-03)))) dss:SignatureObject(dss:Other(XAIP_ERS(AOID-01ab, VID-01, LXAIP_OK_V1_ER_OK), XAIP_ERS(AOID-01ab, VID-02, LXAIP_OK_V2_ER_OK), XAIP_ERS(AOID-01ab, VID-03, LXAIP_OK_V3_ER_OK)))) </pre>	<pre> VerifyResponse(dss:Result(resultmajor#ok), dss:OptionalOutputs(dssvr:VerificationReport(dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:EvidenceRecordReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid) [...]))), dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:EvidenceRecordReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid) [...])))))) </pre>

Verdict of M-EV-01ab

Observations:

Verdict:

3.4.2.4.8 M-EV-02b – Retrieval of Evidence Records for known and unknown AOIDs

Identifier	M-EV-02b
Test Purpose	The test case verifies that that it is possible to retrieve an Evidence Record for known AOIDs and there will be an error in case of unknown AOIDs.
Configuration	CONFIG_S.512
Pre-test conditions	- Authenticated connection to middleware exists. - Make sure that the test cases M-SU-01b and M-RE-02b were successfully performed.

Step	Test sequence	Expected Results
1	Make sure, that the previous chosen AOID-06b is within the possible range and it is still not assigned yet. This is realised by the following call: <pre>RetrievePO (pres:POID(AOID-06b), pres:SubjectOfRetrieval("PO"),)</pre>	<pre>RetrievePOResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/unknownPOID)))</pre>
2	<pre>RetrievePO (pres:POID(AOID-01b), pres:SubjectOfRetrieval("Evidence"))</pre>	<pre>RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="urn:ietf:rfc:4998:EvidenceRecord", pres.xmlData(XAIP_ERS(AOID-01b,VID-03, XAIP_OK_V3_ER_OK))))</pre>
3	<pre>RetrievePO (pres:POID(AOID-02b), pres:SubjectOfRetrieval("Evidence"))</pre>	<pre>RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="urn:ietf:rfc:4998:EvidenceRecord", pres.xmlData(XAIP_ERS(AOID-02b,VID-01, XAIP_OK_SIG_OK_V1_ER_OK))))</pre>
4	<pre>RetrievePO (pres:POID(AOID-06b), pres:SubjectOfRetrieval("PO"),)</pre>	<pre>RetrievePOResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/unknownPOID)))</pre>

Verdict of M-EV-02b
Observations:
Verdict:

3.4.2.4.9 M-EV-03b – Retrieval of Evidence Records for partly known VersionIDs

Identifier	M-EV-03b
Test Purpose	The test case verifies that there is a <code>requestOnlyPartlySuccessful</code> warning, if an Evidence Record for an unknown <code>VersionID</code> is requested.
Configuration	CONFIG_S.512
Pre-test conditions	- Authenticated connection to middleware exists. - Make sure that the test cases M-SU-01b and M-RE-03b were successfully performed.

Step	Test sequence	Expected Results
1	Make sure, that concerning AOID-01b, the chosen version VID-10 is within the possible range but is an unknown <code>VersionID</code>. This is realised by the following call: <pre>RetrievePO (pres:POID(AOID-01b), pres:VersionID(VID-10), pres:SubjectOfRetrieval("PO"),)</pre>	<pre>RetrievePOResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/unknownVersionID)))</pre>
2	<pre>RetrievePO (pres:POID(AOID-01b), pres:SubjectOfRetrieval("Evidence"))</pre>	<pre>RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="urn:ietf:rfc:4998:EvidenceRecord", pres:xmlData(XAIP_ERS(AOID-01b,VID-03, XAIP_OK_V3_ER_OK))))</pre>
3	<pre>RetrievePO (pres:POID(AOID-01b), pres:VersionID(VID-10), pres:SubjectOfRetrieval("Evidence"),)</pre>	<pre>RetrievePOResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/unknownVersionID)))</pre>
4	<pre>RetrievePO (pres:POID(AOID-01b), pres:VersionID(VID-02), pres:VersionID(VID-10), pres:SubjectOfRetrieval("Evidence"),)</pre>	<pre>RetrievePOResponse (dsb:Result(dsb:ResultMajor(resultmajor:Success), dsb:ResultMinor(warning/requestOnlyPartlySuccessful)), pres:PO(@FormatID="urn:ietf:rfc:4998:EvidenceRecord", pres:xmlData(XAIP_ERS(AOID-01b,VID-02, XAIP_OK_V2_ER_OK))))</pre>

Verdict of M-EV-03b
Observations:
Verdict:

3.4.2.4.10 M-EV-04b – Unknown control in OptionalInputs

Identifier	M-EV-04
Test Purpose	The test shall verify that there will be an error, if the request contains unknown controls in the OptionalInputs-element.
Configuration	CONFIG_S.512
Pre-test conditions	- Authenticated connection to middleware exists. - Make sure that the test case M-SU-01b was successfully performed.

Step	Test sequence	Expected Results
1	<pre>RetrievePO (pres:OptionalInputs(dsb:Other(dsb:Value(SomethingUnknown))), pres:POID(AOID-01b), pres:SubjectOfRetrieval("Evidence"),)</pre>	<pre>RetrievePOResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/notSupported)))</pre>

Verdict of M-EV-04b
Observations:
Verdict:

3.4.2.5 Function DeletePO of S.512-interface

3.4.2.5.1 M-DE-01b – Deletion of XAIP without ReasonOfDeletion

Identifier	M-DE-01b
Test Purpose	The test shall verify that it will yield an error, if DeletePO is called without providing a Reason of deletion, if the element retentionPeriod in the XAIP contains a predetermined future date.
Configuration	CONFIG_S.512
Pre-test conditions	- In one test case the element retentionPeriod in the XAIP to be deleted contains a predetermined future date. In the other test case the element retentionPeriod contains a date in the past. - Authenticated connection to middleware exists. - Requirement (A3.5.1-1) of [TR-ESOR-E]: - The ArchiveDeletionRequest of XAIP with AOID-02b with the end of the retentionPeriod in the future must not be successful without a reason for the deletion. - Make sure that the test case M-SU-01b was successfully performed.

Step	Test sequence	Expected Results
1	DeletePO (pres:POID(AOID-02b))	DeletePOResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/parameterError)))
2	RetrievePO (pres:POID(AOID-02b), pres:SubjectOfRetrieval("PO"))	RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID=" http://www.bsi.bund.de/tr-esor/xaip/1.3 ", pres.xmlData(XAIP_OK_SIG_OK(AOID-02b, VID-01))))
3	RetrievePO (pres:POID(AOID-00b) ³⁷ , pres:SubjectOfRetrieval("PO"))	RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID=" http://www.bsi.bund.de/tr-esor/xaip/1.3 ", pres.xmlData(XAIP_OK(AOID-00b, VID-01))))
4	DeletePO (pres:POID(AOID-00b))	DeletePOResponse (dsb:Result(dsb:ResultMajor(resultmajor:Success)))
5	RetrievePO (pres:POID(AOID-00b), pres:SubjectOfRetrieval("PO"))	RetrievePOResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/unknownPOID))))

³⁷ The XAIP with AOID-00b and LXAIP with AOID-00ab were imported some days ago. The retention period is expired.

Verdict of M-DE-01b
Observations:
Verdict:

3.4.2.5.2 M-DE-01ab – Deletion of LXAIP without ReasonOfDeletion

Identifier	M-DE-01ab
Test Purpose	The test shall verify that it will yield an error, if <code>DeletePO</code> is called without providing a <code>Reason of deletion</code> , if the element <code>retentionPeriod</code> in the LXAIP contains a predetermined future date.
Configuration	- CONFIG_S.512 - CONFIG_LXAIP
Pre-test conditions	- In one test case the element <code>retentionPeriod</code> in the LXAIP to be deleted contains a predetermined future date. In the other test case the element <code>retentionPeriod</code> contains a date in the past. - Authenticated connection to middleware exists. - Requirement (A3.5.1-1) of [TR-ESOR-E]. - The <code>ArchiveDeletionRequest</code> of LXAIP with AOID-02ab with the end of the <code>retentionPeriod</code> in the future must not be successful without a reason for the deletion. - Make sure that the test case M-SU-01ab was successfully performed.

Step	Test sequence	Expected Results
1	DeletePO (pres:POID(M-SU-02ab))	DeletePOResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/parameterError)))
2	RetrievePO (pres:POID(AOID-02ab), pres:SubjectOfRetrieval("PO"), pres:POFormat(http://www.bsi.bund.de/tr-esor/lxaip/1.3))	RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID=" http://www.bsi.bund.de/tr-esor/lxaip/1.3 ", pres:xmlData(LXAIP_OK_SIG_OK(AOID-02ab, VID-01))))
3	RetrievePO (pres:POID(AOID-00ab ³⁷), pres:SubjectOfRetrieval("PO"), pres:POFormat(http://www.bsi.bund.de/tr-esor/lxaip/1.3))	RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID=" http://www.bsi.bund.de/tr-esor/lxaip/1.3 ", pres:xmlData(LXAIP_OK(AOID-00ab, VID-01))))
4	DeletePO (pres:POID(AOID-00ab)	DeletePOResponse (dsb:Result(dsb:ResultMajor(resultmajor:Success)))

Step	Test sequence	Expected Results
5	RetrievePO (pres:POID(AOID-00ab), pres:SubjectOfRetrieval("PO"), pres:POFormat(http://www.bsi.bund.de/tr-esor/lxaip/1.3))	RetrievePOResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/unknownPOID)))

Verdict of M-DE-01ab
Observations:
Verdict:

3.4.2.5.3 M-DE-02b – Deletion of XAIP with ReasonOfDeletion

Identifier	M-DE-02b
Test Purpose	The test shall verify that it is possible to delete an XAIP by calling <code>DeletePO</code> with Reason of deletion provided.
Configuration	CONFIG_S.512
Pre-test conditions	- The element <code>retentionPeriod</code> in the XAIP to be deleted contains a predetermined future date. - Make sure that the test case M-SU-01b was successfully performed. - Authenticated connection to middleware exists.

Step	Test sequence	Expected Results
1	DeletePO (pres:POID(AOID-02b), pres:ClaimedRequestorName(SomeName), pres:Reason(ReasonOfDeletion),)	DeletePOResponse (dsb:Result(dsb:ResultMajor(resultmajor:Success)))
2	RetrievePO (pres:POID(AOID-02b), pres:SubjectOfRetrieval("PO"),)	RetrievePOResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/unknownPOID)))
3	Check requirement A3.5.1-2 ³⁸ .	

Verdict of M-DE-02b
Observations:

³⁸ See Annex E (Section 3.5.1).

Verdict of M-DE-02b
Verdict:

3.4.2.5.4 M-DE-02ab – Deletion of LXAIP with ReasonOfDeletion

Identifier	M-DE-02ab
Test Purpose	The test shall verify that it is possible to delete an LXAIP by calling <code>DeletePO</code> with Reason of deletion provided.
Configuration	- CONFIG_S.512 - CONFIG_LXAIP
Pre-test conditions	- The element <code>retentionPeriod</code> in the LXAIP to be deleted contains a predetermined future date. - Make sure that the test case M-SU-01ab was successfully performed. - Authenticated connection to middleware exists.

Step	Test sequence	Expected Results
1	DeletePO (pres:POID(AOID-02ab), pres:ClaimedRequestorName(SomeName), pres:Reason(ReasonOfDeletion),)	DeletePOResponse (dsb:Result(dsb:ResultMajor(resultmajor:Success)))
2	RetrievePO (pres:POID(AOID-02ab), pres:SubjectOfRetrieval("PO"), pres:POFormat(http://www.bsi.bund.de/tr-esor/lxaip/1.3))	RetrievePOResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/unknownPOID)))
3	Check requirement A3.5.1-2 ³⁸ .	

Verdict of M-DE-02ab
Observations:
Verdict:

3.4.2.5.5 M-DE-03b – Deletion of unknown XAIPs without ReasonOfDeletion

Identifier	M-DE-03b
Test Purpose	The test shall verify that it will yield an error, if <code>DeletePO</code> is called without providing a Reason of deletion. As the present call also contains the unknown AOID-06b, the AOID-specific <code>dsb:Result</code> needs to indicate that the AOID is unknown.
Configuration	CONFIG_ArchiSafe_C.2

Identifier	M-DE-03b
Pre-test conditions	- The element <code>retentionPeriod</code> in the XAIP to be deleted contains a predetermined future date. - Make sure that the test cases M-UP-09b and M-RE-02b were successfully performed. - Make sure, that the randomly generated AOID-06b is within the possible range but has not been assigned yet. - Authenticated connection to middleware exists.

Step	Test sequence	Expected Results
1	DeletePO (pres:POID(AOID-05b))	DeletePOResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/parameterError)))
2	DeletePO (pres:POID(AOID-06b))	DeletePOResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/unknownPOID))))
3	RetrievePO (pres:POID(AOID-05b), pres:SubjectOfRetrieval("PO"))	RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID=" http://www.bsi.bund.de/tr-esor/xaip/1.3 ", pres:xmlData(XAIP_OK(AOID-05bVID-03))))

Verdict of M-DE-03b
Observations:
Verdict:

3.4.2.5.6 M-DE-04b – Deletion of unknown XAIPs with `ReasonOfDeletion`

Identifier	M-DE-04b
Test Purpose	The test shall verify that it will yield an error, if <code>DeletePO</code> is called with unknown AOIDs with providing a <code>Reason</code> of deletion.
Configuration	CONFIG_S.512
Pre-test conditions	- The element <code>retentionPeriod</code> in the XAIP to be deleted contains a predetermined future date. - Make sure that the test case M-RE-02b was successfully performed - Make sure, that the randomly generated AOID-06b is within the possible range but has not been assigned yet. - Authenticated connection to middleware exists.

Step	Test sequence	Expected Results
1	DeletePO (pres:POID(AOID-06b), pres:ClaimedRequestorName(SomeName), pres:Reason(ReasonOfDeletion),)	DeletePOResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/unknownPOID)))

Verdict of M-DE-04b

Observations:

Verdict:

3.4.2.5.7 M-DE-05b – Unknown control in OptionalInputs

Identifier	M-DE-05b
Test Purpose	The test shall verify that there will be an error, if the request contains unknown controls in the <code>OptionalInputs</code> -element and the <code>DeletePO</code> request will not be successful.
Configuration	CONFIG_S.512
Pre-test conditions	- Authenticated connection to middleware exists. - The element <code>retentionPeriod</code> in the XAIP to be deleted contains a predetermined future date. - Make sure that the test case M-UP-09b was successfully performed.

Step	Test sequence	Expected Results
1	DeletePO (pres:OptionalInputs(dsb:Other(dsb:Value(SomethingUnknown))), pres:POID(AOID-05b),)	DeletePOResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/notSupported)))
2	RetrievePO (pres:POID(AOID-05b), pres:SubjectOfRetrieval("PO"))	RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID=" http://www.bsi.bund.de/tr-esor/xaip/1.3 ", pres:xmlData(XAIP_OK(AOID-05b, VID-03))))

Verdict of M-DE-05b

Observations:

Verdict:

3.4.2.6 Function ValidateEvidence of S.512-interface

3.4.2.6.1 M-VE-01b –XAIP_OK_SIG_OK and XAIP_OK_SIG_OK_ER

Identifier	M-VE-01b
Test Purpose	The test shall verify that it is possible to verify XAIP_OK_SIG_OK and XAIP_OK_SIG_OK_V1_ER_OK with the TR-ESOR-specific signature policy http://www.bsi.bund.de/tr-esor/sigpolicy/verify-xaip .
Configuration	CONFIG_S.512
Pre-test conditions	Authenticated connection to middleware exists.

Step	Test sequence	Expected Results
1	<pre> PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3", pres:xmlData(XAIP_OK_SIG_OK))) </pre>	<pre> PreservePOResponse (dsb:Result(resultmajor:Success), pres:POID(AOID-100b)) </pre>
2	<pre> RetrievePO (pres:POID(AOID-100b), pres:SubjectOfRetrieval("PO")) </pre>	<pre> RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3", pres:xmlData(XAIP_OK_SIG_OK(AOID-100b, VID-01)))) </pre>
3	<pre> RetrievePO⁷ (pres:POID(AOID-100b), pres:SubjectOfRetrieval("Evidence")) </pre>	<pre> RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="urn:ietf:rfc:4998:EvidenceRecord", pres:xmlData(XAIP_ERS(AOID-100b,VID-01, XAIP_OK_SIG_OK_V1_ER_OK)))) </pre>
4	<pre> ValidateEvidence (pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3", pres:xmlData(XAIP_OK_SIG_OK(AOID-100b, VID-01)))) </pre>	<pre> ValidateEvidenceResponse (dsb:Result(resultmajor:Success)) </pre>

Step	Test sequence	Expected Results
5	<pre> ValidateEvidence (pres:OptionalInputs(dsb:Other(dsb:Value(vr:ReturnVerificationReport))), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3", pres:xmlData(XAIP_OK_SIG_OK(AOID-100b, VID-01)))) </pre>	<pre> ValidateEvidenceResponse (dsb:Result(resultmajor:Success) pres:ValidationReport(pres:xmlData(XAIP_OK_SIG_OK_VR))) </pre>
6	<pre> ValidateEvidence (pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3", pres:binaryData(XAIP_OK_SIG_OK(AOID-100b, VID-01)))) </pre>	<pre> ValidateEvidenceResponse (dsb:Result(resultmajor:Success)) </pre>
7	Prepare XAIP_OK_SIG_OK_ER(AOID-100b, VID-01) by inserting the XAIP_ERS(AOID-100b, VID-01, XAIP_OK_SIG_OK_V1_ER_OK) obtained in step 3 into credentialsSection element of XAIP_OK_SIG_OK(AOID-100b, VID-01) from step 2.	XAIP_OK_SIG_OK_ER(AOID-100b, VID-01)
8	<pre> ValidateEvidence (pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3", pres:xmlData(XAIP_OK_SIG_OK_ER(AOID-100b,VID-01)))) </pre>	<pre> ValidateEvidenceResponse (dsb:Result(resultmajor:Success)) </pre>
9	<pre> ValidateEvidence (pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3", pres:binaryData(XAIP_OK_SIG_OK_ER(AOID-100b,VID-01)))) </pre>	<pre> ValidateEvidenceResponse (dsb:Result(resultmajor:Success)) </pre>

Step	Test sequence	Expected Results
10	<pre> ValidateEvidence (pres:Evidence(@FormatID="urn:ietf:rfc:4998:EvidenceRecord" , @POID="AOID-100b", VersionID="VID-01", pres:xmlData(XAIP_ERS(AOID-100b, VID-01, XAIP_OK_SIG_OK_V1_ER_OK)) pres:PO(@FormatID="http://www.bsi.bund.de/tr- esor/xaip/1.3", pres:xmlData(XAIP_OK_SIG_OK(AOID-100b, VID-01)))) </pre>	<pre> ValidateEvidenceResponse (dsb:Result(resultmajor:Success)) </pre>
11	<pre> ValidateEvidence (pres:Evidence(@FormatID="urn:ietf:rfc:4998:EvidenceRecord" , @POID="AOID-100b", VersionID="VID-01", pres:binaryData(XAIP_ERS(AOID-100b, VID-01, XAIP_OK_SIG_OK_V1_ER_OK)) pres:PO(@FormatID="http://www.bsi.bund.de/tr- esor/xaip/1.3", pres:xmlData(XAIP_OK_SIG_OK(AOID-100b, VID-01)))) </pre>	<pre> ValidateEvidenceResponse (dsb:Result(resultmajor:Success)) </pre>
12	<pre> ValidateEvidence (pres:Evidence(@FormatID="urn:ietf:rfc:4998:EvidenceRecord" , @POID="AOID-100b", VersionID="VID-01", pres:binaryData(XAIP_ERS(AOID-100b, VID-01, XAIP_OK_SIG_OK_V1_ER_OK)) pres:PO(@FormatID="http://www.bsi.bund.de/tr- esor/xaip/1.3", pres:binaryData(XAIP_OK_SIG_OK(AOID-100b, VID-01)))) </pre>	<pre> ValidateEvidenceResponse (dsb:Result(resultmajor:Success)) </pre>

Verdict of M-VE-01b

Observations:

Verdict:

3.4.2.6.2 M-VE-01ab - LXAIP_OK_SIG_OK and LXAIP_OK_SIG_OK_ER

Identifier	M-VE-01ab
Test Purpose	The test shall verify that it is possible to verify LXAIP_OK_SIG_OK and LXAIP_OK_SIG_OK_ER with the TR-ESOR-specific signature policy http://www.bsi.bund.de/tr-esor/sigpolicy/verify-xaip .
Configuration	- CONFIG_S.512 - CONFIG_LXAIP
Pre-test conditions	- Authenticated connection to middleware exists. - Test M-SU-01ab has been performed successfully and the references has been set properly in LXAIP_OK_SIG_OK

Step	Test sequence	Expected Results
1	<pre> PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/lxaip/1.3", pres:xmlData(LXAIP_OK_SIG_OK(REF_TXT_DATA_61b, REF_CADES_DET_TXT_DATA_61b, REF_PDF_DATA_61b, REF_CADES_DET_PDF_DATA_61b, REF_XML_MDO_61b)))) </pre>	<pre> PreservePOResponse (dsb:Result(resultmajor:Success), pres:POID(AOID-101ab)) </pre>
2	<pre> RetrievePO (pres:POID(AOID-101ab), pres:SubjectOfRetrieval("PO"), pres:POFormat(http://www.bsi.bund.de/tr-esor/lxaip/1.3)) </pre>	<pre> RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/lxaip/1.3", pres:xmlData(LXAIP_OK_SIG_OK(AOID-101ab, VID-01)))) </pre>
3	<pre> RetrievePO⁷ (pres:POID(AOID-101ab), pres:SubjectOfRetrieval("Evidence"),) </pre>	<pre> RetrievePOResponse (dss:Result(resultmajor:Success), pres:PO(@FormatID="urn:ietf:rfc:4998:EvidenceRecord", pres:xmlData(XAIP_ERS(AOID-101ab, VID-01, LXAIP_OK_SIG_OK_V1_ER_OK)))) </pre>
4	<pre> ValidateEvidence (pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/lxaip/1.3", pres:xmlData(LXAIP_OK_SIG_OK(AOID-101ab, VID-01)))) </pre>	<pre> ValidateEvidenceResponse (dsb:Result(resultmajor:Success)) </pre>
5	<pre> ValidateEvidence (pres:PO(</pre>	<pre> ValidateEvidenceResponse (dsb:Result(resultmajor:Success)) </pre>

Step	Test sequence	Expected Results
	<pre>@FormatID="http://www.bsi.bund.de/tr- esor/lxaip/1.3", pres:binaryData(LXAIP_OK_SIG_OK(AOID-101ab, VID-01))))</pre>	)
6	Prepare LXAIP_OK_SIG_OK_ER(AOID-101ab, VID-01) by inserting the XAIP_ERS(AOID-101ab, VID-01, LXAIP_OK_SIG_OK_V1_ER_OK) obtained in step 3 into credentialsSection element of LXAIP_OK_SIG_OK(AOID-101ab, VID-01) from step 2.	LXAIP_OK_SIG_OK_ER(AOID-101ab, VID-01)
7	<pre>ValidateEvidence (pres:PO(@FormatID="http://www.bsi.bund.de/tr- esor/lxaip/1.3", pres:xmlData(LXAIP_OK_SIG_OK_ER(AOID-101ab,VID-01)))))</pre>	<pre>ValidateEvidenceResponse (dsb:Result(resultmajor:Success))</pre>
8	<pre>ValidateEvidence (pres:PO(@FormatID="http://www.bsi.bund.de/tr- esor/lxaip/1.3", pres:binaryData(LXAIP_OK_SIG_OK_ER(AOID-101ab,VID-01)))))</pre>	<pre>ValidateEvidenceResponse (dsb:Result(resultmajor:Success))</pre>
09	<pre>ValidateEvidence (pres:Evidence(@FormatID="urn:ietf:rfc:4998:EvidenceRecord" , @POID="AOID-101ab", VersionID="VID-01", pres:xmlData(XAIP_ERS(AOID-101ab,VID-01, LXAIP_OK_SIG_OK_V1_ER_OK))) pres:PO(@FormatID="http://www.bsi.bund.de/tr- esor/lxaip/1.3", pres:xmlData(LXAIP_OK_SIG_OK(AOID-101ab, VID-01)))))</pre>	<pre>ValidateEvidenceResponse (dsb:Result(resultmajor:Success))</pre>
10	<pre>ValidateEvidence (pres:Evidence(@FormatID="urn:ietf:rfc:4998:EvidenceRecord" , @POID="AOID-101ab", VersionID="VID-01", pres:binaryData(XAIP_ERS(AOID-101ab,VID-01, LXAIP_OK_SIG_OK_V1_ER_OK))) pres:PO(@FormatID="http://www.bsi.bund.de/tr- esor/lxaip/1.3", pres:xmlData(LXAIP_OK_SIG_OK(AOID-101ab, VID-01)))))</pre>	<pre>ValidateEvidenceResponse (dsb:Result(resultmajor:Success))</pre>

Step	Test sequence	Expected Results
11	<pre> ValidateEvidence (pres:Evidence(@FormatID="urn:ietf:rfc:4998:EvidenceRecord" , @POID="AOID-101ab", VersionID="VID-01", pres:binaryData(XAIP_ERS(AOID-101ab,VID-01, LXAIP_OK_SIG_OK_V1_ER_OK))) pres:PO(@FormatID="http://www.bsi.bund.de/tr- esor/lxaip/1.3", pres:binaryData(LXAIP_OK_SIG_OK(AOID-101ab, VID-01)))) </pre>	<pre> ValidateEvidenceResponse (dsb:Result(resultmajor:Success)) </pre>

Verdict of M-VE-01ab

Observations:

Verdict:

3.4.2.6.3 M-VE-02b - DXAIP_OK_SIG_OK

Identifier	M-VE-02b
Test Purpose	The test shall verify that it is possible to verify DXAIP_OK_SIG_OK with the TR-ESOR-specific signature policy http://www.bsi.bund.de/tr-esor/sigpolicy/verify-xaip .
Configuration	CONFIG_S.512
Pre-test conditions	- Authenticated connection to middleware exists. - The test M-VE-01b has been performed successfully

Step	Test sequence	Expected Results
1	Prepare DXAIP_OK_SIG_OK by inserting the AOID-100b	DXAIP_OK_SIG_OK(AOID-100b, VID-01, VID-02)
2	<pre> ValidateEvidence (pres:PO(@FormatID="http://www.bsi.bund.de/tr- esor/dxaip/1.3", pres:xmlData(DXAIP_OK_SIG_OK(AOID-100b, VID-01,VID-02)))) </pre>	<pre> ValidateEvidenceResponse (dsb:Result(resultmajor:Success)) </pre>
3	<pre> ValidateEvidence (pres:PO(@FormatID="http://www.bsi.bund.de/tr- esor/dxaip/1.3", pres:binaryData(DXAIP_OK_SIG_OK(AOID-100b, VID-01,VID-02)))) </pre>	<pre> ValidateEvidenceResponse (dsb:Result(resultmajor:Success)) </pre>

Step	Test sequence	Expected Results
	))	

Verdict of M-VE-02b
Observations:
Verdict:

3.4.2.6.4 M-VE-02ab - DLXAIP_OK_SIG_OK

Identifier	M-VE-02ab
Test Purpose	The test shall verify that it is possible to verify DLXAIP_OK_SIG_OK with the TR-ESOR-specific signature policy http://www.bsi.bund.de/tr-esor/sigpolicy/verify-xaip .
Configuration	- CONFIG_S.512 - CONFIG_LXAIP
Pre-test conditions	- Authenticated connection to middleware exists. - The test M-VE-01ab has been performed successfully.

Step	Test sequence	Expected Results
1	Prepare DLXAIP_OK_SIG_OK by inserting the AOID-101ab.	DLXAIP_OK_SIG_OK(AOID-101ab, VID-01, VID-02)
2	<pre> ValidateEvidence (pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/dlxaip/1.3", pres:xmlData(DLXAIP_OK_SIG_OK(AOID-101ab, VID-01, VID-02, REF_XML_DATA_62, REF_CADES_DET_XML_DATA_62)))) </pre>	<pre> ValidateEvidenceResponse (dsb:Result(resultmajor:Success)) </pre>
3	<pre> ValidateEvidence (pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/dlxaip/1.3", pres:binaryData(DLXAIP_OK_SIG_OK(AOID-101ab, VID-01, VID-02, REF_XML_DATA_63, REF_CADES_DET_XML_DATA_63)))) </pre>	<pre> ValidateEvidenceResponse (dsb:Result(resultmajor:Success)) </pre>

Verdict of M-VE-02ab
Observations:
Verdict:

3.4.2.6.5 M-VE-04b - XAIP_OK_SIG_NOK & XAIP_OK_SIG_OK_ER_NOK

Identifier	M-VE-04b
Test Purpose	The test shall verify that there will be a verification error, in case XAIP_OK_SIG_NOK or XAIP_OK_SIG_OK_ER_NOK is submitted.
Configuration	CONFIG_S.512
Pre-test conditions	Authenticated connection to middleware exists.

Step	Test sequence	Expected Results
1	<pre> ValidateEvidence (pres:PO(@FormatID="http://www.bsi.bund.de/tr- esor/xaip/1.3", pres:xmlData(XAIP_OK_SIG_NOK))) </pre>	<pre> ValidateEvidenceResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)), pres:ValidationReport(pres:xmlData(XAIP_OK_SIG_NOK_VR))) </pre> OR <pre> ValidateEvidenceResponse (pres:OptionalOutputs(dsb:Other(dsb:Value(XAIP_OK_SIG_NOK(XAIP_OK_SIG_NOK_VR))), dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)))) </pre>
2	<pre> ValidateEvidence (pres:PO(@FormatID="http://www.bsi.bund.de/tr- esor/xaip/1.3", pres:xmlData(XAIP_OK_SIG_OK_ER_NOK))) </pre>	<pre> ValidateEvidenceResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)), pres:ValidationReport(pres:xmlData(XAIP_OK_SIG_OK_ER_NOK_VR))) </pre> OR <pre> ValidateEvidenceResponse (pres:OptionalOutputs(dsb:Other(dsb:Value(XAIP_OK_SIG_OK_ER_NOK(XAIP_OK_SIG_OK_ER_NOK_VR))), dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)))) </pre>

Step	Test sequence	Expected Results
3	<pre> ValidateEvidence (pres:PO(@FormatID="http://www.bsi.bund.de/tr- esor/xaip/1.3", pres:binaryData(XAIP_OK_SIG_NOK))) </pre>	<pre> ValidateEvidenceResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)), pres:ValidationReport(pres:xmlData(XAIP_OK_SIG_NOK_VR))) </pre> OR <pre> ValidateEvidenceResponse (pres:OptionalOutputs(dsb:Other(dsb:Value(XAIP_OK_SIG_NOK(XAIP_OK_SIG_NOK_VR))), dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)))) </pre>
4	<pre> ValidateEvidence (pres:PO(@FormatID="http://www.bsi.bund.de/tr- esor/xaip/1.3", pres:binaryData(XAIP_OK_SIG_OK_ER_NOK))) </pre>	<pre> ValidateEvidenceResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)), pres:ValidationReport(pres:xmlData(XAIP_OK_SIG_OK_ER_NOK_VR))) </pre> OR <pre> ValidateEvidenceResponse (pres:OptionalOutputs(dsb:Other(dsb:Value(XAIP_OK_SIG_OK_ER_NOK(XAIP_OK_SIG_OK_ER_NOK_VR))), dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)))) </pre>

Verdict of M-EV-04b

Observations:

Verdict:

3.4.2.6.6 M-VE-04ab - LXAIP_OK_SIG_NOK & LXAIP_OK_SIG_OK_ER_NOK

Identifier	M-VE-04ab
Test Purpose	The test shall verify that there will be a verification error, in case of LXAIP_OK_SIG_NOK or LXAIP_OK_SIG_OK_ER_NOK is submitted. The response will include either the VerificationReport-Element only or the LXAIP including the VerificationReport-Element.
Configuration	- CONFIG_S.512 - CONFIG_LXAIP

Identifier	M-VE-04ab
Pre-test conditions	- Authenticated connection to middleware exists. - Test M-SU-01ab has been performed successfully and the references has been set properly in LXAIP_OK_SIG_OK.

Step	Test sequence	Expected Results
1	<pre> ValidateEvidence (pres:PO(@FormatID="http://www.bsi.bund.de/tr- esor/lxaip/1.3", pres:xmlData(LXAIP_OK_SIG_NOK(REF_TXT_DATA_65b, REF_CADES_DET_NOK_TXT_DATA_65b, REF_CADES_DET_PDF_DATA_65b, REF_CADES_DET_PDF_DATA_65b, REF_XML_MDO_65b)))) </pre>	<pre> ValidateEvidenceResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)), pres:ValidationReport(pres:xmlData(LXAIP_OK_SIG_NOK_VR))) OR ValidateEvidenceResponse (pres:OptionalOutputs(dsb:Other(dsb:Value(LXAIP_OK_SIG_NOK(LXAIP_OK_SIG_NOK_VR))), dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)))) </pre>
2	<pre> ValidateEvidence (pres:PO(@FormatID="http://www.bsi.bund.de/tr- esor/lxaip/1.3", pres:xmlData(LXAIP_OK_SIG_OK_ER_NOK(REF_TXT_DATA_66b, REF_CADES_DET_TXT_DATA_66b, REF_CADES_DET_PDF_DATA_66b, REF_CADES_DET_PDF_DATA_66b, REF_XML_MDO_66b, ER_NOK_XAIP_OK_SIG_OK_00)))) </pre>	<pre> ValidateEvidenceResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)), pres:ValidationReport(pres:xmlData(LXAIP_OK_SIG_OK_ER_NOK_VR))) OR ValidateEvidenceResponse (pres:OptionalOutputs(dsb:Other(dsb:Value(LXAIP_OK_SIG_OK_ER_NOK(LXAIP_OK_SIG_OK_ER_NOK_VR))), dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)))) </pre>

Step	Test sequence	Expected Results
3	<pre> ValidateEvidence (pres:PO(@FormatID="http://www.bsi.bund.de/tr- esor/lxaip/1.3", pres:binaryData(LXAIP_OK_SIG_NOK(REF_TXT_DATA_67b, REF_CADES_DET_NOK_TXT_DATA_67b, REF_CADES_DET_PDF_DATA_67b, REF_CADES_DET_PDF_DATA_67b, REF_XML_MDO_67b)))) </pre>	<pre> ValidateEvidenceResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)), pres:ValidationReport(pres:xmlData(LXAIP_OK_SIG_NOK_VR))) OR ValidateEvidenceResponse (pres:OptionalOutputs(dsb:Other(dsb:Value(LXAIP_OK_SIG_NOK(LXAIP_OK_SIG_NOK_VR))), dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)))) </pre>
4	<pre> ValidateEvidence (pres:PO(@FormatID="http://www.bsi.bund.de/tr- esor/lxaip/1.3", pres:binaryData(LXAIP_OK_SIG_OK_ER_NOK(REF_TXT_DATA_68b, REF_CADES_DET_TXT_DATA_68b, REF_CADES_DET_PDF_DATA_68b, REF_CADES_DET_PDF_DATA_68b, REF_XML_MDO_68b, ER_NOK_XAIP_OK_SIG_OK_00)))) </pre>	<pre> ValidateEvidenceResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)), pres:ValidationReport(pres:xmlData(LXAIP_OK_SIG_OK_ER_NOK_VR))) OR ValidateEvidenceResponse (pres:OptionalOutputs(dsb:Other(dsb:Value(LXAIP_OK_SIG_OK_ER_NOK(LXAIP_OK_SIG_OK_ER_NOK_VR))), dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)))) </pre>

Verdict of M-VE-04ab

Observations:

Verdict:

3.4.2.6.7 M-VE-05b - DXAIP_OK_SIG_NOK

Identifier	M-VE-05b
Test Purpose	The test shall verify that there will be a verification error, in case DXAIP_OK_SIG_NOK is submitted.
Configuration	CONFIG_S.512
Pre-test conditions	- Authenticated connection to middleware exists. - Test M-VE-01b has been performed successfully.

Step	Test sequence	Expected Results
1	Prepare DXAIP_OK_SIG_NOK by inserting AOID-100b.	DXAIP_OK_SIG_NOK (AOID-100b, VID-01, VID-02)
2	<pre> ValidateEvidence (pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/dxaip/1.3", pres:xmlData(DXAIP_OK_SIG_NOK (AOID-100b, VID-01, VID-02)))) </pre>	<pre> ValidateEvidenceResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)), pres:ValidationReport(pres:xmlData(DXAIP_OK_SIG_NOK_VR))) </pre> OR <pre> ValidateEvidenceResponse (pres:OptionalOutputs(dsb:Other(dsb:Value(DXAIP_OK_SIG_NOK(DXAIP_OK_SIG_NOK_VR))), dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)))) </pre>
3	<pre> ValidateEvidence (pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/dxaip/1.3", pres:binaryData(DXAIP_OK_SIG_NOK (AOID-100b, VID-01, VID-02)))) </pre>	<pre> ValidateEvidenceResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)), pres:ValidationReport(pres:xmlData(DXAIP_OK_SIG_NOK_VR))) </pre> OR <pre> ValidateEvidenceResponse (pres:OptionalOutputs(dsb:Other(dsb:Value(DXAIP_OK_SIG_NOK(DXAIP_OK_SIG_NOK_VR))), dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)))) </pre>

Verdict of M-VE-05b

Observations:

Verdict:

3.4.2.6.8 M-VE-05ab - DLXAIP_OK_SIG_NOK

Identifier	M-VE-05ab
Test Purpose	The test case shall verify that there will be a verification error, in case DLXAIP_OK_SIG_NOK is submitted.
Configuration	- CONFIG_S.512 - CONFIG_LXAIP

Identifier	M-VE-05ab
Pre-test conditions	- Authenticated connection to middleware exists. - Test M-VE-01ab has been performed successfully.

Step	Test sequence	Expected Results
1	Prepare DLXAIP_OK_SIG_NOK by inserting AOID-101ab.	DLXAIP_OK_SIG_NOK(AOID-101ab, VID-01, VID-02)
2	<pre> ValidateEvidence (pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/dlxaip/1.3", pres:xmlData(DLXAIP_OK_SIG_NOK(AOID-101ab, VID-01, VID-02, REF_XML_DATA_69b, REF_CADES_DET_NOK_XML_DATA_69b)))) </pre>	<pre> ValidateEvidenceResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)), pres:ValidationReport(pres:xmlData(DXAIP_OK_SIG_NOK_VR))) </pre> OR <pre> ValidateEvidenceResponse (pres:OptionalOutputs(dsb:Other(dsb:Value(DLXAIP_OK_SIG_NOK(AOID-101, VID-01, VID-02, DXAIP_OK_SIG_NOK_VR)), dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)),)) </pre>
3	<pre> ValidateEvidence (pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/dlxaip/1.3", pres:binaryData(DLXAIP_OK_SIG_NOK(AOID-101ab, VID-01, VID-02, REF_XML_DATA_70b, REF_CADES_DET_NOK_XML_DATA_70b)))) </pre>	<pre> VerifyResponse (dss:Result(resultmajor#error, resultminor/ar1/DXAIP_NOK_SIG) dss:OptionalOutputs(DXAIP_OK_SIG_NOK_VR)) </pre> OR <pre> ValidateEvidenceResponse (pres:OptionalOutputs(dsb:Other(dsb:Value(DLXAIP_OK_SIG_NOK(AOID-101ab, VID-01, VID-02, DXAIP_OK_SIG_NOK_VR)), dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)),)) </pre>

Verdict of M-VE-05ab
Observations:
Verdict:

3.4.2.6.9 M-VE-07b – Hash algorithms

Identifier	M-VE-07b
Test Purpose	The test shall verify that it is possible to verify XAIP_OK and corresponding evidence record with the TR-ESOR-specific signature policy http://www.bsi.bund.de/tr-esor/sigpolicy/verify-xaip by using different hash algorithms for underlying hash tree. The hash algorithms to be tested are following SHA-1, SHA-224, SHA-256, SHA-384, SHA-512, SHA3-224, SHA3-256, SHA3-384, SHA3-512.
Configuration	CONFIG_S.512
Pre-test conditions	Authenticated connection to middleware exists.

Step	Test sequence	Expected Results
1	Prepare XAIP_OK_SIG_OK by inserting of AOID-999b	XAIP_OK_SIG_OK(AOID-999b, VID-01)
2	<pre> ValidateEvidence (pres:Evidence(@FormatID= "urn:ietf:rfc:4998:EvidenceRecord", @RelatedObjects="XAIP-01", @POID="AOID-999b", @VersionID="VID-01", pres:binaryData(ER_OK_SHA-1_XAIP_OK_SIG_OK), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3", @ID="XAIP-01", pres:xmlData(XAIP_OK_SIG_OK(AOID-999b,VID-01))))) </pre>	<pre> ValidateEvidenceResponse (dsb:Result(resultmajor:Success)) OR ValidateEvidenceResponse (dsb:Result(resultmajor:RequesterError)), dsb:ResultMinor(error/POFormatError)), pres:ValidationReport(pres:binaryData(XAIP_OK_SIG_OK_VR(AOID-999b, VID-01, ER_OK_SHA-1_XAIP_OK_SIG_OK_VR))) OR ValidateEvidenceResponse (dsb:Result(resultmajor:RequesterError)), dsb:ResultMinor(error/POFormatError)), pres:ValidationReport(pres:xmlData(XAIP_OK_SIG_OK_VR(AOID-999b, VID-01, ER_OK_SHA-1_XAIP_OK_SIG_OK_VR))) OR ValidateEvidenceResponse (pres:OptionalOutputs(dsb:Other(dsb:Value(XAIP_OK_SIG_OK(AOID-999b, VID-01, ER_OK_SHA-1_XAIP_OK_SIG_OK_VR))), dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)))) </pre>

Step	Test sequence	Expected Results
3	<pre> ValidateEvidence (pres:Evidence(@FormatID= "urn:ietf:rfc:4998:EvidenceRecord", @RelatedObjects="XAIP-01", @POID=AOID-999b, @VersionID=VID-01 pres:xmlData(ER_NOK_SHA-1_XAIP_OK_SIG_OK), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3" @ID="XAIP-01", pres:xmlData(XAIP_OK_SIG_OK(AOID-999b,VID-01))))) </pre>	<pre> ValidateEvidenceResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)), pres:ValidationReport(pres:xmlData(ER_NOK_SHA-1_XAIP_OK_SIG_OK_VR))) </pre> OR <pre> ValidateEvidenceResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)), pres:ValidationReport(pres:binaryData(ER_NOK_SHA-1_XAIP_OK_SIG_OK_VR))) </pre> OR <pre> ValidateEvidenceResponse (pres:OptionalOutputs(dsb:Other(dsb:Value(XAIP_OK_SIG_OK(AOID-999b, VID-01, ER_NOK_SHA-1_XAIP_OK_SIG_OK_VR))), dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)))) </pre>

Step	Test sequence	Expected Results
4	<pre> ValidateEvidence (pres:Evidence(@FormatID= "urn:ietf:rfc:4998:EvidenceRecord", @RelatedObjects="XAIP-01", @POID=AOID-999b, @VersionID=VID-01 pres:xmlData(ER_OK_SHA-224_XAIP_OK_SIG_OK)), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3" @ID="XAIP-01", pres:xmlData(XAIP_OK_SIG_OK(AOID-999b,VID-01)))) </pre>	<pre> ValidateEvidenceResponse (dsb:Result(resultmajor:Success)) </pre> OR <pre> ValidateEvidenceResponse (dsb:Result(resultmajor:RequesterError)), dsb:ResultMinor(error/POFormatError)), pres:ValidationReport(pres:binaryData(XAIP_OK_SIG_OK_VR(AOID-999b, VID-01, ER_OK_SHA-224_XAIP_OK_SIG_OK_VR)))) </pre> OR <pre> ValidateEvidenceResponse (dsb:Result(resultmajor:RequesterError)), dsb:ResultMinor(error/POFormatError)), pres:ValidationReport(pres:xmlData(XAIP_OK_SIG_OK_VR(AOID-999b, VID-01, ER_OK_SHA-224_XAIP_OK_SIG_OK_VR)))) </pre> OR <pre> ValidateEvidenceResponse (pres:OptionalOutputs(dsb:Other(dsb:Value(XAIP_OK_SIG_OK(AOID-999b, VID-01, ER_OK_SHA-224_XAIP_OK_SIG_OK_VR))), dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)))) </pre>

Step	Test sequence	Expected Results
5	<pre> ValidateEvidence (pres:Evidence(@FormatID= "urn:ietf:rfc:4998:EvidenceRecord", @RelatedObjects="XAIP-01", @POID=AOID-999b, @VersionID=VID-01 pres:xmlData(ER_NOK_SHA-224_XAIP_OK_SIG_OK)), pres:PO(@FormatID="http://www.bsi.bund.de/tr- esor/xaip/1.3" @ID="XAIP-01", pres:xmlData(XAIP_OK_SIG_OK(AOID-999b,VID-01)))) </pre>	<pre> ValidateEvidenceResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)), pres:ValidationReport(pres:xmlData(ER_NOK_SHA-224_XAIP_OK_SIG_OK_VR))) OR ValidateEvidenceResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)), pres:ValidationReport(pres:binaryData(ER_NOK_SHA-224_XAIP_OK_SIG_OK_VR))) OR ValidateEvidenceResponse (pres:OptionalOutputs(dsb:Other(dsb:Value(XAIP_OK_SIG_OK(AOID-999b, VID-01, ER_NOK_SHA-224_XAIP_OK_SIG_OK_VR))), dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)))) </pre>
6	<pre> ValidateEvidence (pres:Evidence(@FormatID= "urn:ietf:rfc:4998:EvidenceRecord", @RelatedObjects="XAIP-01", @POID=AOID-999b, @VersionID=VID-01 pres:xmlData(XAIP_ERS(AOID-999b,VID-01, ER_OK_SHA-256_XAIP_OK_SIG_OK))), pres:PO(@FormatID="http://www.bsi.bund.de/tr- esor/xaip/1.3" @ID="XAIP-01", pres:xmlData(XAIP_OK_SIG_OK(AOID-999b,VID-01)))) </pre>	<pre> ValidateEvidenceResponse (dsb:Result(resultmajor:Success)) </pre>

Step	Test sequence	Expected Results
7	<pre> ValidateEvidence (pres:Evidence(@FormatID= "urn:ietf:rfc:4998:EvidenceRecord", @RelatedObjects="XAIP-01", @POID=AOID-999b, @VersionID=VID-01 pres:xmlData(ER_NOK_SHA-256_XAIP_OK_SIG_OK)), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3" @ID="XAIP-01", pres:xmlData(XAIP_OK_SIG_OK(AOID-999b,VID-01)))) </pre>	<pre> ValidateEvidenceResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)), pres:ValidationReport(pres:xmlData(ER_NOK_SHA-256_XAIP_OK_SIG_OK_VR))) OR ValidateEvidenceResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)), pres:ValidationReport(pres:binaryData(ER_NOK_SHA-256_XAIP_OK_SIG_OK_VR))) OR ValidateEvidenceResponse (pres:OptionalOutputs(dsb:Other(dsb:Value(XAIP_OK_SIG_OK(AOID-999b, VID-01, ER_NOK_SHA-256_XAIP_OK_SIG_OK_VR))), dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)))) </pre>
8	<pre> ValidateEvidence (pres:Evidence(@FormatID= "urn:ietf:rfc:4998:EvidenceRecord", @RelatedObjects="XAIP-01", @POID=AOID-999b, @VersionID=VID-01 pres:xmlData(ER_OK_SHA-384_XAIP_OK_SIG_OK)), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3" @ID="XAIP-01", pres:xmlData(XAIP_OK_SIG_OK(AOID-999b,VID-01)))) </pre>	<pre> ValidateEvidenceResponse (dsb:Result(resultmajor:Success)) </pre>

Step	Test sequence	Expected Results
9	<pre> ValidateEvidence (pres:Evidence(@FormatID= "urn:ietf:rfc:4998:EvidenceRecord", @RelatedObjects="XAIP-01", @POID=AOID-999b, @VersionID=VID-01 pres:xmlData(ER_NOK_SHA-384_XAIP_OK_SIG_OK)), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3" @ID="XAIP-01", pres:xmlData(XAIP_OK_SIG_OK(AOID-999b,VID-01)))) </pre>	<pre> ValidateEvidenceResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)), pres:ValidationReport(pres:xmlData(ER_NOK_SHA-384_XAIP_OK_SIG_OK_VR))) </pre> OR <pre> ValidateEvidenceResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)), pres:ValidationReport(pres:binaryData(ER_NOK_SHA-384_XAIP_OK_SIG_OK_VR))) </pre> OR <pre> ValidateEvidenceResponse (pres:OptionalOutputs(dsb:Other(dsb:Value(XAIP_OK_SIG_OK(AOID-999b, VID-01, ER_NOK_SHA-384_XAIP_OK_SIG_OK_VR))), dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)))) </pre>
10	<pre> ValidateEvidence (pres:Evidence(@FormatID= "urn:ietf:rfc:4998:EvidenceRecord", @RelatedObjects="XAIP-01", @POID=AOID-999b, @VersionID=VID-01 pres:xmlData(ER_OK_SHA-512_XAIP_OK_SIG_OK)), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3" @ID="XAIP-01", pres:xmlData(XAIP_OK_SIG_OK(AOID-999b,VID-01)))) </pre>	<pre> ValidateEvidenceResponse (dsb:Result(resultmajor:Success)) </pre>

Step	Test sequence	Expected Results
11	<pre> ValidateEvidence (pres:Evidence(@FormatID= "urn:ietf:rfc:4998:EvidenceRecord", @RelatedObjects="XAIP-01", @POID=AOID-999b, @VersionID=VID-01 pres:xmlData(ER_NOK_SHA-512_XAIP_OK_SIG_OK)), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3" @ID="XAIP-01", pres:xmlData(XAIP_OK_SIG_OK(AOID-999b,VID-01)))) </pre>	<pre> ValidateEvidenceResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)), pres:ValidationReport(pres:xmlData(ER_NOK_SHA-512_XAIP_OK_SIG_OK_VR))) OR ValidateEvidenceResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)), pres:ValidationReport(pres:binaryData(ER_NOK_SHA-512_XAIP_OK_SIG_OK_VR))) OR ValidateEvidenceResponse (pres:OptionalOutputs(dsb:Other(dsb:Value(XAIP_OK_SIG_OK(AOID-999b, VID-01, ER_NOK_SHA-512_XAIP_OK_SIG_OK_VR))), dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)))) </pre>

Step	Test sequence	Expected Results
12	<pre> ValidateEvidence (pres:Evidence(@FormatID= "urn:ietf:rfc:4998:EvidenceRecord", @RelatedObjects="XAIP-01", @POID=AOID-999b, @VersionID=VID-01 pres:xmlData(ER_OK_SHA3-224_XAIP_OK_SIG_OK)), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3" @ID="XAIP-01", pres:xmlData(XAIP_OK_SIG_OK(AOID-999b,VID-01)))) </pre>	<pre> ValidateEvidenceResponse (dsb:Result(resultmajor:Success)) </pre> OR <pre> ValidateEvidenceResponse (dsb:Result(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError), pres:ValidationReport(pres:binaryData(XAIP_OK_SIG_OK_VR(AOID-999b, VID-01, ER_OK_SHA3-224_XAIP_OK_SIG_OK_VR)))) </pre> OR <pre> ValidateEvidenceResponse (dsb:Result(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError), pres:ValidationReport(pres:xmlData(XAIP_OK_SIG_OK_VR(AOID-999b, VID-01, ER_OK_SHA3-224_XAIP_OK_SIG_OK_VR)))) </pre> OR <pre> ValidateEvidenceResponse (pres:OptionalOutputs(dsb:Other(dsb:Value(XAIP_OK_SIG_OK(AOID-999b, VID-01, ER_OK_SHA3- 224_XAIP_OK_SIG_OK_VR))), dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)))) </pre>

Step	Test sequence	Expected Results
13	<pre> ValidateEvidence (pres:Evidence(@FormatID= "urn:ietf:rfc:4998:EvidenceRecord", @RelatedObjects="XAIP-01", @POID=AOID-999b, @VersionID=VID-01 pres:xmlData(ER_NOK_SHA3-224_XAIP_OK_SIG_OK)), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3" @ID="XAIP-01", pres:xmlData(XAIP_OK_SIG_OK(AOID-999b,VID-01)))) </pre>	<pre> ValidateEvidenceResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)), pres:ValidationReport(pres:xmlData(ER_NOK_SHA3-224_XAIP_OK_SIG_OK_VR))) OR ValidateEvidenceResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)), pres:ValidationReport(pres:binaryData(ER_NOK_SHA3-224_XAIP_OK_SIG_OK_VR))) OR ValidateEvidenceResponse (pres:OptionalOutputs(dsb:Other(dsb:Value(XAIP_OK_SIG_OK(AOID-999b, VID-01, ER_NOK_SHA3-224_XAIP_OK_SIG_OK_VR))), dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)))) </pre>
14	<pre> ValidateEvidence (pres:Evidence(@FormatID= "urn:ietf:rfc:4998:EvidenceRecord", @RelatedObjects="XAIP-01", @POID=AOID-999b, @VersionID=VID-01 pres:xmlData(ER_OK_SHA3-256_XAIP_OK_SIG_OK)), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3" @ID="XAIP-01", pres:xmlData(XAIP_OK_SIG_OK(AOID-999b,VID-01)))) </pre>	<pre> ValidateEvidenceResponse (dsb:Result(resultmajor:Success)) </pre>

Step	Test sequence	Expected Results
15	<pre> ValidateEvidence (pres:Evidence(@FormatID= "urn:ietf:rfc:4998:EvidenceRecord", @RelatedObjects="XAIP-01", @POID=AOID-999b, @VersionID=VID-01 pres:xmlData(ER_NOK_SHA3-256_XAIP_OK_SIG_OK)), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3" @ID="XAIP-01", pres:xmlData(XAIP_OK_SIG_OK(AOID-999b,VID-01)))) </pre>	<pre> ValidateEvidenceResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)), pres:ValidationReport(pres:xmlData(ER_NOK_SHA3-256_XAIP_OK_SIG_OK_VR))) </pre> OR <pre> ValidateEvidenceResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)), pres:ValidationReport(pres:binaryData(ER_NOK_SHA3-256_XAIP_OK_SIG_OK_VR))) </pre> OR <pre> ValidateEvidenceResponse (pres:OptionalOutputs(dsb:Other(dsb:Value(XAIP_OK_SIG_OK(AOID-999b, VID-01, ER_NOK_SHA3-256_XAIP_OK_SIG_OK_VR))), dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)))) </pre>
16	<pre> ValidateEvidence (pres:Evidence(@FormatID= "urn:ietf:rfc:4998:EvidenceRecord", @RelatedObjects="XAIP-01", @POID=AOID-999b, @VersionID=VID-01 pres:xmlData(ER_OK_SHA3-384_XAIP_OK_SIG_OK)), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3" @ID="XAIP-01", pres:xmlData(XAIP_OK_SIG_OK(AOID-999b,VID-01)))) </pre>	<pre> ValidateEvidenceResponse (dsb:Result(resultmajor:Success)) </pre>

Step	Test sequence	Expected Results
17	<pre> ValidateEvidence (pres:Evidence(@FormatID= "urn:ietf:rfc:4998:EvidenceRecord", @RelatedObjects="XAIP-01", @POID=AOID-999b, @VersionID=VID-01 pres:xmlData(ER_NOK_SHA3-384_XAIP_OK_SIG_OK)), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3" @ID="XAIP-01", pres:xmlData(XAIP_OK_SIG_OK(AOID-999b,VID-01)))) </pre>	<pre> ValidateEvidenceResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)), pres:ValidationReport(pres:xmlData(ER_NOK_SHA3-384_XAIP_OK_SIG_OK_VR))) OR ValidateEvidenceResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)), pres:ValidationReport(pres:binaryData(ER_NOK_SHA3-384_XAIP_OK_SIG_OK_VR))) OR ValidateEvidenceResponse (pres:OptionalOutputs(dsb:Other(dsb:Value(XAIP_OK_SIG_OK(AOID-999b, VID-01, ER_NOK_SHA3-384_XAIP_OK_SIG_OK_VR))), dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)))) </pre>
18	<pre> ValidateEvidence (pres:Evidence(@FormatID= "urn:ietf:rfc:4998:EvidenceRecord", @RelatedObjects="XAIP-01", @POID=AOID-999b, @VersionID=VID-01 pres:xmlData(ER_OK_SHA3-512_XAIP_OK_SIG_OK)), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3" @ID="XAIP-01", pres:xmlData(XAIP_OK_SIG_OK(AOID-999b,VID-01)))) </pre>	<pre> ValidateEvidenceResponse (dsb:Result(resultmajor:Success)) </pre>

Step	Test sequence	Expected Results
19	<pre> ValidateEvidence (pres:Evidence(@FormatID= "urn:ietf:rfc:4998:EvidenceRecord", @RelatedObjects="XAIP-01", @POID=AOID-999b, @VersionID=VID-01 pres:xmlData(ER_NOK_SHA3-512_XAIP_OK_SIG_OK)), pres:PO(@FormatID="http://www.bsi.bund.de/tr- esor/xaip/1.3" @ID="XAIP-01", pres:xmlData(XAIP_OK_SIG_OK(AOID-999b,VID-01)))) </pre>	<pre> ValidateEvidenceResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)), pres:ValidationReport(pres:xmlData(ER_NOK_SHA3-512_XAIP_OK_SIG_OK_VR))) OR ValidateEvidenceResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)), pres:ValidationReport(pres:binaryData(ER_NOK_SHA3-512_XAIP_OK_SIG_OK_VR))) OR ValidateEvidenceResponse (pres:OptionalOutputs(dsb:Other(dsb:Value(XAIP_OK_SIG_OK(AOID-999b, VID-01, ER_NOK_SHA3-512_XAIP_OK_SIG_OK_VR))), dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError)))) </pre>

Verdict of M-VE-07b

Observations:

Verdict:

3.4.2.7 Additional tests of S.512-interface

3.4.2.7.1 M-ADD-01b – XAIP_OK and Resigning

Identifier	M-ADD-01b
Test Purpose	The test shall verify that it is possible to retrieve previously submitted and updated XAIP_OK after the time-stamp resigning procedure has been successfully applied.
Configuration	CONFIG_S.512
Pre-test conditions	- Authenticated connection to middleware exists. - The time-stamp resigning procedure has been successfully performed - Make sure that the test case M-UP-01b was successfully performed.

Step	Test sequence	Expected Results
1	<pre>RetrievePO (pres:POID(AOID-01b), pres:VersionID("all"), pres:SubjectOfRetrieval("PO"))</pre>	<pre>RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3", pres:xmlData(XAIP_OK(AOID-01b, VID-01, VID-02, VID-03))))</pre>
2	<pre>RetrievePO (pres:POID(AOID-01b), pres:VersionID("all"), pres:SubjectOfRetrieval("Evidence"))</pre>	<pre>RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="urn:ietf:rfc:4998:EvidenceRecord", pres:xmlData(XAIP_ERS(AOID-01b, VID-01, XAIP_OK_V1_ER_OK_RESG))), pres:PO(@FormatID="urn:ietf:rfc:4998:EvidenceRecord", pres:xmlData(XAIP_ERS(AOID-01b, VID-02, XAIP_OK_V2_ER_OK_RESG))), pres:PO(@FormatID="urn:ietf:rfc:4998:EvidenceRecord", pres:xmlData(XAIP_ERS(AOID-01b, VID-03, XAIP_OK_V3_ER_OK_RESG))))</pre>

Step	Test sequence	Expected Results
3	Test of correctness of the obtained: • XAIP_OK_V1_ER_OK_RESG, • XAIP_OK_V2_ER_OK_RESG and • XAIP_OK_V3_ER_OK_RESG by using ERVerifyTool. <pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:InlineXML(XAIP_OK(AOID-01b, VID-01, VID-02, VID-03))))) dss:SignatureObject(dss:Other(XAIP_ERS(AOID-01b, VID-01, XAIP_OK_V1_ER_OK_RESG), XAIP_ERS(AOID-01b, VID-02, XAIP_OK_V2_ER_OK_RESG), XAIP_ERS(AOID-01b, VID-03, XAIP_OK_V3_ER_OK_RESG)))) </pre>	<pre> VerifyResponse(dss:Result(resultmajor#ok), dss:OptionalOutputs(dssvr:VerificationReport(dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:EvidenceRecordReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid) [...]))), dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:EvidenceRecordReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid) [...]))), dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:EvidenceRecordReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid) [...])))))) </pre>

Verdict of M-ADD-01b
Observations:
Verdict:

3.4.2.7.2 M-ADD-01ab – LXAIP_OK and Resigning

Identifier	M-ADD-01ab
Test Purpose	The test shall verify that it is possible to retrieve previously submitted and updated LXAIP_OK after the time-stamp resigning procedure has been successfully applied.

Identifier	M-ADD-01ab
Configuration	- CONFIG_S.512 - CONFIG_LXAIP
Pre-test conditions	- Authenticated connection to middleware exists. - The time-stamp resigning procedure has been successfully performed - Make sure that the test case M-UP-01ab was successfully performed.

Step	Test sequence	Expected Results
1	RetrievePO (pres:POID(AOID-01ab), pres:VersionID("all"), pres:SubjectOfRetrieval("PO"), pres:POFormat(" http://www.bsi.bund.de/tr-esor/lxaip/1.3 "))	RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID=" http://www.bsi.bund.de/tr-esor/lxaip/1.3 ", pres:xmlData(LXAIP_OK(AOID-01ab, VID-01, VID-02, VID-03))))
2	RetrievePO (pres:POID(AOID-01ab), pres:VersionID("all"), pres:SubjectOfRetrieval("Evidence"),)	RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="urn:ietf:rfc:4998:EvidenceRecord", pres:xmlData(XAIP_ERS(AOID-01ab, VID-01, LXAIP_OK_V1_ER_OK_RESG))), pres:PO(@FormatID="urn:ietf:rfc:4998:EvidenceRecord", pres:xmlData(XAIP_ERS(AOID-01ab, VID-02, LXAIP_OK_V2_ER_OK_RESG))), pres:PO(@FormatID="urn:ietf:rfc:4998:EvidenceRecord", pres:xmlData(XAIP_ERS(AOID-01ab, VID-03, LXAIP_OK_V3_ER_OK_RESG))))

Step	Test sequence	Expected Results
3	Test of correctness of the obtained: • LXAIP_OK_V1_ER_OK_RESg, • LXAIP_OK_V2_ER_OK_RESg and • LXAIP_OK_V3_ER_OK_RESg by using ERVerifyTool. The particular object references in the obtained LXAIP_OK will be replaced by the references compatible with the ERVerifyTool in advance. <pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:InlineXML(LXAIP_OK(AOID-01ab, VID-01, VID-02, VID-03))))) dss:SignatureObject(dss:Other(XAIP_ERS(AOID-01ab, VID-01, LXAIP_OK_V1_ER_OK_RESg), XAIP_ERS(AOID-01ab, VID-02, LXAIP_OK_V2_ER_OK_RESg), XAIP_ERS(AOID-01ab, VID-03, LXAIP_OK_V3_ER_OK_RESg)))) </pre>	<pre> VerifyResponse(dss:Result(resultmajor#ok), dss:OptionalOutputs(dssvr:VerificationReport(dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:EvidenceRecordReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid)) [...])), dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:EvidenceRecordReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid)) [...])), dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:EvidenceRecordReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid)) [...]))))) </pre>

Verdict of M-ADD-01ab

Observations:

Verdict:

3.4.2.7.3 M-ADD-02b – XAIP_OK and Rehashing

Identifier	M-ADD-02b
Test Purpose	The test shall verify that it is possible to retrieve previously submitted and updated XAIP_OK after the hash tree renewal procedure (rehashing) has been successfully applied.
Configuration	CONFIG_S.512
Pre-test conditions	- Authenticated connection to middleware exists. - The rehashing procedure has been successfully performed - Make sure that the test case M-ADD-01b was successfully performed.

Step	Test sequence	Expected Results
1	RetrievePO (pres:POID(AOID-01b), pres:VersionID("all"), pres:SubjectOfRetrieval("PO"))	RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID=" http://www.bsi.bund.de/tr-esor/xaip/1.3 ", pres:xmlData(XAIP_OK(AOID-01b, VID-01, VID-02, VID-03)))))
2	RetrievePO (pres:POID(AOID-01b), pres:VersionID("all"), pres:SubjectOfRetrieval("Evidence"))	RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="urn:ietf:rhc:4998:EvidenceRecord", pres:xmlData(XAIP_ERS(AOID-01b, VID-01, XAIP_OK_V1_ER_OK_RESG_REH))), pres:PO(@FormatID="urn:ietf:rhc:4998:EvidenceRecord", pres:xmlData(XAIP_ERS(AOID-01b, VID-02, XAIP_OK_V2_ER_OK_RESG_REH))), pres:PO(@FormatID="urn:ietf:rhc:4998:EvidenceRecord", pres:xmlData(XAIP_ERS(AOID-01b, VID-03, XAIP_OK_V3_ER_OK_RESG_REH)))))

Step	Test sequence	Expected Results
3	Test of correctness of the obtained: • XAIP_OK_V1_ER_OK_RESG_REH, • XAIP_OK_V2_ER_OK_RESG_REH and • XAIP_OK_V3_ER_OK_RESG_REH by using ERVerifyTool. <pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:InlineXML(XAIP_OK(AOID-01b, VID-01, VID-02, VID-03)))) dss:SignatureObject(dss:Other(XAIP_ERS(AOID-01b, VID-01, XAIP_OK_V1_ER_OK_RESG_REH), XAIP_ERS(AOID-01b, VID-02, XAIP_OK_V2_ER_OK_RESG_REH), XAIP_ERS(AOID-01b, VID-03, XAIP_OK_V3_ER_OK_RESG_REH)))) </pre>	<pre> VerifyResponse(dss:Result(resultmajor#ok), dss:OptionalOutputs(dssvr:VerificationReport(dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:EvidenceRecordReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid) [...]))), dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:EvidenceRecordReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid) [...]))), dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:EvidenceRecordReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid) [...])))))) </pre>

Verdict of M-ADD-02b

Observations:

Verdict:

3.4.2.7.4 M-ADD-02ab – LXAIP_OK and Resigning

Identifier	M-ADD-02ab
Test Purpose	The test shall verify that it is possible to retrieve previously submitted and updated LXAIP_OK after the hash tree renewal procedure (rehashing) has been successfully applied.

Identifier	M-ADD-02ab
Configuration	- CONFIG_S.512 - CONFIG_LXAIP
Pre-test conditions	- Authenticated connection to middleware exists. - The rehashing procedure has been successfully performed - Make sure that the test case M-ADD-01ab was successfully performed.

Step	Test sequence	Expected Results
1	<pre>RetrievePO (pres:POID(AOID-01ab), pres:VersionID("all"), pres:SubjectOfRetrieval("PO"), pres:POFormat("http://www.bsi.bund.de/tr-esor/lxaip/1.3"))</pre>	<pre>RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/lxaip/1.3", pres:xmlData(LXAIP_OK(AOID-01ab, VID-01, VID-02, VID-03))))</pre>
2	<pre>RetrievePO (pres:POID(AOID-01ab), pres:VersionID("all"), pres:SubjectOfRetrieval("Evidence"),)</pre>	<pre>RetrievePOResponse (dsb:Result(resultmajor:Success), pres:PO(@FormatID="urn:ietf:rfc:4998:EvidenceRecord", pres:xmlData(XAIP_ERS(AOID-01ab, VID-01, LXAIP_OK_V1_ER_OK_RESG_REH))), pres:PO(@FormatID="urn:ietf:rfc:4998:EvidenceRecord", pres:xmlData(XAIP_ERS(AOID-01ab, VID-02, LXAIP_OK_V2_ER_OK_RESG_REH))), pres:PO(@FormatID="urn:ietf:rfc:4998:EvidenceRecord", pres:xmlData(XAIP_ERS(AOID-01ab, VID-03, LXAIP_OK_V3_ER_OK_RESG_REH))))</pre>

Step	Test sequence	Expected Results
3	Test of correctness of the obtained: • LXAIP_OK_V1_ER_OK_RESg_REH, • LXAIP_OK_V2_ER_OK_RESg_REH and • LXAIP_OK_V3_ER_OK_RESg_REH by using ERVerifyTool. The particular object references in the obtained LXAIP_OK will be replaced by the references compatible with the ERVerifyTool in advance. <pre> VerifyRequest (dss:InputDocuments(dss:Document(dss:InlineXML(LXAIP_OK (AOID-01ab, VID-01, VID-02, VID-03))))) dss:SignatureObject(dss:Other(XAIP_ERS(AOID-01ab, VID-01, LXAIP_OK_V1_ER_OK_RESg_REH), XAIP_ERS(AOID-01ab, VID-02, LXAIP_OK_V2_ER_OK_RESg_REH), XAIP_ERS(AOID-01ab, VID-03, LXAIP_OK_V3_ER_OK_RESg_REH)))) </pre>	<pre> VerifyResponse(dss:Result(resultmajor#ok), dss:OptionalOutputs(dssvr:VerificationReport(dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:EvidenceRecordReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid)) [...])), dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:EvidenceRecordReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid)) [...])), dssvr:IndividualReport(dss:Result(urn:oasis:names:tc:dss:1.0:detail:valid) dssvr:Details(vr:EvidenceRecordReport(vr:FormatOK(urn:oasis:names:tc:dss:1.0:detail:valid)) [...]))))) </pre>

Verdict of M-ADD-02ab

Observations:

Verdict:

3.4.2.7.5 M-ADD-03ab – Relation between upload data and LXAIP/DLXAIP

Identifier	M-ADD-03ab
Test Purpose	The test shall verify that 1:1-relation between the uploaded data and corresponding LXAIP/DLXAIP is adhered to. It is not possible to use a single reference in two different LXAIP/DLXAIP instances and in such case, an error will be returned.
Configuration	- CONFIG_S.512 - CONFIG_LXAIP
Pre-test conditions	- Authenticated connection to middleware exists. - Test case M-SU-01a has been performed successfully.

Step	Test sequence	Expected Results
1	<pre> PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID="http://www.bsi.bund.de/tr- esor/lxaip/1.3", pres:xmlData(LXAIP_OK(REF_TXT_DATA_71b, REF_PDF_DATA_71b, REF_XML_MDO_71b))) </pre>	<pre> PreservePOResponse (dsb:Result(resultmajor:Success), pres:POID(AOID-300ab)) </pre>
2	<pre> PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID="http://www.bsi.bund.de/tr- esor/lxaip/1.3", pres:xmlData(LXAIP_OK(REF_TXT_DATA_71b, REF_PDF_DATA_71b, REF_XML_MDO_71b))) </pre>	<pre> PreservePOResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/parameterError))) </pre>
3	<pre> PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID="http://www.bsi.bund.de/tr- esor/lxaip/1.3", pres:xmlData(LXAIP_OK_ER_NOK(REF_TXT_DATA_72b, REF_PDF_DATA_72b, REF_XML_MDO_72b, ER_NOK_LXAIP_OK_72)))) </pre>	<pre> PreservePOResponse (pres:OptionalOutputs(dsb:Other(dsb:Value(LXAIP_OK_ER_NOK_VR))), dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/POFormatError))) </pre>

Step	Test sequence	Expected Results
4	<pre> PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/lxaip/1.3", pres:xmlData(LXAIP_OK_ER_NOK(REF_TXT_DATA_72b, REF_PDF_DATA_72b, REF_XML_MDO_72b, ER_NOK_LXAIP_OK_72)))) </pre>	<pre> PreservePOResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/parameterError))) </pre>
5	<pre> UpdatePOC (pres:POID(AOID-01ab), pres:DeltaPOC(@FormatID("http://www.bsi.bund.de/tr-esor/dlxaip/1.3"), pres:xmlData(DLXAIP_OK(AOID-300ab,VID-01,VID-02, REF_CADES_ATT_TXT_DATA_73b)))) </pre>	<pre> UpdatePOCResponse (dsb:Result(resultmajor:Success), VersionID(VID-02)) </pre>
6	<pre> PreservePO (pres:Profile(ACTIVE_PROFILE_URI), pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/lxaip/1.3", pres:xmlData(LXAIP_OK(REF_TXT_DATA_74b, REF_PDF_DATA_74b, REF_XML_MDO_74b))) </pre>	<pre> PreservePOResponse (dsb:Result(resultmajor:Success), pres:POID(AOID-301ab)) </pre>
7	<pre> UpdatePOC (pres:POID(AOID-01ab), pres:DeltaPOC(@FormatID("http://www.bsi.bund.de/tr-esor/dlxaip/1.3"), pres:xmlData(DLXAIP_OK(AOID-301ab,VID-01,VID-02, REF_CADES_ATT_TXT_DATA_73b)))) </pre>	<pre> PreservePOResponse (dsb:Result(dsb:ResultMajor(resultmajor:RequesterError), dsb:ResultMinor(error/parameterError))) </pre>

Verdict of M-ADD-03ab

Observations:

Verdict:

3.4.2.7.6 M-ADD-04b – Evidence Record with single hash value

Identifier	M-ADD-04b
Test Purpose	The test shall verify that it is possible to verify every variant of the evidence record computed on a single data object: - ER(TSP(H(DO-01))) – hash value of the data object is directly included in the message imprint of the timestamp (ER_SO-1_OK_XAIP_OK_SO), - ER(H(DO-01), TSP(H(DO-01))) – the hash value of the data object is placed in the single reduced hash tree and the same hash value is included in the message imprint of the timestamp (ER_SO-2_OK_XAIP_OK_SO), - ER(H(DO-01), TSP(H(H(DO-01)))) – the hash value of the data object is placed in the single reduced hash tree and the hash value of those hash value (computed hash tree root) is included in the message imprint of the timestamp (ER_SO-3_OK_XAIP_OK_SO).
Configuration	CONFIG_S.512
Pre-test conditions	Authenticated connection to middleware exists.

Step	Test sequence	Expected Results
1	<pre> ValidateEvidence (pres:Evidence(@FormatID= "urn:ietf:rfc:4998:EvidenceRecord", @RelatedObjects="XAIP-01", @POID="AOID-999b", @VersionID="VID-01", pres: binaryData(ER_SO-1_OK_XAIP_OK_SO)) pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3", @ID="XAIP-01", pres:xmlData(XAIP_OK_SIG_OK(AOID-100b, VID-01)))) </pre>	<pre> PreservePOResponse (dsb:Result(resultmajor:Success)) </pre>
2	<pre> ValidateEvidence (pres:Evidence(@FormatID= "urn:ietf:rfc:4998:EvidenceRecord", @RelatedObjects="XAIP-01", @POID="AOID-999b", @VersionID="VID-01", pres: binaryData(ER_SO-2_OK_XAIP_OK_SO)) pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3", @ID="XAIP-01", pres:xmlData(XAIP_OK_SIG_OK(AOID-100b, VID-01)))) </pre>	<pre> PreservePOResponse (dsb:Result(resultmajor:Success)) </pre>

Step	Test sequence	Expected Results
3	<pre>ValidateEvidence (pres:Evidence(@FormatID= "urn:ietf:rfc:4998:EvidenceRecord", @RelatedObjects="XAIP-01", @POID="AOID-999b", @VersionID="VID-01", pres: binaryData(ER_SO-3_OK_XAIP_OK_SO)) pres:PO(@FormatID="http://www.bsi.bund.de/tr-esor/xaip/1.3", @ID="XAIP-01", pres:xmlData(XAIP_OK_SIG_OK(AOID-100b, VID-01))))</pre>	<pre>PreservePOResponse (dsb:Result(resultmajor:Success))</pre>

Verdict of M-ADD-04b**Observations:****Verdict:**

4 TR-ESOR-C.2-Testbed how-to

The following chapter will give a short description, how the TR-ESOR-C.2-Testbed (C.2-Testbed) can be set up and used.

4.1 TR-ESOR-C.2-Testbed Prerequisites

In order to use the C.2-Testbed following tools have to be provided:

- Soap-UI - <https://www.soapui.org/downloads/soapui/>

The setup and generation of Soap-UI-configuration can be done either by using directly Microsoft Excel Product (optional requirement), or by editing the provided initial configuration file (cf. chapter 4.2.4).

In order to run the XSVT and ERVT a docker environment is required, e.g.:

- Desktop Docker - <https://www.docker.com/products/docker-desktop/> (please check de license conditions), or
- Rancher Desktop - <https://rancherdesktop.io/>, or
- any other compatible docker installation.

4.2 TR-ESOR-C.2-Testbed introduction

The last version of C.2-Testbed can be obtained from the corresponding GIT repository:

- <https://github.com/de-bund-bsi-tr-esor/TR-ESOR-C.2-Testbed>.

The C.2-Testbed can be cloned by using the GIT functionality:

- `git clone https://github.com/de-bund-bsi-tr-esor/TR-ESOR-C.2-Testbed.git`, or

can be downloaded as a ZIP-archive and unzip locally.

Notice: The used test material is dealing with cryptographic functions such electronic signatures timestamps etc. Therefore, this is important, that the checked out/cloned sources **must not** be altered in any way. This refer also to the CR and CRLF issue. Before clone the repository, please make sure the original alignment of the end lines, especially of xml- and txt-based files will not be altered. You can do that e.g. by switching off this transformation global in your git environment: `git config --global core.autocrlf false`.

The obtained project directory will have the structured depicted in the Figure 4 below.

Name	Änderungsdatum	Typ	Größe
DOCKERS	15.04.2025 09:27	Dateiordner	
ERVT-UPLOADs	15.04.2025 09:27	Dateiordner	
LXAIP-UPLOADs	15.04.2025 09:27	Dateiordner	
PROPERTIES	15.04.2025 09:27	Dateiordner	
RUN-DATA	15.04.2025 09:27	Dateiordner	
TESTDATA	15.04.2025 09:27	Dateiordner	
XSVT-UPLOADs	15.04.2025 09:27	Dateiordner	
.gitignore	15.04.2025 09:27	GITIGNORE-Datei	1 KB
BSI TR-03125_C.2_V1.3.pdf	15.04.2025 09:27	Adobe Acrobat-D...	1.364 KB
LICENSE	15.04.2025 09:27	Datei	12 KB
README.md	15.04.2025 09:27	MD-Datei	12 KB
TR-ESOR-1.3.0-C2-Testbed-soapui-proje...	15.04.2025 09:27	XML-Datei	29.349 KB

Figure 4: TR-ESOR-C.2-Testbed – project directory structure.

Following chapters describe the content of the directories and the important files as well as their role in the C.2-Testbed. For short version please refer to <https://github.com/de-bund-bsi-tr-esor/TR-ESOR-C.2-Testbed/blob/main/README.md>.

4.2.1 Subdirectory DOCKERS

This directory does contain the Docker definitions in order to setup and start the both BSI test tools:

- xsv-srv – BSI AIP signature validator (cf. chapter 3.1.1)
- erv-srv – BSI Evidence Record validator (cf. chapter 3.1.1).

4.2.2 Subdirectory ERVT-UPLOADS

Notice: Only relevant in case the TOT is supporting LXAIP.

This directory does contain all the files, which are necessary to execute the corresponding test cases linked to Evidence Record Verify Tool (ERVT). The references to those files are placed in the corresponding section of a given LXAIP and will be adjusted automatically by the C.2-Testbed during the execution of the test cases. All the files from this directory have to be copied to the dedicated LXAIP-directory of the ERVT.

4.2.3 Subdirectory LXAIP-UPLOADS

Notice: Only relevant in case the TOT is supporting LXAIP.

This directory does contain all files required to perform the LXAIP test cases, which have to be uploaded to the TOT by using the Upload/Download module. The returned references have to be noted in the configuration file in order to instrument the C.2-Testbed for test execution (cf. chapter 4.2.4). The C.2-Testbed will automatically adjust the configured references in the test data during the test execution.

Please refer to step 1 in M-SU-01a or in M-SU-01ab test case in the chapter 1.1.1.1.1 or chapter 3.4.2.2.2 for further information

4.2.4 Subdirectory PROPERTIES

This Subdirectory does contain only two files:

- *TR-C2-Soap-UI-List-of-Project-Properties.xlsm* – which defines all the configuration properties have to be set in advance of the test execution. After the necessary properties has been set, a configuration file can be exported and after that imported into the C.2-Testbed test suite; please refer to the “introduction” tab of the excel-file for further information on that.
- *Initial-Soap-UI-configuration.props* – which does contain the initial version of SoapUI-Project configuration; the file can be directly edited and after that imported into the SoapUI-Project as an alternative solution.

4.2.5 Subdirectory RUN-DATA

This subdirectory does contain the environment to be used in order to run the final tests and collect the required documentation on test execution and results. In the “bin” subdirectory can be find e set of scripts, which can be used to execute either a chosen test cases, or whole tests, to clean up the documented results etc.

The details how to use the environment and the particular scripts please find in the section 4.6.

4.2.6 Subdirectory TESTDATA

This Subdirectory does contain the test data which is directly used by the C.2-Testbed in order to execute the test cases, e.g. XAIP-, LXAIP- ER-objects etc. (cf. chapter 3.3).

4.2.7 Subdirectory XSVT-UPLOADs

Notice: Only relevant in case the TOT is supporting LXAIP.

This directory does contain all the files, which are necessary to execute the corresponding test cases linked to XAIP Signature Validation Tool (XSVT). The references to those files are placed in the corresponding section of a given LXAIP and will be adjusted automatically by the C.2-Testbed during the execution of the test cases. All the files from this directory have to be copied to the dedicated LXAIP-directory of the XSVT.

4.2.8 File TR-ESOR-1.3.0-C2-Testbed-soapui-project.xml

This file does contain all the test cases implemented by C.2-Testbed – SoapUI-Project file.

4.2.9 File BSI TR-03125_C.2_V1.3.pdf

This file does contain the current version of the [TR-ESOR-C.2].

4.3 TR-ESOR-C.2-Testbed configuration

Notice: XAIP shall be supported, LXAIP may be supported (c.f. [TR-ESOR-F], A3-1).

Notice: At least one of the both interfaces TR-S.4 or TR-S.512 shall be supported (c.f. [TR-ESOR-E], A2.0-1).

Notice: The terms target of test (TOT) and target of evaluation/examination (TOE) are used as synonyms.

Following Table 3 gives overview of steps have to be done in order to configure the C.2-Testbed. The column “Container” says, corresponding action should be performed in every case “All”, or only if LXAIP is supported by the TOT. The column “Action” describes the activity to be done.

#	Container	Action
1	All	Do create a test tenant in the TOT with chosen client-id e.g. "Bsi" (cf. <xaip:submissionInfoType> in [TR-ESOR-F], page 17) and set the value of the corresponding property CLIENT_ID_TO_BE_SET.
2	All	If TR-S.4 is supported by TOT, do set the value of S.4_SUPPORTED property to “YES” and do insert the TOT endpoint address of the TR-S.4 interface into the corresponding property TOE_S4_ENDPOINT in the props-file. If TR-S.512 is supported by TOT, do set the value of the S.512_SUPPORTED property to “YES” and do insert the TOT endpoint address of the TR-S.512 interface into the corresponding property TOE_S512_ENDPOINT in the props-file. Please notice the C.2-Testbed does support the http communication as default one. Anything supplementary has to be configured according to the SoapUI documentation.
3	All	Do check and adjust the value of the ERVT_ENDPOINT property in the props-file, as needed (cf. chapter 4.4).
4	All	Do check and adjust the value of the XSVT_ENDPOINT property in the props-file, as needed (cf. chapter 4.4).
5	All	Do configure the behaviour of TOT in case of signature verification errors by enabling one (the other one has to be disabled - set to NO) of the following properties: • set value of OPT_VERIFY_VR_STANDALONE property to YES, if only verification report used to be returned, • set value of OPT_VERIFY_VR_EMB_IN_CONTAINER property to YES, if the verification report embedded into a given XAIP or LXAIP used to be returned.

#	Container	Action
6	All	Do configure the behaviour of the TOT in case an invalid XML structure is used at S.4 by enabling one of the following properties (the other one has to be set to NO): - set value of OPT_INVALID_XML_AS_SOAP_FAULT to YES in case a soap fault used to be returned, - set value of OPT_INVALID_XML_AS_S4_ERROR to YES in case an application level status code used to be returned. If TR-S.4 is supported by TOT, do configure, if the TOT is supporting passing by of the XML-data as native XML (e.g. XAdES): - set value of OPT_SUPPORT_VERIFY_XML_DATA_PARAMETER to YES if passing by XML data as native XML is supported by the TOT while calling Verify function of TR.S.4 → the CONDITIONAL test steps in test cases M-VE-03 and M-VE-06 will be executed in a such case.
7	All	Do check and adjust as needed the value of SUBMISSION_TIME_RANGE_IN_DAYS property (in days), which does describe the maximal deviation of the given <xaip:submissionTime> and the current time.
8	All	Do configure the supported hash algorithms by checking and adjusting (as needed) the properties {SHA1, SHA224, SHA256, SHA384, SHA512, SHA3-224, SHA3-256, SHA3-384, SHA3-512}_SUPPORTED (enabled = YES, disabled - NO).
9	All	Do configure the acceptable hash algorithms for hash tree verification in the evidence records by checking and adjusting (as needed) the properties {SHA1, SHA224, SHA256, SHA384, SHA512, SHA3-224, SHA3-256, SHA3-384, SHA3-512}_TREE_ACCEPTABLE (enabled = YES, disabled = NO).
10	All	Do configure an inactive profile in the TOT and set the value of the URI_PROFILE_V1.3_S.4_V1.0-inactive property as the corresponding profile id.
11	All	Do define the support of LXAIP by proper setting the value of property CONFIG_LXAIP (possible values are ENABLED or DISABLED).
12	All	Do define the not existed AOIDs for the XAIP test cases by setting a value of AOID-03 and AOID-06 (if TR-s.4 is supported) or AOID-03b and AOID-06b (if TR-S.512 is supported) properties.
13	All	Do submit XAIP in order to have the belonging retention period expired at the time of test execution and document the corresponding AOID and VersionID in properties: - if TR-S.4 is supported: in the properties AOID-00 and AOID-00.V01 - if TR-S.512 is supported: in the properties AOID-00b and AOID-00b.V1
14	LXAIP	Do check and adjust (as needed) the location for the referenced data objects for XSVT, property XSV_LXAIP_DIR_URI (a valid directory path).
15	LXAIP	Do check and adjust (as needed) the location for the referenced data objects for ERVT, property ERVT_LXAIP_DIR_URI (a valid directory path).
16	LXAIP	Do copy all the files from directory XSVT-UPLOADs into the location specified in step 14.
17	LXAIP	Do copy all the files from directory ERVT-UPLOADs into the location specified in step 15.
18	LXAIP	Do submit an LXAIP in order to have the belonging retention period expired at the time of test execution and document the corresponding AOID and Version ID in properties: - if TR-S.4 is supported: in the properties AOID-00a and AOID-00a.V01 - if TR-S.512 is supported: in the properties AOID-00ab and AOID-00ab.V1

#	Container	Action
19	LXAIP	Do upload all the files from LXAIP-UPLOADs directory into TOE by using the Upload/Download-Interface and document the given references in the corresponding property in the props-file. Example: Uploading of PDF_DATA_50.pdf file did produce uuid:1122-2343-5634-4545 as reference. Set the test value of REF_PDF_DATA_50 property to uuid:1122-2343-5634-4545.
20	All	Do generate the property file used by the SoapUI-Project by pressing the “Generate SoapUI props” button placed in the “Introduction” tab of the file <i>TR-C2-Soap-UI-List-of-Project-Properties.xlsm</i> (cf. chapter 4.2.4). Do provide the path and name of the SoapUI-Project property file and confirm the export.

Table 3: TR-ESOR-C.2-Testbed list of steps to be done during configuration.

4.4 TR-ESOR-C.2-Testbed: configuration and start of XSVT and ERVT

Following steps have to be performed in order to create the necessary docker images and containers:

S.1 Change into DOCKERS directory

Notice: The following usage of scripts means, the scripts with postfix “.cmd” can be used on the Windows and the script with postfix “.sh” can be used on Linux/Apple.

S.2 Call the script `dbuild.cmd/dbuild.sh` – the necessary docker images will be built:

- `alpine-j11:1.0.0` – foundation for further images,
- `xsv-srv:1.0.0` – the image for XSVT
- `erv-srv:1.0.0` – the image for ERVT

and following or similar output will be displayed:

```
[i] list of known containers:

REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
xsv-srv       1.0.0     827cb319b72b  About an hour ago  335MB
erv-srv       1.0.0     137205968755  6 hours ago    345MB
alpine-j11    1.0.0     ee1705519e0b  22 hours ago    335MB
```

Figure 5: Example output of `dbuild.cmd/dbuild.sh`.

S.3 Call the script `drun.cmd/drun.sh` – the both necessary containers will be created and started:

- `xsv-srv` – container built on base of `xsv-srv:1.0.0` docker image, will host the running XSVT,
- `erv-srv` – container built on base of `erv-srv:1.0.0` docker image, will host the running ERVT.

Please notice, that this task will take some time, because the both tools will be built from the sources obtained from the official BSI repository on the GitHub (cf. <https://github.com/de-bund-bsi-tr-esor>).

Only Windows: In order to monitor the progress, two extra shell windows will start up automatically, which have assigned following titles:

- `xsv-srv log` – progress of XSVT in `xsv-srv` container,
- `erv-srv log` – progress of ERVT in `erv-srv` container.

The creation and initialization of the `xsv-srv` is finished as soon as following statement will be displayed in `xsv-srv log` window:



```

[INFO] -----
[!] starting XSV as xsv-srv:19090
Dec 13, 2022 3:01:48 PM org.apache.cxf.wsdl.service.factory.ReflectionServiceFactoryBean buildServiceFromWSDL
INFO: Creating Service {http://www.bsi.bund.de/tr-esor/api/1.3}S4 from WSDL: /wsdl/tr-esor-S-4-v1.3.wsdl
Dec 13, 2022 3:01:51 PM org.apache.cxf.endpoint.ServerImpl initDestination
INFO: Setting the server's publish address to be http://xsv-srv:19090/xaip-validate
SLF4J: Failed to load class "org.slf4j.impl.StaticLoggerBinder".
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.
[de.bund.bsi.tresor.aip.validator.soap.Server] published server on address: http://xsv-srv:19090/xaip-validate

```

Figure 6: Final statement of initialization of XSVT.

The creation and initialization of the erv-srv is finished as soon as following statement will be displayed in erv-srv log window:



```

BUILD SUCCESSFUL in 3m 25s
69 actionable tasks: 66 executed, 3 up-to-date
[!] starting ERV on port 9090 und with config /opt/config-rfc4998-online.xml, /opt/erverifytool-1.3.3
Running S4 webservice on address http://erv-srv:9090/ErVerifyTool/esor13/exec

```

Figure 7: Final statement of initialization of ERVT.

The both Tools are up and running.

Notice: The both log windows can be closed at any time.

Linux/Apple: A log output of the erv-srv docker container will be displayed in the terminal. As soon as the statement depicted in the Figure 7 will appear, the erv-srv ist up and running. In order to check the status of the xsv-srv docker container please use following command: “docker logs xsv-srv”. If the statement depicted in the Figure 6 does appear, the xsv-srv docker container is up and running.

S.4 In order to access the both test tools by using the hostnames xsv-srv and erv-srv the following files have to be adjusted:

- Windows: c:\Windows\System32\drivers\etc\hosts
- Linux/Apple: /etc/hosts.

Please edit the above-named file as administrator/root and insert following line:

- “127.0.0.1 xsv-srv erv-srv”.

The availability of the both test tools could be tested by inserting following URLs in your browser:

- xsv-srv:
 - <http://xsv-srv:19090/xaip-validate?wsdl> or
 - <http://localhost:19090/xaip-validate?wsdl>,
- erv-srv:
 - <http://erv-srv:9090/ErVerifyTool/esor13/exec?wsdl>, or
 - <http://localhost:9090/ErVerifyTool/esor13/exec?wsdl>.

the output should be the underlying WSDL.

There are some more scripts, which could be used to control the both test tools:

- dstop.cmd/dstop.sh – by using this script the both running containers of test tools XSVT and ERVT can be stopped,
- dstatus.cmd/dstatus.sh – execution of this script will display current status of known docker containers and images, below a sample output of the script,

```
[i] current status of the testbed

current status of images:

REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
xsv-srv        1.0.0     827cb319b72b   About an hour ago   335MB
erv-srv        1.0.0     137205968755   7 hours ago        345MB
alpine-j11     1.0.0     ee1705519e0b   23 hours ago        335MB

[i] current status of container:

CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS              PORTS          NAMES
9f644d340492   erv-srv:1.0.0  "/opt/run-srv.sh"       7 minutes ago   Exited (137) 59 seconds ago           erv-srv
7c591b8fa190   xsv-srv:1.0.0  "/opt/run-srv.sh"       7 minutes ago   Exited (137) About a minute ago           xsv-srv
```

Figure 8: Sample output of `dstatus.cmd/dstatus.sh`

- `dstart.cmd/dstart.sh` – (re)starts the both docker containers.

Notice: This script can only be used after the containers have been successfully created with the script `drun.cmd/drunk.sh`.

Below a sample of output of the script:

```
[i] starting all needed container: xsv-srv and erv-srv

xsv-srv
erv-srv
[i] status of the container:

CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS              PORTS          NAMES
9f644d340492   erv-srv:1.0.0  "/opt/run-srv.sh"       11 minutes ago   Up Less than a second   0.0.0.0:9090->9090/tcp   erv-srv
7c591b8fa190   xsv-srv:1.0.0  "/opt/run-srv.sh"       11 minutes ago   Up 1 second            0.0.0.0:19090->19090/tcp   xsv-srv
```

Figure 9: Sample output of `dstart.cmd/dstart.sh`

- `dclean-container.cmd/dclean-container.sh` – removes the both docker containers (the containers have to stopped in advance). The containers can be recreated by using the script `drun.cmd/drunk.sh`.
- `dclean-all.cmd/dclean-all.sh` – removes the both docker containers (the containers have to be stopped in advance) and the three docker images (only if the containers have been removed successfully). The images can be recreated by using the script `dbuild.cmd/dbuild.sh`.

4.5 TR-ESOR-C.2-Testbed usage

Following Table 4 describes the action steps to be done in order to perform the test cases.

#	Action
1	Do start SoapUI application (cf. chapter 4.1) • Notice: The provided C.2-Testbed project (cf. chapter 4.2.8) has been created with the open source version of the SoapUI tool and hasn't been tested with the commercial version "ReadyAPI" of the software.
2	Do load the SoapUI project into the started SoapUI tool: - Do chose "File->Import Project" or use the shortcut "STRG+I" - Do chose the file TR-ESOR-1.3.0-C2-Testbed-soapui-project.xml (cf. chapter 4.2.8) from the root directory of this project and press the "Open" button. The Project with name "TR-C2-1.3-u2" will appear in the "Navigator" panel of SoapUI. The project consists of 15 test suites with 122 test cases in total.
3	Do set up the necessary security settings for the sommuniaction between C2-Testbed and TOT. Please consult the SoapUI documentation for further advice, how to do that.

#	Action
4	Do load the generated SoapUI project property file: - Do double click on the "TR-C2-1.3-u2" project in the "Navigator" panel. - Do make sure, the "Overview" tab at the top and the "Properties" tab at the bottom of the project configuration dialog are activated. - Do click the next-to-last icon in the icon bar of the "Properties" tab -> the "Load Properties"-Dialog will appear. - Do browse to the generated SoapUI project property file, as result of step 20 in the Table 3 and do activate both checkboxes "Creates Missing Properties" as well as "Deletes properties not in file". - Do press "OK" button
5	Do execute the test suites. Notice: depending of your test strategy you can try to perform all the tests at once, doing it suite by suite, execute them test case by test case or even execute the single steps in the particular test cases manually. It is important to keep the execution, which is top-down, beginning by chapter 3.4.1.1 and ending by chapter 3.4.1.8, to preserve the interdependencies of the particular test cases. Notice: In order to create the final documentation of the achieved test results please refer to the following chapter 4.6. This documentation <u>shall</u> be created and presented to the BSI in parts or as a whole, if required.

Table 4: TR-ESOR-C.2-Testbed list of steps to be performed during the test execution.

4.6 TR-ESOR-C.2-Testbed – final documentation

In order to create a sufficient documentation of the execution and the results of the relevant test cases following steps have to be performed.

Please **notice**, it is recommended to execute those steps as a final task, after it has been made sure, the relevant test cases will deliver positive results. In the preparation phase, it is more comfortable to use the graphical interface of SoapUI tool.

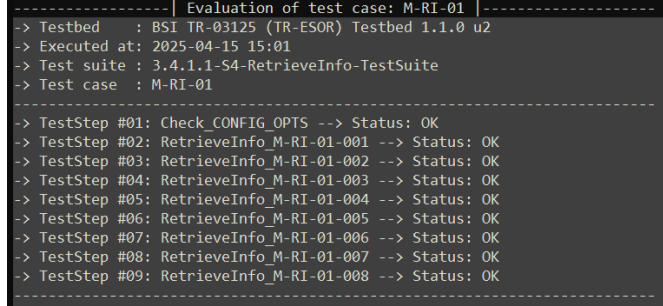
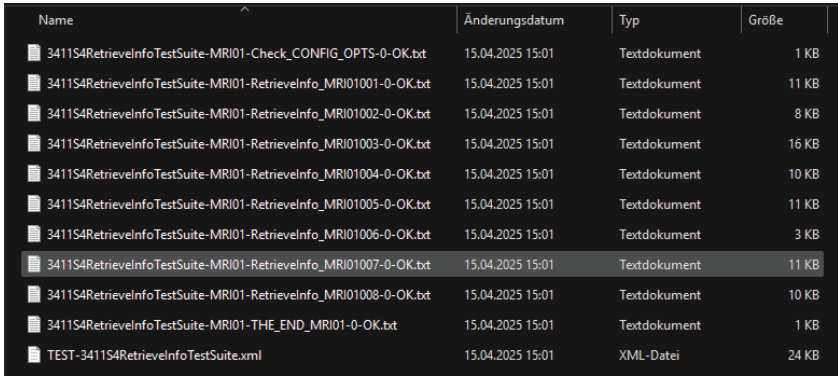
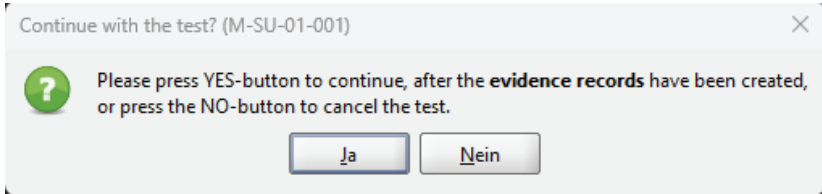
It is assumed, that the steps defined in sections 4.1, 4.3, 4.4 and 4.5 have been positive finished.

#	Action
0	Please take care to have an environment variable <code>JAVA_HOME</code> properly set. It requires at least a version 1.8 of Java. The given test environment has been tested with Java version 17.
1	Do change to directory <code>RUN-DATA/bin</code> and edit the file: <code>env.cmd</code>³⁹ or <code>env</code>⁴⁰. Check and set the following variables: <code>SOAPUI_HOME</code> – shall contain the path pointing to the SoapUI installation folder. After that has been done, call the script <code>clean-all.cmd</code> or <code>clean-all.sh</code>⁴¹. This will take care of all the potentially remained results.

³⁹ The scripts with the postfix „cmd“ are to be used on Windows.

⁴⁰ In case of using Linux or Apple.

⁴¹ The scripts with the postfix „sh“ are to be used on Linux and Apple.

#	Action
2	Do execute the script <code>run-all-tests.cmd</code> or <code>run-all-tests.sh</code>. It will start the execution of one test case after another, according to the configuration stored in the subfolder <code>data</code>⁴². The configuration has been made in the section 4.3 will control the execution flow and skip the not supported testcases. At the beginning, there will be created two folders: a) <code>RUN-TEST/TEST-RUN-EVALUATION</code> – this folder will contain all evaluation reports of the tests execution in form of kind of summary as depicted in the following Figure 10. It allows to gain a rapid overview if a given test case has been executed or skipped and which result has it produced.  Figure 10: Example of evaluation summary for test case M-RI-01. For every existing test suite will be created a separate folder, which contains the summaries of the particular test cases, e.g.: <code>RUN-TEST/TEST-RUN-EVALUATION/3.4.1.1-S4-RetrieveInfo-TestSuite/M-RI-01/M-RI-01-Evaluation.txt</code> b) <code>RUN-TEST/TEST-RUN-REPORTS</code> – this folder contains the execution documentation incl. the corresponding requests and responses. For every test case will be created a separate folder, e.g. for M-RI-01: <code>RUN-DATA\TEST-RUN-REPORTS\REPORTS\3.4.1.1-S4-RetrieveInfo-TestSuite\M-RI-01</code>. The directory does contain a detailed execution report for every step of the particular test case. For M-RI-01 it would be:  Figure 11: Example of a reports folder for test case M-RI-01. During the test execution, there is a user interaction necessary at some steps. E.g. M-SU-01 request the permission to continue after the evidence records creation process in TOE has been successfully initiated and finished (c.f. following Figure 12):  Figure 12: Example of the user interaction dialog while executing the test cases.

⁴² Please do not make any changes on this configuration.

#	Action
3	In case an execution of a one test case has to be repeated, it can be done by executing the script <code>run-chosen-test.cmd</code> or <code>run-chosen-test.sh</code>, e.g. for repeating the test case M-RI-01 please call: • <code>run-chosen-test.cmd -TS 3.4.1.1-S4-RetrieveInfo-TestSuite -TC M-RI-01, or</code> • <code>run-chosen-test.sh -TS 3.4.1.1-S4-RetrieveInfo-TestSuite -TC M-RI-01.</code> Please notice, the potentially already existing results will be overwritten!
4.	Please preserve the whole content of the following artefacts as a documentation: a) SoapUI-configuration used (c.f. section 4.3) b) The whole content of the folder: <code>RUN-TEST/TEST-RUN-EVALUATION</code> c) The whole content of the folder: <code>RUN-TEST/TEST-RUN-REPORTS</code>

4.7 TR-ESOR-C.2-Testbed results

In order to get the [TR-ESOR-C.2] passed, following tests have to be performed successfully:

1. In case the TOT does support **TR-S.4**-interface, all tests specified in the chapters 3.4.1.1 - 3.4.1.8 which id doesn't end with an "a", e.g. M-SU-01 or M-DE-02 etc.
2. In case the TOT does support the additional **S.512**-interface, all tests specified in the chapters 3.4.2.1 - 3.4.2.7, which id doesn't carry an "a" at next to last position e.g. M-SU-01b or M-DE-02b.
3. In case the TOT does support **LXAIP** container and **TR-S.4**-interface, all tests specified in chapters 3.4.1.1 - 3.4.1.8 which id does end with an "a", e.g. M-DE-01a or M-SU-06a.
4. In case the TOT does support LXAIP container and **TR-S.512**-interface, all tests specified in chapters 3.4.2.1 - 3.4.2.7, which id does carry an "a" at next to last position e.g. M-SU-01ab or M-DE-02ab.

The final documentation shall be preserved as a confirmation of the certification test cases have been performed and shall be presented to the BSI, if required.

In order to protected the integrity of the documentation, it is recommended to (at least) timestamp the documentation.